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<table><tr><th>PAGE</th><th>CSA</th><th>CONTENTS</th><th>SYNC</th><th>DATE</th></tr><tr><td>1</td><td>1</td><td>Table of Contents</td><td></td><td></td></tr><tr><td>2</td><td>2</td><td>BOM Configuration</td><td>SHART_J44</td><td>11/27/2012</td></tr><tr><td>3</td><td>3</td><td>BOM Configuration</td><td>J79_JACK</td><td>04/07/2016</td></tr><tr><td>4</td><td>4</td><td>PD Parts</td><td>LDUNN_J44</td><td>01/13/2013</td></tr><tr><td>5</td><td>5</td><td>CPU GFX</td><td>J130_DEV_MLB_U</td><td>04/29/2015</td></tr><tr><td>6</td><td>6</td><td>CPU MISC/JTAG/CFG/RSVD</td><td>J130_DEV_MLB_U</td><td>04/28/2015</td></tr><tr><td>7</td><td>7</td><td>CPU LPDDR3 Interface</td><td>J52_MLB</td><td>05/12/2015</td></tr><tr><td>8</td><td>8</td><td>CPU & PCH Power</td><td>J79_JSHAO</td><td>03/14/2016</td></tr><tr><td>9</td><td>9</td><td>CPU & PCH Grounds</td><td>J79_ALFRED</td><td>05/12/2015</td></tr><tr><td>10</td><td>10</td><td>CPU Core Decoupling</td><td>J79_JSHAO</td><td>08/28/2015</td></tr><tr><td>11</td><td>11</td><td>CPU GT Decoupling</td><td>J79_JSHAO</td><td>08/28/2015</td></tr><tr><td>12</td><td>12</td><td>PCH Decoupling</td><td>J79_JSHAO</td><td>03/14/2016</td></tr><tr><td>13</td><td>13</td><td>PCH Audio/LPC/SPI/SMBus</td><td>J130_MLB</td><td>02/22/2016</td></tr><tr><td>14</td><td>14</td><td>PCH Power Management</td><td>J130_MLB</td><td>05/04/2016</td></tr><tr><td>15</td><td>15</td><td>PCH PCIE/USB/CLKS</td><td>J130_MLB</td><td>06/23/2015</td></tr><tr><td>16</td><td>16</td><td>PCH SPI/UART/GPIO</td><td>J130_MLB</td><td>12/08/2015</td></tr><tr><td>17</td><td>18</td><td>CPU/PCH Merged XDP</td><td>J130_MLB</td><td>12/08/2015</td></tr><tr><td>18</td><td>19</td><td>Chipset Support 1</td><td>J79_GREG</td><td>09/09/2015</td></tr><tr><td>19</td><td>20</td><td>Chipset Support 2</td><td>J79_GREG</td><td>07/05/2016</td></tr><tr><td>20</td><td>22</td><td>LPDDR3 VREF Margining</td><td>J52_MLB</td><td>05/12/2015</td></tr><tr><td>21</td><td>23</td><td>LPDDR3 DRAM Channel A (00-31)</td><td>J52_MLB</td><td>05/12/2015</td></tr><tr><td>22</td><td>24</td><td>LPDDR3 DRAM Channel A (32-63)</td><td>J52_MLB</td><td>05/12/2015</td></tr><tr><td>23</td><td>25</td><td>LPDDR3 DRAM Channel B (00-31)</td><td>J52_MLB</td><td>05/12/2015</td></tr><tr><td>24</td><td>26</td><td>LPDDR3 DRAM Channel B (32-63)</td><td>J52_MLB</td><td>05/12/2015</td></tr><tr><td>25</td><td>27</td><td>LPDDR3 DRAM Termination</td><td>J52_MLB</td><td>05/12/2015</td></tr><tr><td>26</td><td>28</td><td>USB-C HIGH SPEED 1</td><td>J79_GREG</td><td>07/27/2015</td></tr><tr><td>27</td><td>29</td><td>USB-C HIGH SPEED 2</td><td>J79_GREG</td><td>09/09/2015</td></tr><tr><td>28</td><td>30</td><td>USB-C Support</td><td>J79_GREG</td><td>07/05/2016</td></tr><tr><td>29</td><td>31</td><td>USB-C PORT CONTROLLER A</td><td>J79_GREG</td><td>02/28/2016</td></tr><tr><td>30</td><td>32</td><td>USB-C PORT CONTROLLER B</td><td>J79_GREG</td><td>02/28/2016</td></tr><tr><td>31</td><td>33</td><td>USB-C CONNECTOR A</td><td>J79_GREG</td><td>07/05/2016</td></tr><tr><td>32</td><td>34</td><td>USB-C CONNECTOR B</td><td>J79_GREG</td><td>03/24/2016</td></tr><tr><td>33</td><td>35</td><td>TBT 5V REGULATOR</td><td>J79_JSHAO</td><td>12/18/2015</td></tr><tr><td>34</td><td>36</td><td>Display Mux</td><td>J79_GREG</td><td>02/28/2016</td></tr><tr><td>35</td><td>37</td><td>WIFI/BT: MODULE 1</td><td>J79_METE</td><td>05/17/2016</td></tr><tr><td>36</td><td>38</td><td>WIFI/BT: MODULE 2</td><td>J79_METE</td><td>03/02/2016</td></tr><tr><td>37</td><td>39</td><td>Camera/DFR 1</td><td>J80_MLB_BAFFIN</td><td>07/22/2016</td></tr><tr><td>38</td><td>40</td><td>Camera/DFR 2</td><td>J79_ANDREW</td><td>03/22/2016</td></tr><tr><td>39</td><td>41</td><td>Camera/DFR 3</td><td>J79_ANDREW</td><td>04/25/2016</td></tr><tr><td>40</td><td>42</td><td>Berkelium - 1</td><td>J79_ANDREW</td><td>03/14/2016</td></tr><tr><td>41</td><td>43</td><td>Berkelium - 2</td><td>J79_ANDREW</td><td>02/01/2016</td></tr><tr><td>42</td><td>44</td><td>T208 Support</td><td>J79_ANDREW</td><td>07/01/2016</td></tr><tr><td>43</td><td>45</td><td>Connectors&ESD</td><td>J79_GAREN</td><td>11/21/2015</td></tr><tr><td>44</td><td>46</td><td>Empty</td><td>J79_DAYU</td><td>05/26/2015</td></tr><tr><td>45</td><td>47</td><td>Empty</td><td>J79_DAYU</td><td>05/26/2015</td></tr><tr><td>46</td><td>48</td><td>Empty</td><td>J79_DAYU</td><td>05/12/2015</td></tr><tr><td>47</td><td>49</td><td>MESA</td><td>J79_ANDREW</td><td>01/06/2016</td></tr><tr><td>48</td><td>50</td><td>SMC</td><td>J79_JACK</td><td>04/11/2016</td></tr><tr><td>49</td><td>51</td><td>SMC Shared Support</td><td>J79_JACK</td><td>04/14/2016</td></tr><tr><td>50</td><td>52</td><td>SMC Project Support</td><td>J79_JACK</td><td>04/11/2016</td></tr><tr><td>51</td><td>53</td><td>SMBus Connections</td><td>J79_JACK</td><td>03/31/2016</td></tr><tr><td>52</td><td>54</td><td>Power Sensors: High Side</td><td>J79_JACK</td><td>12/07/2015</td></tr><tr><td>53</td><td>55</td><td>Power Sensors: Load Side</td><td>J79_JACK</td><td>04/03/2016</td></tr><tr><td>54</td><td>56</td><td>Power Sensors: Extended</td><td>J79_JACK</td><td>01/08/2016</td></tr><tr><td>55</td><td>57</td><td>Power Sensors: Extended 2</td><td>J79_JACK</td><td>05/10/2016</td></tr><tr><td>56</td><td>58</td><td>Thermal Sensors</td><td>J79_JACK</td><td>09/24/2015</td></tr><tr><td>57</td><td>59</td><td>Power Sensors:Extended 3</td><td>J79_JACK</td><td>04/14/2016</td></tr><tr><td>58</td><td>60</td><td>Fans</td><td>J79_JACK</td><td>08/21/2015</td></tr><tr><td>59</td><td>61</td><td>SPI Debug Connector</td><td>J52_MLB</td><td>05/12/2015</td></tr><tr><td>60</td><td>62</td><td>HDA Bridge</td><td>J79_JCURCIO</td><td>03/24/2016</td></tr></table>												PAGE	CSA	CONTENTS	SYNC	DATE	1	1	Table of Contents			2	2	BOM Configuration	SHART_J44	11/27/2012	3	3	BOM Configuration	J79_JACK	04/07/2016	4	4	PD Parts	LDUNN_J44	01/13/2013	5	5	CPU GFX	J130_DEV_MLB_U	04/29/2015	6	6	CPU MISC/JTAG/CFG/RSVD	J130_DEV_MLB_U	04/28/2015	7	7	CPU LPDDR3 Interface	J52_MLB	05/12/2015	8	8	CPU & PCH Power	J79_JSHAO	03/14/2016	9	9	CPU & PCH Grounds	J79_ALFRED	05/12/2015	10	10	CPU Core Decoupling	J79_JSHAO	08/28/2015	11	11	CPU GT Decoupling	J79_JSHAO	08/28/2015	12	12	PCH Decoupling	J79_JSHAO	03/14/2016	13	13	PCH Audio/LPC/SPI/SMBus	J130_MLB	02/22/2016	14	14	PCH Power Management	J130_MLB	05/04/2016	15	15	PCH PCIE/USB/CLKS	J130_MLB	06/23/2015	16	16	PCH SPI/UART/GPIO	J130_MLB	12/08/2015	17	18	CPU/PCH Merged XDP	J130_MLB	12/08/2015	18	19	Chipset Support 1	J79_GREG	09/09/2015	19	20	Chipset Support 2	J79_GREG	07/05/2016	20	22	LPDDR3 VREF Margining	J52_MLB	05/12/2015	21	23	LPDDR3 DRAM Channel A (00-31)	J52_MLB	05/12/2015	22	24	LPDDR3 DRAM Channel A (32-63)	J52_MLB	05/12/2015	23	25	LPDDR3 DRAM Channel B (00-31)	J52_MLB	05/12/2015	24	26	LPDDR3 DRAM Channel B (32-63)	J52_MLB	05/12/2015	25	27	LPDDR3 DRAM Termination	J52_MLB	05/12/2015	26	28	USB-C HIGH SPEED 1	J79_GREG	07/27/2015	27	29	USB-C HIGH SPEED 2	J79_GREG	09/09/2015	28	30	USB-C Support	J79_GREG	07/05/2016	29	31	USB-C PORT CONTROLLER A	J79_GREG	02/28/2016	30	32	USB-C PORT CONTROLLER B	J79_GREG	02/28/2016	31	33	USB-C CONNECTOR A	J79_GREG	07/05/2016	32	34	USB-C CONNECTOR B	J79_GREG	03/24/2016	33	35	TBT 5V REGULATOR	J79_JSHAO	12/18/2015	34	36	Display Mux	J79_GREG	02/28/2016	35	37	WIFI/BT: MODULE 1	J79_METE	05/17/2016	36	38	WIFI/BT: MODULE 2	J79_METE	03/02/2016	37	39	Camera/DFR 1	J80_MLB_BAFFIN	07/22/2016	38	40	Camera/DFR 2	J79_ANDREW	03/22/2016	39	41	Camera/DFR 3	J79_ANDREW	04/25/2016	40	42	Berkelium - 1	J79_ANDREW	03/14/2016	41	43	Berkelium - 2	J79_ANDREW	02/01/2016	42	44	T208 Support	J79_ANDREW	07/01/2016	43	45	Connectors&ESD	J79_GAREN	11/21/2015	44	46	Empty	J79_DAYU	05/26/2015	45	47	Empty	J79_DAYU	05/26/2015	46	48	Empty	J79_DAYU	05/12/2015	47	49	MESA	J79_ANDREW	01/06/2016	48	50	SMC	J79_JACK	04/11/2016	49	51	SMC Shared Support	J79_JACK	04/14/2016	50	52	SMC Project Support	J79_JACK	04/11/2016	51	53	SMBus Connections	J79_JACK	03/31/2016	52	54	Power Sensors: High Side	J79_JACK	12/07/2015	53	55	Power Sensors: Load Side	J79_JACK	04/03/2016	54	56	Power Sensors: Extended	J79_JACK	01/08/2016	55	57	Power Sensors: Extended 2	J79_JACK	05/10/2016	56	58	Thermal Sensors	J79_JACK	09/24/2015	57	59	Power Sensors:Extended 3	J79_JACK	04/14/2016	58	60	Fans	J79_JACK	08/21/2015	59	61	SPI Debug Connector	J52_MLB	05/12/2015	60	62	HDA Bridge	J79_JCURCIO	03/24/2016	<table><tr><th>PAGE</th><th>CSA</th><th>CONTENTS</th><th>SYNC</th><th>DATE</th></tr><tr><td>61</td><td>63</td><td>AUDIO JACK CODEC</td><td>J79_JCURCIO</td><td>05/13/2016</td></tr><tr><td>62</td><td>64</td><td>Left Speaker Amps & Conn</td><td>J79_JCURCIO</td><td>11/18/2015</td></tr><tr><td>63</td><td>65</td><td>Right Speaker Amps & Conn</td><td>J79_JCURCIO</td><td>12/03/2015</td></tr><tr><td>64</td><td>66</td><td>AUDIO JACK CONNECTOR</td><td>J79_JCURCIO</td><td>12/18/2015</td></tr><tr><td>65</td><td>69</td><td>DC-In & Battery Connectors</td><td>J79_JSHAO</td><td>12/03/2015</td></tr><tr><td>66</td><td>70</td><td>PBUS Supply & Battery Charger</td><td>J79_JSHAO</td><td>12/03/2015</td></tr><tr><td>67</td><td>71</td><td>CORE & SA IMVP IC</td><td>J79_JSHAO</td><td>03/02/2016</td></tr><tr><td>68</td><td>72</td><td>CORE & SA IMVP POWER BLOCK</td><td>J79_JSHAO</td><td>12/03/2015</td></tr><tr><td>69</td><td>73</td><td>Empty</td><td>J79_SILUCHEN</td><td>04/02/2015</td></tr><tr><td>70</td><td>74</td><td>GT & GTX IMVP POWER BLOCK</td><td>J79_JSHAO</td><td>09/25/2015</td></tr><tr><td>71</td><td>75</td><td>Empty</td><td>J79_SILUCHEN</td><td>03/27/2015</td></tr><tr><td>72</td><td>76</td><td>Power - 5V 3.3V Supply</td><td>J79_JSHAO</td><td>03/23/2016</td></tr><tr><td>73</td><td>77</td><td>Power - EOPIO EDRAM Supply</td><td>J79_JSHAO</td><td>04/12/2016</td></tr><tr><td>74</td><td>78</td><td>PMIC-1 & Power Control</td><td>J79_JSHAO</td><td>09/09/2015</td></tr><tr><td>75</td><td>79</td><td>PMIC-1 1.2V 0.6V VCCIO</td><td>J79_JSHAO</td><td>12/03/2015</td></tr><tr><td>76</td><td>80</td><td>PMIC-1 1V 1.8V VCCPCH</td><td>J79_JSHAO</td><td>12/03/2015</td></tr><tr><td>77</td><td>81</td><td>PMIC-1 Aliases & TPs</td><td>J79_SILUCHEN</td><td>07/17/2015</td></tr><tr><td>78</td><td>82</td><td>Power FETs</td><td>J79_JSHAO</td><td>03/14/2016</td></tr><tr><td>79</td><td>84</td><td>LCD Backlight Driver</td><td>J79_RUENJOU</td><td>09/09/2015</td></tr><tr><td>80</td><td>85</td><td>eDP Display Connector</td><td>J79_RUENJOU</td><td>09/12/2015</td></tr><tr><td>81</td><td>86</td><td>S3X CORE PCIE</td><td>J79_RUENJOU</td><td>08/20/2015</td></tr><tr><td>82</td><td>87</td><td>S3X POWER</td><td>J79_RIO</td><td>06/18/2015</td></tr><tr><td>83</td><td>88</td><td>S3X GND</td><td>J79_RIO</td><td>06/18/2015</td></tr><tr><td>84</td><td>89</td><td>Connector</td><td>J79_RUENJOU</td><td>09/09/2015</td></tr><tr><td>85</td><td>90</td><td>NAND VR, I2C ROM, TEMP SENSORS</td><td>J79_RUENJOU</td><td>09/12/2015</td></tr><tr><td>86</td><td>91</td><td>ANI[3:0]</td><td>J79_RUENJOU</td><td>09/25/2015</td></tr><tr><td>87</td><td>92</td><td>ANI[7:4]</td><td>J79_RUENJOU</td><td>09/25/2015</td></tr><tr><td>88</td><td>93</td><td>PICCOLO PMIC</td><td>J79_RUENJOU</td><td>09/24/2015</td></tr><tr><td>89</td><td>94</td><td>SSD NAND VR</td><td>J79_JSHAO</td><td>12/18/2015</td></tr><tr><td>90</td><td>95</td><td>Empty</td><td>J14</td><td>10/23/2012</td></tr><tr><td>91</td><td>96</td><td>LIFEBOAT</td><td>J79_RUENJOU</td><td>09/09/2015</td></tr><tr><td>92</td><td>110</td><td>USB-C HIGH SPEED 1</td><td>J79_GREG</td><td>07/28/2015</td></tr><tr><td>93</td><td>111</td><td>USB-C HIGH SPEED 2</td><td>J79_GREG</td><td>08/28/2015</td></tr><tr><td>94</td><td>112</td><td>USB-C Support</td><td>J79_GREG</td><td>07/05/2016</td></tr><tr><td>95</td><td>113</td><td>USB-C PORT CONTROLLER A</td><td>J79_GREG</td><td>02/28/2016</td></tr><tr><td>96</td><td>114</td><td>USB-C PORT CONTROLLER B</td><td>J79_GREG</td><td>02/28/2016</td></tr><tr><td>97</td><td>115</td><td>USB-C CONNECTOR A</td><td>J79_GREG</td><td>07/05/2016</td></tr><tr><td>98</td><td>116</td><td>USB-C CONNECTOR B</td><td>J79_GREG</td><td>03/24/2016</td></tr><tr><td>99</td><td>117</td><td>TBT 5V REGULATOR</td><td>J79_JSHAO</td><td>12/18/2015</td></tr><tr><td>100</td><td>120</td><td>Power Aliases - 1</td><td>J79_ALFRED</td><td>06/17/2015</td></tr><tr><td>101</td><td>121</td><td>Power Aliases - 2</td><td>J79_ALFRED</td><td>06/18/2015</td></tr><tr><td>102</td><td>122</td><td>Signal Aliases</td><td>SHART_J44</td><td>11/19/2012</td></tr><tr><td>103</td><td>123</td><td>LPDDR3 Bit & Byte Swizzle</td><td>AHARTMAN_J52</td><td>10/29/2013</td></tr><tr><td>104</td><td>124</td><td>ICT FCT 1</td><td>YHARTANTO_J44</td><td>12/18/2012</td></tr><tr><td>105</td><td>125</td><td>ICT FCT 2</td><td>YHARTANTO_J44</td><td>12/18/2012</td></tr><tr><td>106</td><td>127</td><td>Desense Capacitors</td><td>YHARTANTO_J44</td><td>01/09/2013</td></tr><tr><td>107</td><td>129</td><td>Empty</td><td>J79_RIO</td><td>06/18/2015</td></tr><tr><td>108</td><td>130</td><td>PCB Rule Definitions</td><td>YHARTANTO_J44</td><td>12/14/2012</td></tr><tr><td>109</td><td>131</td><td>CPU Constraints</td><td>YHARTANTO_J44</td><td>01/13/2013</td></tr><tr><td>110</td><td>132</td><td>PCH Constraints</td><td>YHARTANTO_J44</td><td>01/08/2013</td></tr><tr><td>111</td><td>133</td><td>Memory Constraints</td><td>YHARTANTO_J44</td><td>01/02/2013</td></tr><tr><td>112</td><td>134</td><td>TBT DP HDMI Constraints</td><td>J79_JACK</td><td>05/19/2015</td></tr><tr><td>113</td><td>135</td><td>PCiE Constraints</td><td>J79_JACK</td><td>05/19/2015</td></tr><tr><td>114</td><td>136</td><td>USB Constraints</td><td>J79_JACK</td><td>05/21/2015</td></tr><tr><td>115</td><td>137</td><td>SMC Constraints</td><td>YHARTANTO_J44</td><td>01/02/2013</td></tr><tr><td>116</td><td>138</td><td>Camera Constraints</td><td>YHARTANTO_J44</td><td>01/09/2013</td></tr><tr><td>117</td><td>139</td><td>Sensors & Audio Constraints</td><td>YHARTANTO_J44</td><td>01/04/2013</td></tr><tr><td>118</td><td>140</td><td>References</td><td>J79_RUENJOU_CONSTRAINTS</td><td>01/02/2015</td></tr><tr><td>119</td><td>145</td><td>Alternates BOM Table</td><td>J80_MLB</td><td>12/12/2015</td></tr></table>				PAGE	CSA	CONTENTS	SYNC	DATE	61	63	AUDIO JACK CODEC	J79_JCURCIO	05/13/2016	62	64	Left Speaker Amps & Conn	J79_JCURCIO	11/18/2015	63	65	Right Speaker Amps & Conn	J79_JCURCIO	12/03/2015	64	66	AUDIO JACK CONNECTOR	J79_JCURCIO	12/18/2015	65	69	DC-In & Battery Connectors	J79_JSHAO	12/03/2015	66	70	PBUS Supply & Battery Charger	J79_JSHAO	12/03/2015	67	71	CORE & SA IMVP IC	J79_JSHAO	03/02/2016	68	72	CORE & SA IMVP POWER BLOCK	J79_JSHAO	12/03/2015	69	73	Empty	J79_SILUCHEN	04/02/2015	70	74	GT & GTX IMVP POWER BLOCK	J79_JSHAO	09/25/2015	71	75	Empty	J79_SILUCHEN	03/27/2015	72	76	Power - 5V 3.3V Supply	J79_JSHAO	03/23/2016	73	77	Power - EOPIO EDRAM Supply	J79_JSHAO	04/12/2016	74	78	PMIC-1 & Power Control	J79_JSHAO	09/09/2015	75	79	PMIC-1 1.2V 0.6V VCCIO	J79_JSHAO	12/03/2015	76	80	PMIC-1 1V 1.8V VCCPCH	J79_JSHAO	12/03/2015	77	81	PMIC-1 Aliases & TPs	J79_SILUCHEN	07/17/2015	78	82	Power FETs	J79_JSHAO	03/14/2016	79	84	LCD Backlight Driver	J79_RUENJOU	09/09/2015	80	85	eDP Display Connector	J79_RUENJOU	09/12/2015	81	86	S3X CORE PCIE	J79_RUENJOU	08/20/2015	82	87	S3X POWER	J79_RIO	06/18/2015	83	88	S3X GND	J79_RIO	06/18/2015	84	89	Connector	J79_RUENJOU	09/09/2015	85	90	NAND VR, I2C ROM, TEMP SENSORS	J79_RUENJOU	09/12/2015	86	91	ANI[3:0]	J79_RUENJOU	09/25/2015	87	92	ANI[7:4]	J79_RUENJOU	09/25/2015	88	93	PICCOLO PMIC	J79_RUENJOU	09/24/2015	89	94	SSD NAND VR	J79_JSHAO	12/18/2015	90	95	Empty	J14	10/23/2012	91	96	LIFEBOAT	J79_RUENJOU	09/09/2015	92	110	USB-C HIGH SPEED 1	J79_GREG	07/28/2015	93	111	USB-C HIGH SPEED 2	J79_GREG	08/28/2015	94	112	USB-C Support	J79_GREG	07/05/2016	95	113	USB-C PORT CONTROLLER A	J79_GREG	02/28/2016	96	114	USB-C PORT CONTROLLER B	J79_GREG	02/28/2016	97	115	USB-C CONNECTOR A	J79_GREG	07/05/2016	98	116	USB-C CONNECTOR B	J79_GREG	03/24/2016	99	117	TBT 5V REGULATOR	J79_JSHAO	12/18/2015	100	120	Power Aliases - 1	J79_ALFRED	06/17/2015	101	121	Power Aliases - 2	J79_ALFRED	06/18/2015	102	122	Signal Aliases	SHART_J44	11/19/2012	103	123	LPDDR3 Bit & Byte Swizzle	AHARTMAN_J52	10/29/2013	104	124	ICT FCT 1	YHARTANTO_J44	12/18/2012	105	125	ICT FCT 2	YHARTANTO_J44	12/18/2012	106	127	Desense Capacitors	YHARTANTO_J44	01/09/2013	107	129	Empty	J79_RIO	06/18/2015	108	130	PCB Rule Definitions	YHARTANTO_J44	12/14/2012	109	131	CPU Constraints	YHARTANTO_J44	01/13/2013	110	132	PCH Constraints	YHARTANTO_J44	01/08/2013	111	133	Memory Constraints	YHARTANTO_J44	01/02/2013	112	134	TBT DP HDMI Constraints	J79_JACK	05/19/2015	113	135	PCiE Constraints	J79_JACK	05/19/2015	114	136	USB Constraints	J79_JACK	05/21/2015	115	137	SMC Constraints	YHARTANTO_J44	01/02/2013	116	138	Camera Constraints	YHARTANTO_J44	01/09/2013	117	139	Sensors & Audio Constraints	YHARTANTO_J44	01/04/2013	118	140	References	J79_RUENJOU_CONSTRAINTS	01/02/2015	119	145	Alternates BOM Table	J80_MLB	12/12/2015
PAGE	CSA	CONTENTS	SYNC	DATE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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5	5	CPU GFX	J130_DEV_MLB_U	04/29/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
6	6	CPU MISC/JTAG/CFG/RSVD	J130_DEV_MLB_U	04/28/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
7	7	CPU LPDDR3 Interface	J52_MLB	05/12/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
8	8	CPU & PCH Power	J79_JSHAO	03/14/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
9	9	CPU & PCH Grounds	J79_ALFRED	05/12/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
10	10	CPU Core Decoupling	J79_JSHAO	08/28/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
11	11	CPU GT Decoupling	J79_JSHAO	08/28/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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13	13	PCH Audio/LPC/SPI/SMBus	J130_MLB	02/22/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
14	14	PCH Power Management	J130_MLB	05/04/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
15	15	PCH PCIE/USB/CLKS	J130_MLB	06/23/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
16	16	PCH SPI/UART/GPIO	J130_MLB	12/08/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
17	18	CPU/PCH Merged XDP	J130_MLB	12/08/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
18	19	Chipset Support 1	J79_GREG	09/09/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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20	22	LPDDR3 VREF Margining	J52_MLB	05/12/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
21	23	LPDDR3 DRAM Channel A (00-31)	J52_MLB	05/12/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
22	24	LPDDR3 DRAM Channel A (32-63)	J52_MLB	05/12/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
23	25	LPDDR3 DRAM Channel B (00-31)	J52_MLB	05/12/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
24	26	LPDDR3 DRAM Channel B (32-63)	J52_MLB	05/12/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
25	27	LPDDR3 DRAM Termination	J52_MLB	05/12/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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35	37	WIFI/BT: MODULE 1	J79_METE	05/17/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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37	39	Camera/DFR 1	J80_MLB_BAFFIN	07/22/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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52	54	Power Sensors: High Side	J79_JACK	12/07/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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54	56	Power Sensors: Extended	J79_JACK	01/08/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
55	57	Power Sensors: Extended 2	J79_JACK	05/10/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
56	58	Thermal Sensors	J79_JACK	09/24/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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58	60	Fans	J79_JACK	08/21/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
59	61	SPI Debug Connector	J52_MLB	05/12/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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66	70	PBUS Supply & Battery Charger	J79_JSHAO	12/03/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
67	71	CORE & SA IMVP IC	J79_JSHAO	03/02/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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70	74	GT & GTX IMVP POWER BLOCK	J79_JSHAO	09/25/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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85	90	NAND VR, I2C ROM, TEMP SENSORS	J79_RUENJOU	09/12/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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118	140	References	J79_RUENJOU_CONSTRAINTS	01/02/2015																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
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Schematic / PCB #'s <table><tr><th>PART NUMBER</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DES</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>051-00515</td><td>1</td><td>SCHEM,MLB,X362</td><td>SCH</td><td>CRITICAL</td><td></td></tr><tr><td>820-00239</td><td>1</td><td>PCBF,MLB,X362</td><td>PCB</td><td>CRITICAL</td><td></td></tr></table>												PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION	051-00515	1	SCHEM,MLB,X362	SCH	CRITICAL		820-00239	1	PCBF,MLB,X362	PCB	CRITICAL		? <table><tr><td colspan="3">DRAWING TITLE</td></tr><tr><td colspan="3">SCHEM,MLB,X362</td></tr><tr><td rowspan="2"> Apple Inc.</td><td>DRAWING NUMBER</td><td>051-00515</td></tr><tr><td>REVISION</td><td>7.0.0</td></tr><tr><td rowspan="3">NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED</td><td>BRANCH</td><td>dvt1</td></tr><tr><td>PAGE</td><td>1 OF 145</td></tr><tr><td>SHEET</td><td>1 OF 119</td></tr></table>				DRAWING TITLE			SCHEM,MLB,X362			 Apple Inc.	DRAWING NUMBER	051-00515	REVISION	7.0.0	NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	dvt1	PAGE	1 OF 145	SHEET	1 OF 119																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
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8		7		6		5		4		3		2		1	
D	BOM Groups														
	BOM GROUP		BOM OPTIONS												
	X362_COMMON		ALTERNATE,COMMON,X362_COMMON1,X362_COMMON2,X362_COMMON3,X362_PROGPARTS												
	X362_COMMON1		SOC:SAMSUNG,SE:PROD,BOARD_ID:6,T208_PROG:REV5,CPUHYS,AP_TEMP,TBTTHRM_SNS,S3XCLK:INT												
	X362_COMMON2		EDP_ENABLE,XDP:YES,PCH_CLK:GRMCLK,TBT_DBG,SANCONN,SKIP_S3V3:AUDIBLE,SOC_BOOT:SPI												
	X362_COMMON3		CPUTHRM:ALRT,TBTTHRM:ALRT,LOADRC:YES,CUMULUS_IPD,S3_STATE:YES,WIFI_SAK:NO,VCCPLLLOC:S3												
	X362_PROGPARTS		BOOTROM_PROG,SMC_PROG,AR_LT_PROG,AR_RT_PROG,WIFI_PROG,BTROM_PROG												
	X362_DEVEL:ENG		ALTERNATE,DBGLED,USBC_DBG,XDP_CONN:YES,WIFI_DBG,S3X_DBG,DEBUG_BUTTON,LOADISNS												
	X362_DEVEL:DVT		ALTERNATE												
	X362_DEVEL:PVT		ALTERNATE												
C	Module Parts														
	PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION									
	337S00266	1	CPU,SKLU,SR2JK,PRQ,2.9,28W,1.05,B1356	U0500	CRITICAL	CPU_SKL23:2.9G									
	337S00267	1	CPU,SKLU,SR2JJ,PRQ,3.1,28W,1.1,B1356	U0500	CRITICAL	CPU_SKL23:3.1G									
	337S00268	1	CPU,SKLU,SR2JH,PRQ,3.3,28W,1.1,B1356	U0500	CRITICAL	CPU_SKL23:3.3G									
	998-04195	1	INTERPOSER,VTT ADAPTER,SKL-U,BGA1356	U0500	CRITICAL	CPU_SOCKET									
	338S00254	2	IC,TBT,ALPINE RIDGE,QSTY,PRQ,C1,CSP337	U2800,UB000	CRITICAL										
	353S00961	4	IC,CD3215,ACE,C00,USB PWR SW,BLANK,BGA96	U3100,U3200,UB300,UB400	CRITICAL										
	998-06736	1	IC,S3X CNTLR,B1,DEV,CSP900	U8600	CRITICAL										
	333S00055	1	IC,LPODR3-1600,4GBIT,25NM,A,276B	POP8600	CRITICAL	POP_4GBIT									
B	333S00056	1	IC,LPODR3-1600,8GBIT,25NM,A,276B	POP8600	CRITICAL	POP_8GBIT									
	343S00147	1	IC,SL64AP41172,PAK3,STQFN20	U3620	CRITICAL										
	338S00221	1	IC,PMU,SN650839,7X7MM,BGA168	U7800	CRITICAL										
	353S01016	1	IC,ISL9239HIZ,PMU,TUBA,WCSF40,2.1X3.3MM	U7000	CRITICAL										
	338S00227	1	IC,PMU,PICCOLO,D2231A0,OTF-AK,WLCSP96	U9300	CRITICAL										
	Programmables (All Builds)														
	EFI ROM														
	341S00698	1	IC,EFI ROM (V0193) DVT,X362	U6100	CRITICAL	BOOTROM_PROG									
	SMC ROM														
	341S00697	1	IC,SMC-B1,EXT (VXXXX) DVT,X362	U5000	CRITICAL	SMC_PROG									
TBT ROMs															
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION										
341S00692	1	T29,AR1 (VXXX) DVT,X362	U2890	CRITICAL	AR_LT_PROG										
341S00693	1	T29,AR2 (VXXX) DVT,X362	UB090	CRITICAL	AR_RT_PROG										
WIFI/BT ROM															
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION										
341S00695	1	IC,BT ROM (VXXXXXX) DVT,X362/X363	U3750	CRITICAL	BTROM_PROG										
341S00696	1	WIFI ROM (PXXX) DVT,WW1,X362/X363	U3710	CRITICAL	WIFI_PROG										
Variable BOM Groups Development/Base BOMs															
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION										
685-00055	1	COMMON PARTS,MLB,X362	BASE	CRITICAL	BASE_BOM										
985-00070	1	DEV,MLB,X362	DEVEL	CRITICAL	DEVEL_BOM										
Main DRAM Parts															
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION										
333S00069	4	1C,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	8G_MICRON_2133										
333S00070	4	1C,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_MICRON_2133										
333S00068	4	1C,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	8G_SAMSUNG_2133										
333S00050	4	1C,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_SAMSUNG_2133										
Main DRAM SPD Straps															
BOM GROUP		BOM OPTIONS													
RAM_8G_MICRON_2133		8G_MICRON_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG2_L,RAMCFG1_L													
RAM_16G_MICRON_2133		16G_MICRON_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG1_L													
RAM_8G_SAMSUNG_2133		8G_SAMSUNG_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG2_L,RAMCFG0_L													
RAM_16G_SAMSUNG_2133		16G_SAMSUNG_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG0_L													
NAND Parts															
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION										
335S00124	4	NAND,12NM,64GB,TOGO,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_256GB										
335S00125	4	NAND,12NM,128GB,TOGO,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_512GB										
335S00126	4	NAND,12NM,256GB,TOGO,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_1TB										
335S00135	4	NAND,TGDDR2,128GX4,15NM,HP,USHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_256GB										
335S00136	4	NAND,TGDDR2,128GX8,15NM,HP,USHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_512GB										
335S00137	4	NAND,TGDDR2,128GX16,15NM,HP,UHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_1TB										
NAND Straps															
BOM GROUP		BOM OPTIONS													
SAND_256G		ALTERNATE,NAND_SDISK_256GB,POP_4GBIT,CAPACITY1,CAPACITY0													
SAND_512G		ALTERNATE,NAND_SDISK_512GB,POP_4GBIT,CAPACITY2													
SAND_1T		ALTERNATE,NAND_SDISK_1TB,POP_8GBIT,CAPACITY2,CAPACITY0													
TOSH_256G		ALTERNATE,NAND_TSHBA_256GB,POP_4GBIT,CAPACITY1,CAPACITY0													
TOSH_512G		ALTERNATE,NAND_TSHBA_512GB,POP_4GBIT,CAPACITY2													
TOSH_1T		ALTERNATE,NAND_TSHBA_1TB,POP_8GBIT,CAPACITY2,CAPACITY0													
Strategic Silicon															
PART#	STRATEGIC VALUE	COMMENT	PART#	STRATEGIC VALUE	COMMENT										
337S00266	08	CPU	338S00227	02	PICCOLO										
337S00267	08	CPU	343S00095	10	T208										
337S00268	08	CPU	343S00096	10	T208										
333S00050	07	MEMORY	343S00097	10	T208										
333S00068	07	MEMORY	338S00193	09	BERKELIUM										
333S00069	07	MEMORY	353S0978	02	MOJAVE										
333S00070	07	MEMORY	338S00147	02	SECURE ELEMENT										
335S00124	02	NAND	338S00254	08	ALPINE RIDGE										
335S00125	02	NAND	353S00961	09	ACE										
335S00126	02	NAND	338S00142	09	CLIPDEN										
335S00135	02	NAND	353S00685	07	AUDIO AMP										
335S00136	02	NAND	353S4316	08	BAYSIDE										
335S00137	02	NAND	338S00221	08	BANJO										
333S00025	02	S3X DRAM	353S01016	09	TUBA										
333S00026	02	S3X DRAM	339S00056	05	ICEBOCK										
333S00055	02	S3X DRAM	343S00147	08	PAK										
333S00056	02	S3X DRAM	359S00006	08	GREEN CLOCK										
333S00107	02	S3X DRAM	353S00795	09	DEBUG MUX										
333S00108	02	S3X DRAM													
998-06736	02	S3X CONTROLLER													
CPU DRAM CFG Chart															
DIE REV		CFG 4													
A		0													
B		1													
SPEED		CFG 3													
2133		0													
1866		1													
CAPACITY		CFG 2													
8GB		0													
16GB		1													
VENDOR		CFG 1		CFG 0											
HYNIX		0		0											
MICRON		0		1											
SAMSUNG		1		0											
N/A		1		1											
BOM Configuration															
DRAWING NUMBER		051-00515		SIZE		D									
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PAGE		2 OF 145													
SHEET		2 OF 119													
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Variable BOM Groups Development/Base BOMs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-00055	1	COMMON PARTS,MLB,X362	BASE	CRITICAL	BASE_BOM
985-00070	1	DEV,MLB,X362	DEVEL	CRITICAL	DEVEL_BOM

Main DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S00069	4	IC,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	8G_MICRON_2133
333S00070	4	IC,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_MICRON_2133
333S00068	4	IC,SDRAM,LPODR3-2133,16GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	8G_SAMSUNG_2133
333S00050	4	IC,SDRAM,LPODR3-2133,32GBIT,20NM,BGA178	U2300,U2400,U2500,U2600	CRITICAL	16G_SAMSUNG_2133

Main DRAM SPD Straps

BOM GROUP	BOM OPTIONS
RAM_8G_MICRON_2133	8G_MICRON_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG2_L,RAMCFG1_L
RAM_16G_MICRON_2133	16G_MICRON_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG1_L
RAM_8G_SAMSUNG_2133	8G_SAMSUNG_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG2_L,RAMCFG0_L
RAM_16G_SAMSUNG_2133	16G_SAMSUNG_2133,RAMCFG4_L,RAMCFG3_L,RAMCFG0_L

NAND Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
335S00124	4	NAND,12NM,64GB,T0GG,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_256GB
335S00125	4	NAND,12NM,128GB,T0GG,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_512GB
335S00126	4	NAND,12NM,256GB,T0GG,HPN,128G,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_SDISK_1TB
335S00135	4	NAND,TGDDR2,128GX4,15NM,HP,USHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_256GB
335S00136	4	NAND,TGDDR2,128GX8,15NM,HP,USHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_512GB
335S00137	4	NAND,TGDDR2,128GX16,15NM,HP,UHD,T2,LGA60	U9100,U9120,U9200,U9220	CRITICAL	NAND_TSHBA_1TB

NAND Straps

BOM GROUP	BOM OPTIONS
SAND_256G	ALTERNATE,NAND_SDISK_256GB,POP_4GBIT,CAPACITY1,CAPACITY0
SAND_512G	ALTERNATE,NAND_SDISK_512GB,POP_4GBIT,CAPACITY2
SAND_1T	ALTERNATE,NAND_SDISK_1TB,POP_8GBIT,CAPACITY2,CAPACITY0
TOSH_256G	ALTERNATE,NAND_TSHBA_256GB,POP_4GBIT,CAPACITY1,CAPACITY0
TOSH_512G	ALTERNATE,NAND_TSHBA_512GB,POP_4GBIT,CAPACITY2
TOSH_1T	ALTERNATE,NAND_TSHBA_1TB,POP_8GBIT,CAPACITY2,CAPACITY0

Strategic Silicon

PART#	STRATEGIC VALUE	COMMENT
337S00266	08	CPU
337S00267	08	CPU
337S00268	08	CPU
333S00050	07	MEMORY
333S00068	07	MEMORY
333S00069	07	MEMORY
333S00070	07	MEMORY
335S00124	02	NAND
335S00125	02	NAND
335S00126	02	NAND
335S00135	02	NAND
335S00136	02	NAND
335S00137	02	NAND
333S00025	02	S3X DRAM
333S00026	02	S3X DRAM
333S00055	02	S3X DRAM
333S00056	02	S3X DRAM
333S00107	02	S3X DRAM
333S00108	02	S3X DRAM
998-06736	02	S3X CONTROLLER

PART#	STRATEGIC VALUE	COMMENT
338S00227	02	PICCOLO
343S00095	10	T208
343S00096	10	T208
343S00097	10	T208
338S00193	09	BERKELIUM
353S3978	02	MOJAVE
338S00147	02	SECURE ELEMENT
338S00254	08	ALPINE RIDGE
353S00961	09	ACE
338S00142	09	CLIFDEN
353S00685	07	AUDIO AMP
353S4316	08	BAYSIDE
338S00221	08	BANJO
353S01016	09	TUBA
339S00056	05	ICEBOCK
343S00147	08	PAK
359S00006	08	GREEN CLOCK
353S00795	09	DEBUG MUX

CPU DRAM CFG Chart

DIE REV	CFG 4
A	0
B	1

SPEED	CFG 3
2133	0
1866	1

CAPACITY	CFG 2
8GB	0
16GB	1

VENDOR	CFG 1	CFG 0
HYNIX	0	0
MICRON	0	1
SAMSUNG	1	0
N/A	1	1

BOM Configuration

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051-00515

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PAGE
2 OF 145

SHEET
2 OF 119

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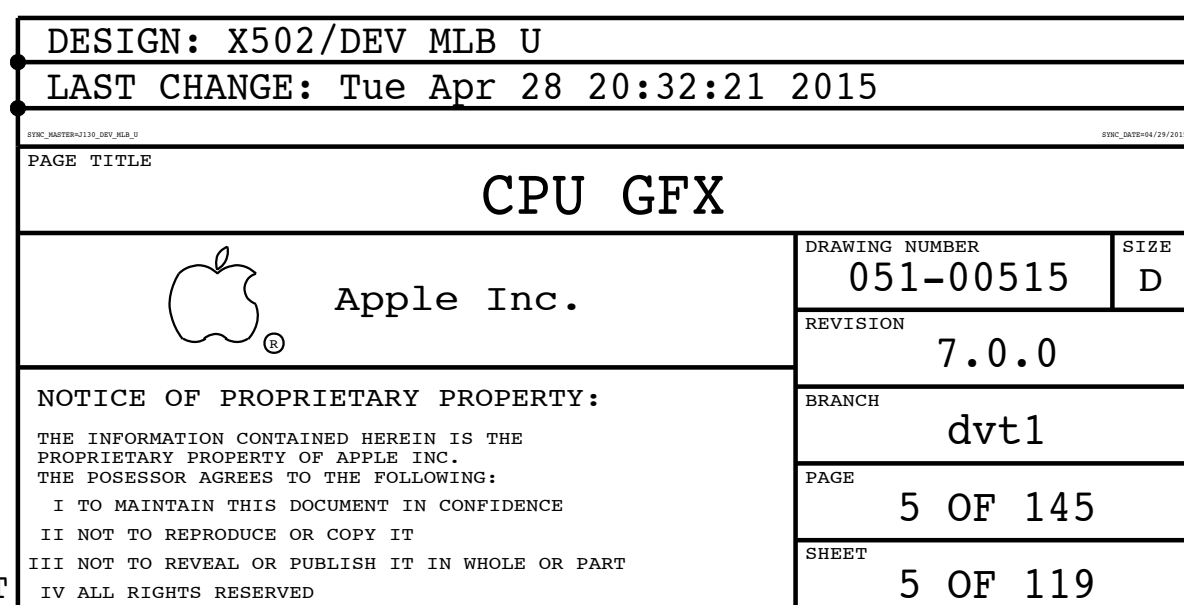
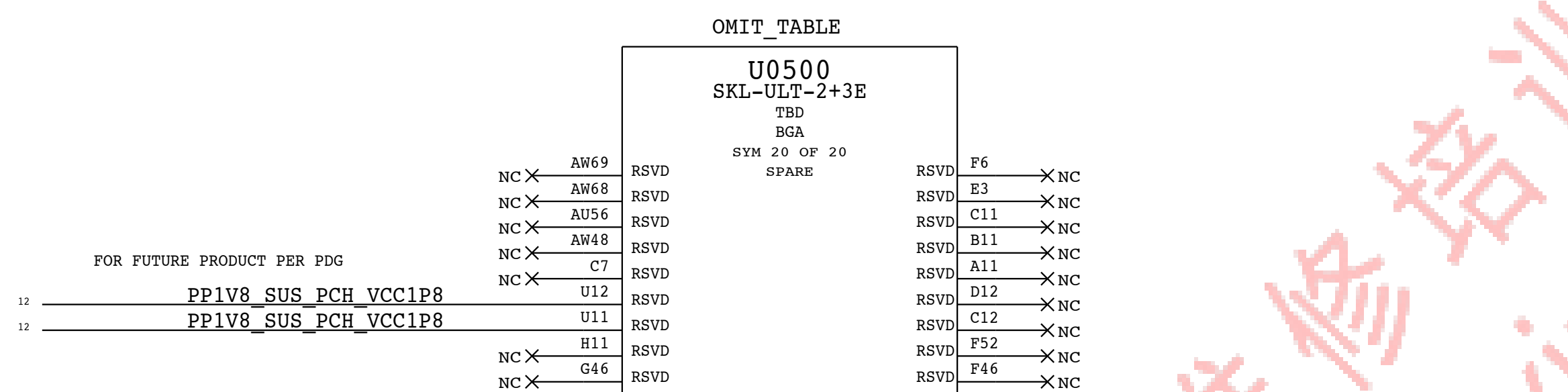
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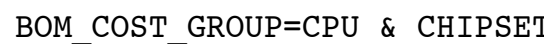
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		PAGE	
		6 OF 145	
		SHEET	
		6 OF 119	



D

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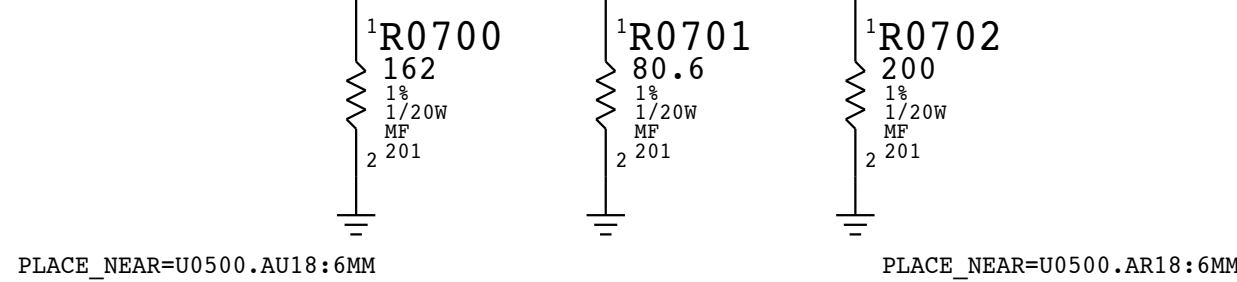
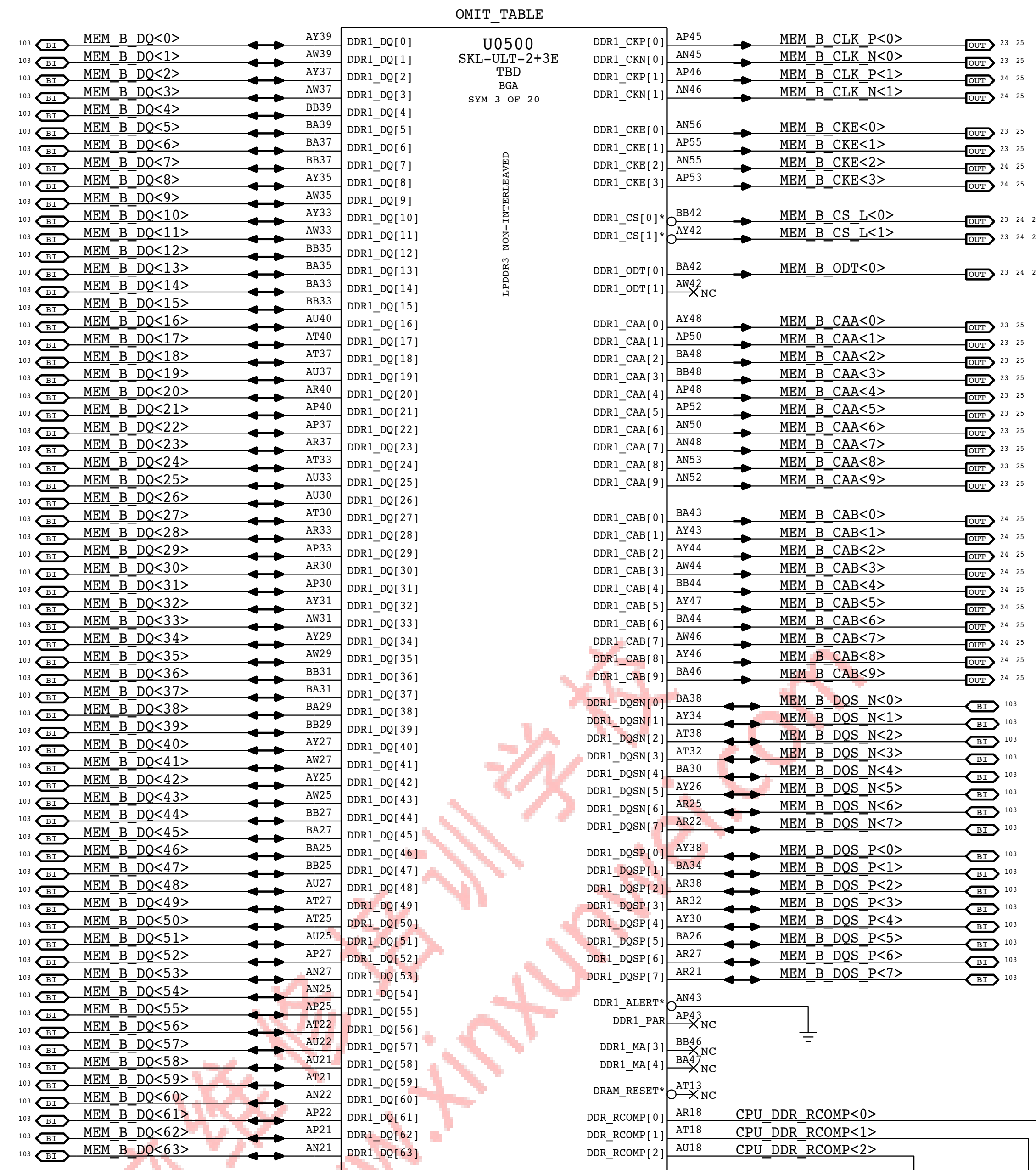
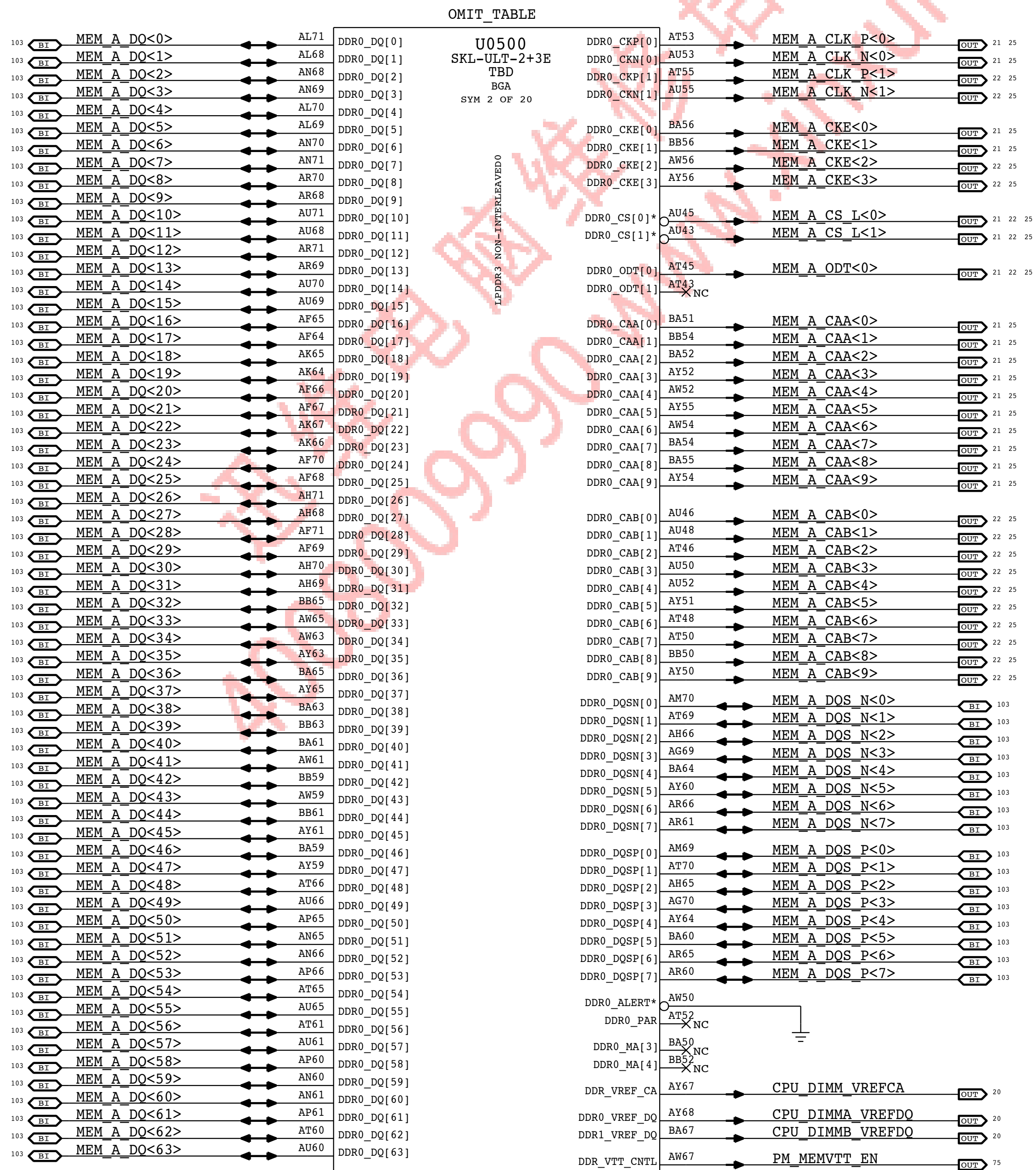
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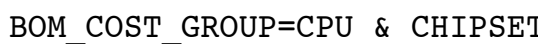
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CPU LPDDR3 Interface			
 Apple Inc.	DRAWING NUMBER 051-00515		STEP D
	REVISION 7.0.0		
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		PAGE 7 OF 145	
		SHEET 7 OF 119	

BOM_COST_GROUP=CPU & CHIPSET



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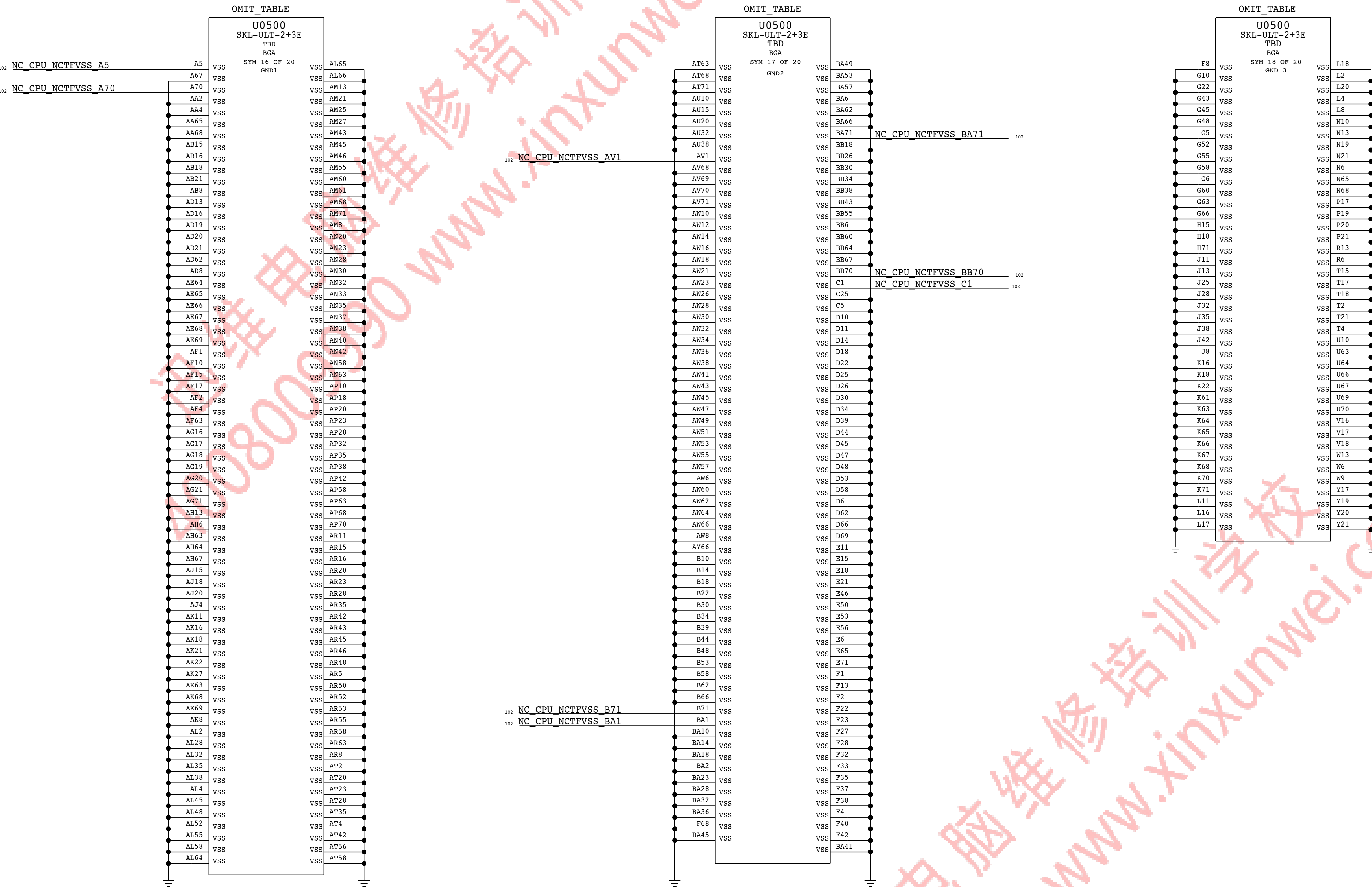
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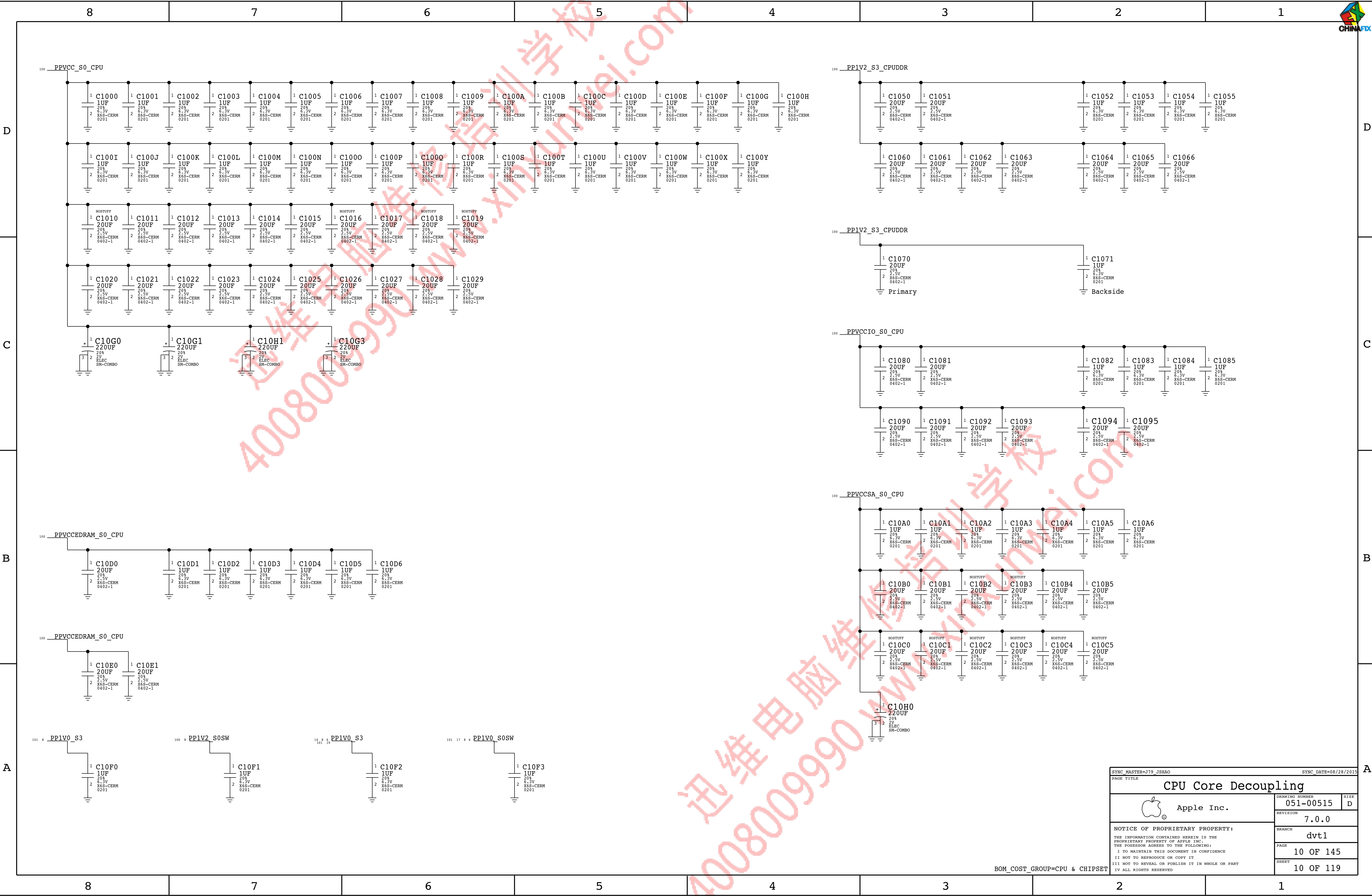
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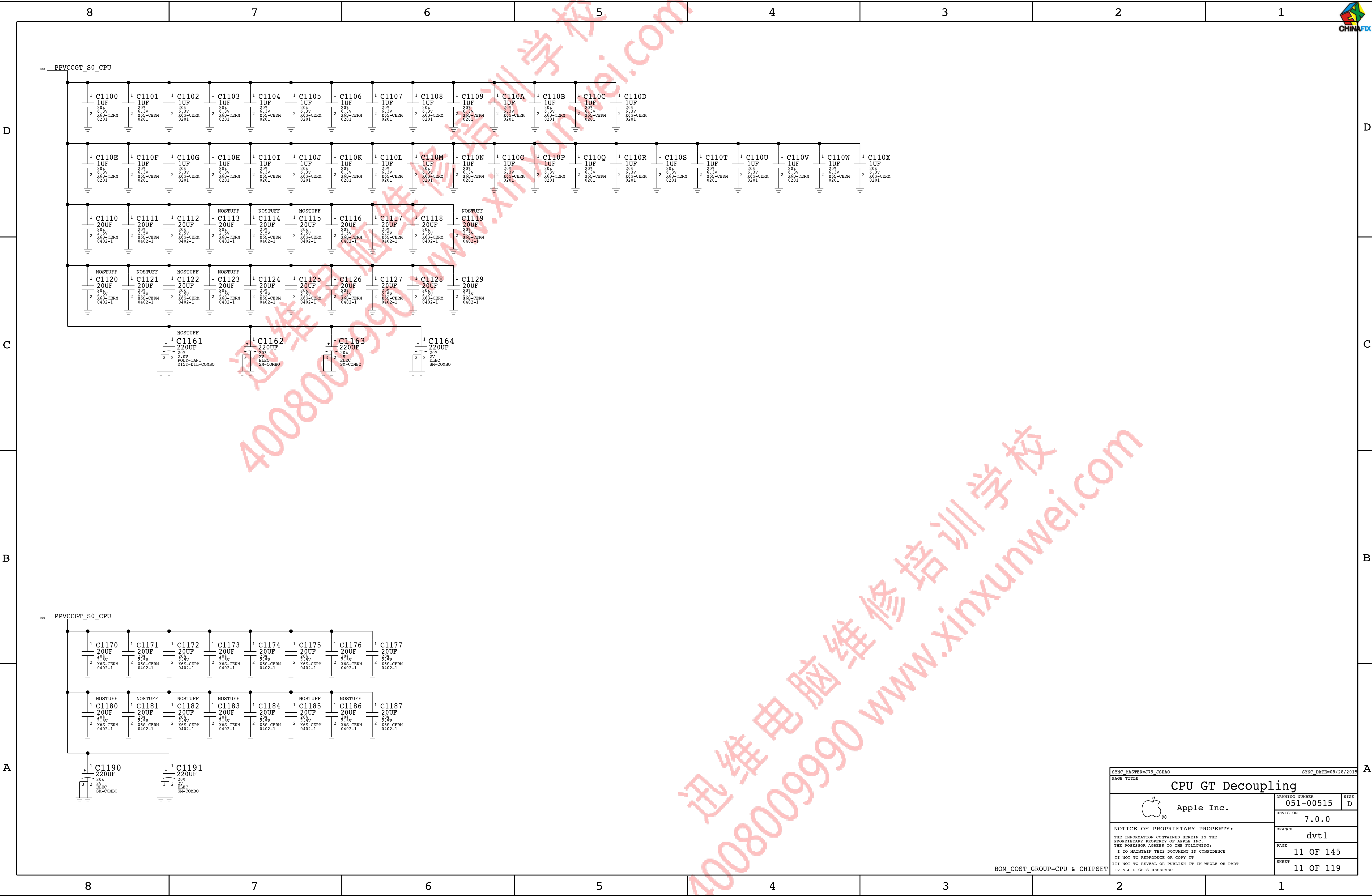
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PAGE TITLE			
CPU & PCH Grounds			
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		REVISON	7.0.0
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		PAGE	9 OF 145
		SHEET	9 OF 119

BOM_COST_GROUP=CPU & CHIPSET



SYNC_MASTER=J79 JSHAO		SYNC_DATE=08/28/2015	
PAGE TITLE			
CPU Core Decoupling			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
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	7.0.0		
	BRANCH		
	dvt1		
	PAGE	10 OF 145	
	SHEET	10 OF 119	

BOM_COST_GROUP=CPU & CHIPSET



SYNC_MASTER=J79_JSHAO		SYNC_DATE=08/28/2015	
PAGE TITLE			
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 Apple Inc.	DRAWING NUMBER	051-00515	SIZE
	REVISION	7.0.0	D
	NOTICE OF PROPRIETARY PROPERTY:		
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		SHEET	11 OF 119

BOM_COST_GROUP=CPU & CHIPSET

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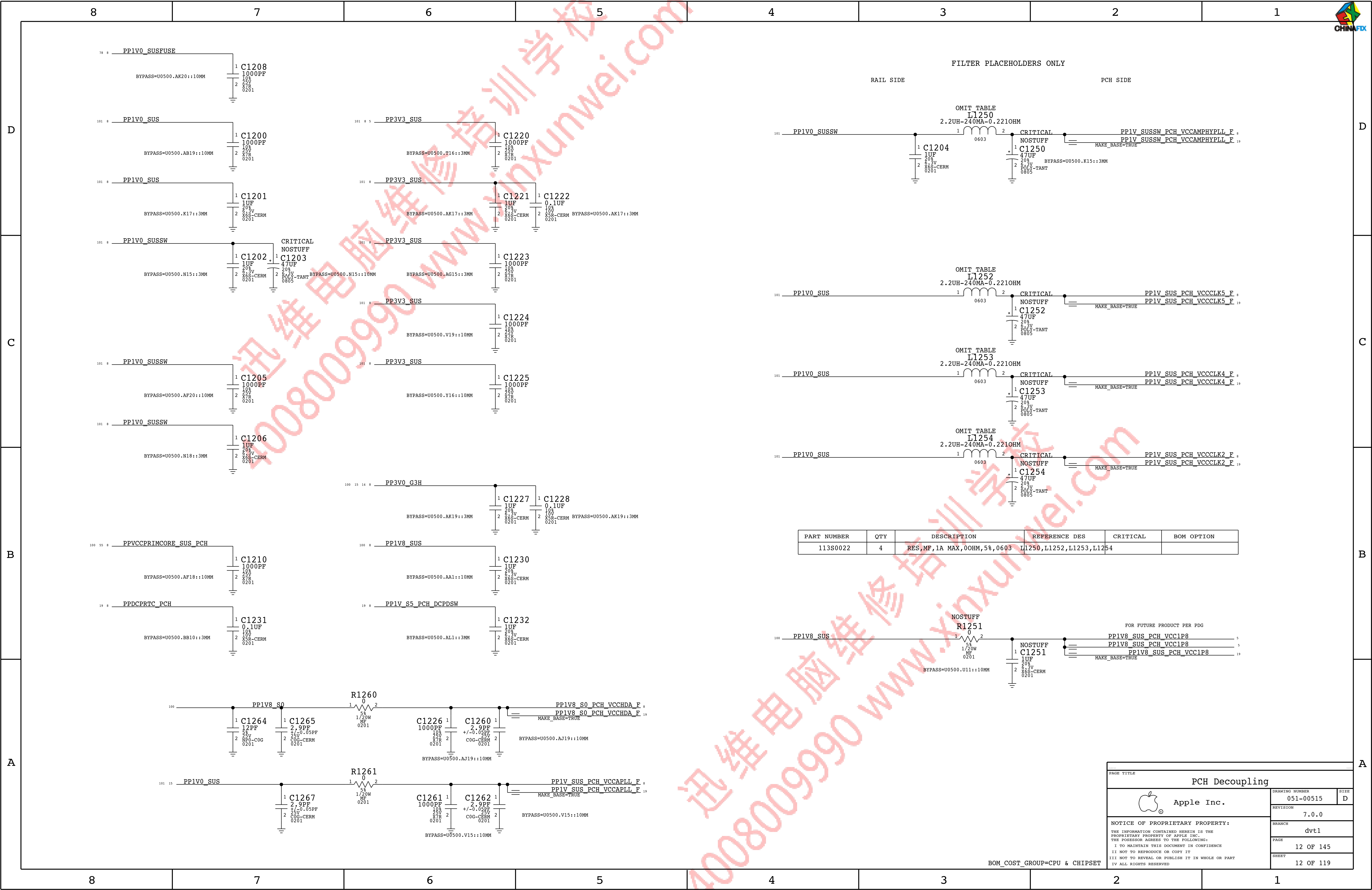
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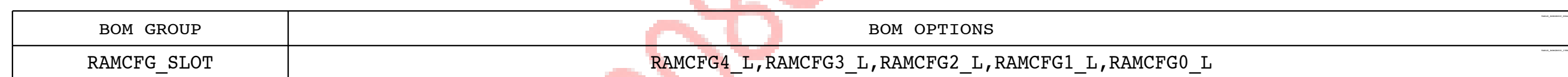
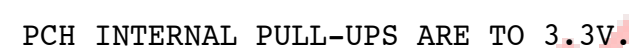
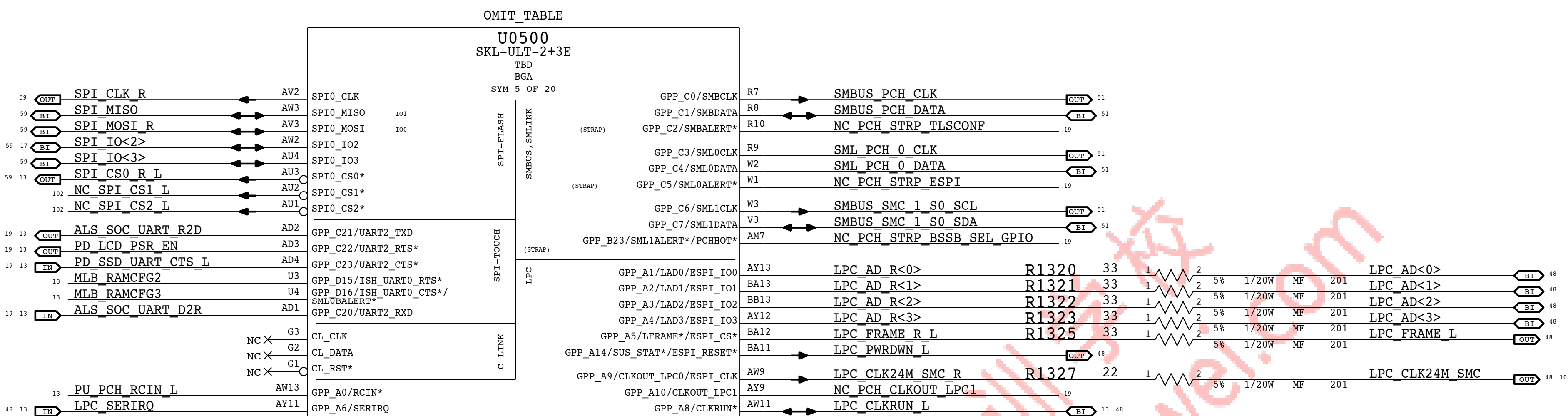
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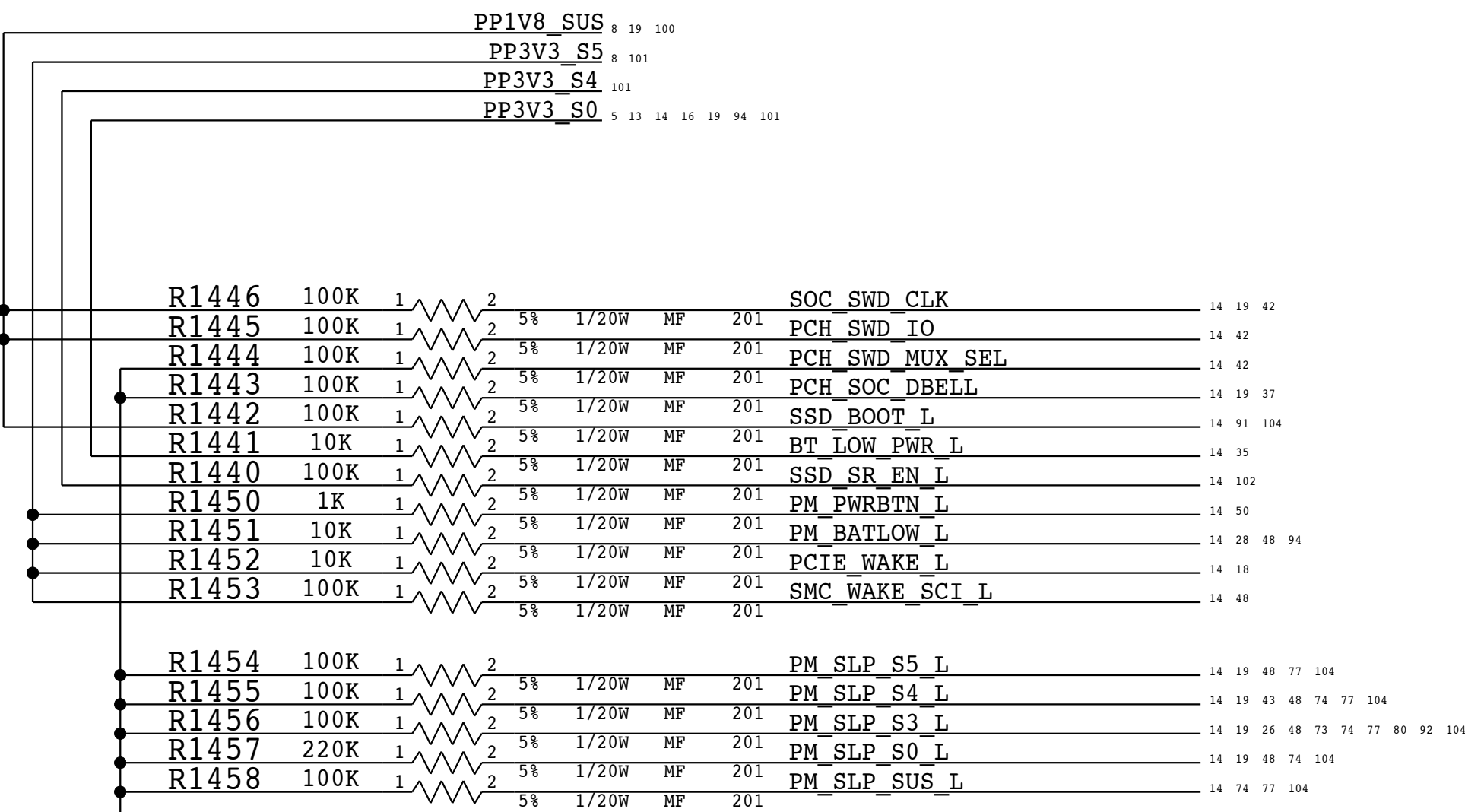
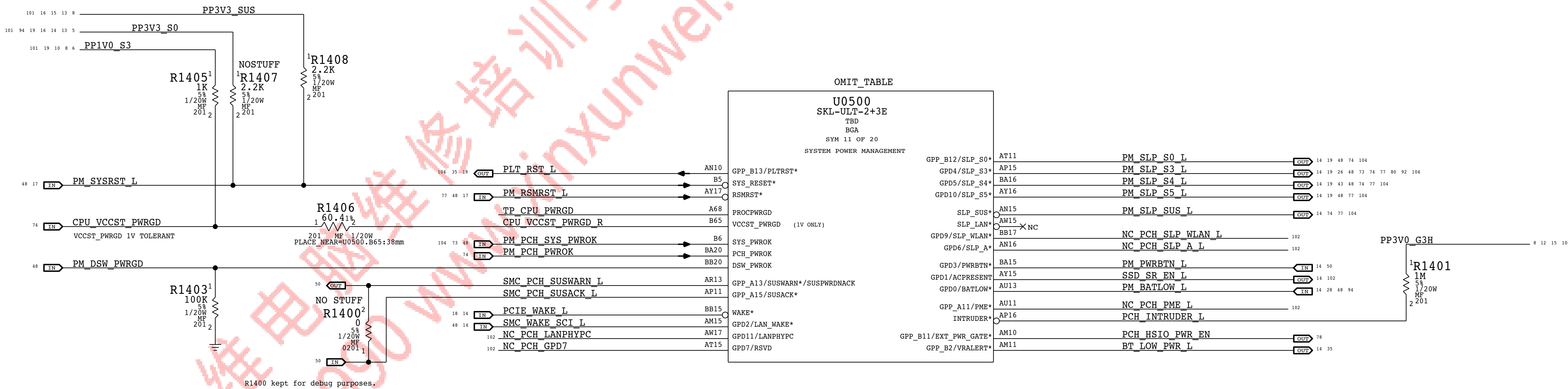


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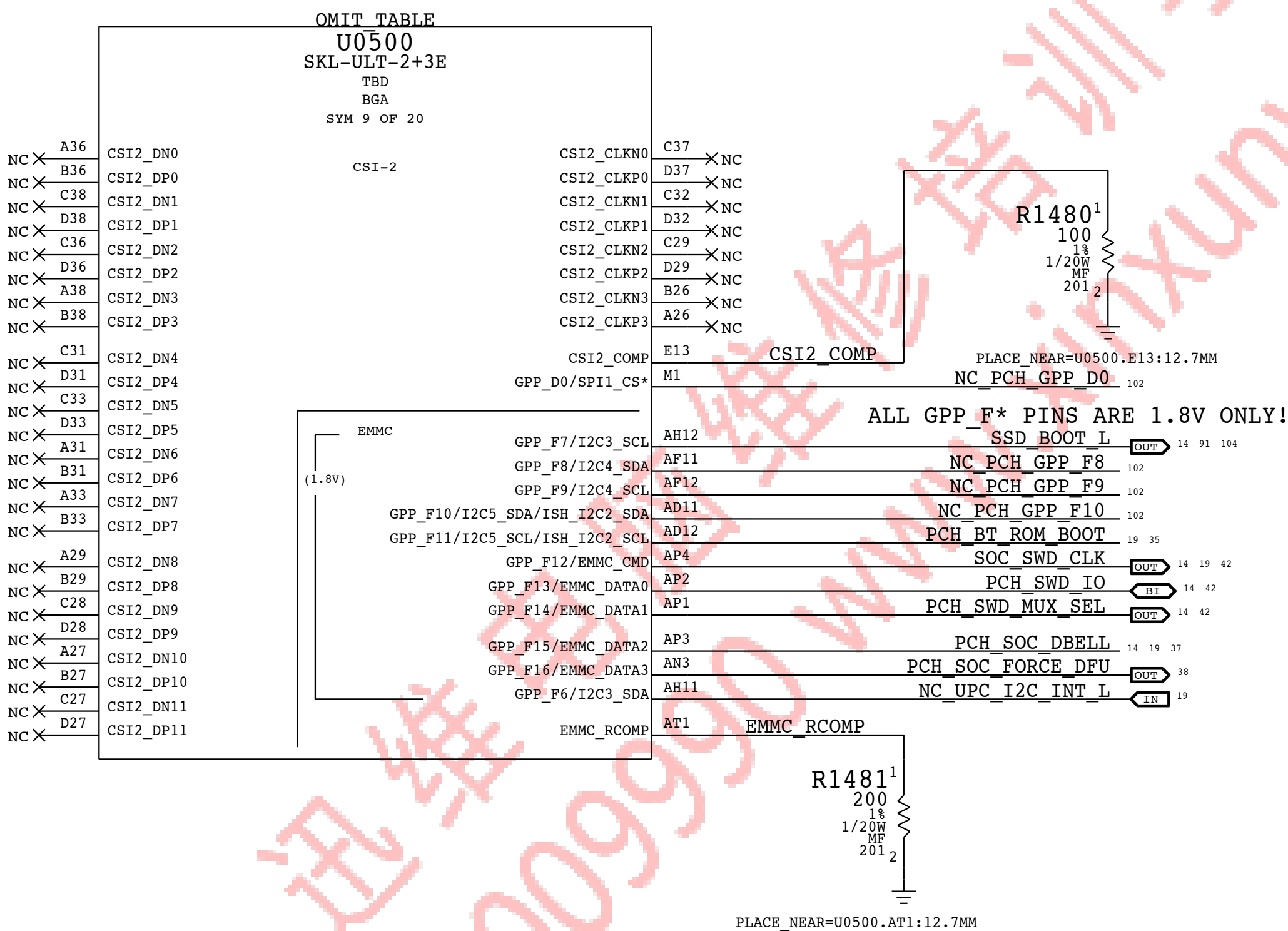
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LAST CHANGE: Tue Feb 2 13:18:21 2016			
PAGE TITLE			
PCH Audio/LPC/SPI/SMBus			
 Apple Inc.		DRAWING NUMBER	051-00515
		SIZE	D
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		BRANCH	dvt1
		PAGE	13 OF 145
		SHEET	13 OF 119



PCH Reset Button



NOTE: PM_SLP_S0_L HAS INTERNAL PULL-UP BEFORE RSMRST_L IS RELEASED.
THIS CAUSES A VOLTAGE DIVIDER WITH THE PULL-DOWN HERE.
THE SIGNAL IS DRIVEN HI AFTER RSMRST_L IS RELEASED.

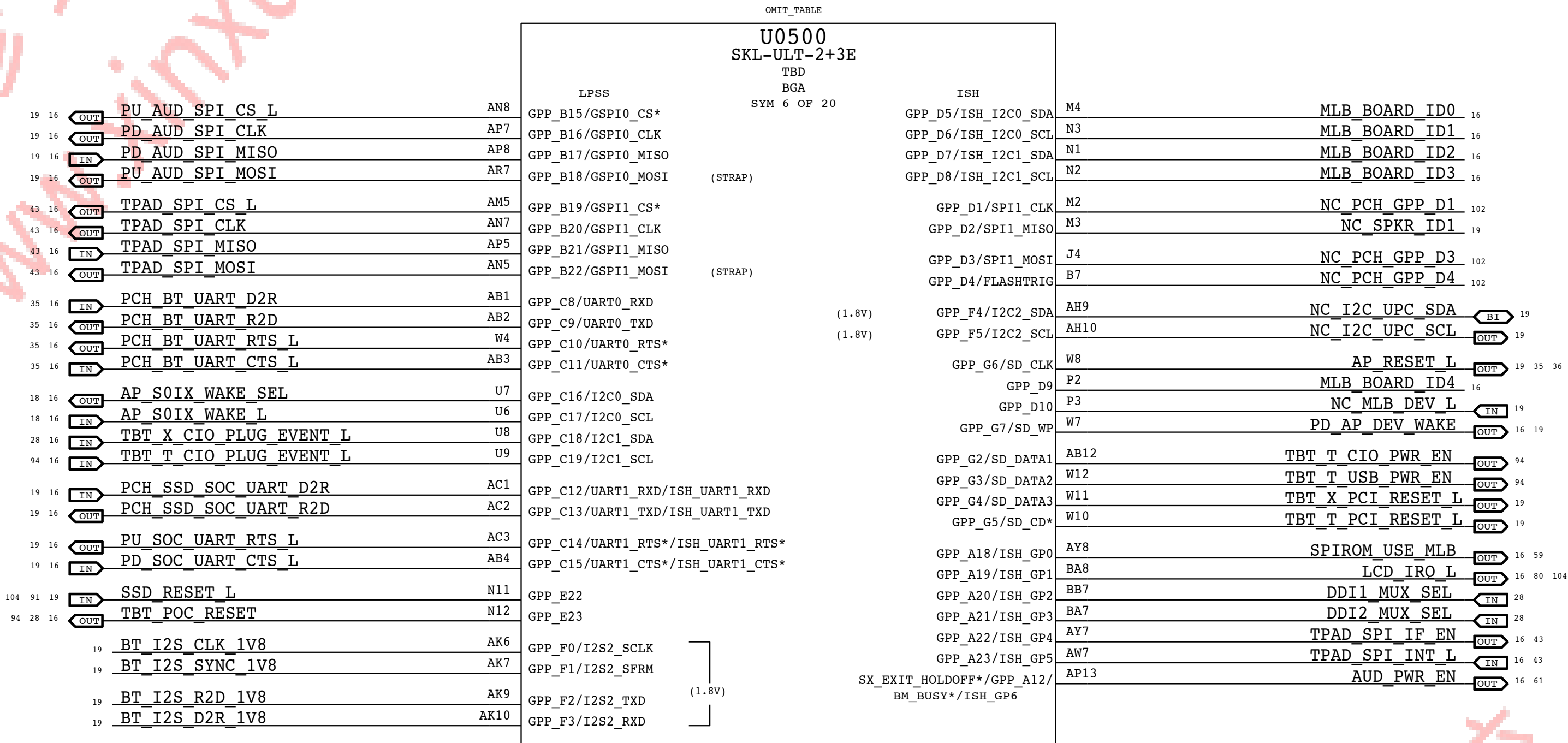


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LAST CHANGE: Tue May 3 17:45:28 2016		
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PCH Power Management		
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	REVISION	7.0.0
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	PAGE	14 OF 145
	SHEET	14 OF 119

BOM_COST_GROUP=CPU & CHIPSET

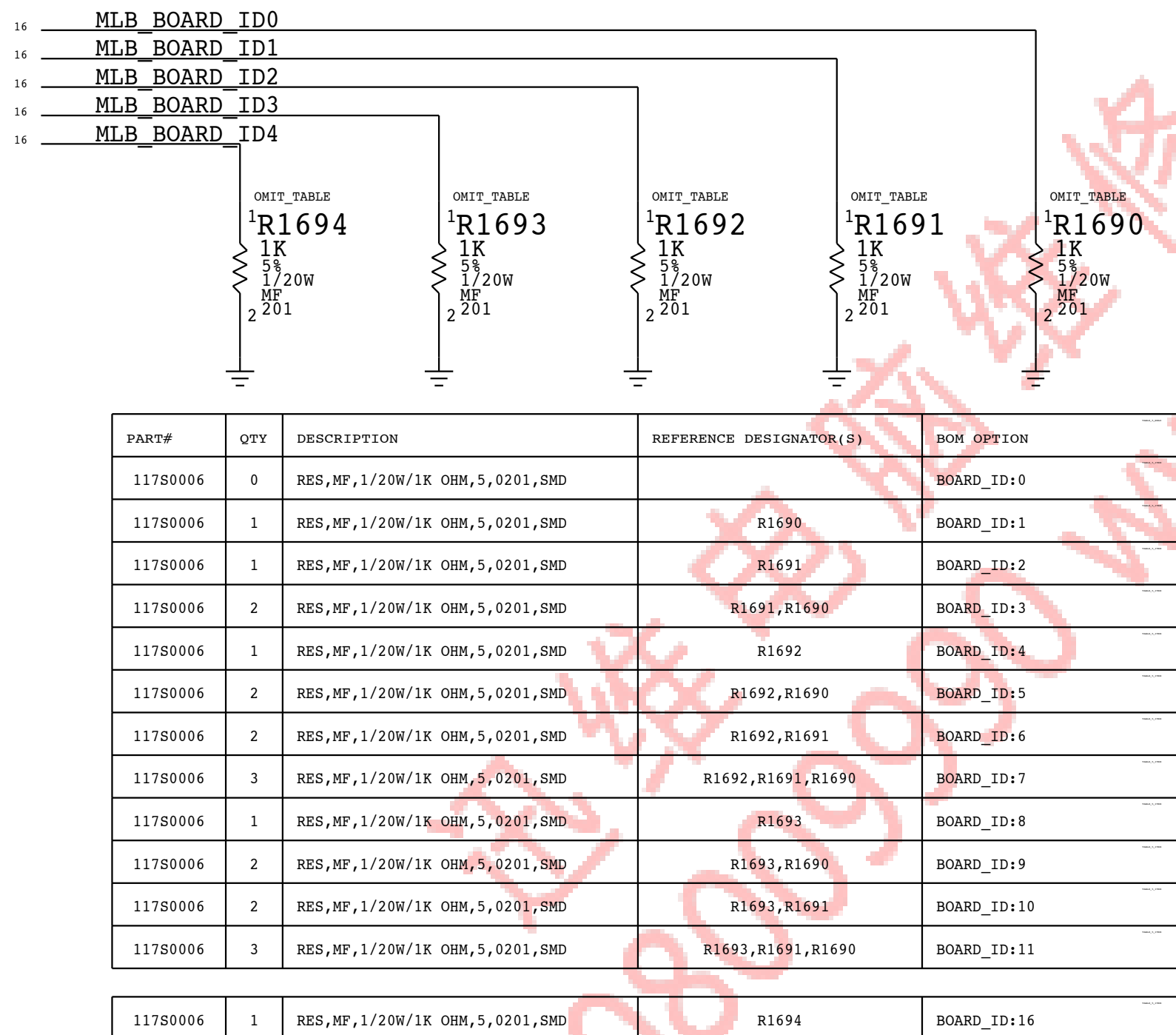


ALL GPP_F* PINS ARE 1.8V ONLY!



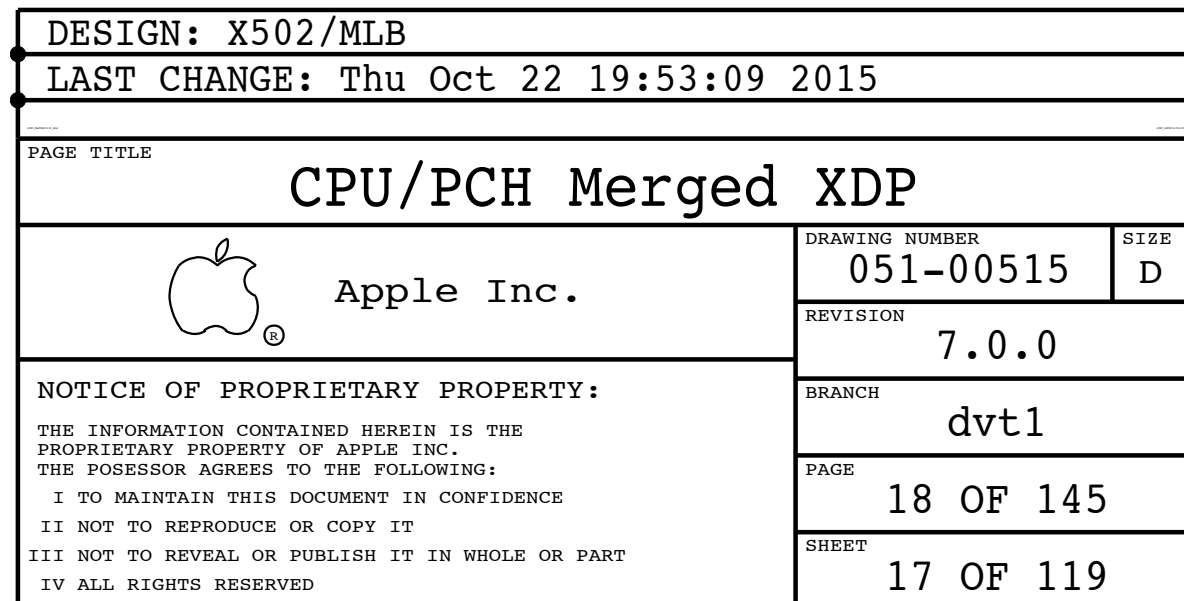
MLB ID STRAPS.

PCH INTERNAL PULL-UPS ARE TO VCCGPPD = 3.3V.



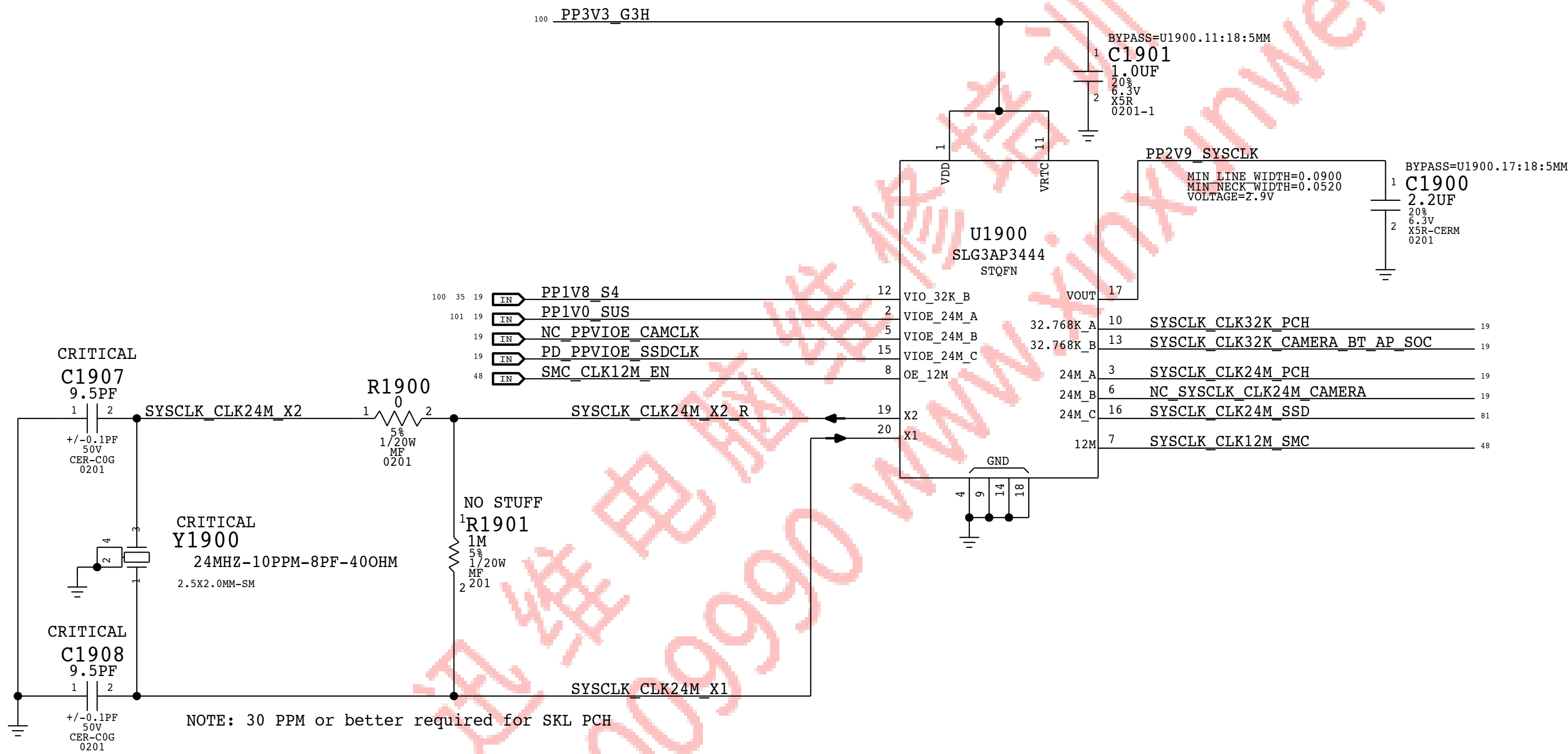
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PCH SPI/UART/GPIO		
Apple Inc.		DRAWING NUMBER 051-00515
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		REVISION 7.0.0
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		PAGE 16 OF 145
		SHEET 16 OF 119

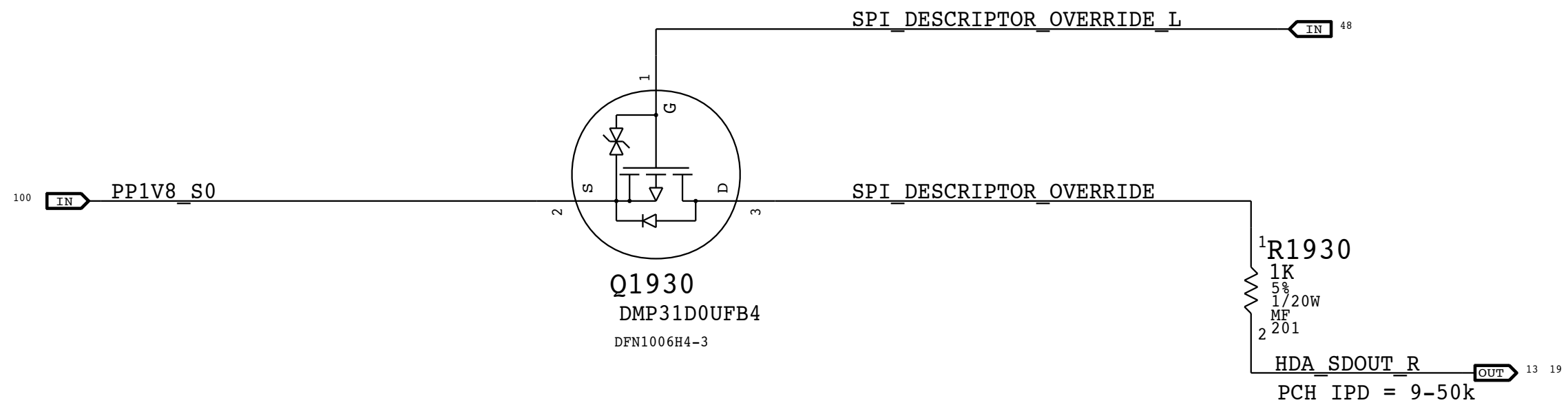


BOM COST GROUP=DEBUG

System 32kHz / 12MHz / 24MHz Clock Generator

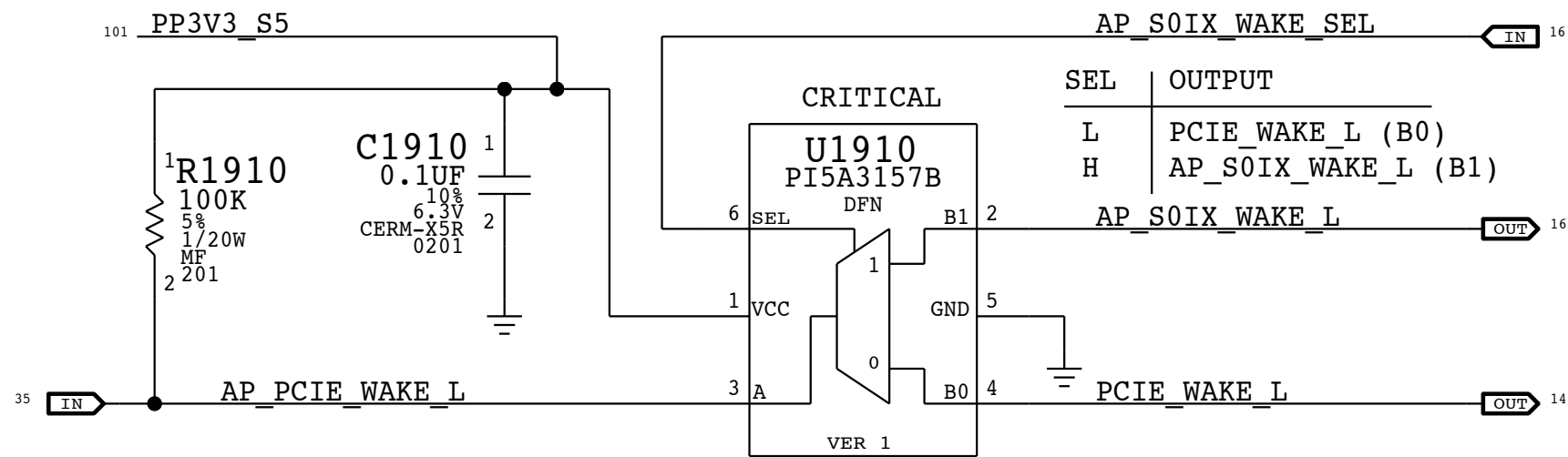


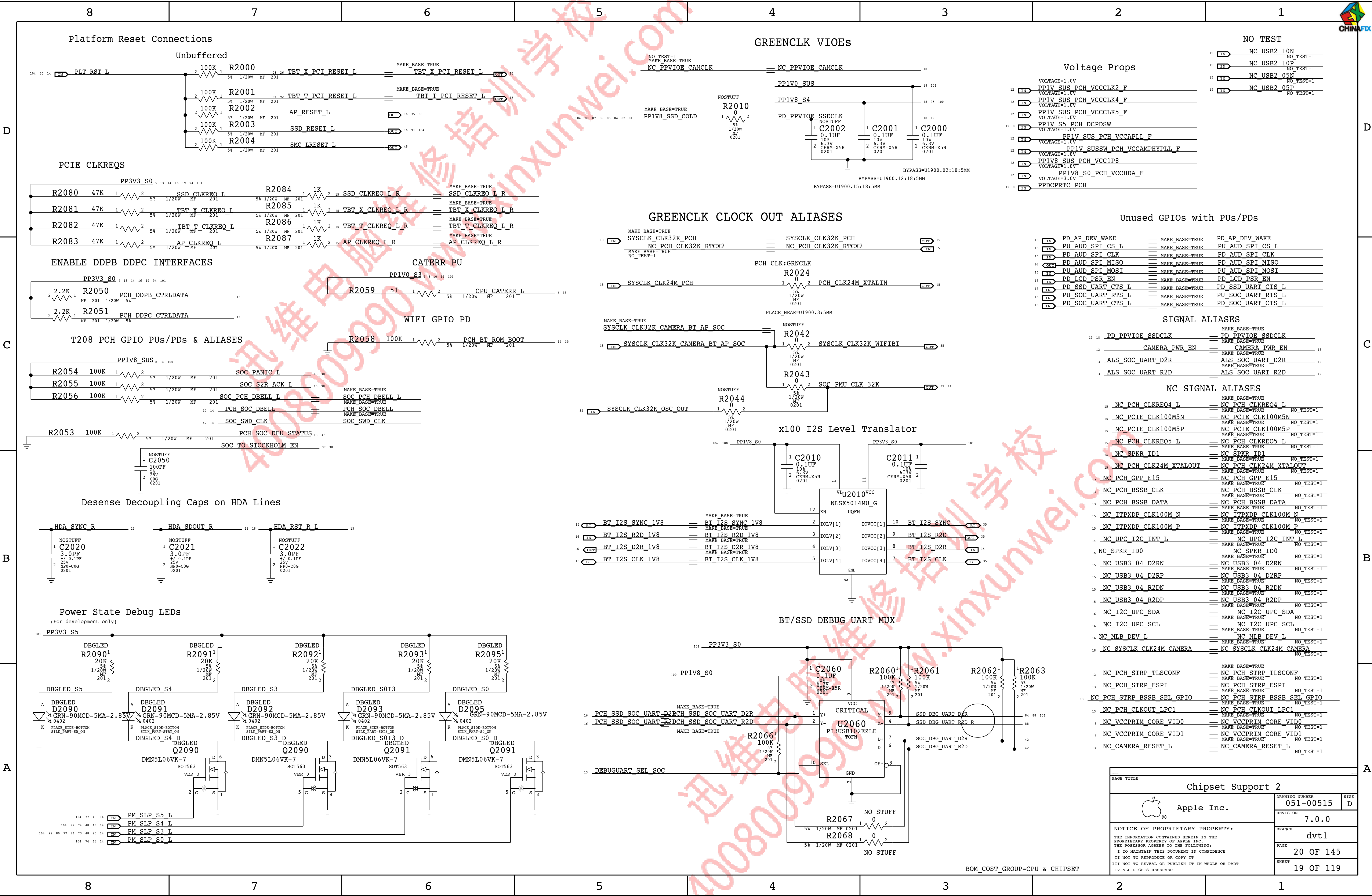
PCH ME Disable Strap



PCH uses HDA SDO as a power-up strap. If low, ME functions normally. If high, ME is disabled. This allows for full re-flashing of SPI ROM. SMC controls strap enable to allow in-field control of strap setting. ***** Circuit does not support HDA voltage >3.3V.

PCIe Wake Muxing

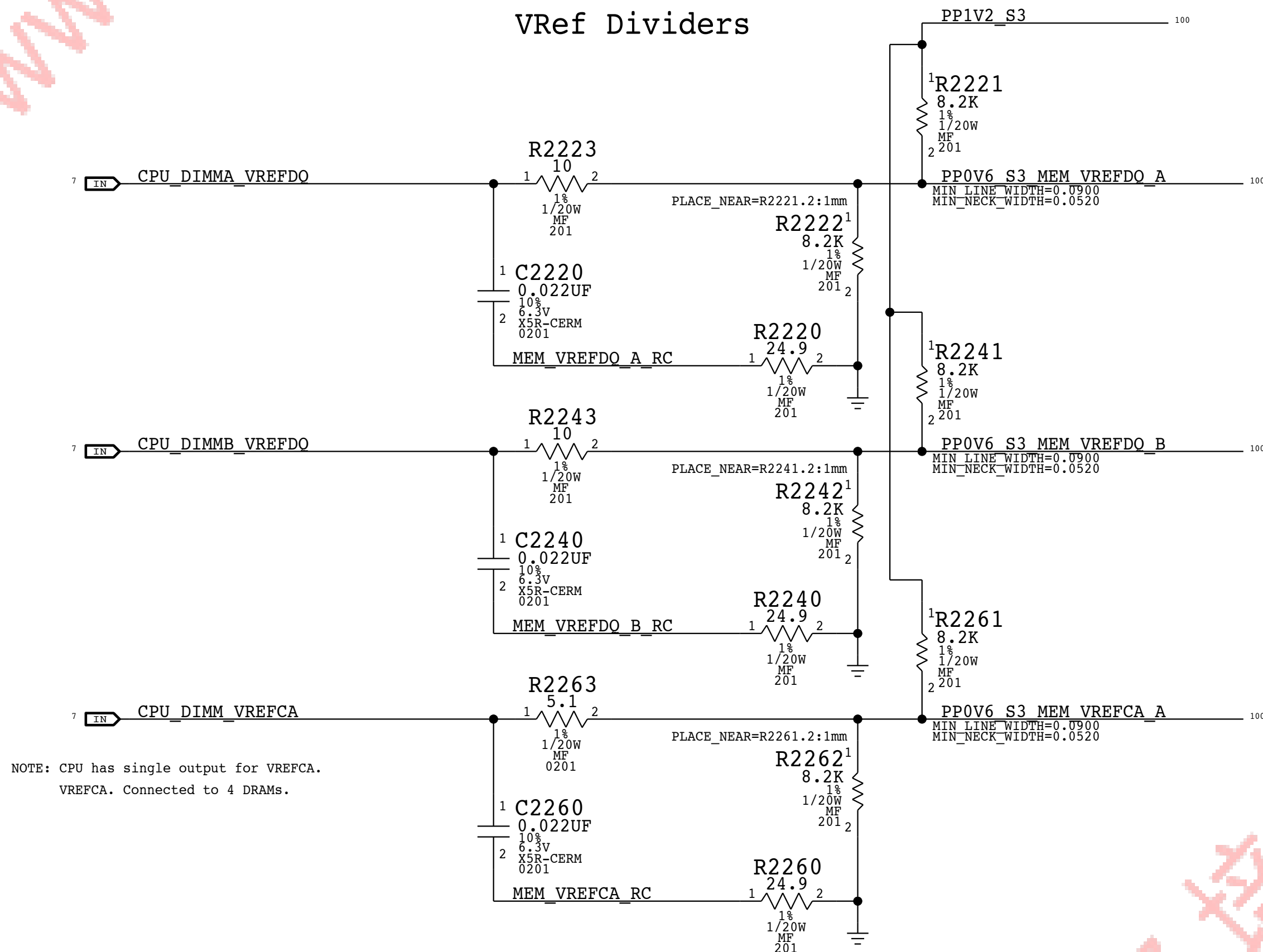




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1 TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE
11 NOT TO REPRODUCE OR COPY IT		20 OF 145
111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET
114 ALL RIGHTS RESERVED		19 OF 119

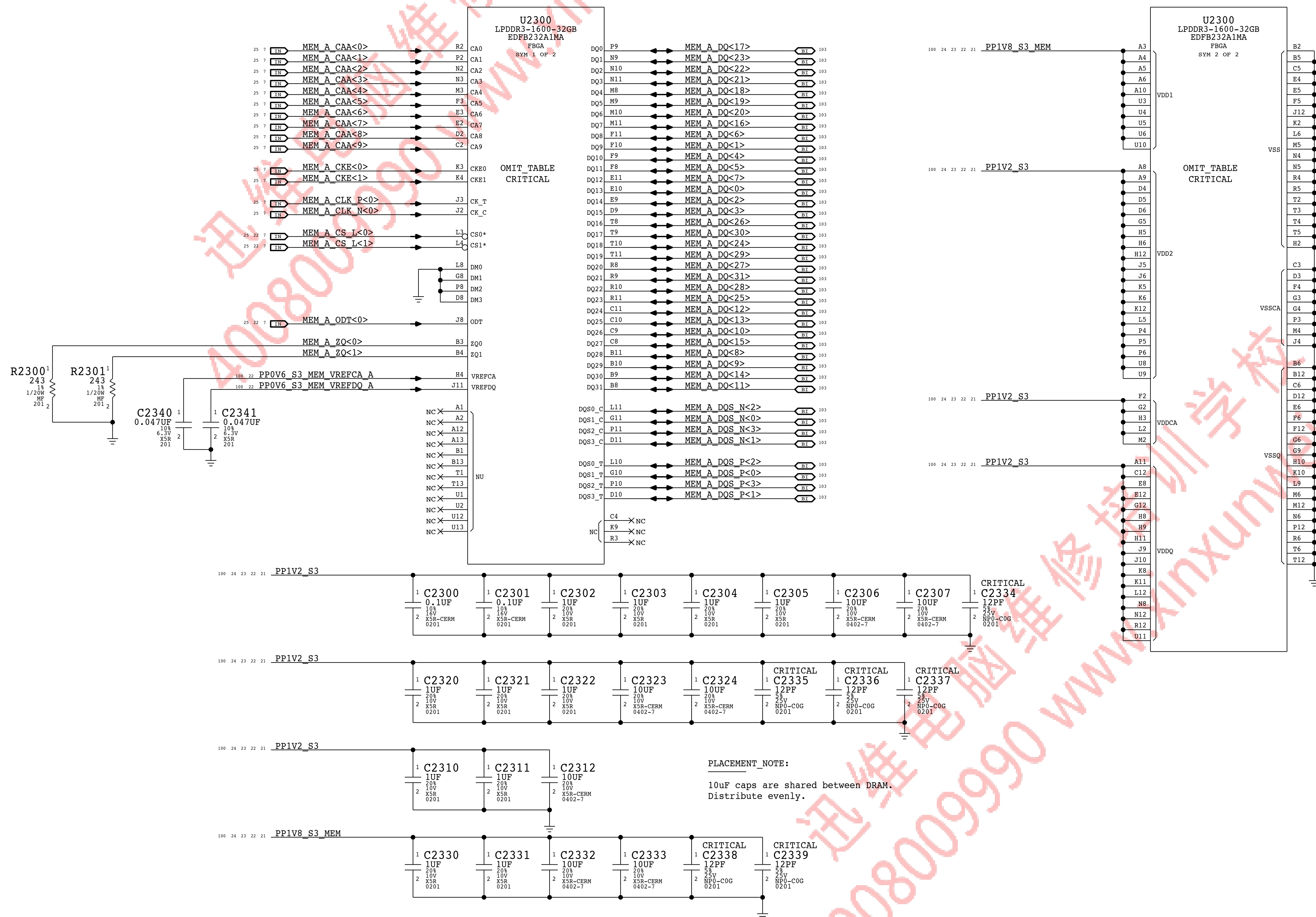
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CPU-Based Margining



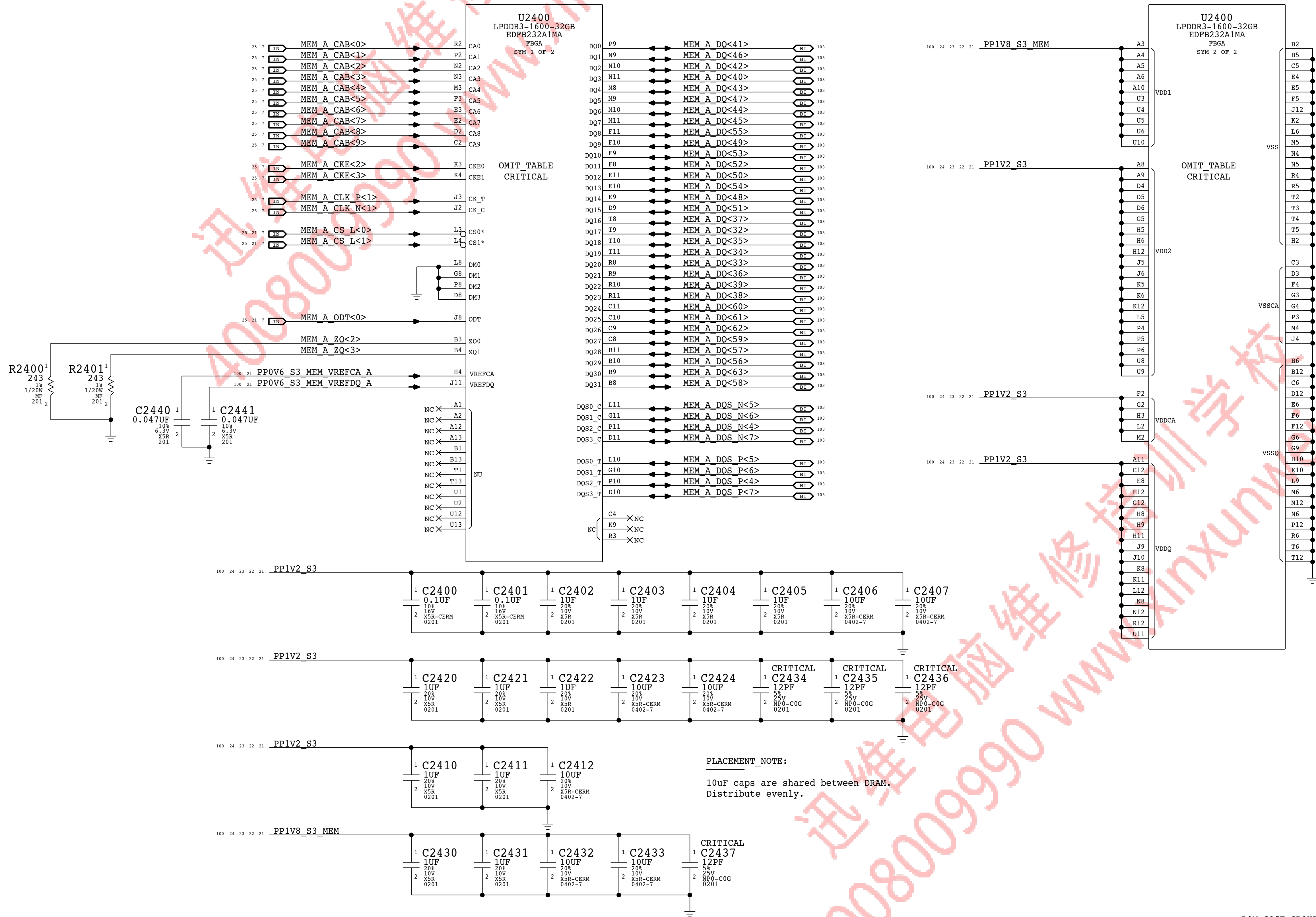
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LPDDR3 VREF Margining			
 Apple Inc.	DRAWING NUMBER		SIZE
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	REVISION		
	7.0.0		
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		PAGE	
		22 OF 145	
		SHEET	
	20 OF 119		

LPDDR3 CHANNEL A (0-31)



SYNC MASTER=J52 MLB		SYNC DATE=05/12/2013	
PAGE TITLE			
LPDDR3 DRAM Channel A (00-31)			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-00515	D
		REVISION	
		7.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
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		PAGE	23 OF 145
		SHEET	21 OF 119

LPDDR3 CHANNEL A (32-63)



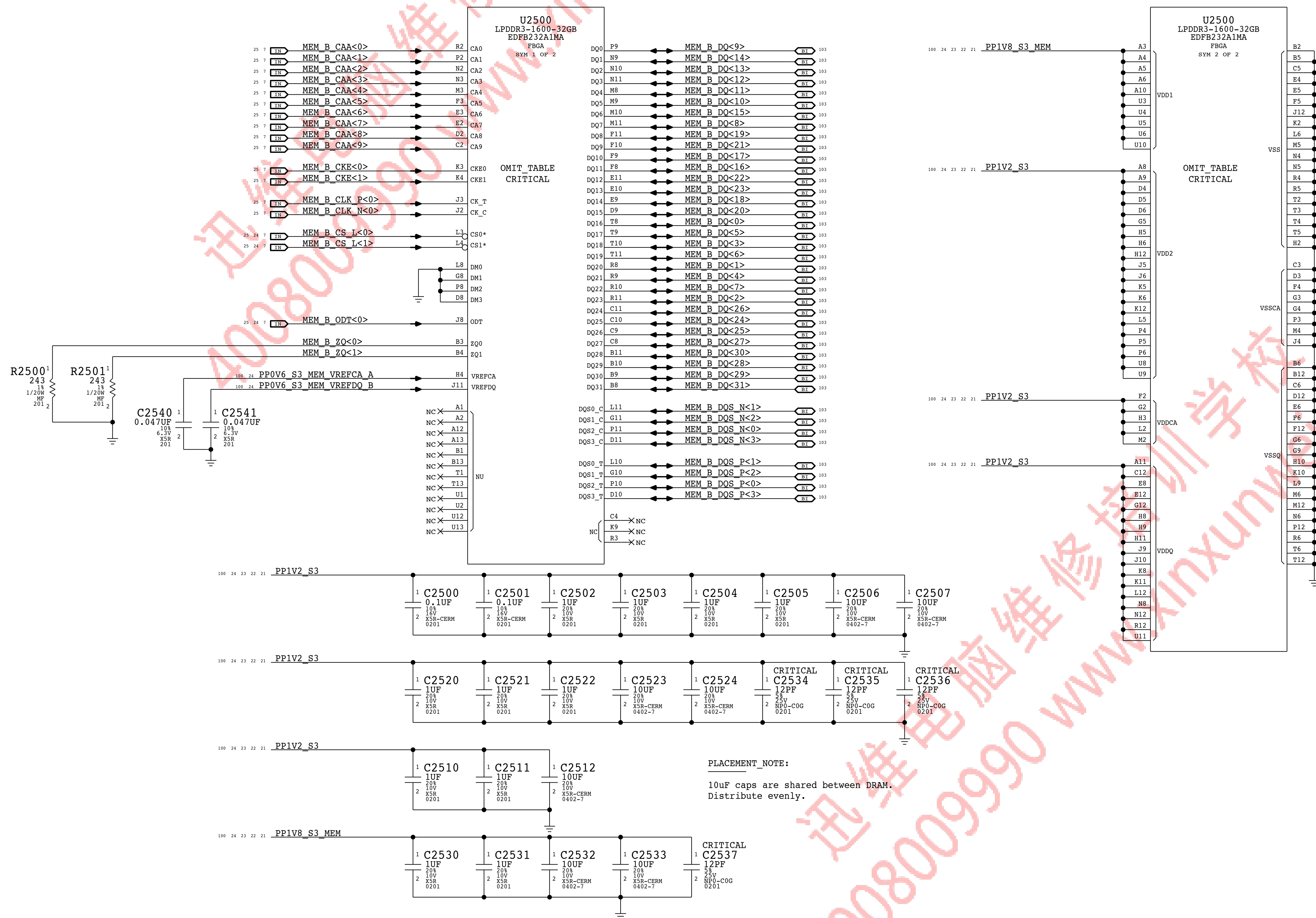
PLACEMENT NOTE:

10uF caps are shared between DRAM.
Distribute evenly.

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PAGE TITLE			
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	DRAWING NUMBER		STEP
	051-00515		D
	REVISION		
		7.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
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I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		PAGE	
II NOT TO REPRODUCE OR COPY IT		24 OF 145	
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		SHEET	
IV ALL RIGHTS RESERVED		22 OF 119	

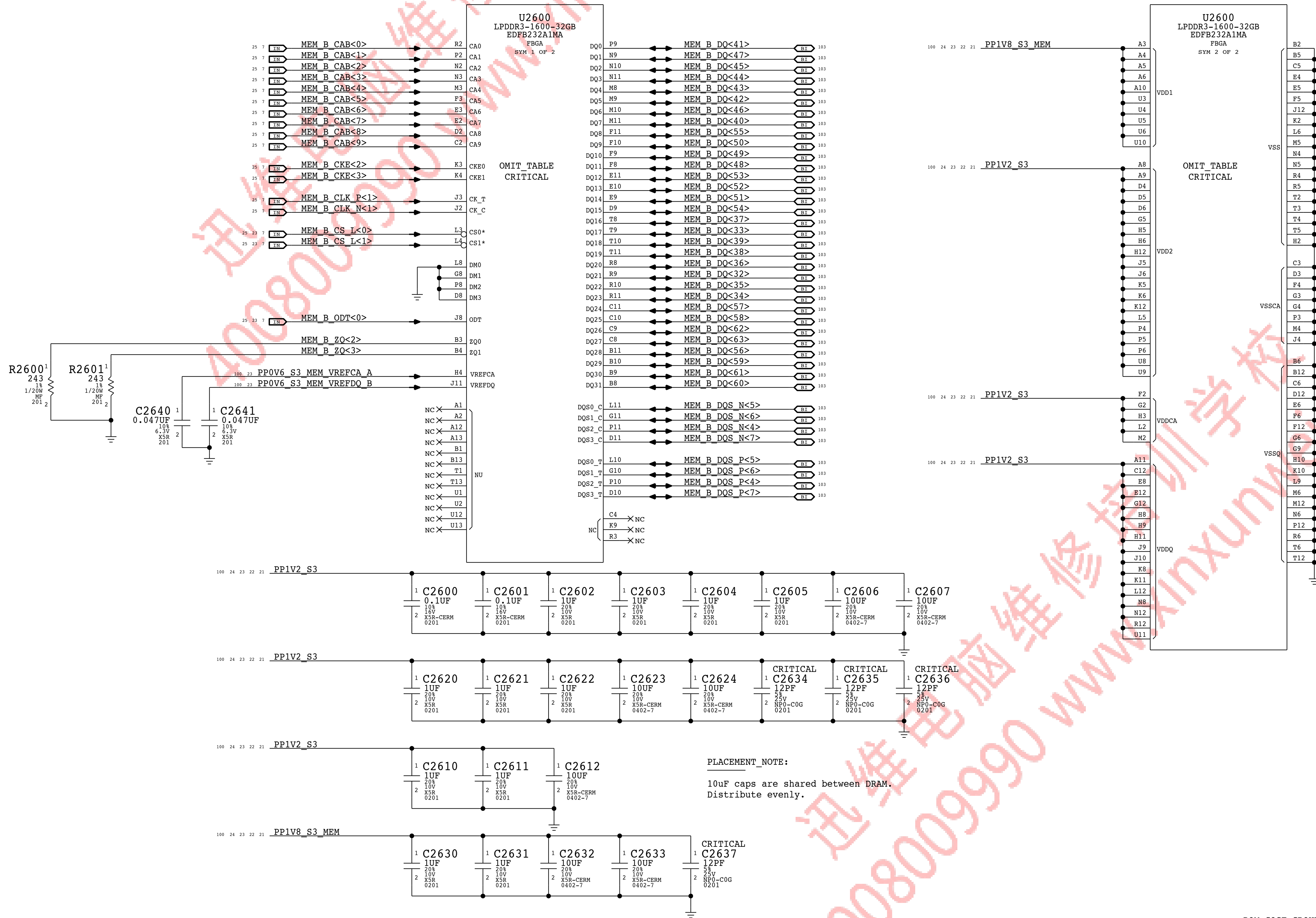
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LPDDR3 CHANNEL B (0-31)



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LPDDR3 DRAM Channel B (00-31)			
 Apple Inc.		DRAWING NUMBER 051-00515	SITE D
		REVISION 7.0.0	
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LPDDR3 CHANNEL B (32-63)



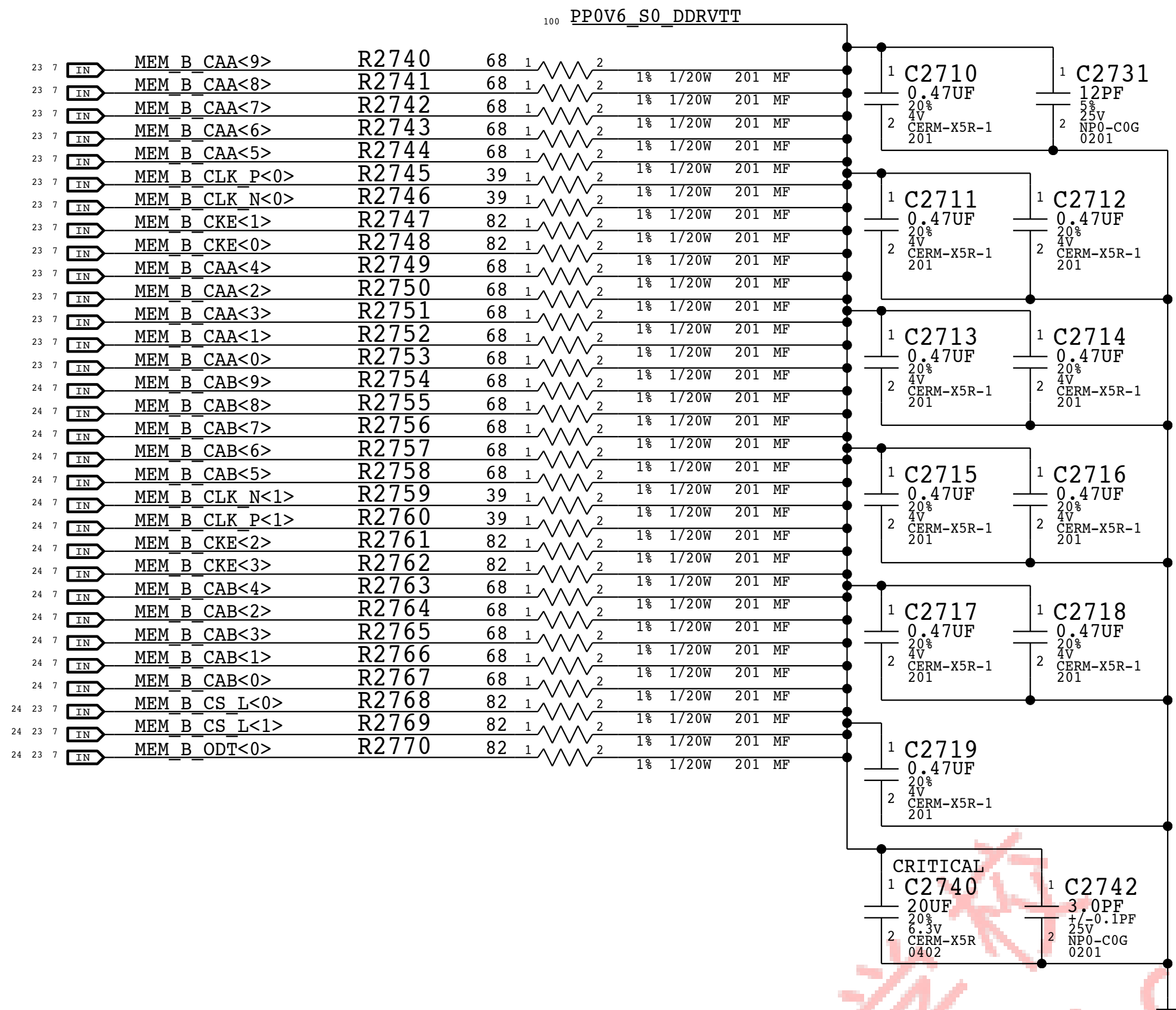
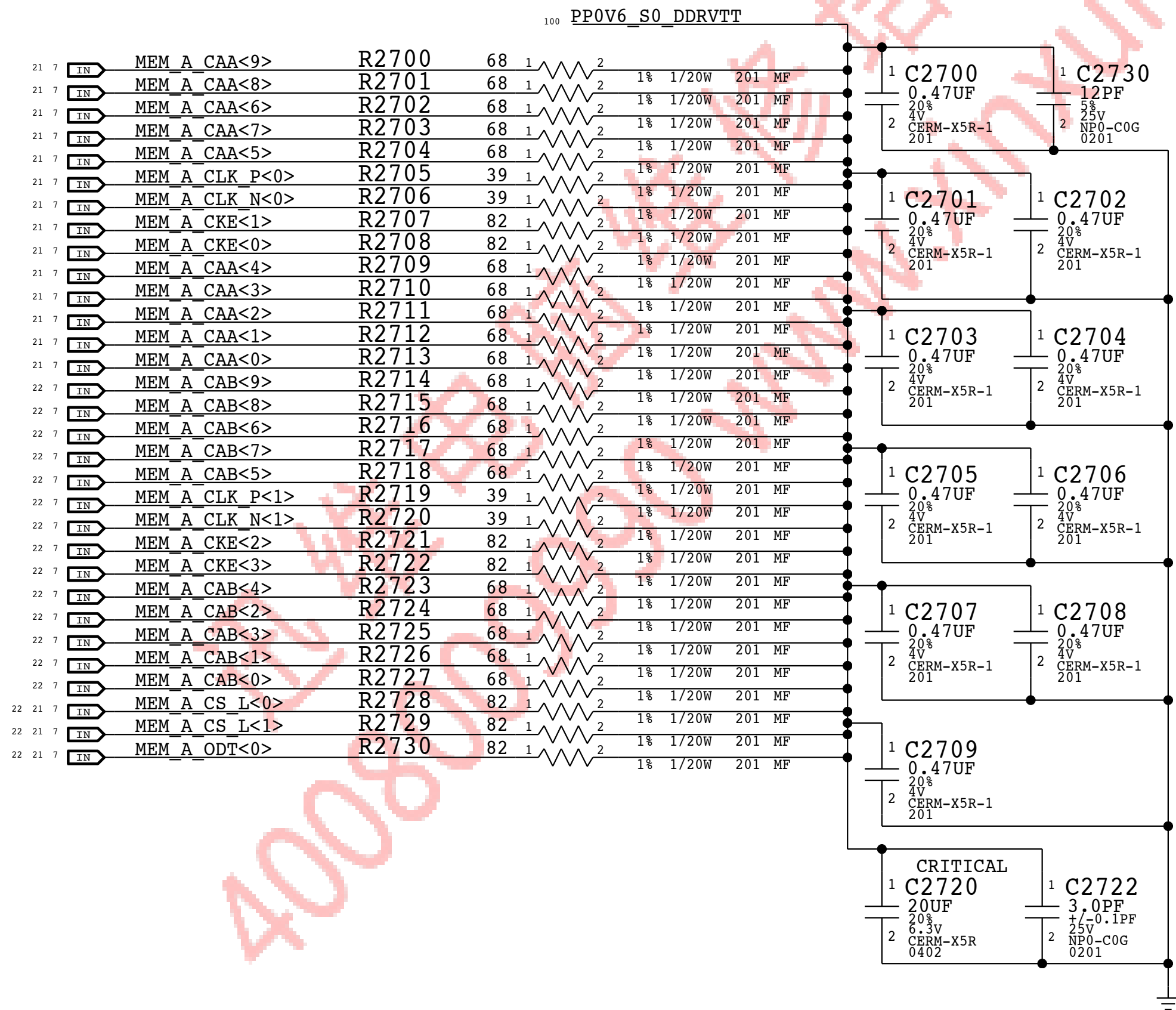
PLACEMENT NOTE:
10uF caps are shared between DRAM.
Distribute evenly.

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PAGE_TITLE			
LPDDR3 DRAM Channel B (32-63)			
	DRAWING NUMBER	051-00515	STEP D
	REVISION	7.0.0	
	BRANCH	dvt1	
	PAGE	26 OF 145	
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BOM_COST_GROUP=DRAM

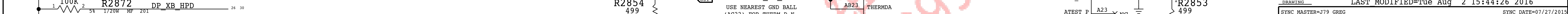


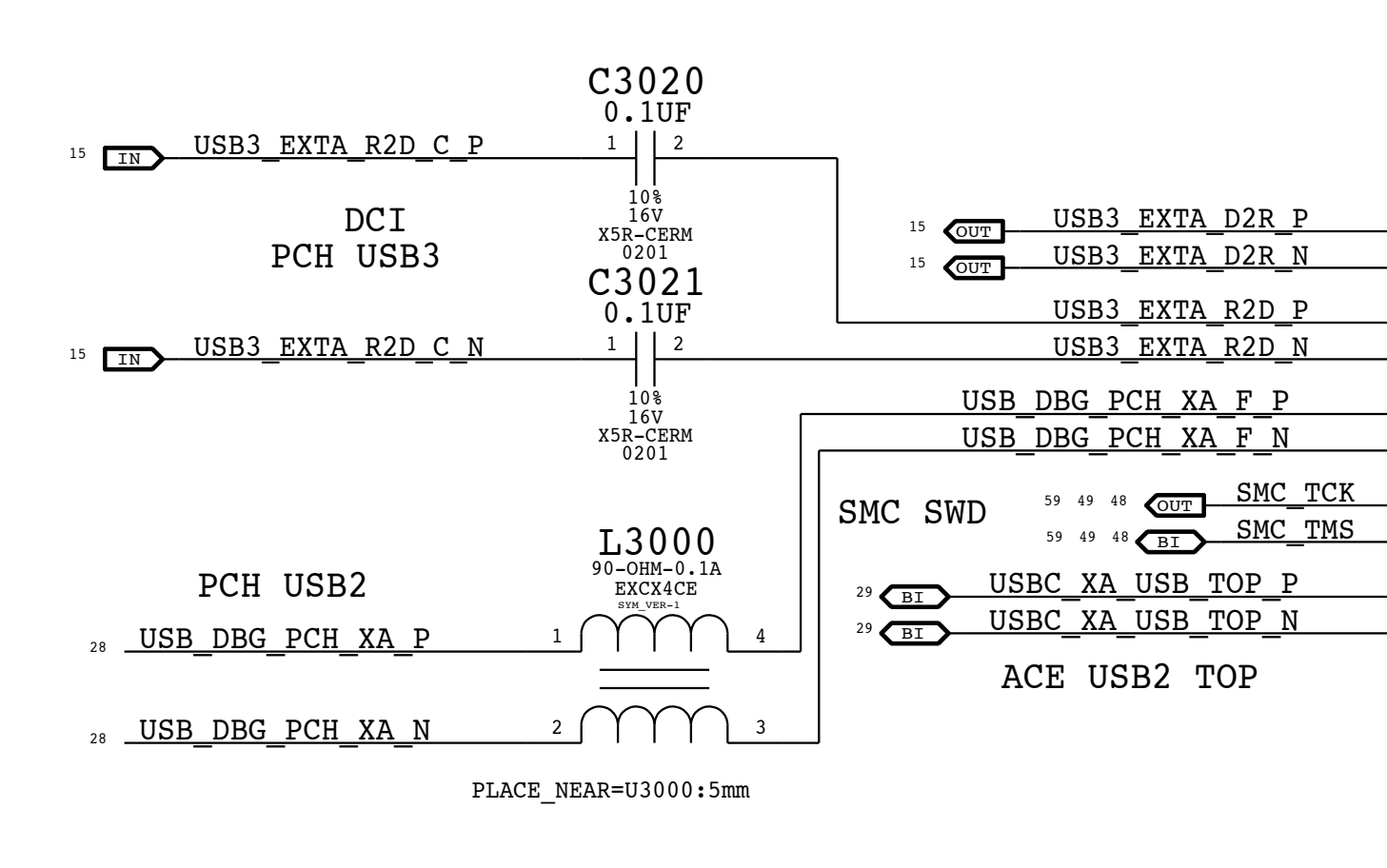
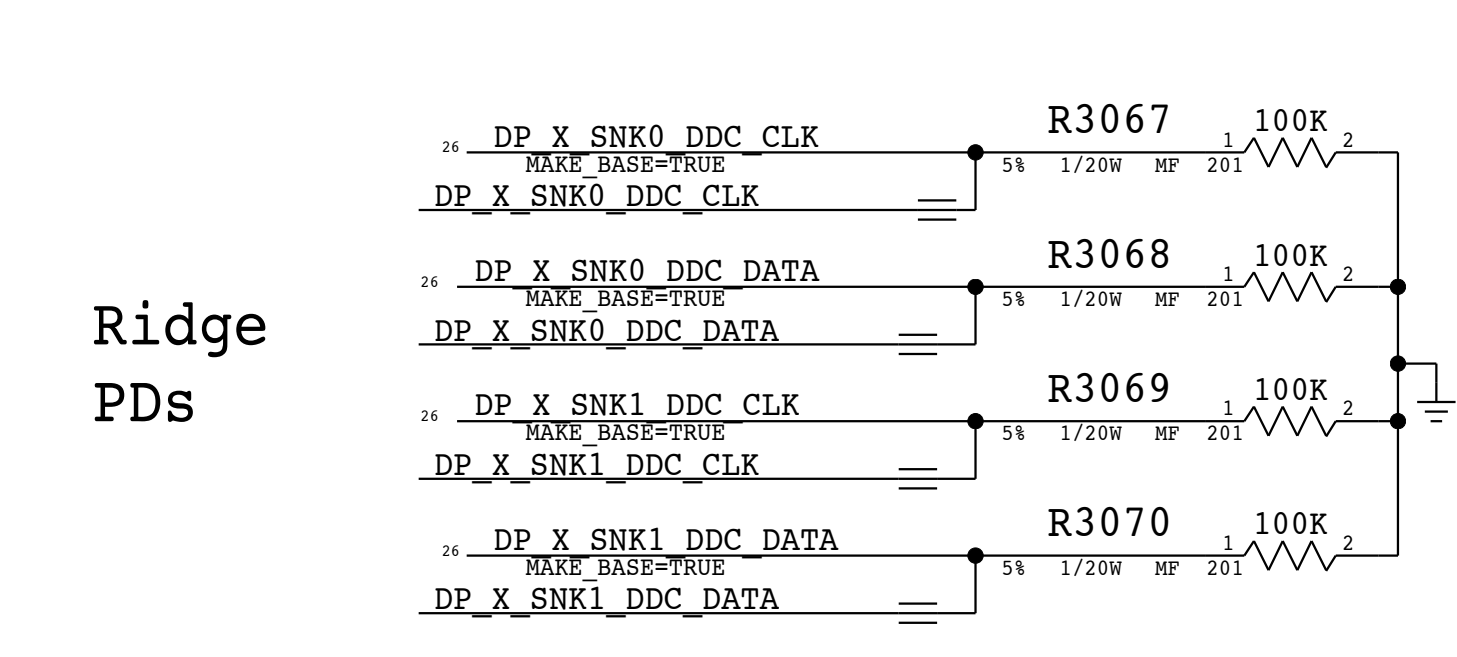
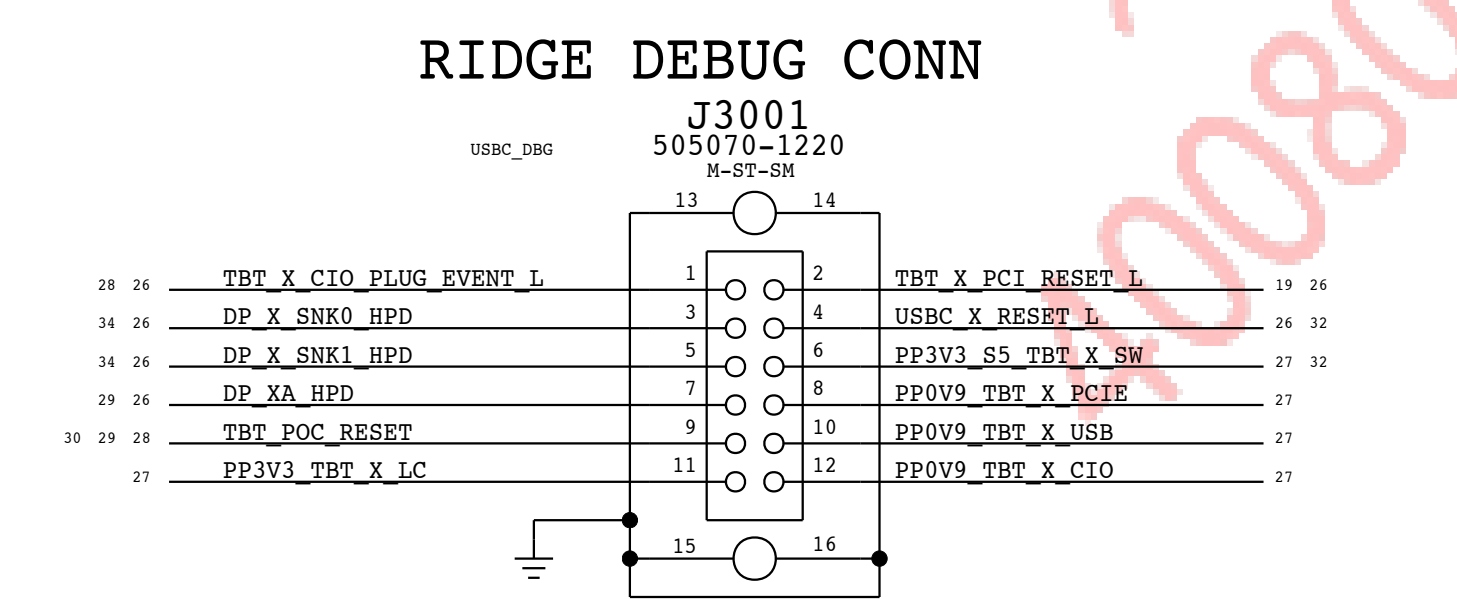
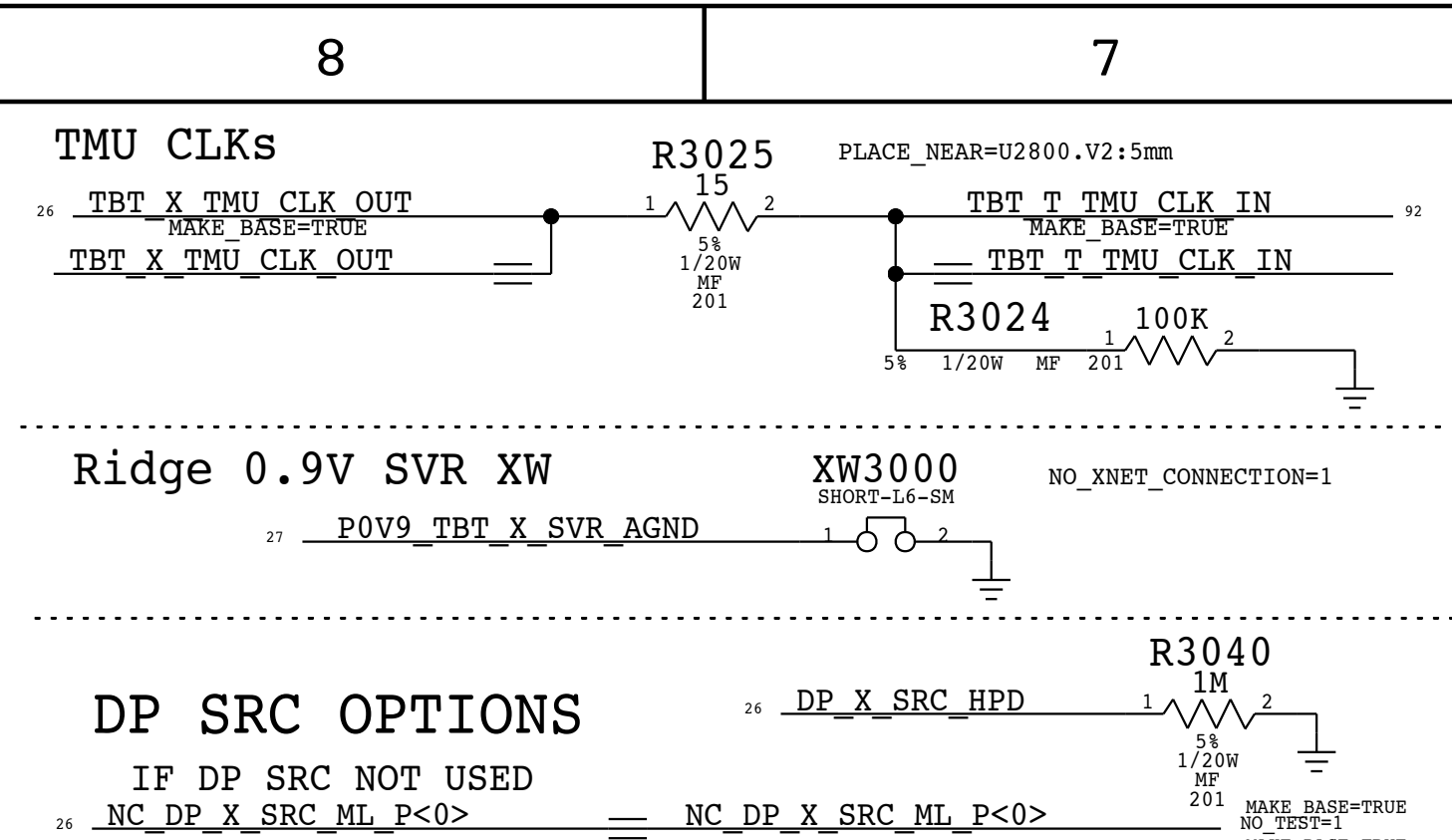
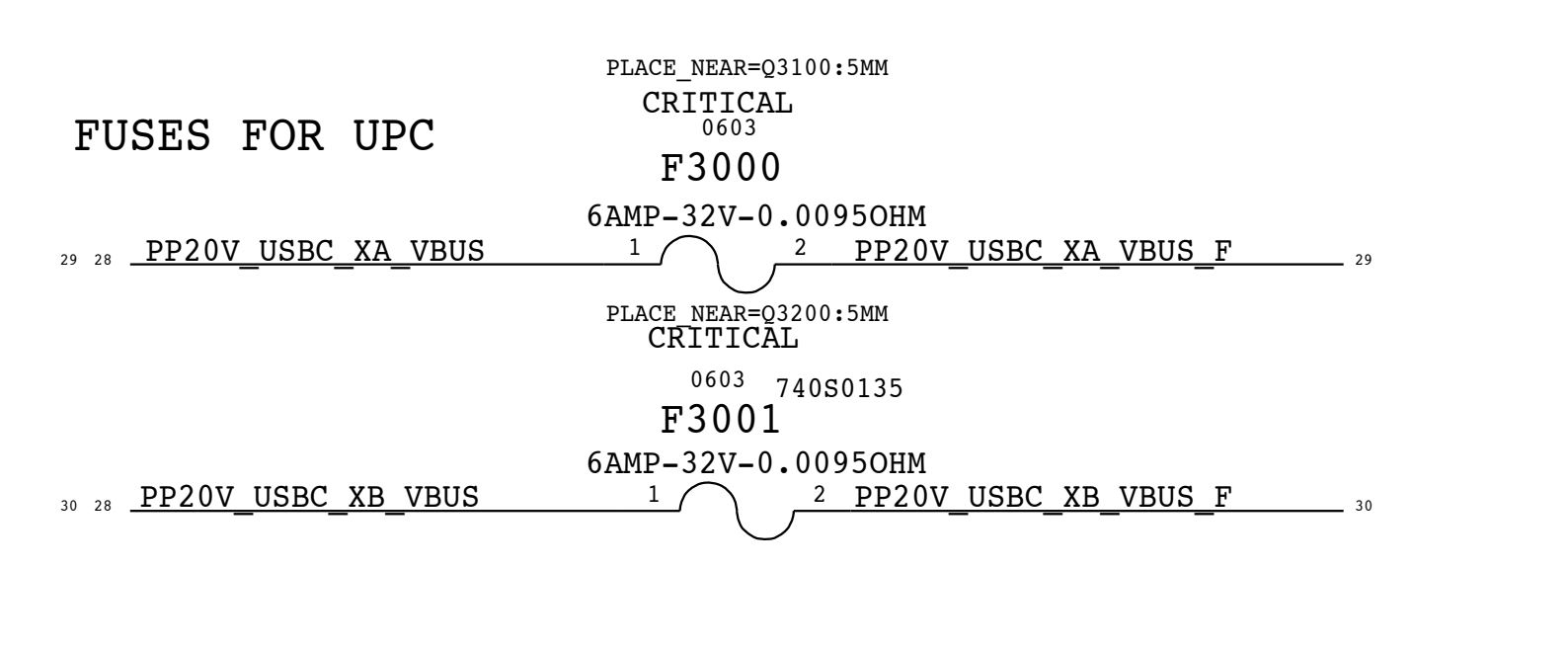
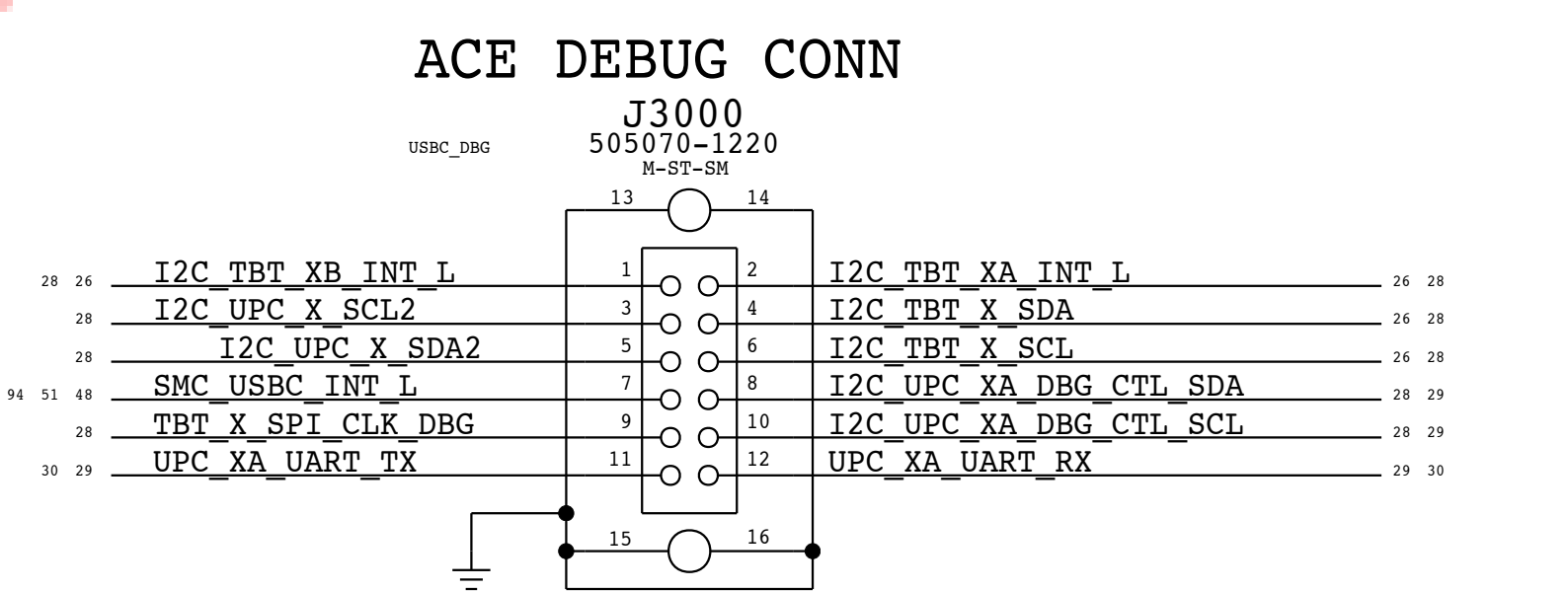
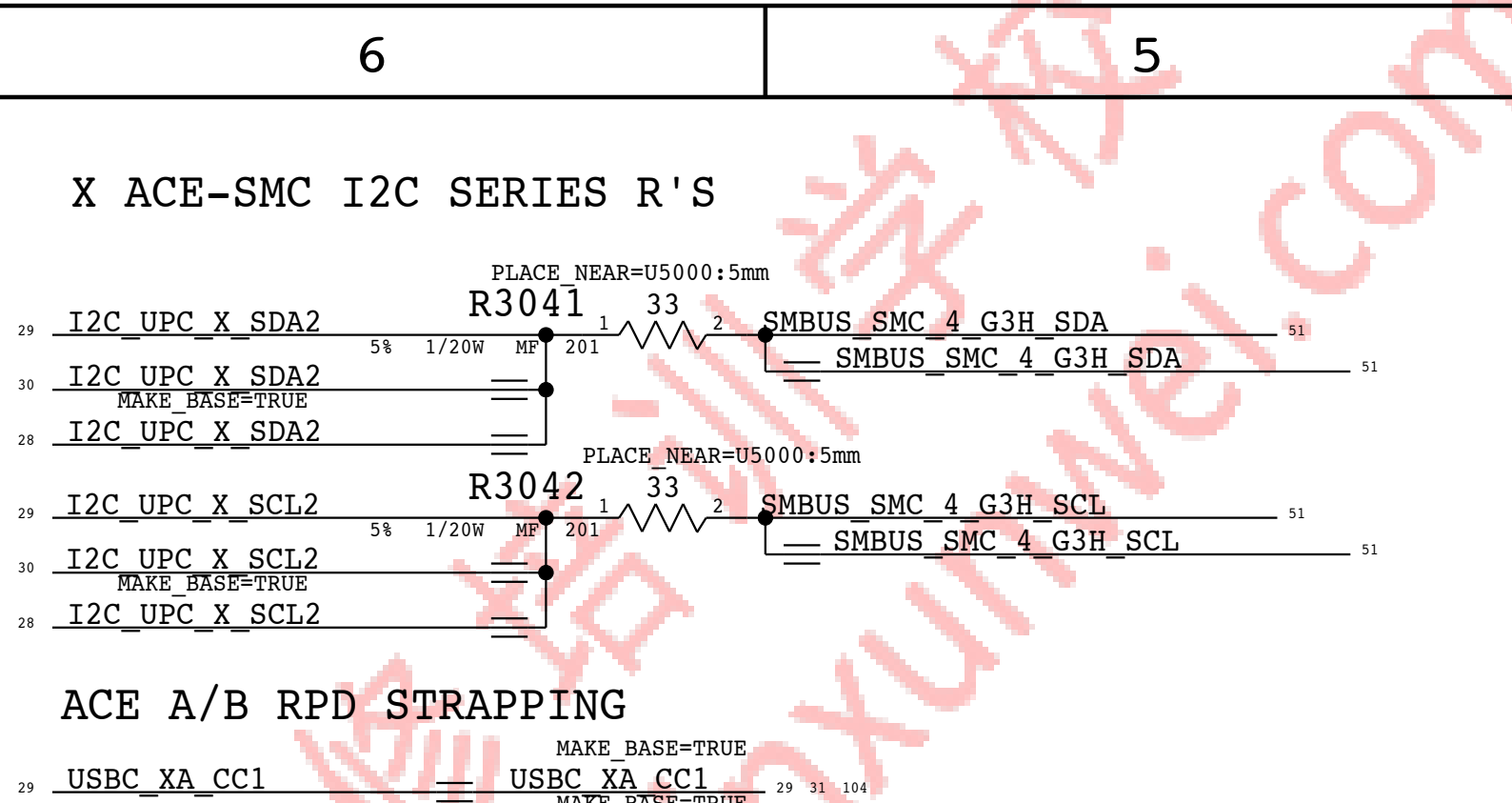
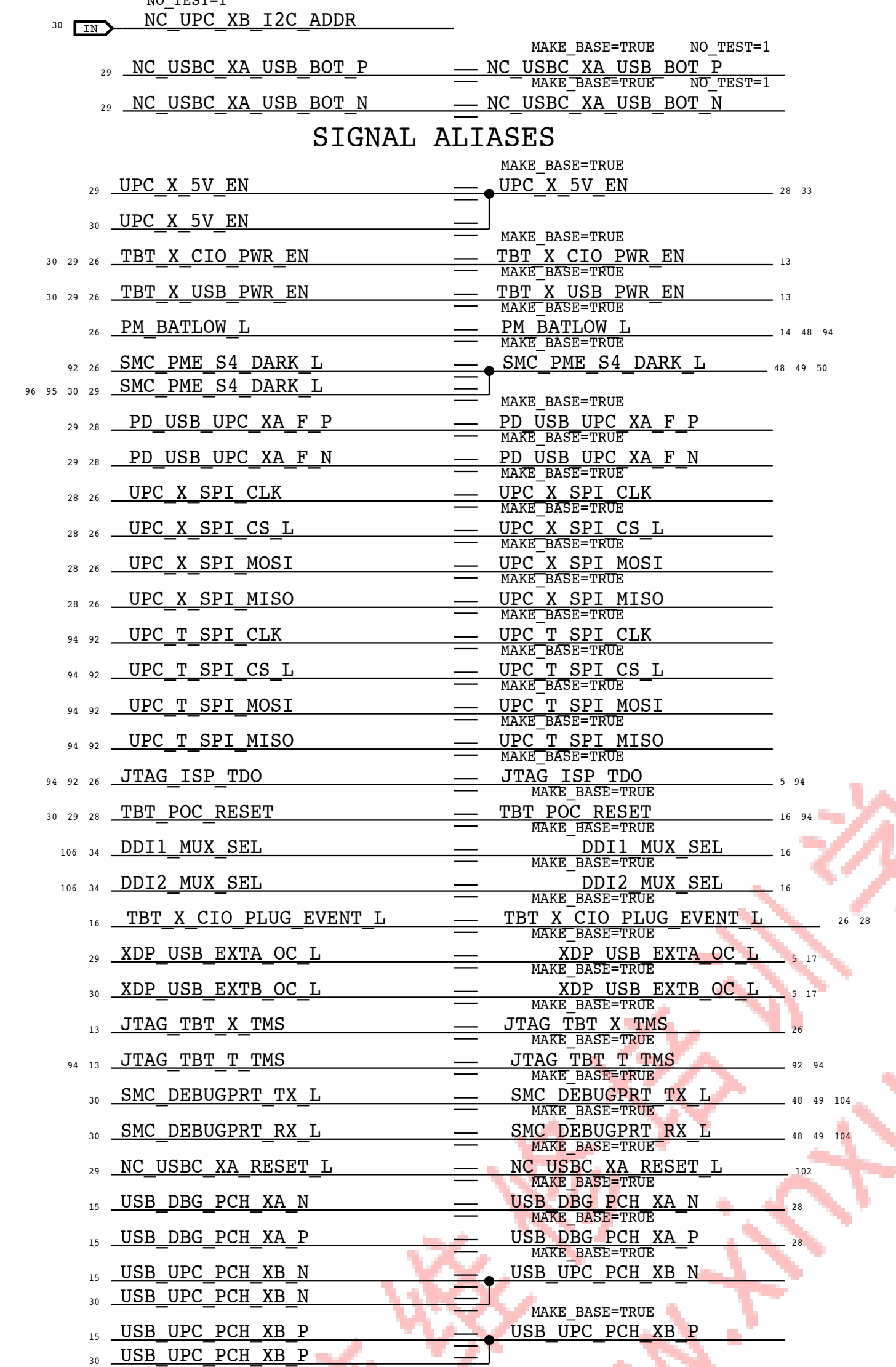
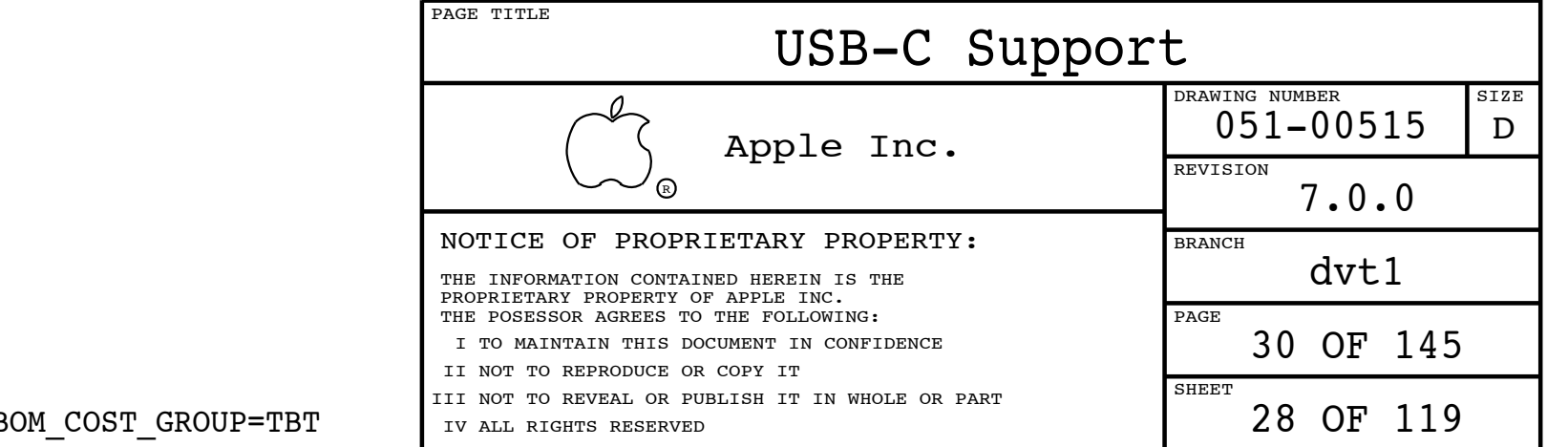
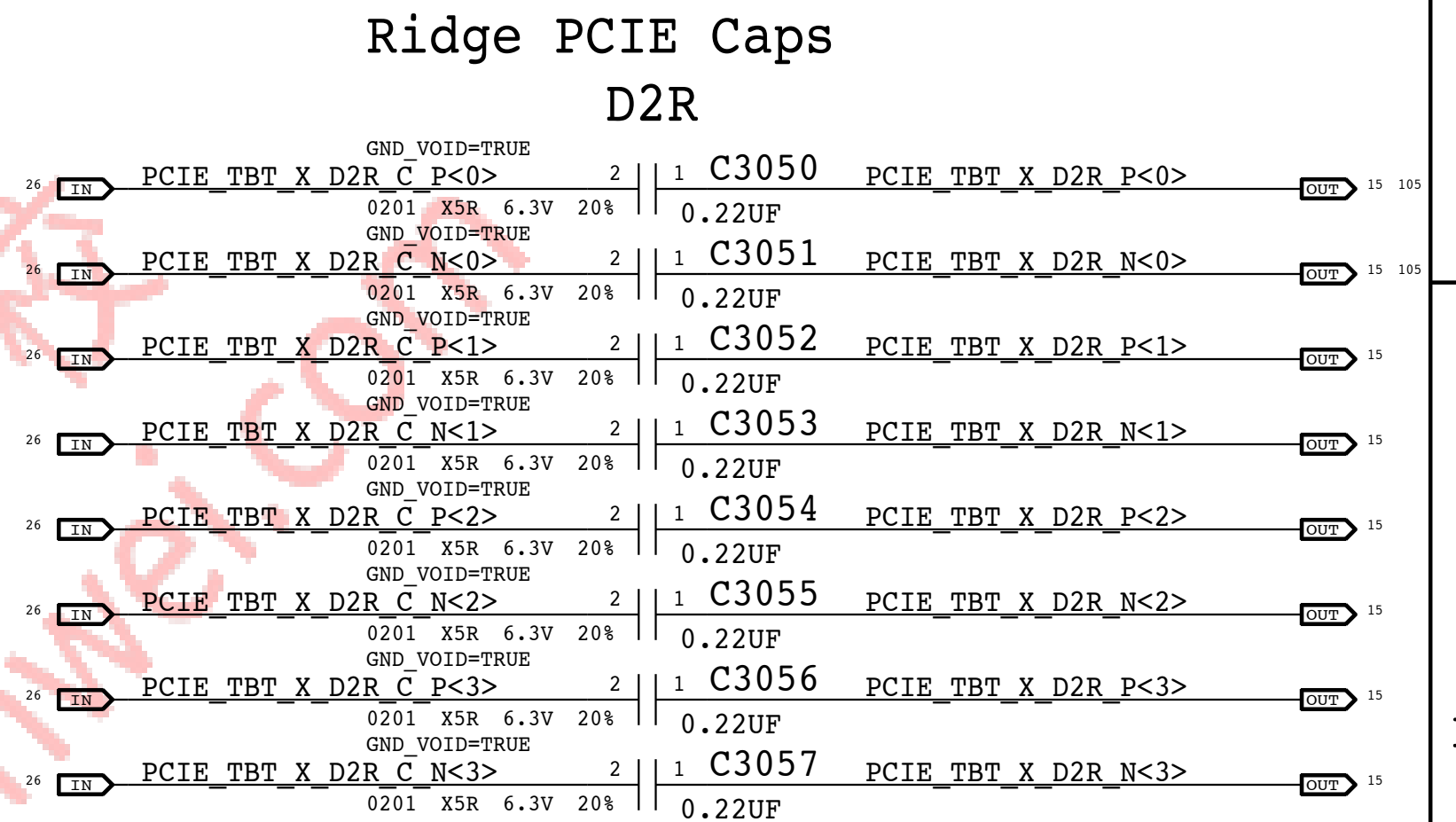
Intel recommends 68 Ohm for CMD/ADDR, 80 Ohm for CTRL/CKE, 38 Ohm for CLK



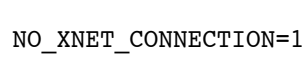
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		BRANCH	dvt1
		PAGE	27 OF 145
		SHEET	25 OF 119

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BOM_COST_GROUP=USB-C



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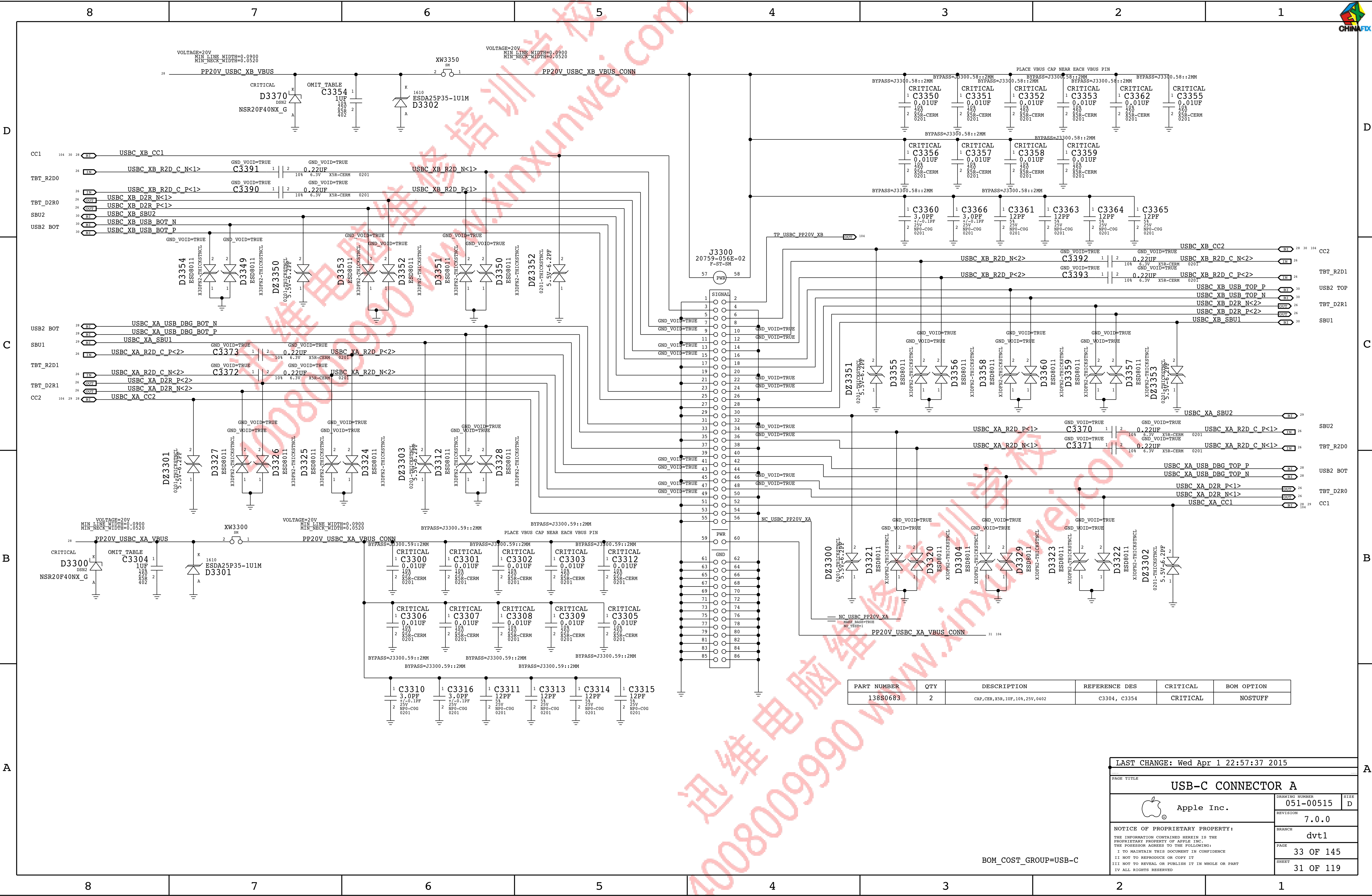
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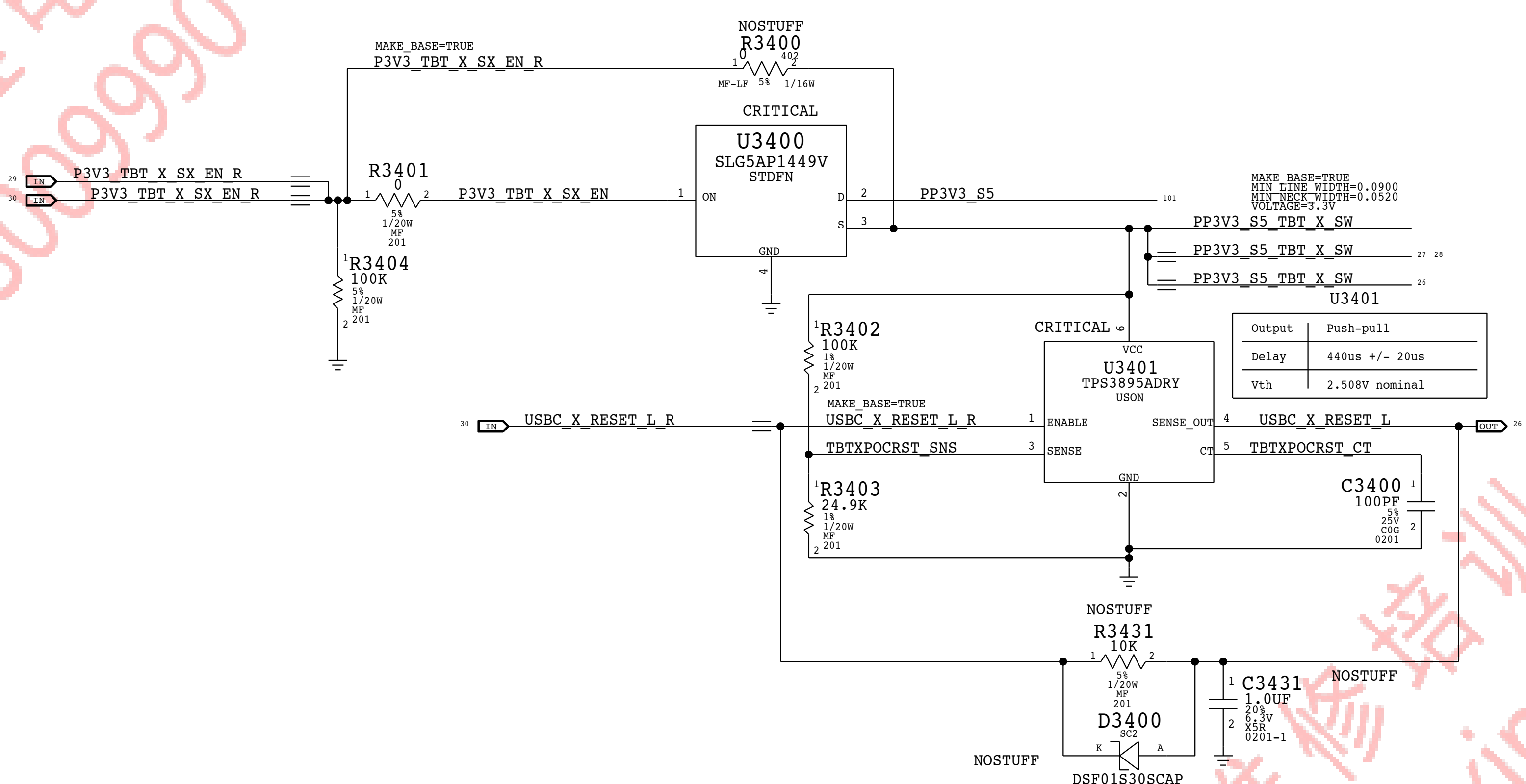
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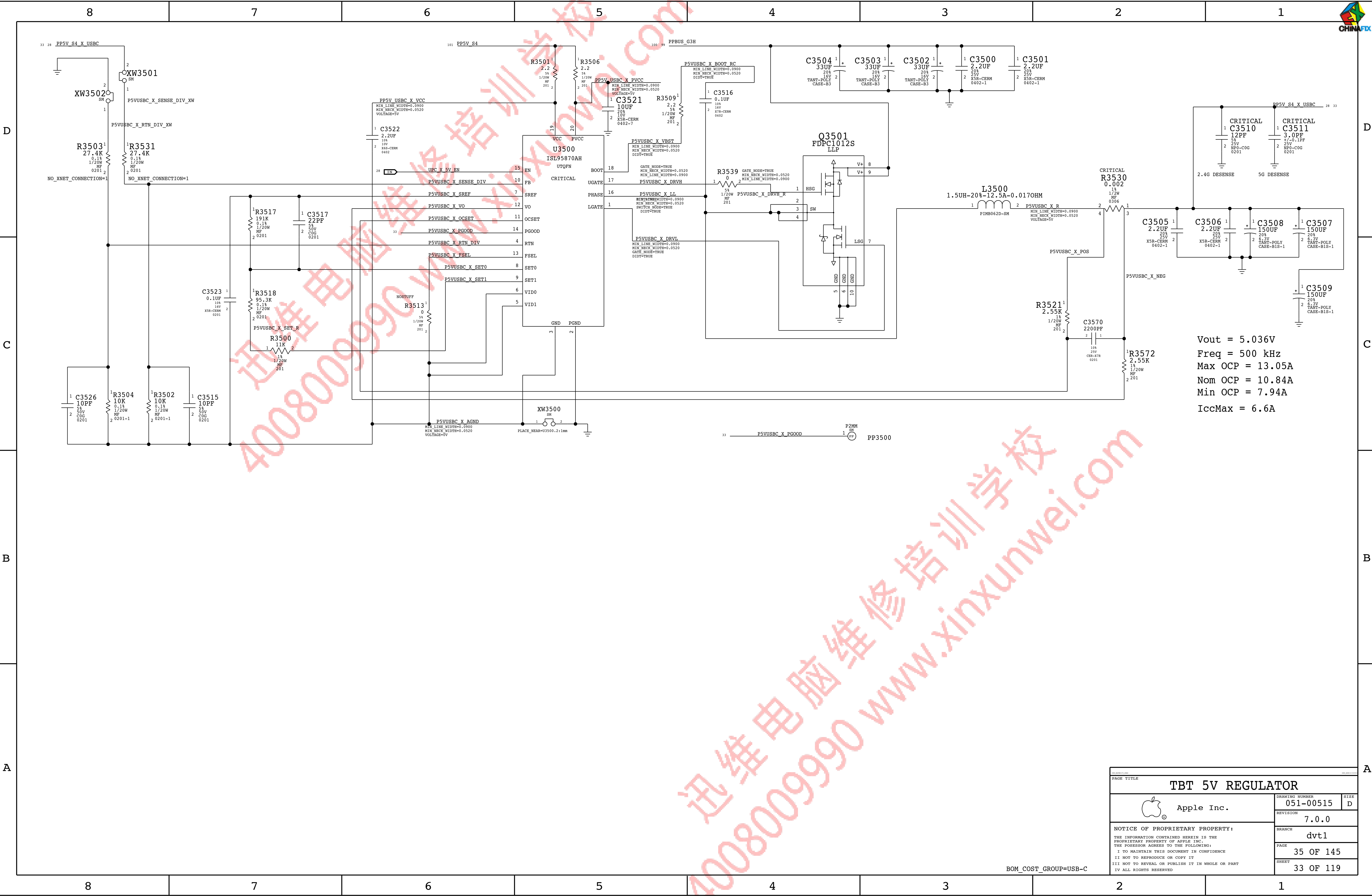
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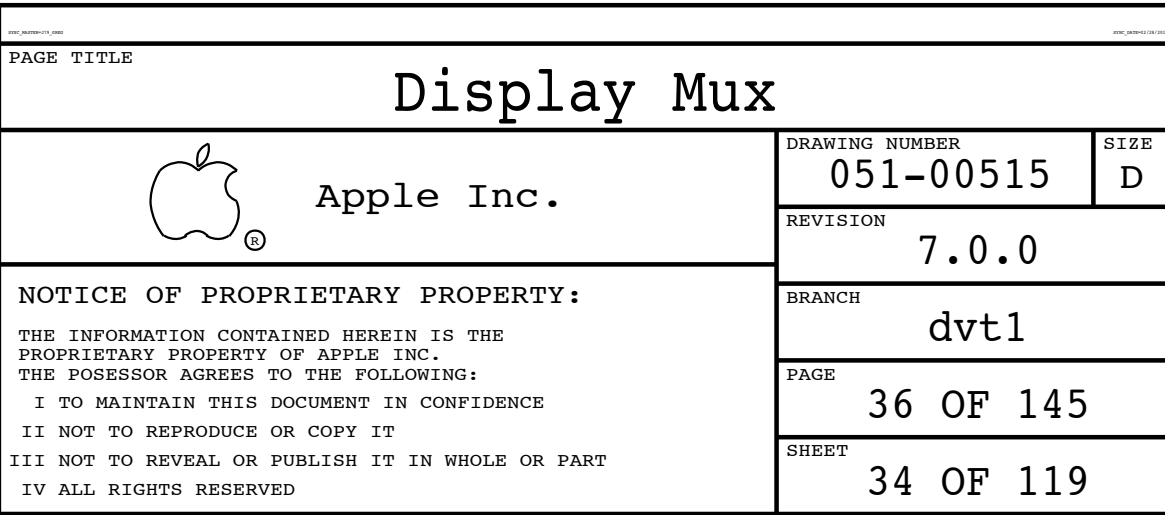


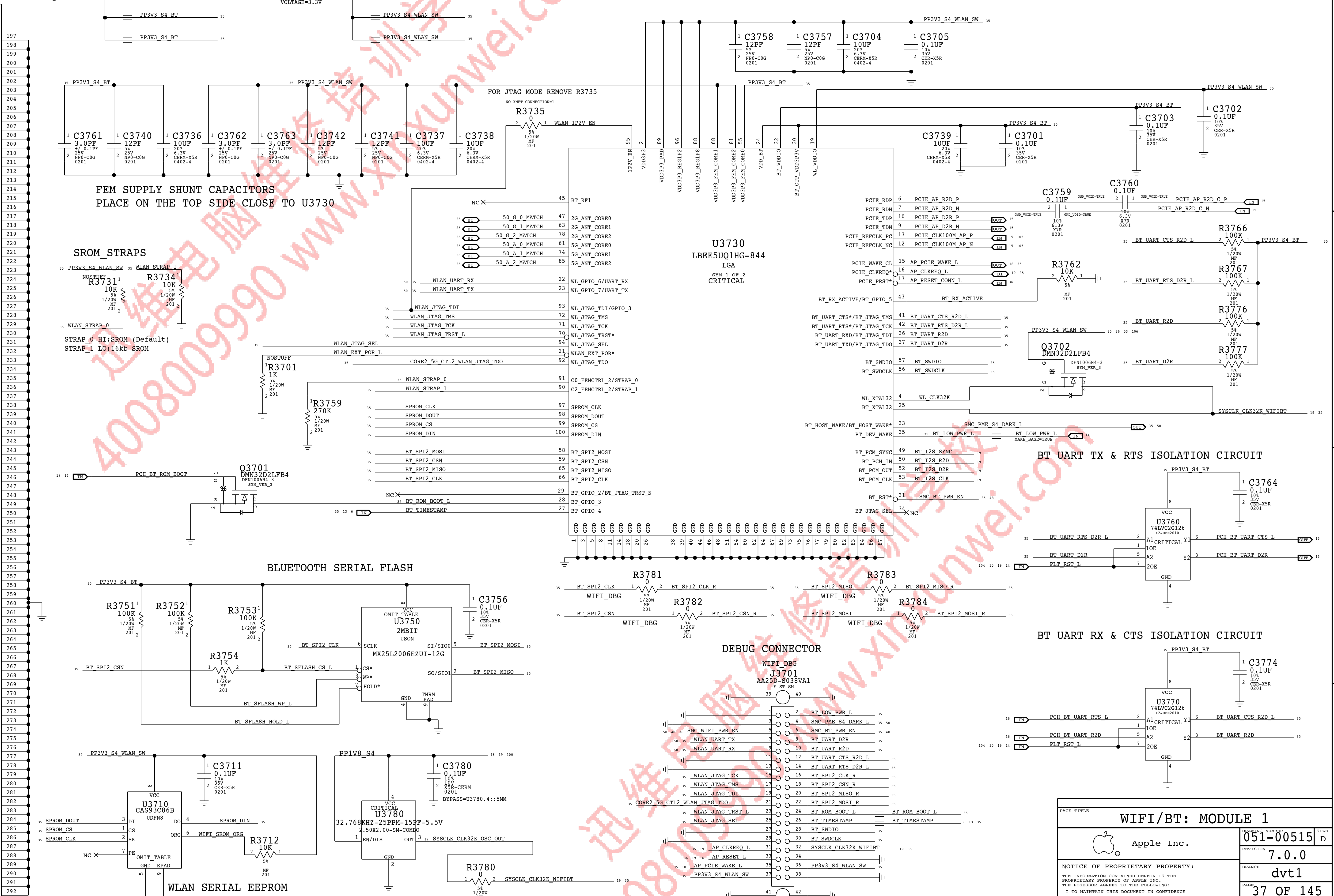
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LAST CHANGE: Wed Feb 18 17:12:24 2015			
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		REVISION 7.0.0	
		BRANCH dvt1	
		PAGE 34 OF 145	
		SHEET 32 OF 119	



PAGE TITLE		
TBT 5V REGULATOR		
 Apple Inc.	DRANTING NUMBER	051-00515
	REVISION	7.0.0
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	PAGE	35 OF 145
	SHEET	33 OF 119

BOM_COST_GROUP=USB-C





PAGE TITLE		
WIFI/BT: MODULE 1		
 Apple Inc.	DRAWING NUMBER	051-00515
	SIZE	D
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	BRANCH	dvt1
	PAGE	37 OF 145
	SHEET	35 OF 119

BOM COST GROUP=WIRELESS

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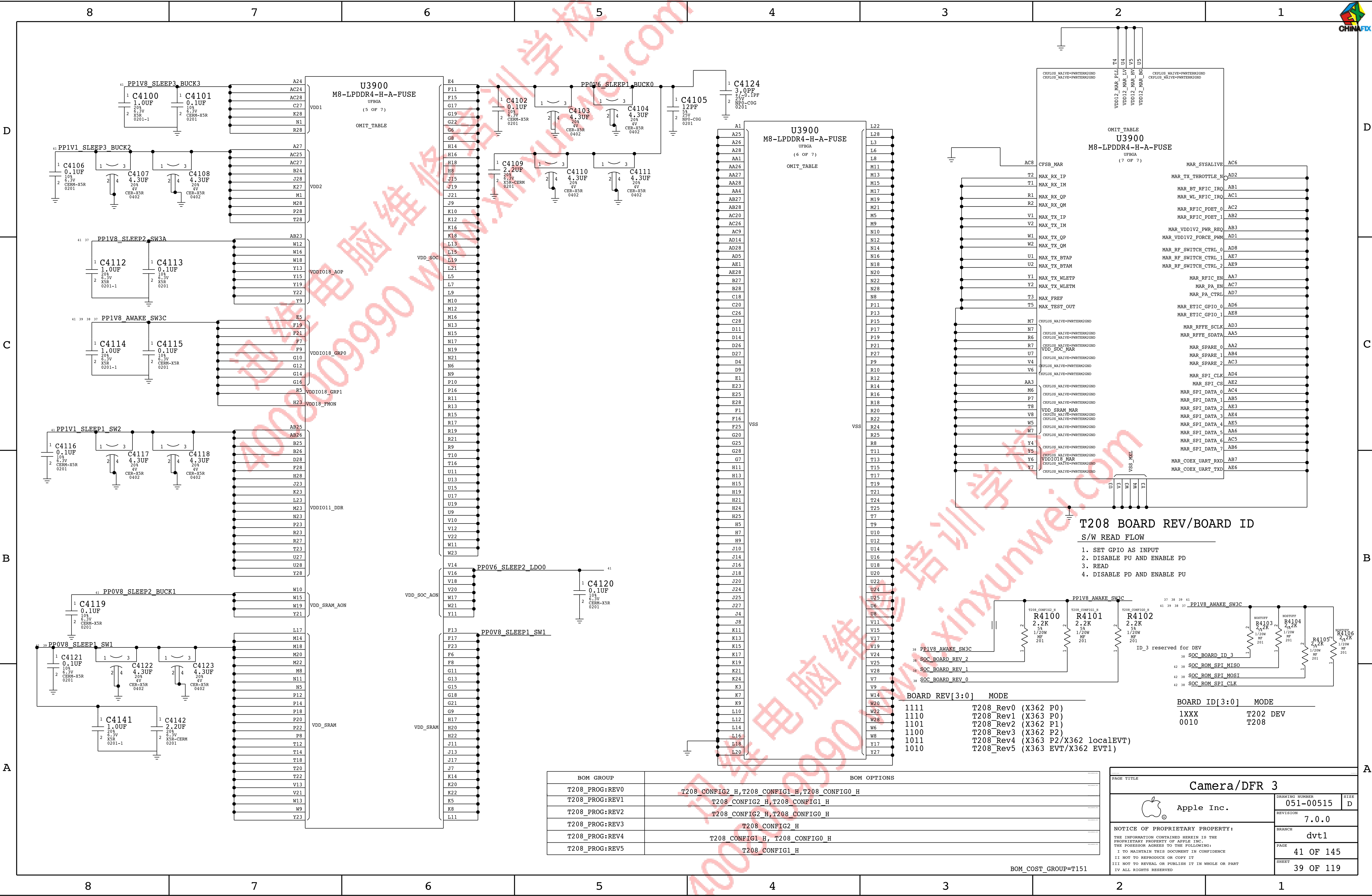
8 SEP ROM



STOCKHOLM DWLD REQ

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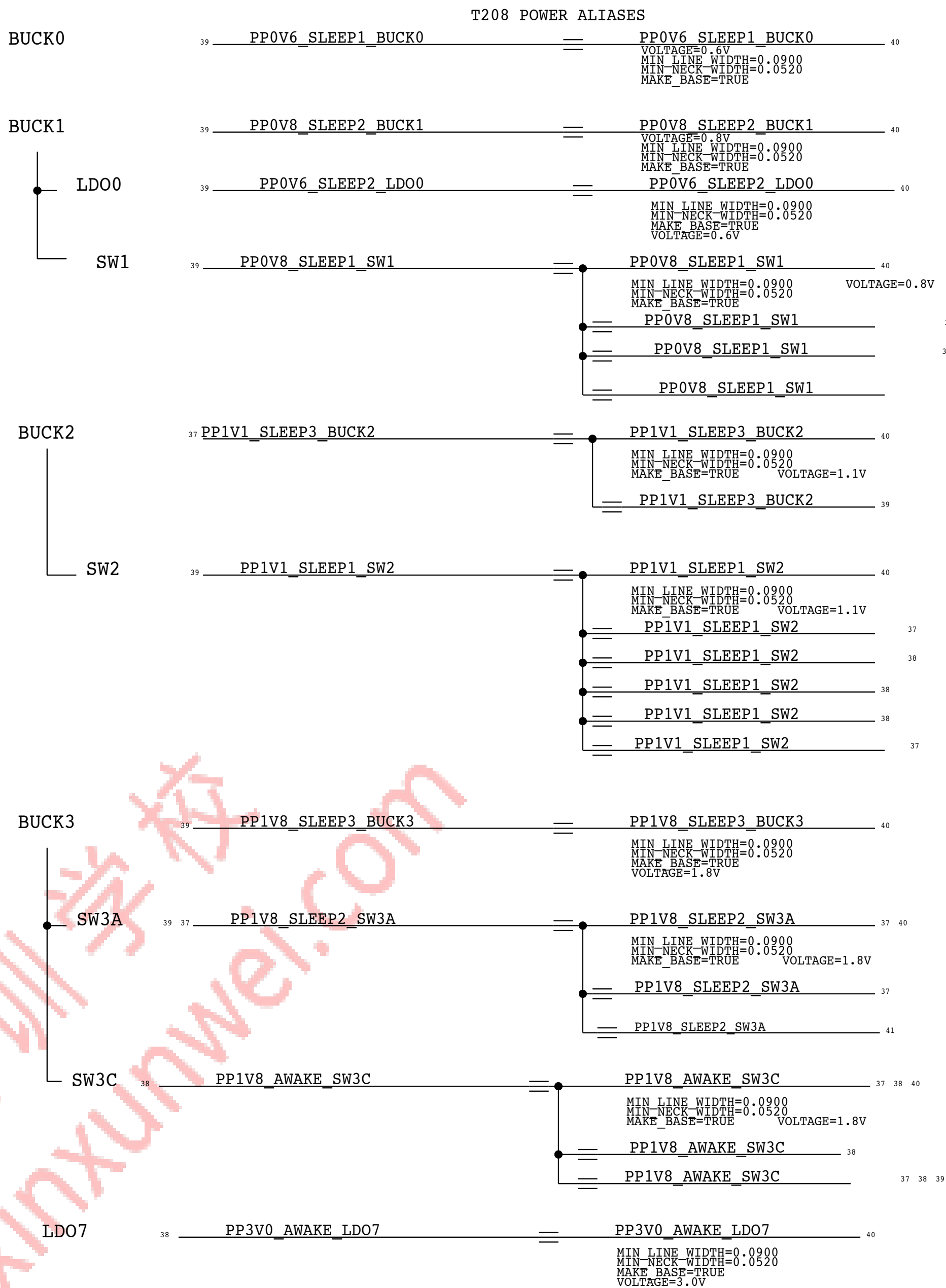
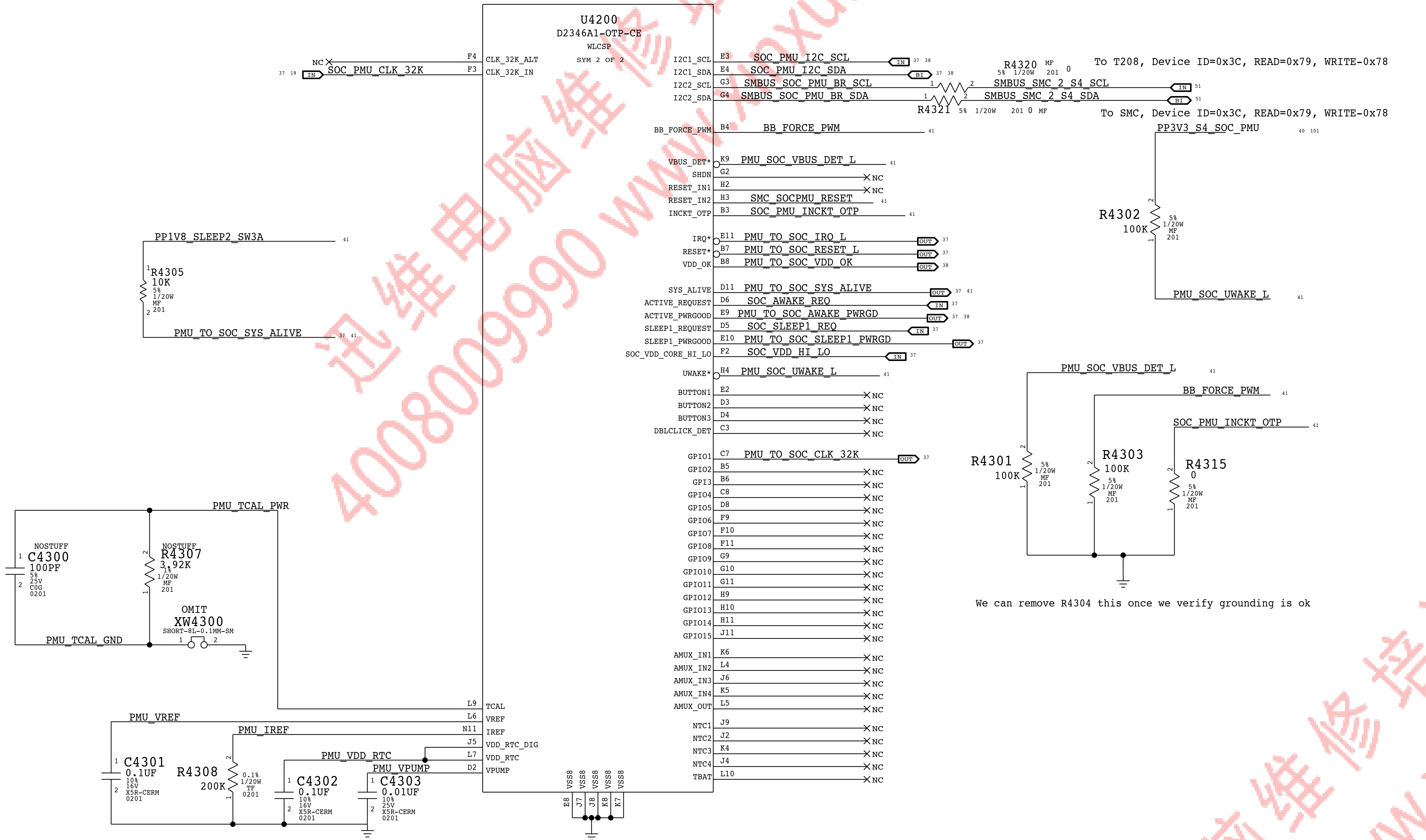




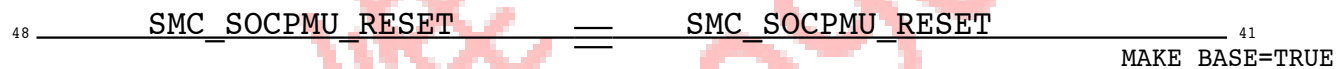


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Berkelium - 2



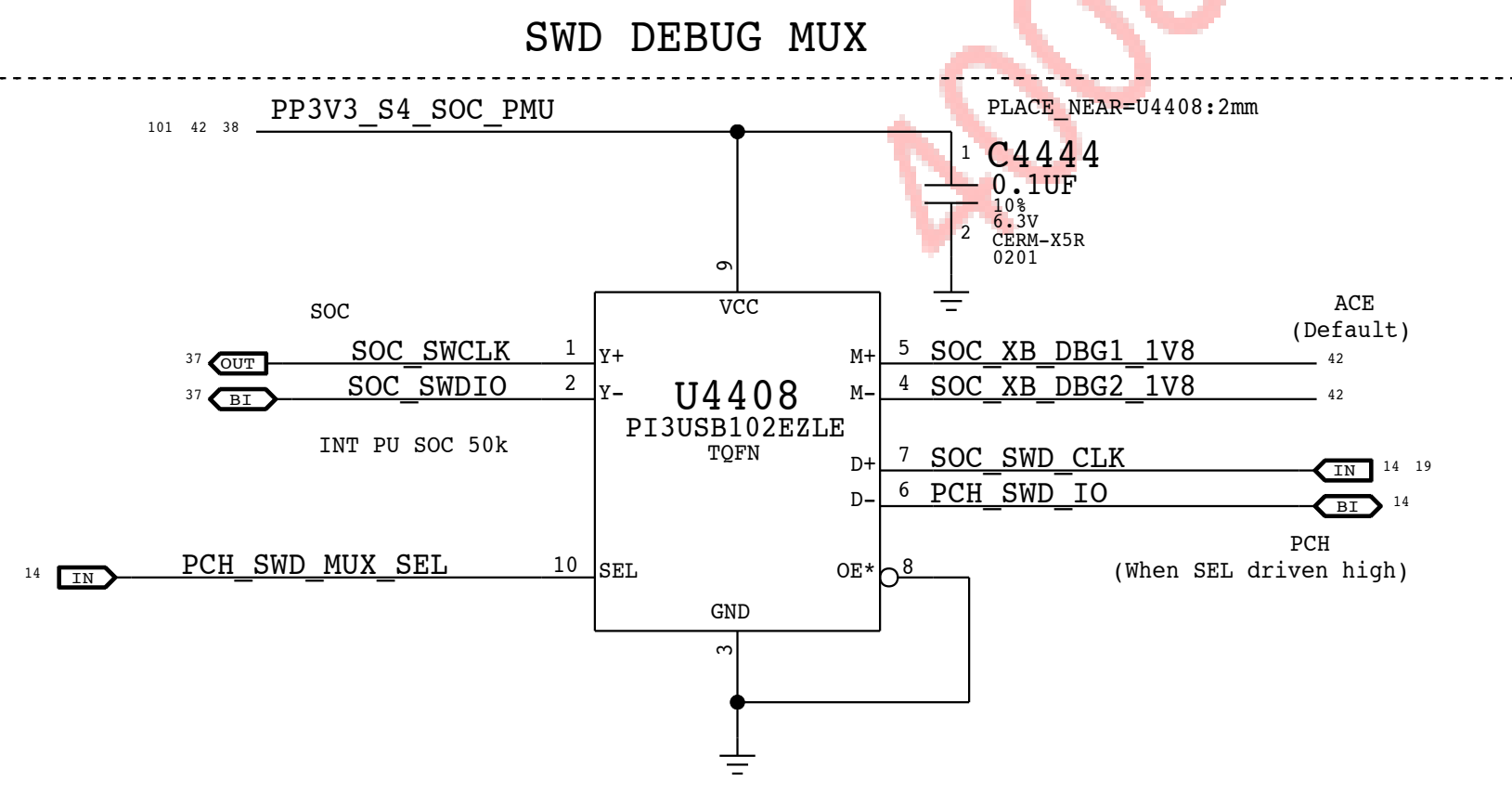
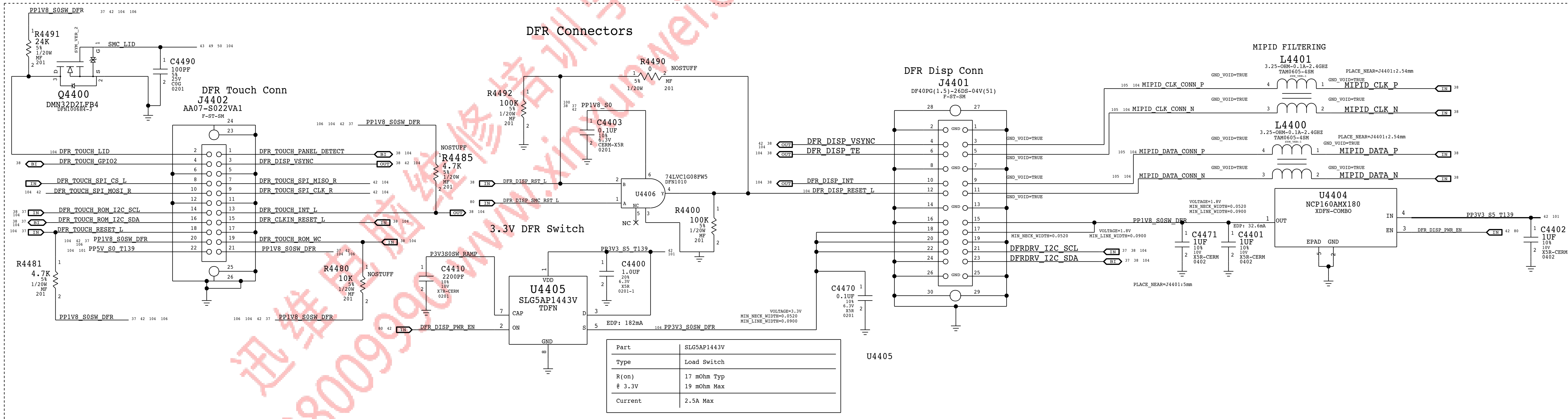
Signal Aliases



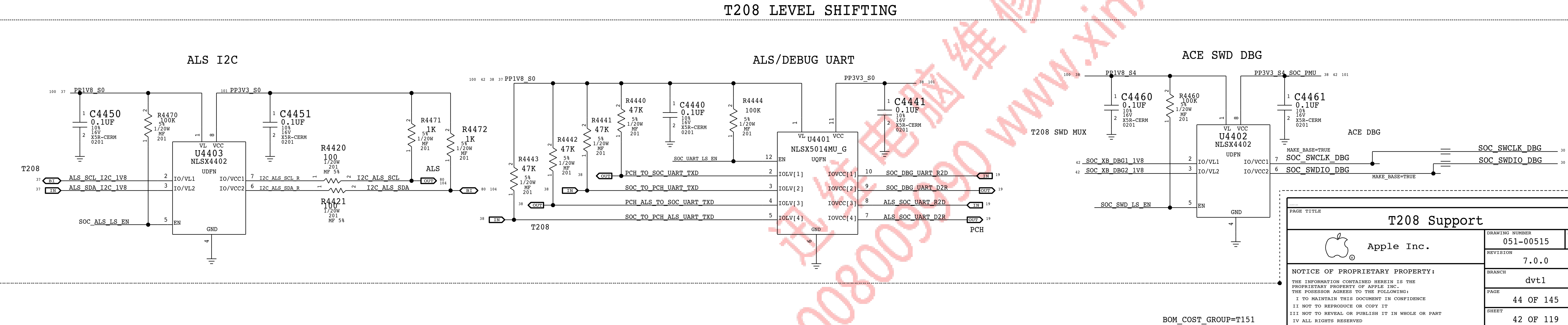
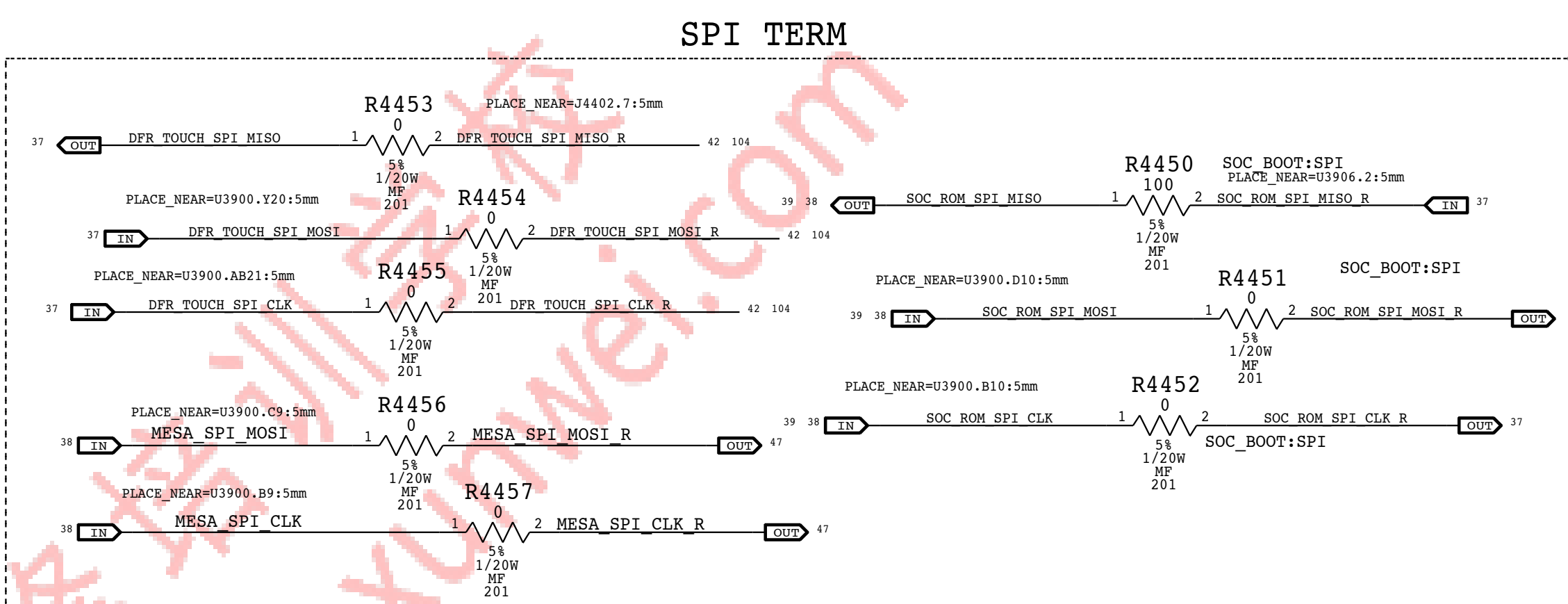
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	REVISION	7.0.0
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	PAGE	43 OF 145
	SHEET	41 OF 119

BOM_COST_GROUP=T151

T208 Support

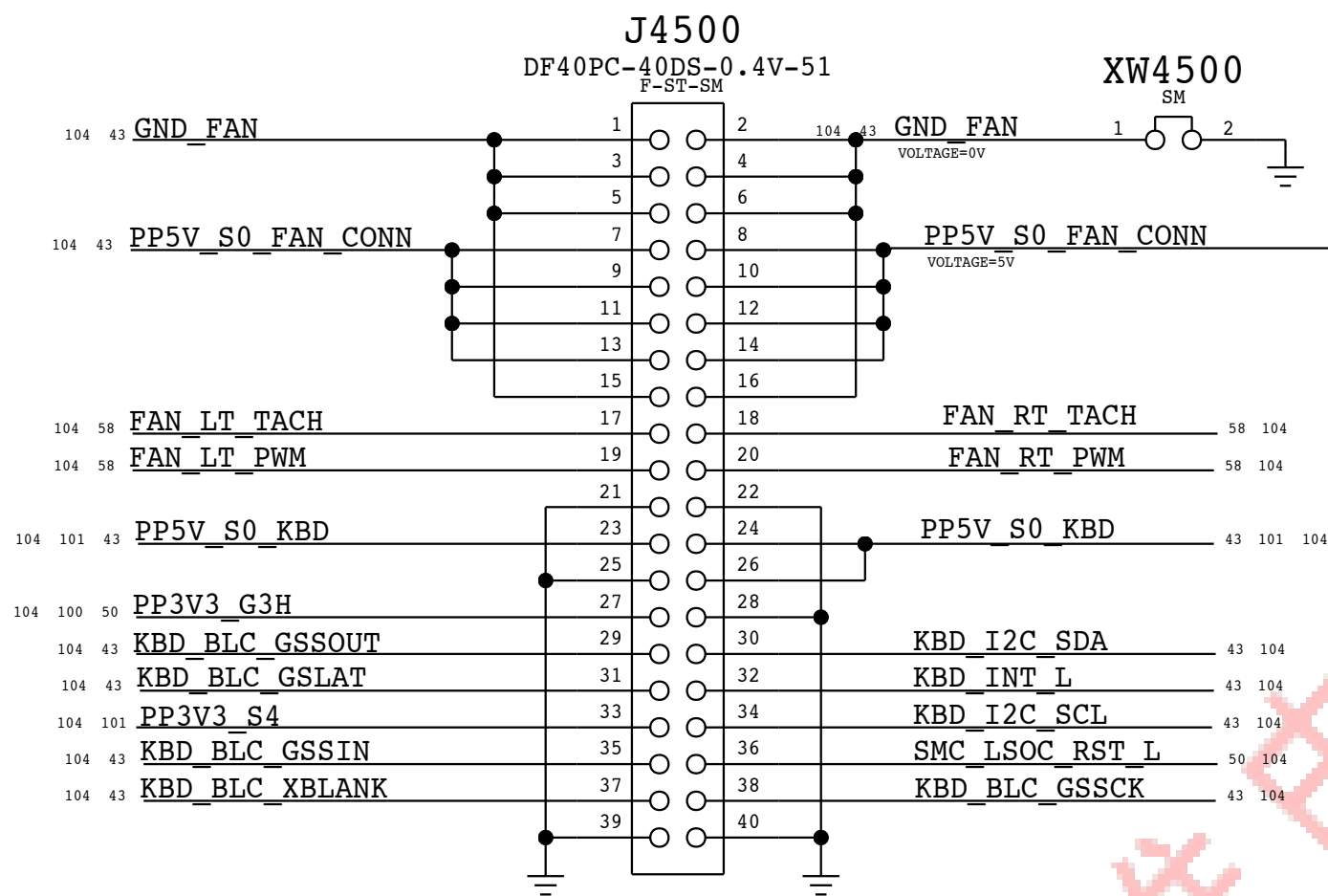


T208 I2C Mapping					
Bus	Device	7-bit Address	8-bit Address		
			Read	Write	
AP0	PMU	0011110 (0x3C)	0x79	0x78	
AP1	Touch EEPROM	1010000 (0x50)	0xA1	0xA0	
AP2_0	Tesla	1010100 (0x4C)	0x99	0x98	
AOP0	Mesa EEPROM	101000x (0x50/0x51)	0xA1/A3	0xA0/A2	
AOP1	ALS	0111001 (0x39)	0x73	0x72	
SEP	M34128 EEPROM	1010001 (0x51)	0xA3	0xA2	



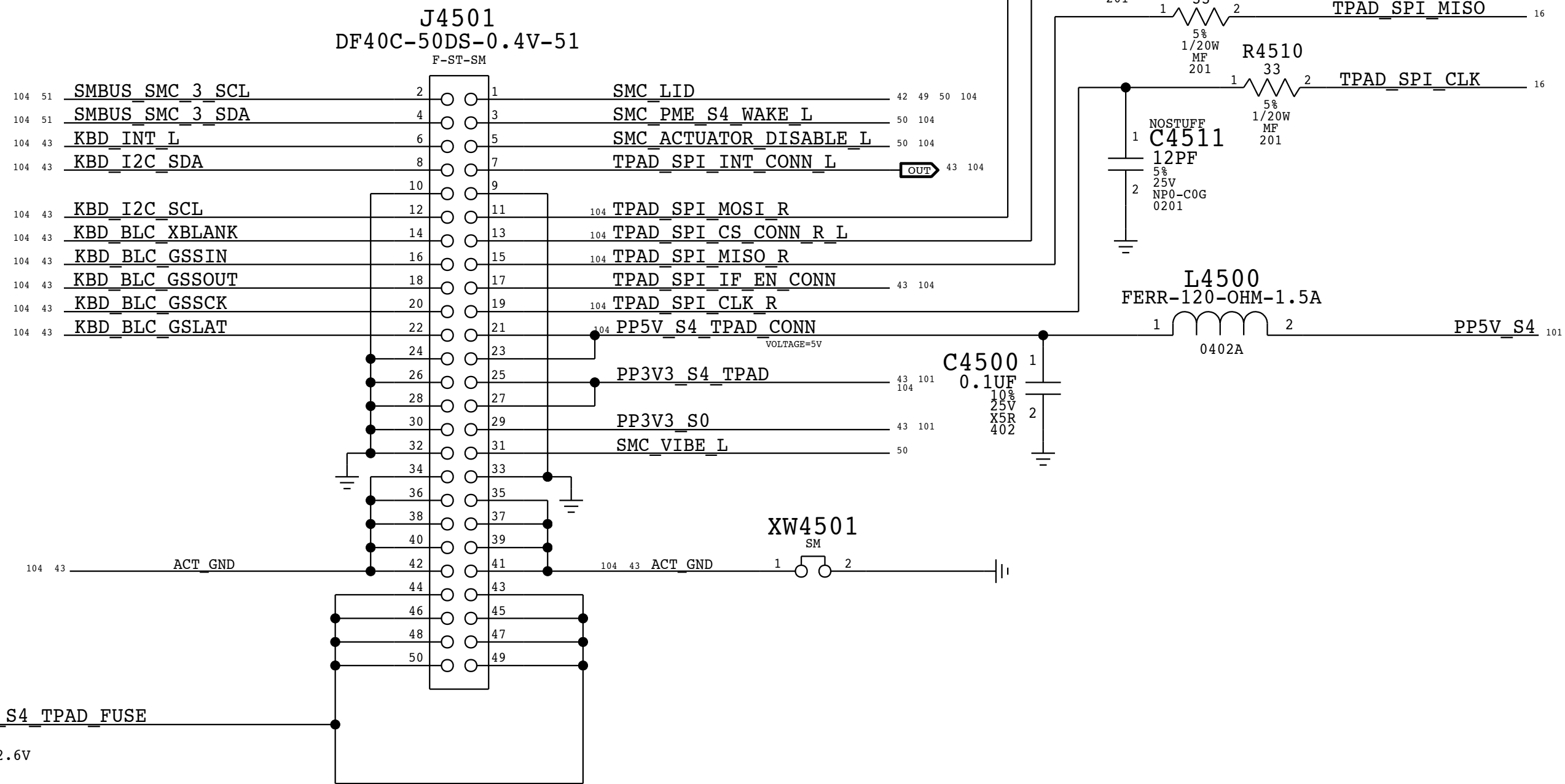


KBD CONNECTOR



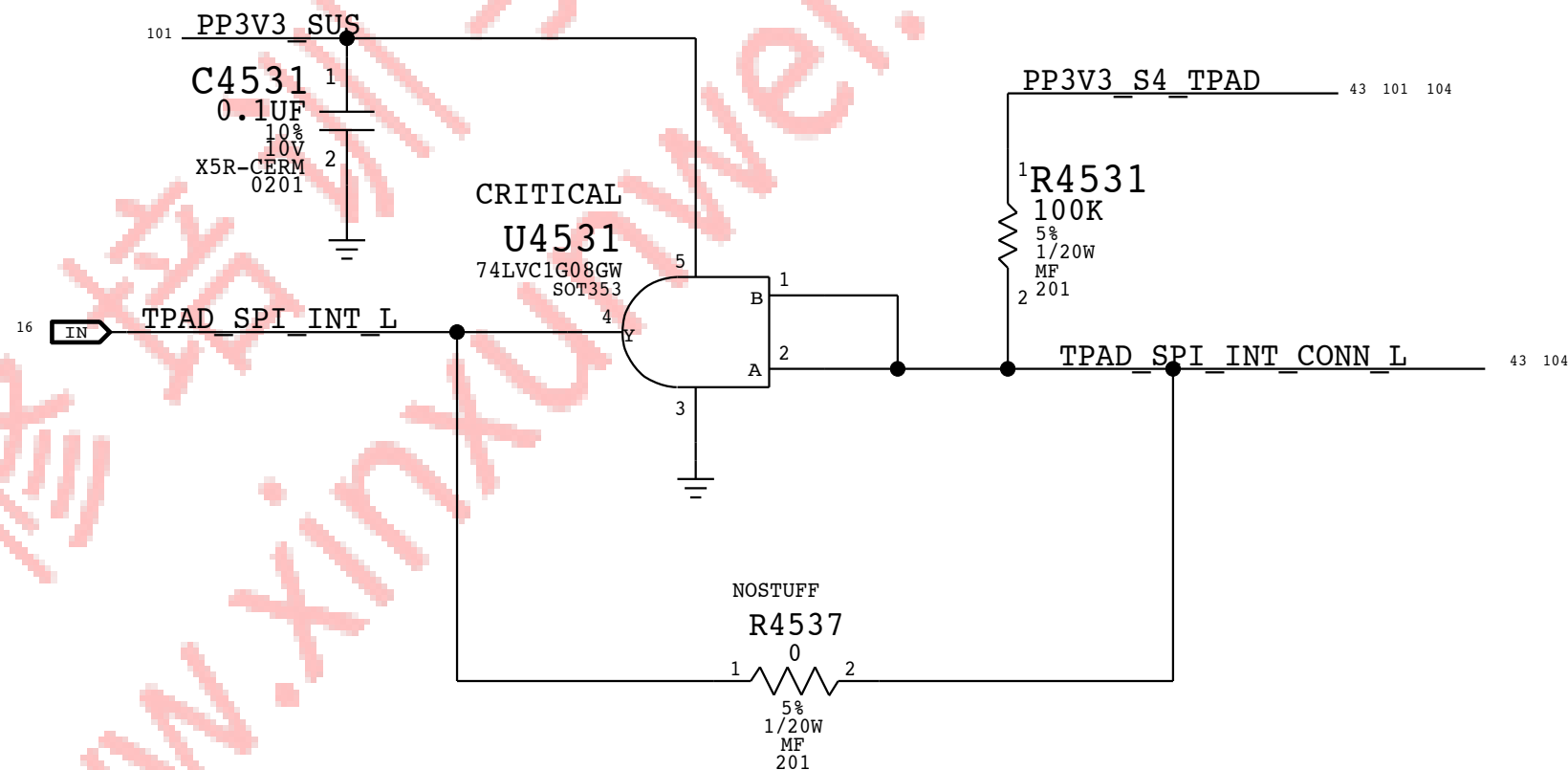
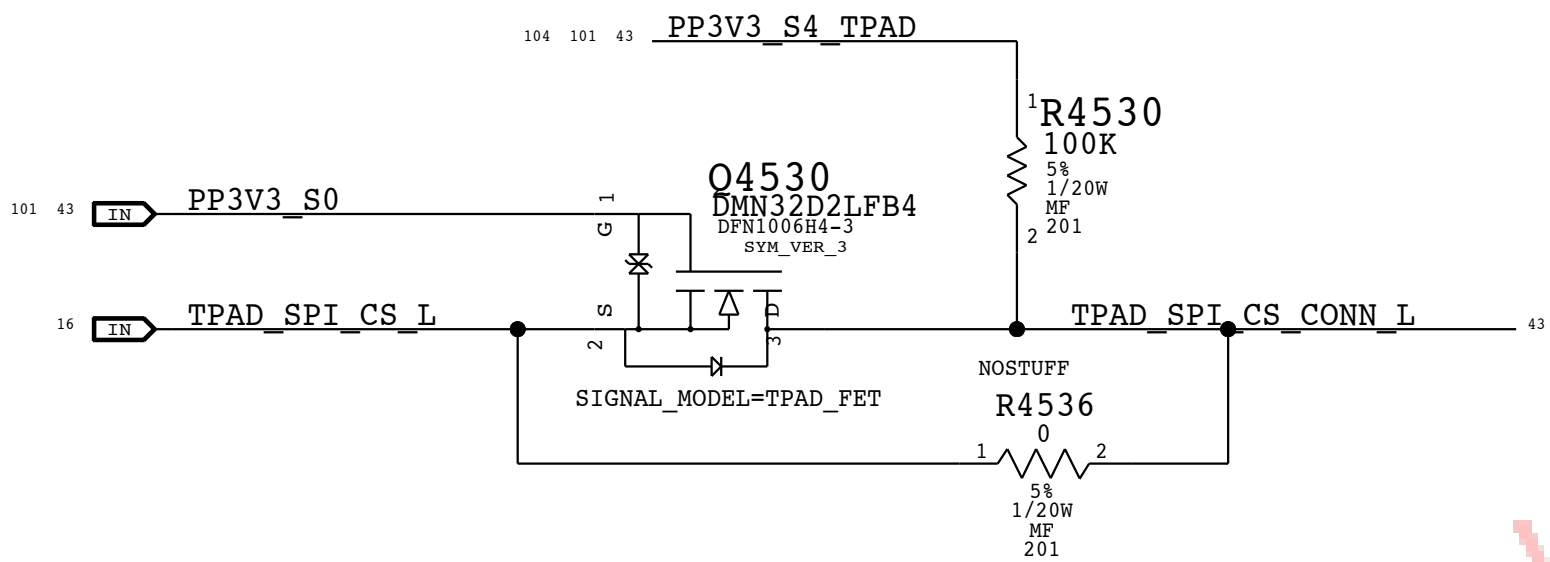
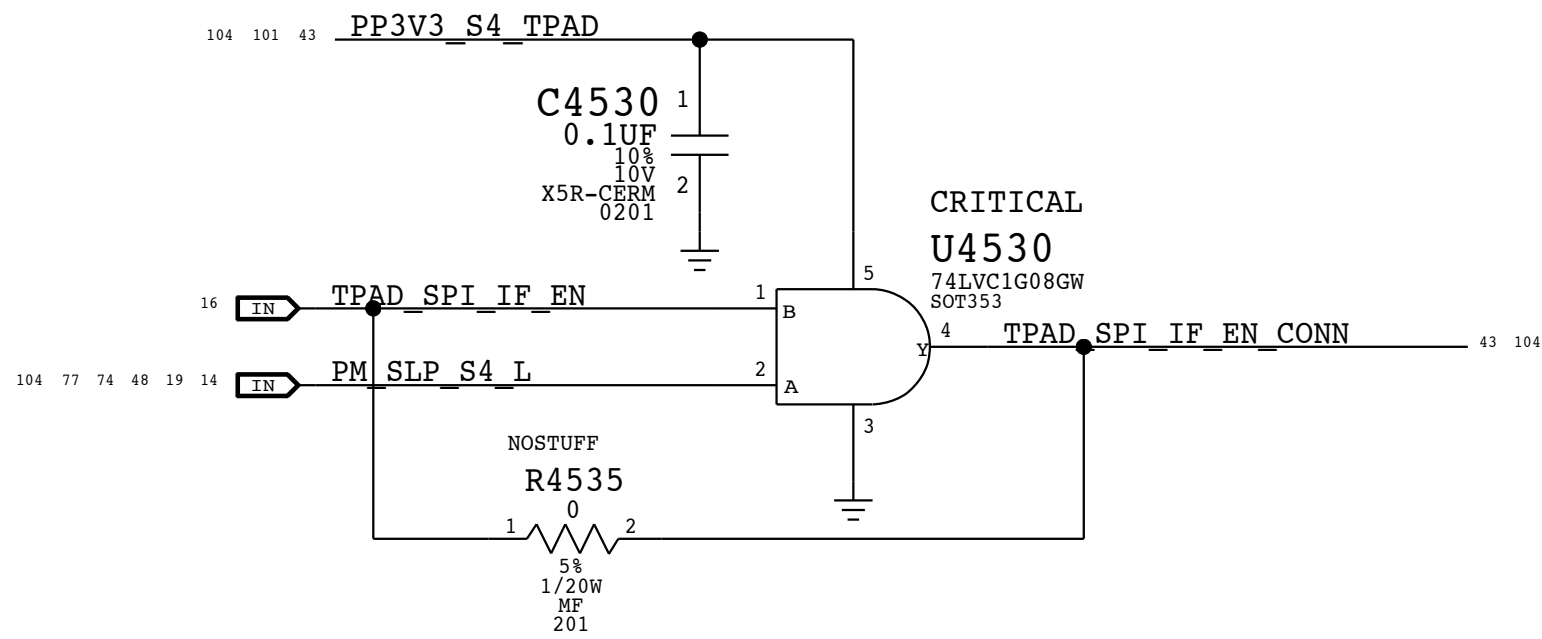
516S00177 (RCPT, 0.3A per pin)
MATE WITH PLUG 516S00054

TPAD CONNECTOR



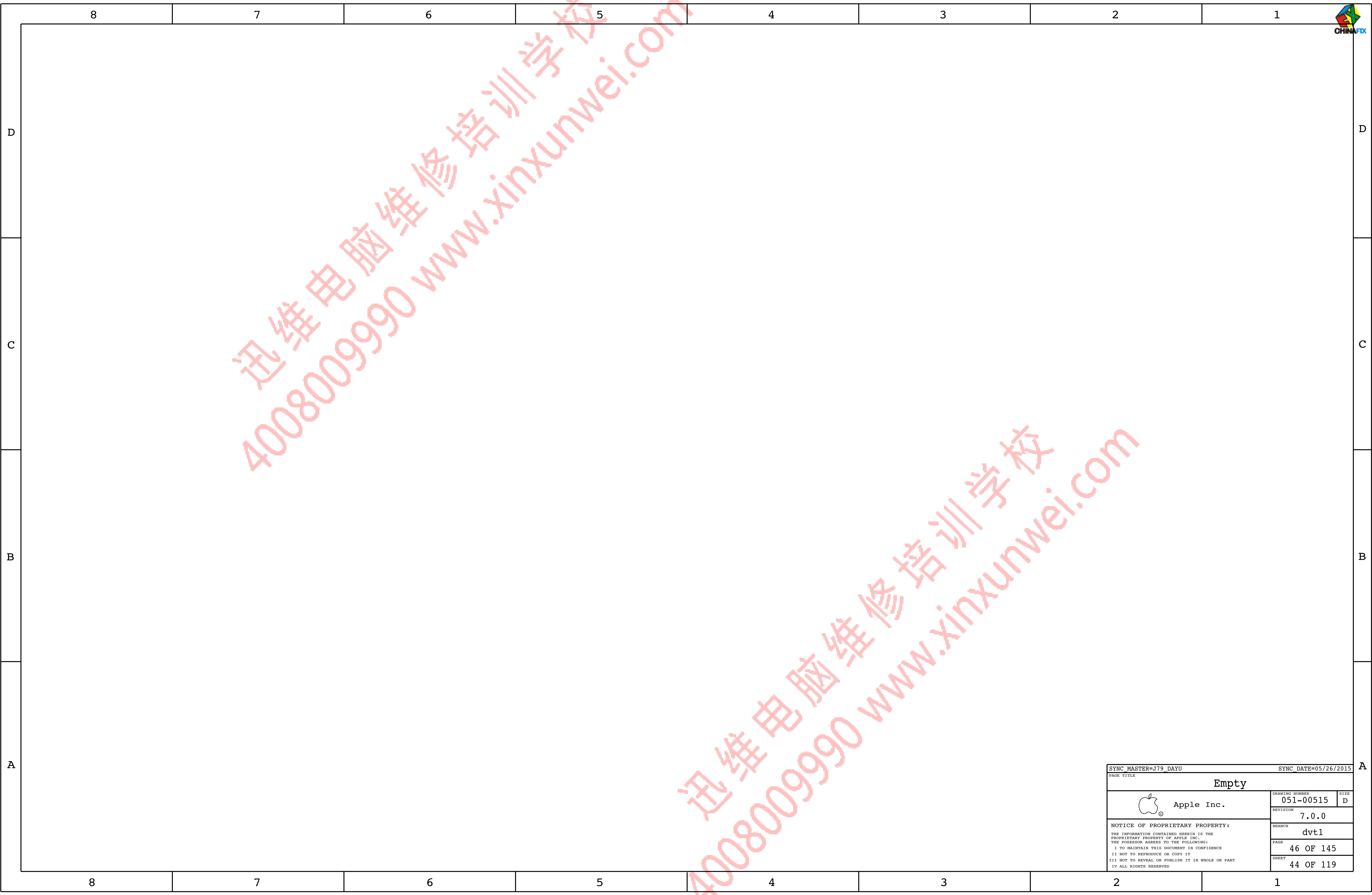
516S00187, MATE WITH 516S00188

TRACKPAD ISOLATION

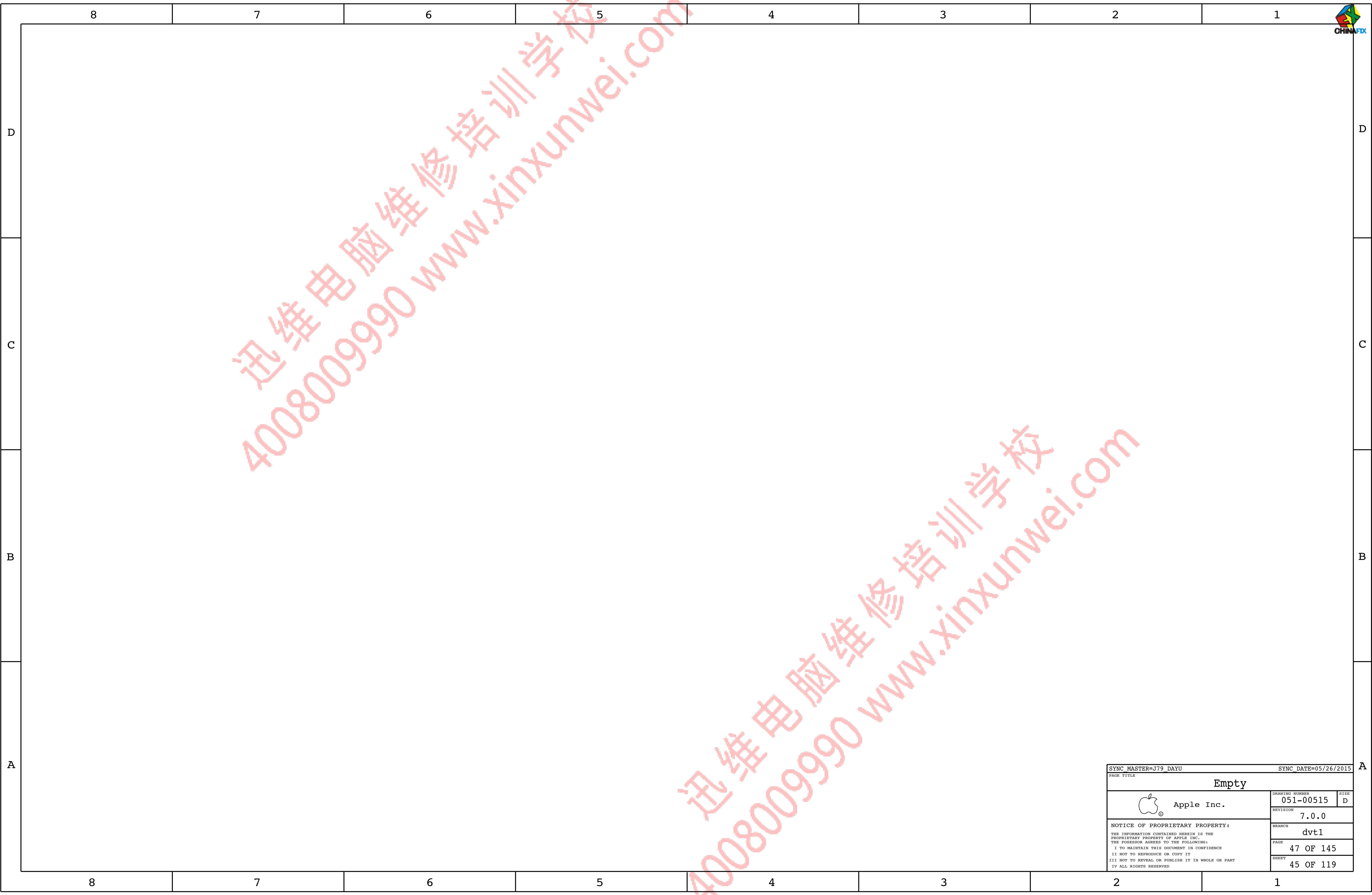


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		SHEET	43 OF 119

BOM_COST_GROUP=KEYBOARD



SYNC_MASTER=J79 DAYU		SYNC_DATE=05/26/2015	
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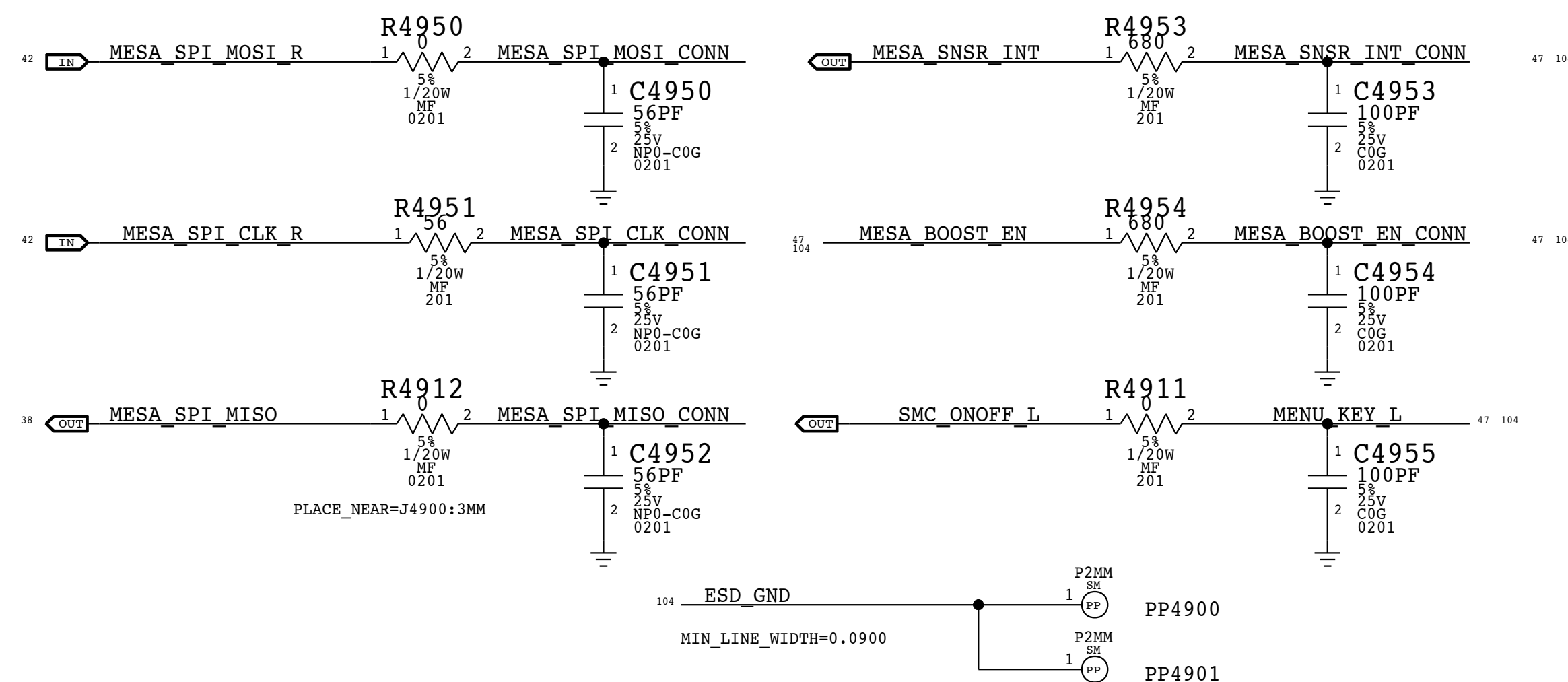


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	PAGE		47 OF 145
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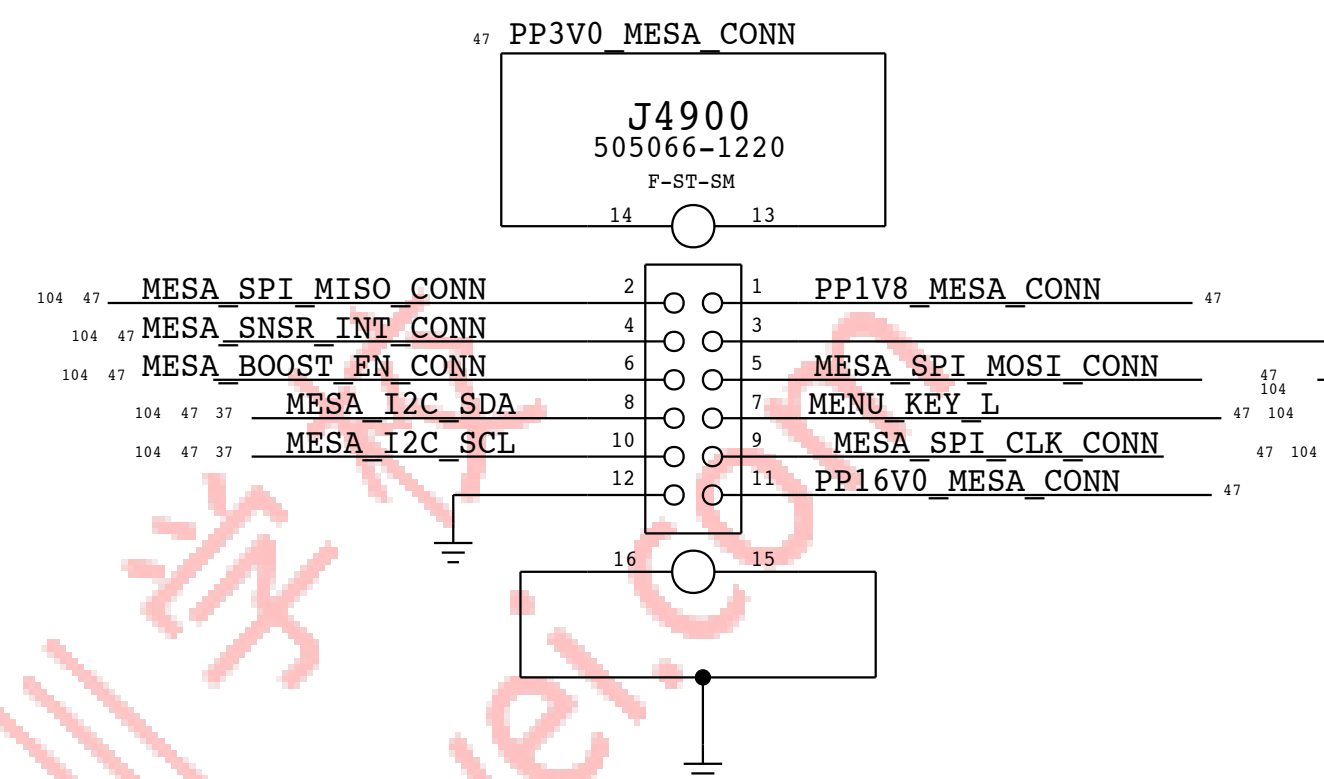
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C															
B															
A															
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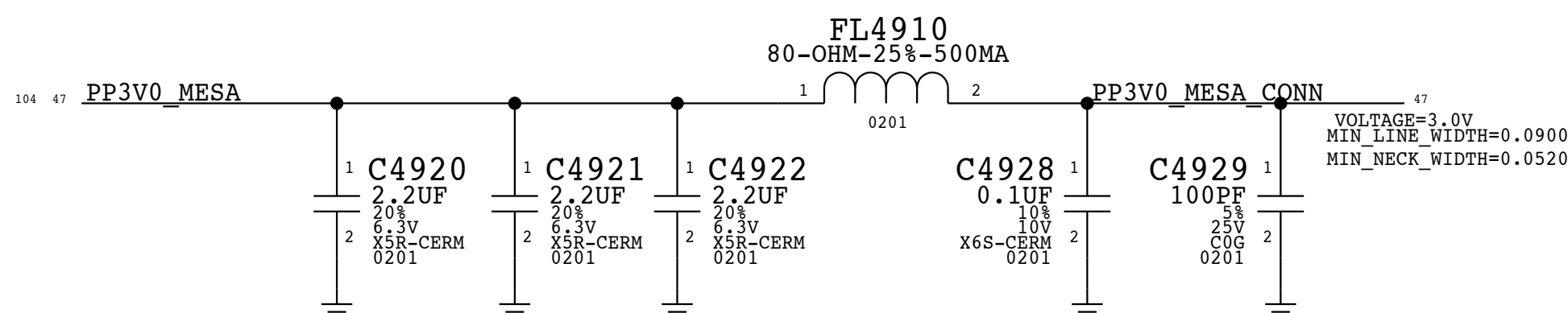
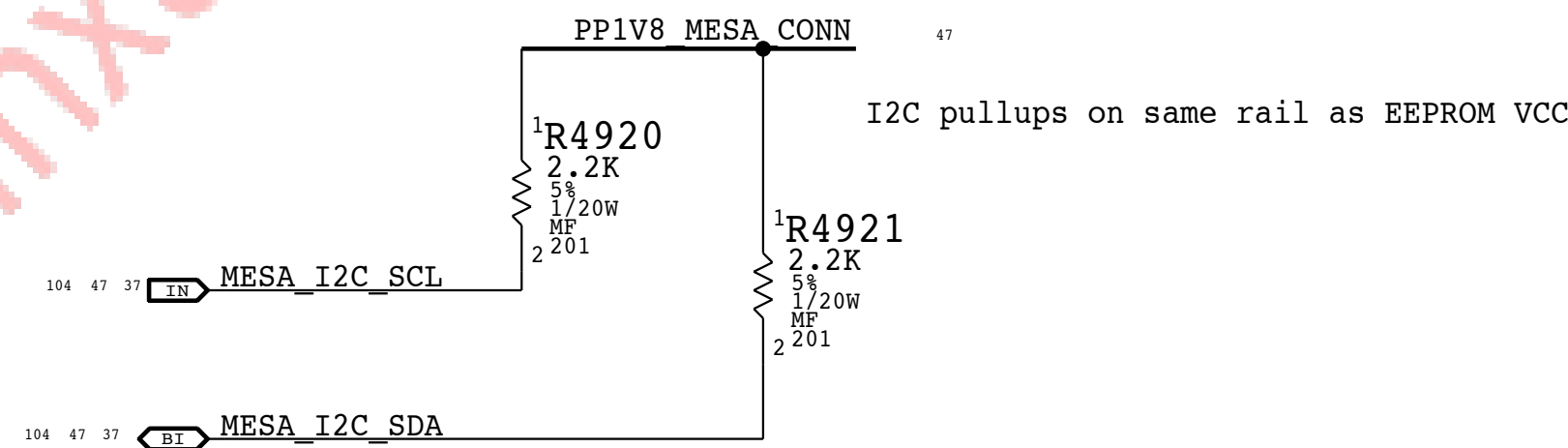
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										PAGE					48 OF 145									
										SHEET					46 OF 119									



Protol Connector for X434/X435 Support
PLUG (516S00115) - X434/ X435 Jumper
Recptacle (516S00203) - X362/X363 MLB



Power On: 1V8 -> 3V3 -> 16V0

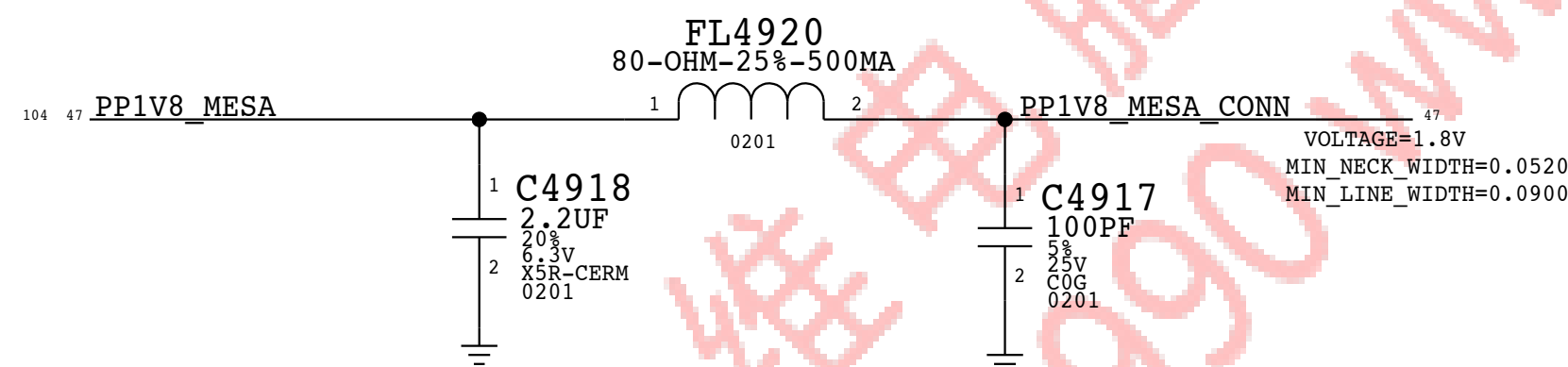


U4920
LP5907SNX-1.825
X2SON

VIN 4
VOUT 1
EN 3
GND EPAD

PP3V3_S4 MESA
C4912 1uF
MESA_PWR_EN
PP1V8 MESA
C4914 1uF
EDP:1.5mA

VOLTAGE=1.8V
MIN_NECK_WIDTH=0.0520
MIN_LINE_WIDTH=0.0900



PAGE TITLE		DRAWING NUMBER		SIZE	
MESA		051-00515		D	
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D

D

C

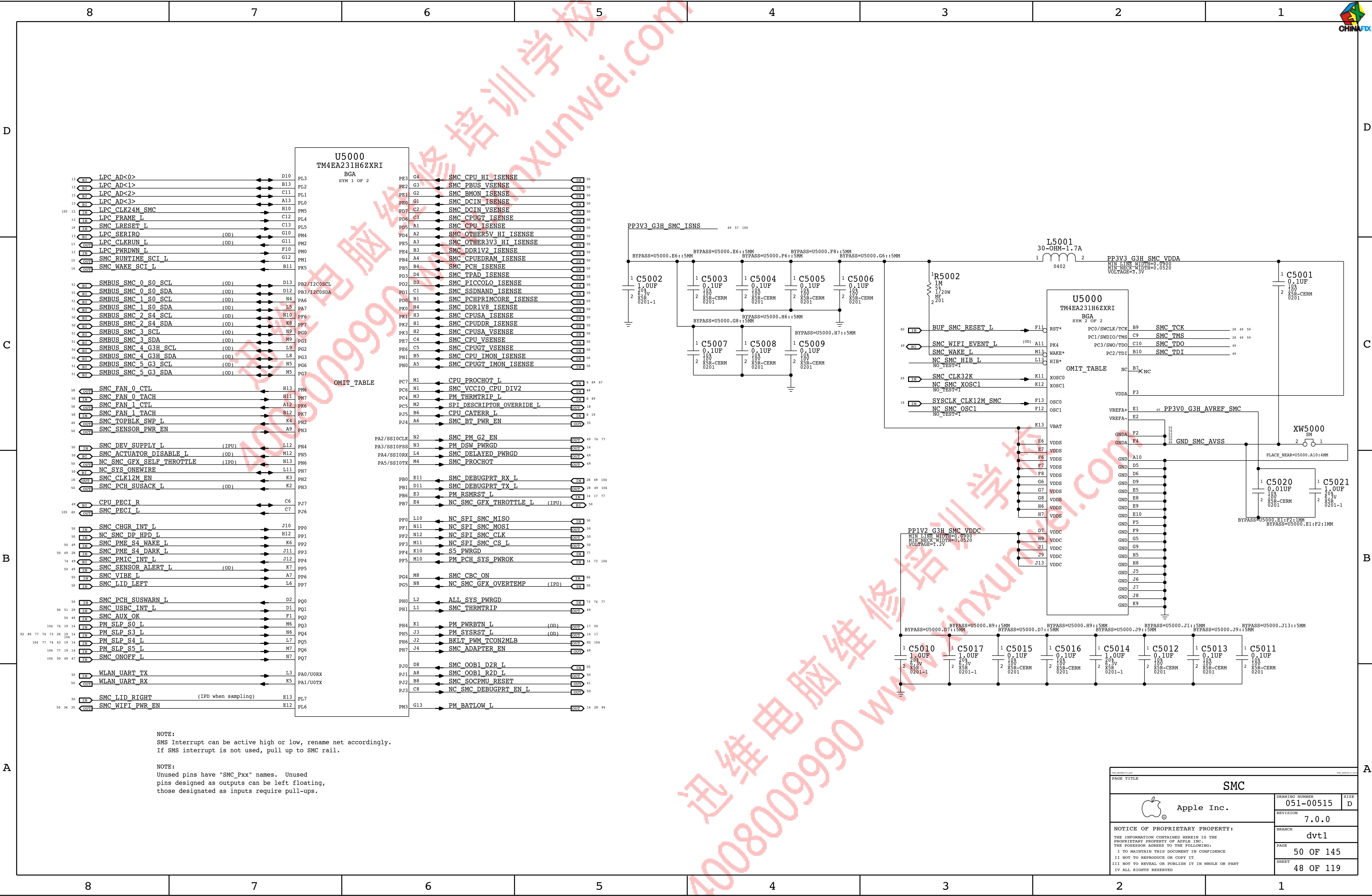
C

B

B

A

A



NOTE:
SMS Interrupt can be active high or low, rename net accordingly.
If SMS interrupt is not used, pull up to SMC rail.

NOTE:
Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

SMC			
 Apple Inc.		DRAWING NUMBER	051-00515
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		PAGE	50 OF 145
		SHEET	48 OF 119



SMC12 ADC Assignments

OUT	SMC_CPU_HI_ISENSE	==	SMC_CPU_HI_ISENSE	IN	32
OUT	SMC_PBUS_VSENSE	==	MAKE_BASE=TRUE	IN	32
OUT	SMC_BMON_ISENSE	==	SMC_BMON_ISENSE	IN	32
OUT	SMC_DCIN_ISENSE	==	SMC_DCIN_ISENSE	IN	32
OUT	SMC_DCIN_VSENSE	==	SMC_DCIN_VSENSE	IN	32
OUT	SMC_CPUGT_ISENSE	==	SMC_CPUGT_ISENSE	IN	35
OUT	SMC_CPU_ISENSE	==	SMC_CPU_ISENSE	IN	33
OUT	SMC_OTHER5V_HI_ISENSE	==	SMC_OTHER5V_HI_ISENSE	IN	32
OUT	SMC_OTHER3V3_HI_ISENSE	==	SMC_OTHER3V3_HI_ISENSE	IN	32
OUT	SMC_DDRIV2_ISENSE	==	SMC_DDRIV2_ISENSE	IN	33
OUT	SMC_CPUEDRAM_ISENSE	==	SMC_CPUEDRAM_ISENSE	IN	35
OUT	SMC_PCH_ISENSE	==	SMC_PCH_ISENSE	IN	33
OUT	SMC_TPAD_ISENSE	==	SMC_TPAD_ISENSE	IN	32
OUT	SMC_PICCOLO_ISENSE	==	SMC_PICCOLO_ISENSE	IN	35
OUT	SMC_SSDNAND_ISENSE	==	SMC_SSDNAND_ISENSE	IN	35
OUT	SMC_PCHPRIMCORE_ISENSE	==	SMC_PCHPRIMCORE_ISENSE	IN	35
OUT	SMC_DDRIV8_ISENSE	==	SMC_DDRIV8_ISENSE	IN	35
OUT	SMC_CPUSA_ISENSE	==	SMC_CPUSA_ISENSE	IN	35
OUT	SMC_CPUDDR_ISENSE	==	SMC_CPUDDR_ISENSE	IN	33
OUT	SMC_CPUSA_VSENSE	==	SMC_CPUSA_VSENSE	IN	35
OUT	SMC_CPU_VSENSE	==	SMC_CPU_VSENSE	IN	34
OUT	SMC_CPUGT_VSENSE	==	SMC_CPUGT_VSENSE	IN	34
OUT	SMC_CPU_IMON_ISENSE	==	SMC_CPU_IMON_ISENSE	IN	34
OUT	SMC_CPUGT_IMON_ISENSE	==	SMC_CPUGT_IMON_ISENSE	IN	34

SMC12 Pin Assignments

OUT	NC_SMC_GFX_THROTTLE_L	==	NC_SMC_GFX_THROTTLE_L	NO_TEST=1	
OUT	NC_SMC_GFX_OVERTEMP	==	NC_SMC_GFX_OVERTEMP	NO_TEST=1	
OUT	NC_SMC_GFX_SELF_THROTTLE	==	NC_SMC_GFX_SELF_THROTTLE	NO_TEST=1	
OUT	NC_SMC_DP_HPD_L	==	NC_SMC_DP_HPD_L	NO_TEST=1	
OUT	NC_SYS_ONEWIRE	==	NC_SYS_ONEWIRE	NO_TEST=1	
OUT	NC_SMC_DEBUGPRT_EN_L	==	NC_SMC_DEBUGPRT_EN_L	NO_TEST=1	

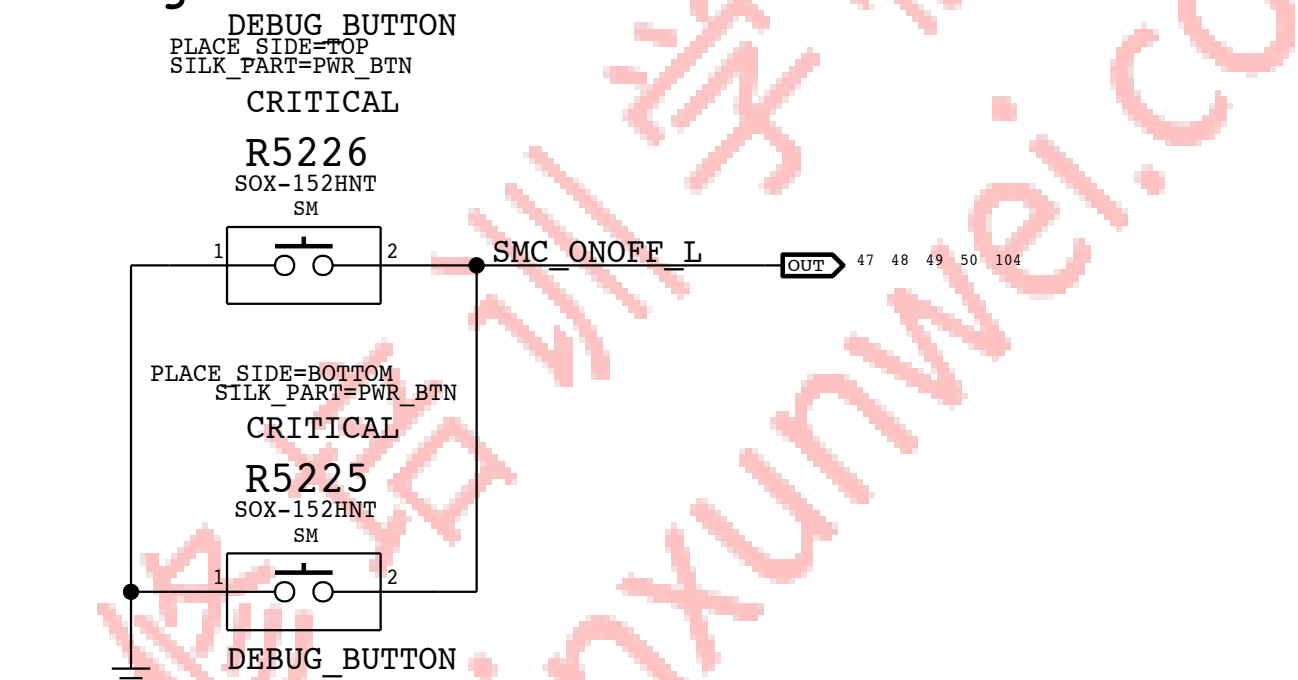
OUT	SMC_LID_RIGHT	==	SMC_LID_RIGHT		50
OUT	NC_SMC_SPI_SMC_MISO	==	NC_SMC_SPI_SMC_MISO	NO_TEST=1	TRUE
OUT	NC_SMC_SPI_SMC_MOSI	==	NC_SMC_SPI_SMC_MOSI	NO_TEST=1	TRUE
OUT	NC_SMC_SPI_SMC_CLK	==	NC_SMC_SPI_SMC_CLK	NO_TEST=1	TRUE
OUT	NC_SMC_SPI_SMC_CS_L	==	NC_SMC_SPI_SMC_CS_L	NO_TEST=1	TRUE

OUT	SMC_VIBE_L	==	SMC_VIBE_L		43
OUT	SMC_PCH_SUSWARN_L	==	SMC_PCH_SUSWARN_L		14
IN	SMC_PCH_SUSACK_L	==	SMC_PCH_SUSACK_L		14

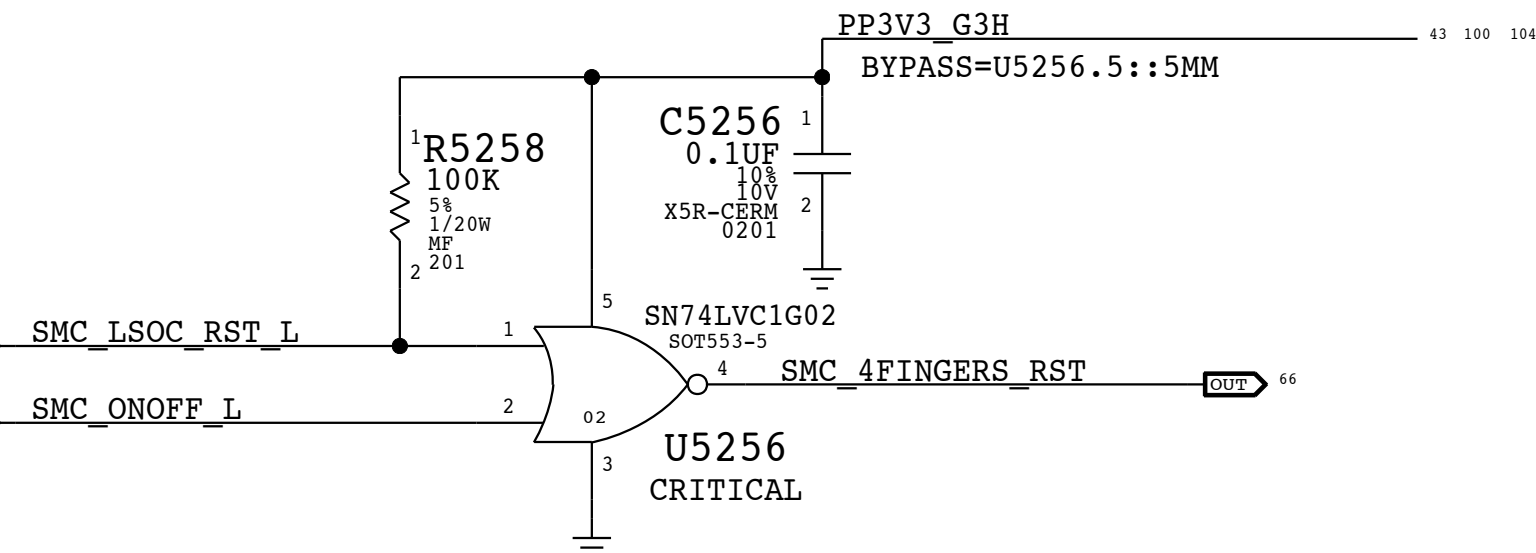
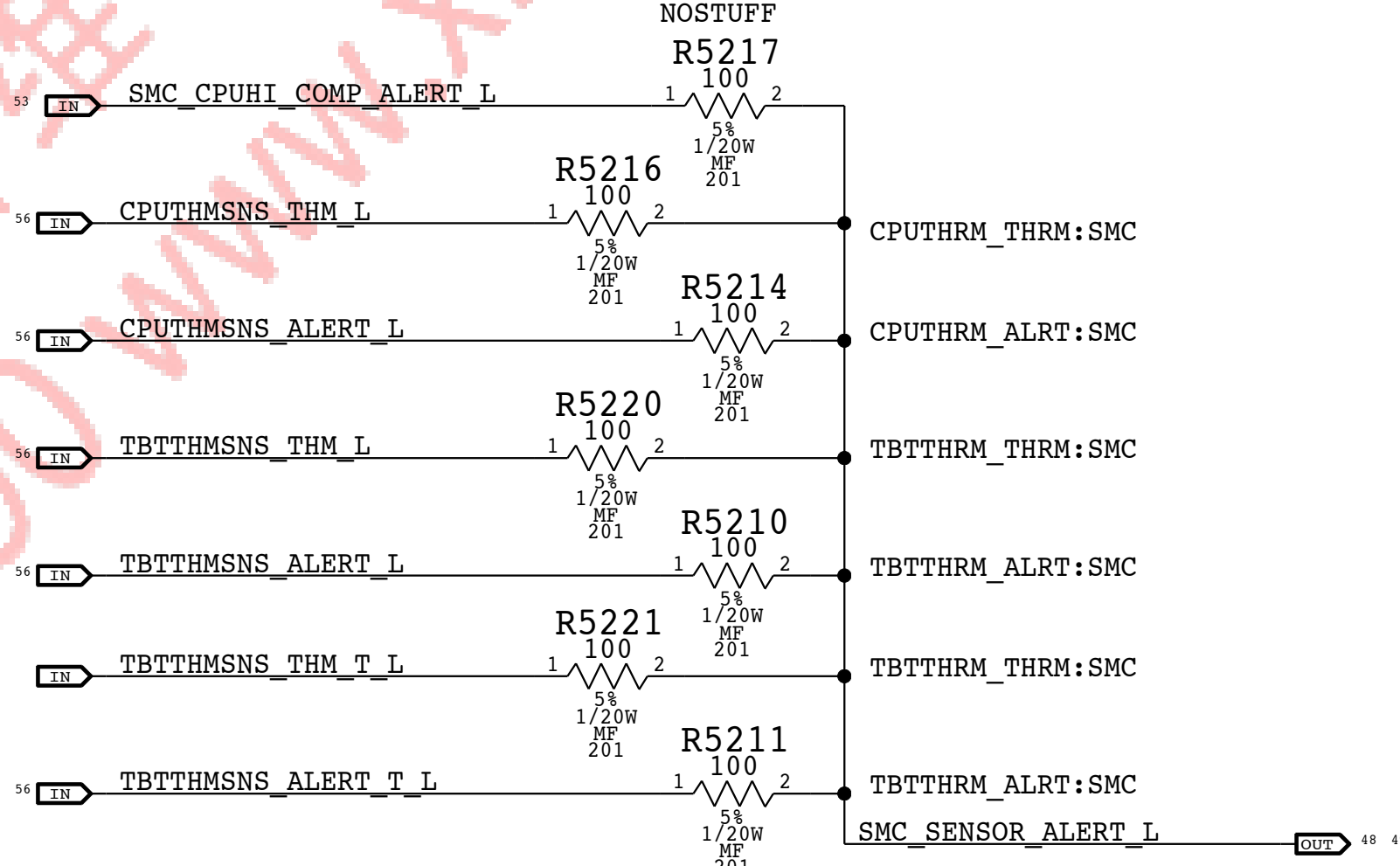
OUT	SMC_SENSOR_PWR_EN	==	SMC_SENSOR_PWR_EN		78
OUT	WLAN_UART_RX	==	WLAN_UART_RX		35
OUT	WLAN_UART_TX	==	WLAN_UART_TX		35
OUT	SMC_AUX_OK	==	SMC_AUX_OK		49
OUT	PM_PWRBTN_L	==	PM_PWRBTN_L		14
OUT	SMC_AUX_OK	==	SMC_AUX_OK		52
OUT	SMBUS_SMC_4_G3H_SCL	==	SMBUS_SMC_4_G3H_SCL		48
OUT	SMBUS_SMC_4_G3H_SDA	==	SMBUS_SMC_4_G3H_SDA		48
OUT	SMBUS_SMC_2_S4_SCL	==	SMBUS_SMC_2_S4_SCL		48
OUT	SMBUS_SMC_2_S4_SDA	==	SMBUS_SMC_2_S4_SDA		48
OUT	SMC_OOB1_D2R_L	==	SMC_OOB1_D2R_L		88
OUT	SMC_OOB1_R2D_L	==	SMC_OOB1_R2D_L		88

OUT	SMC_ACTUATOR_DISABLE_L	==	SMC_ACTUATOR_DISABLE_L		43 104
OUT	SMC_CHGR_INT_L	==	SMC_CHGR_INT_L		66

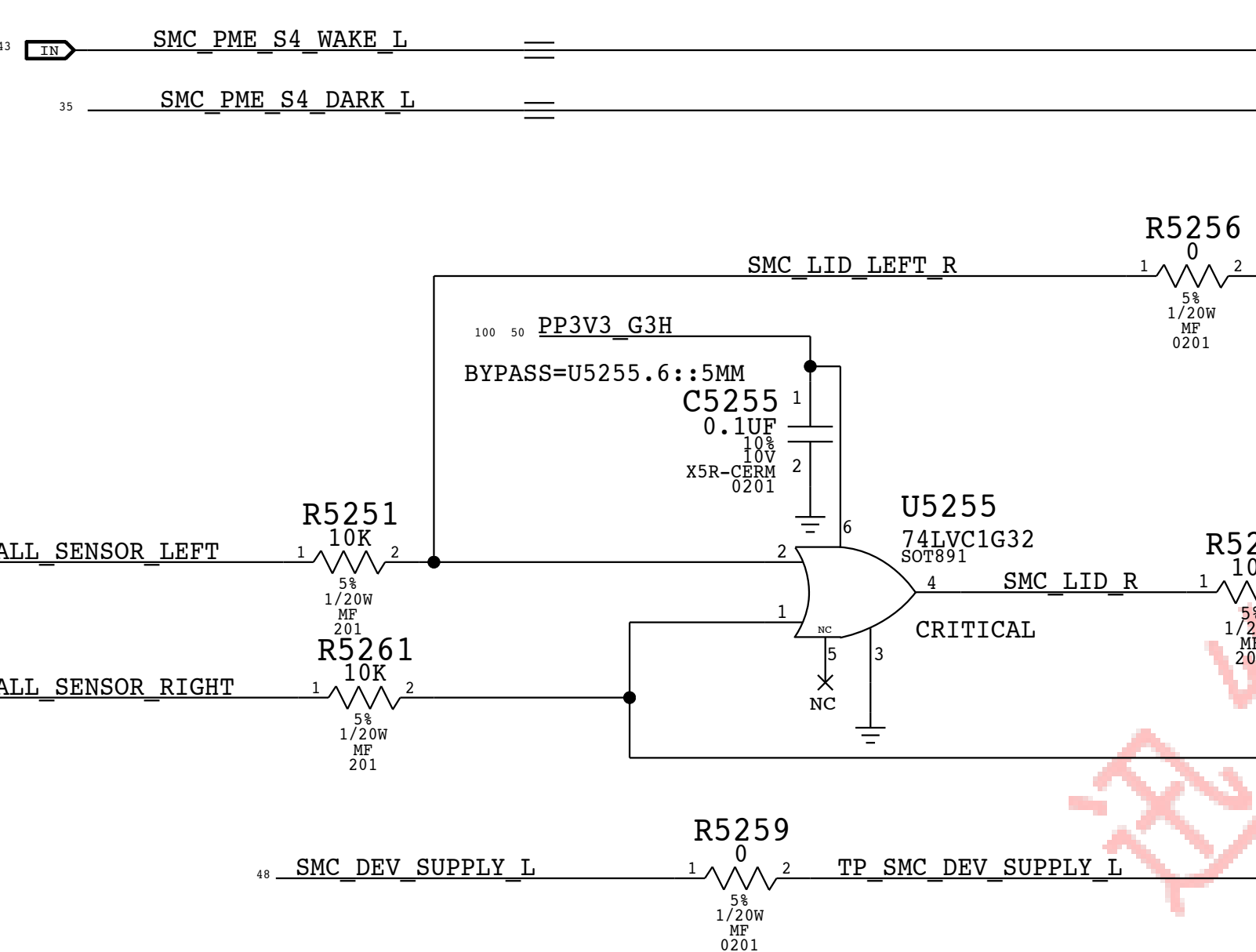
Debug Power "Buttons"



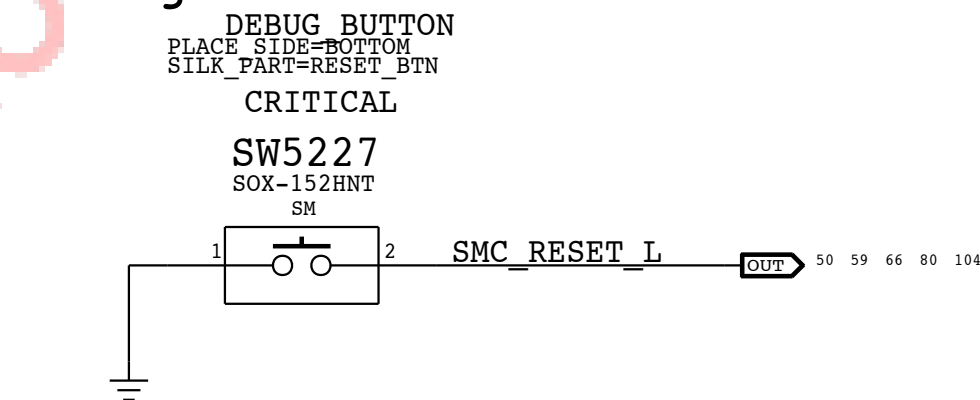
Thermal Alerts



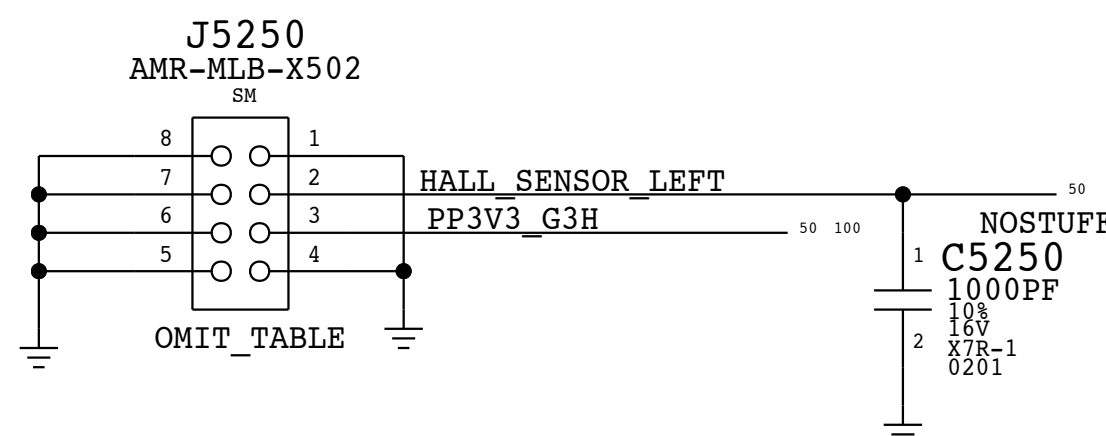
S4 SMC Wake Sources

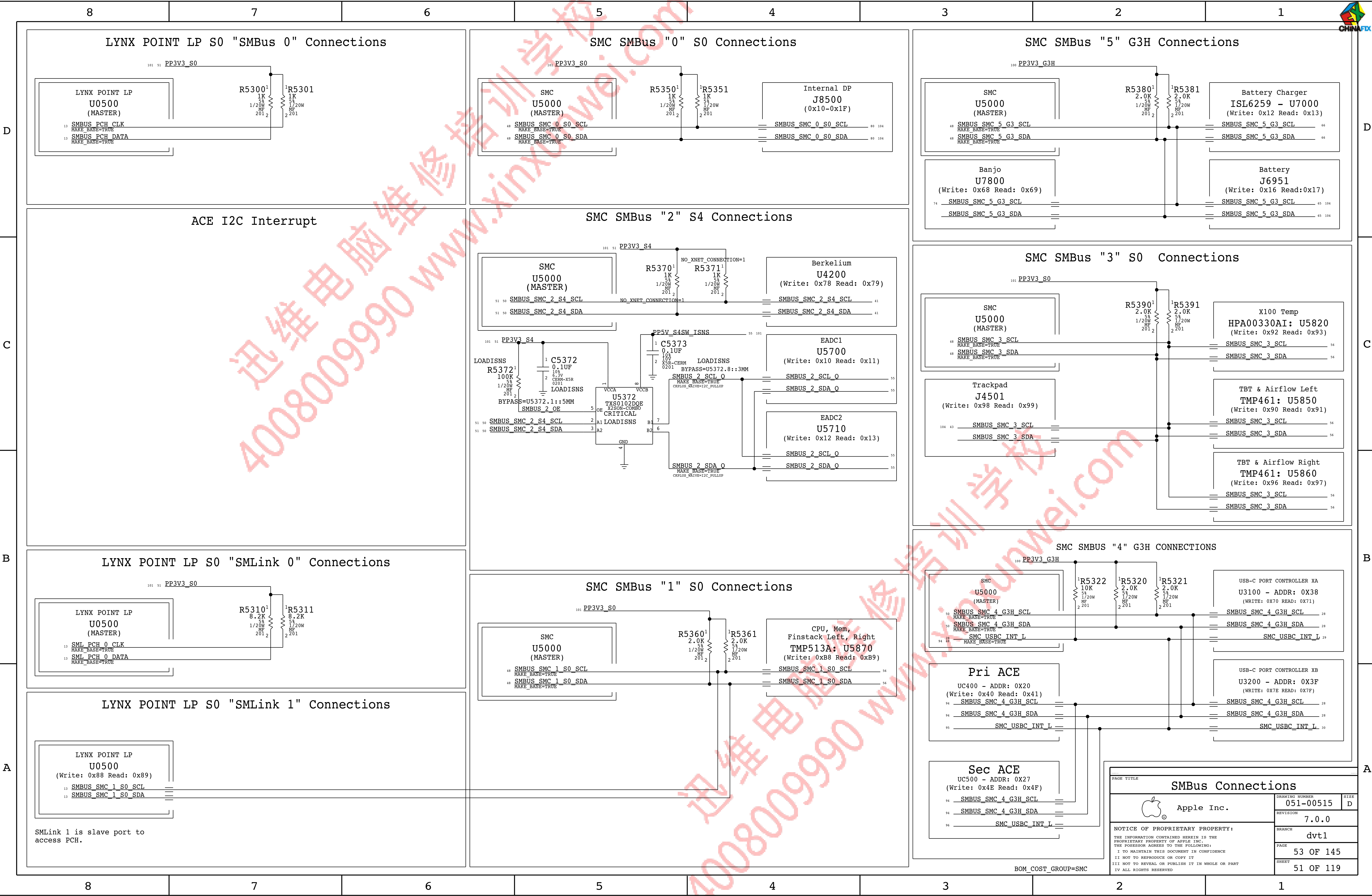


Debug RESET "Buttons"



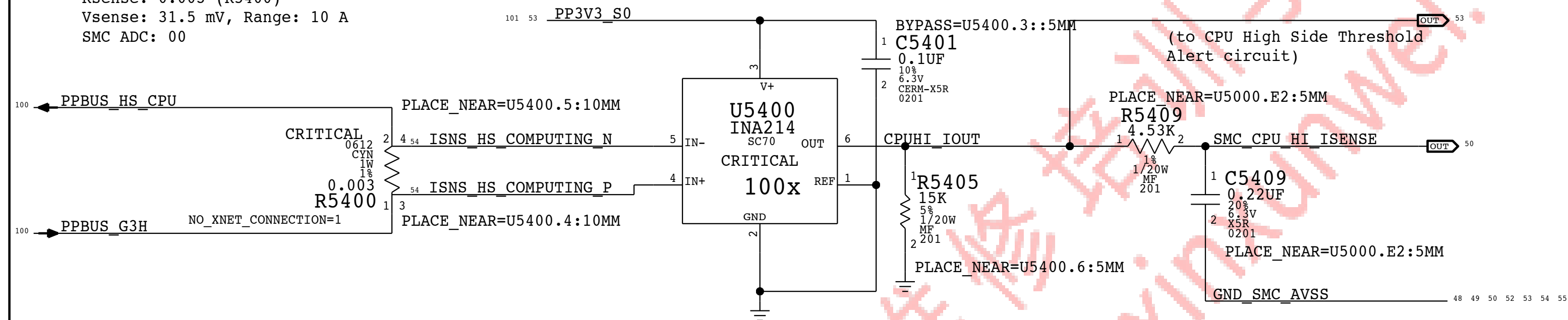
Hall Effect Pads - Left





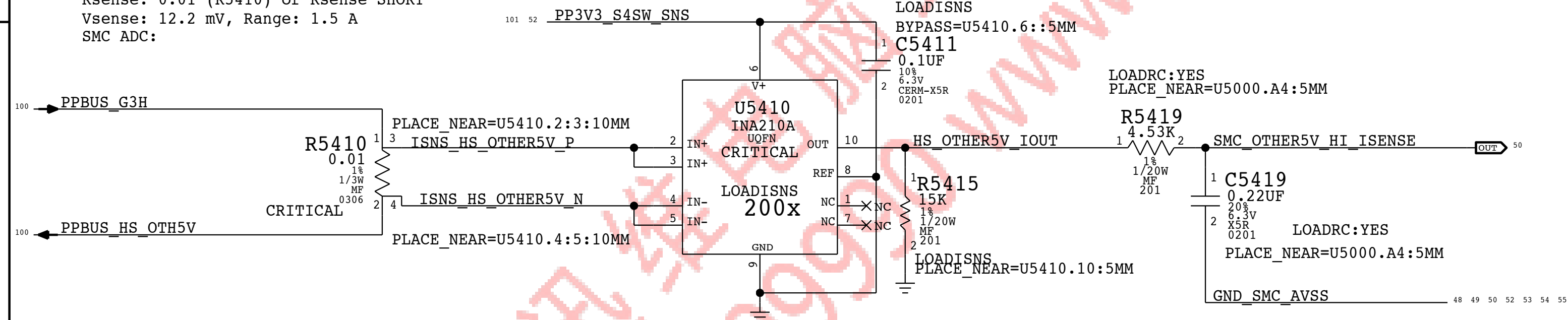
CPU High Side Current Sense (IC0R)

Gain: 100x, EDP: 10.5 A
Rsense: 0.003 (R5400)
Vsense: 31.5 mV, Range: 10 A
SMC ADC: 00



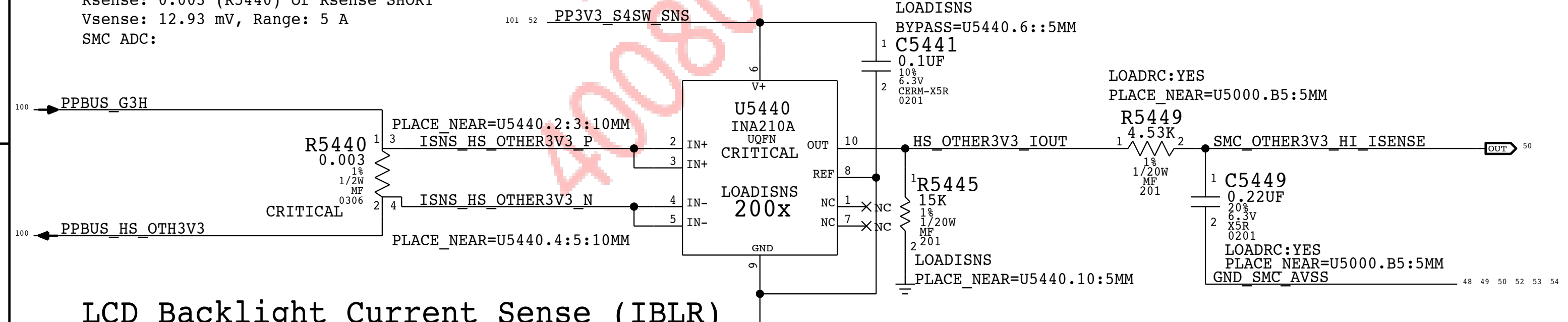
OTHER 5V High Side Current Sense (IO5R)

Gain: 200x, EDP: 1.22 A
Rsense: 0.01 (R5410) or Rsense SHORT
Vsense: 12.2 mV, Range: 1.5 A
SMC ADC:



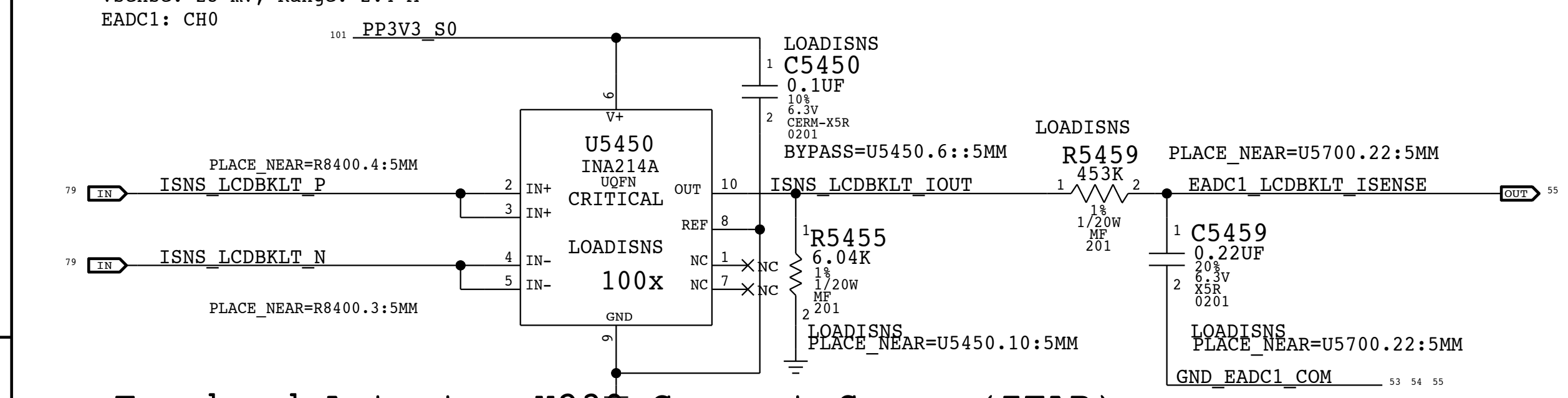
OTHER 3.3V High Side Current Sense (IO3R)

Gain: 200x, EDP: 4.31 A
Rsense: 0.003 (R5440) or Rsense SHORT
Vsense: 12.93 mV, Range: 5 A
SMC ADC:



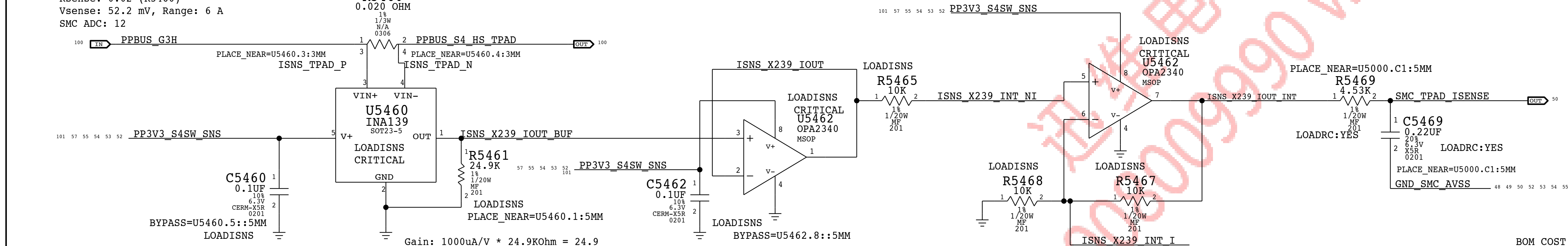
LCD Backlight Current Sense (IBLR)

Gain: 100x. EDP: 1 A
Rsense: 0.025 (R8400)
Vsense: 25 mV, Range: 2.4 A
EADC1: CH0



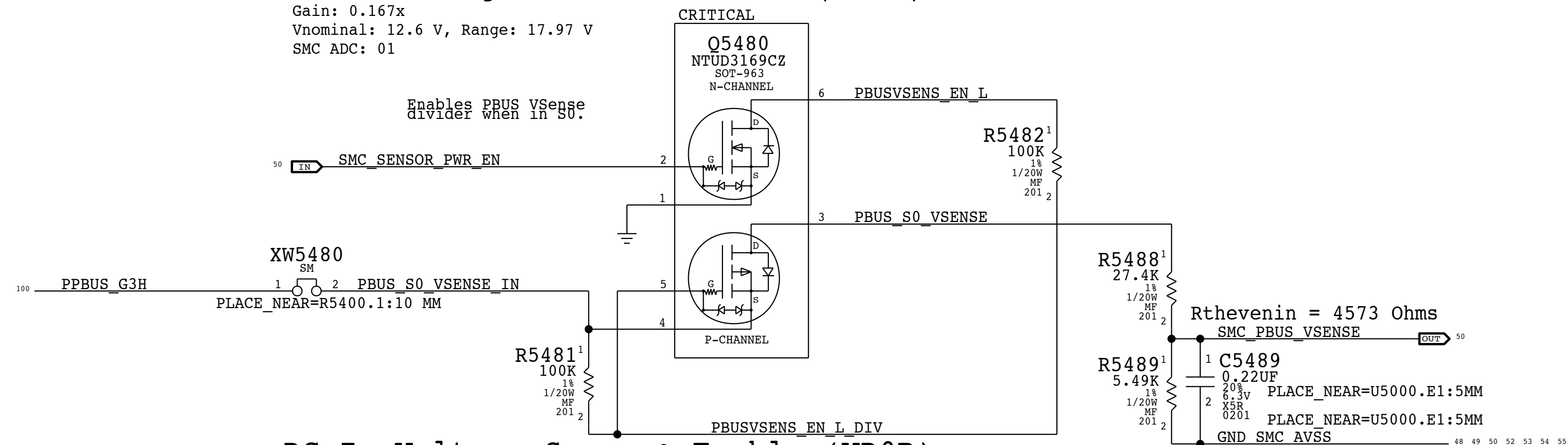
Trackpad Actuator X239 Current Sense (ITAR)

Gain: 24.9x, EDP: 2.61 A (Transient)
Rsense: 0.02 (R5460)
Vsense: 52.2 mV, Range: 6 A
SMC ADC: 12



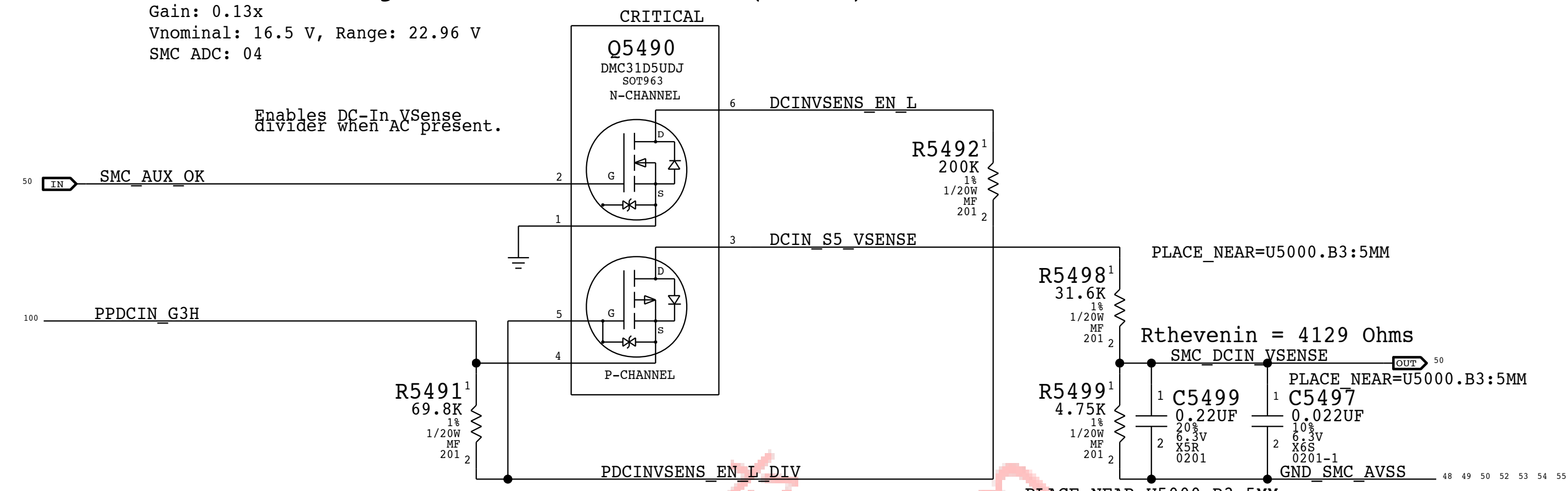
PBUS Voltage Sense & Enable (VP0R)

Gain: 0.167x
Vnominal: 12.6 V, Range: 17.97 V
SMC ADC: 01



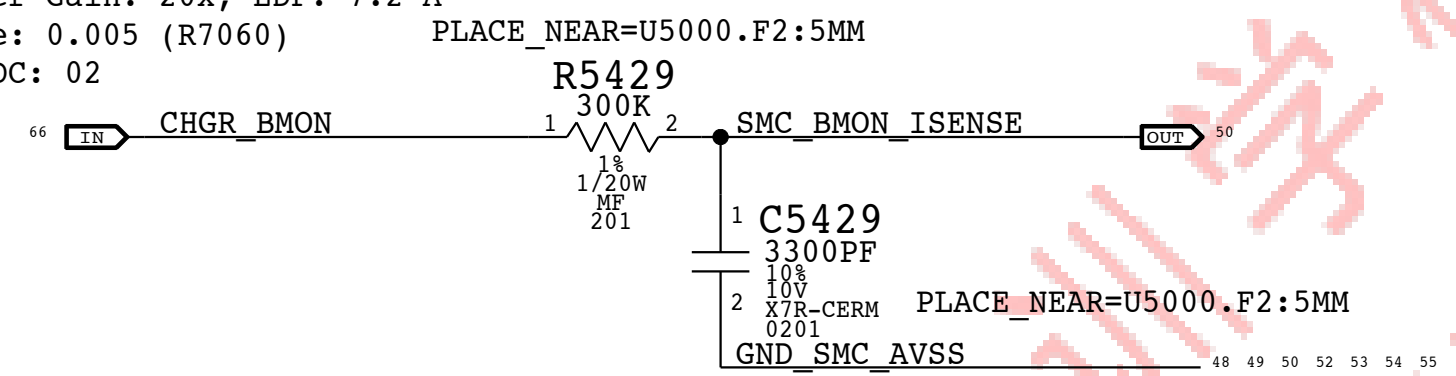
DC In Voltage Sense & Enable (VD0R)

Gain: 0.13x
Vnominal: 16.5 V, Range: 22.96 V
SMC ADC: 04



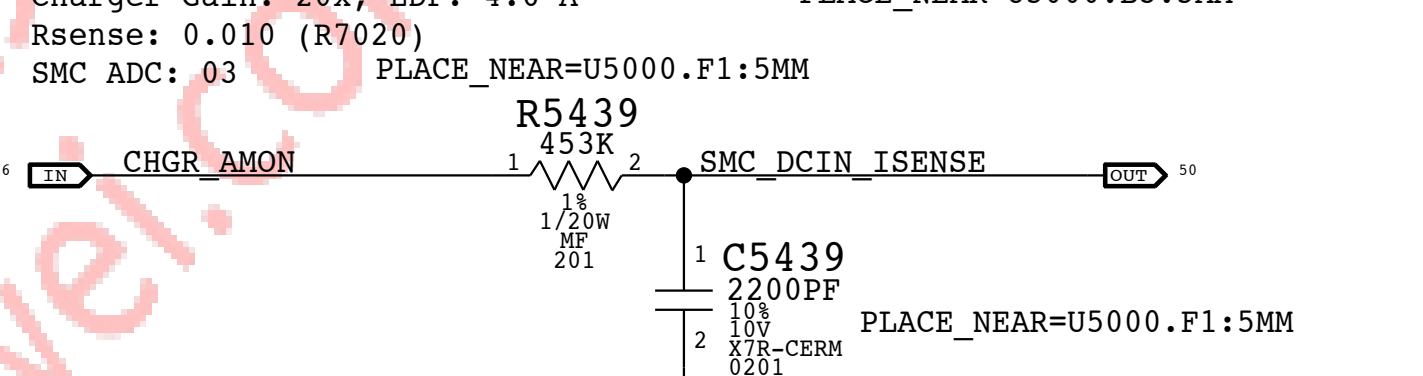
Charger (BMON) Current Sense (IPBR)

Charger Gain: 20x, EDP: 7.2 A
Rsense: 0.005 (R7060)
SMC ADC: 02



DC-IN (AMON) Current Sense (ID0R)

Charger Gain: 20x, EDP: 4.6 A
Rsense: 0.010 (R7020)
SMC ADC: 03



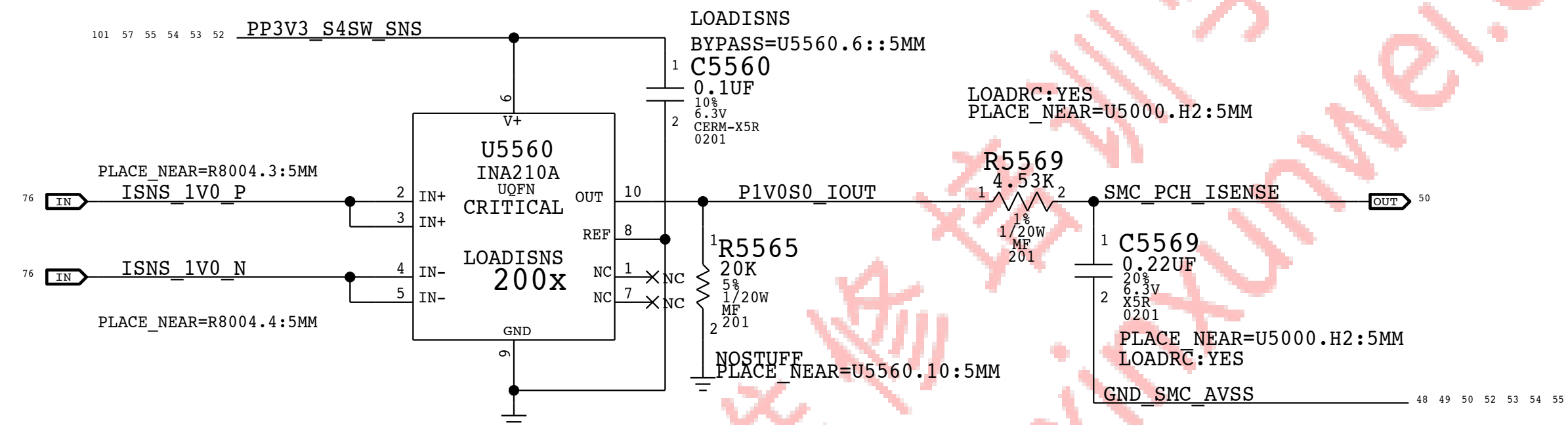
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5419,C5449		LOADRC:NO
117S0008	1	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5469		LOADRC:NO

Power Sensors: High Side		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
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	PAGE	54 OF 145
BOM_COST_GROUP=SENSORS		SHEET 52 OF 119



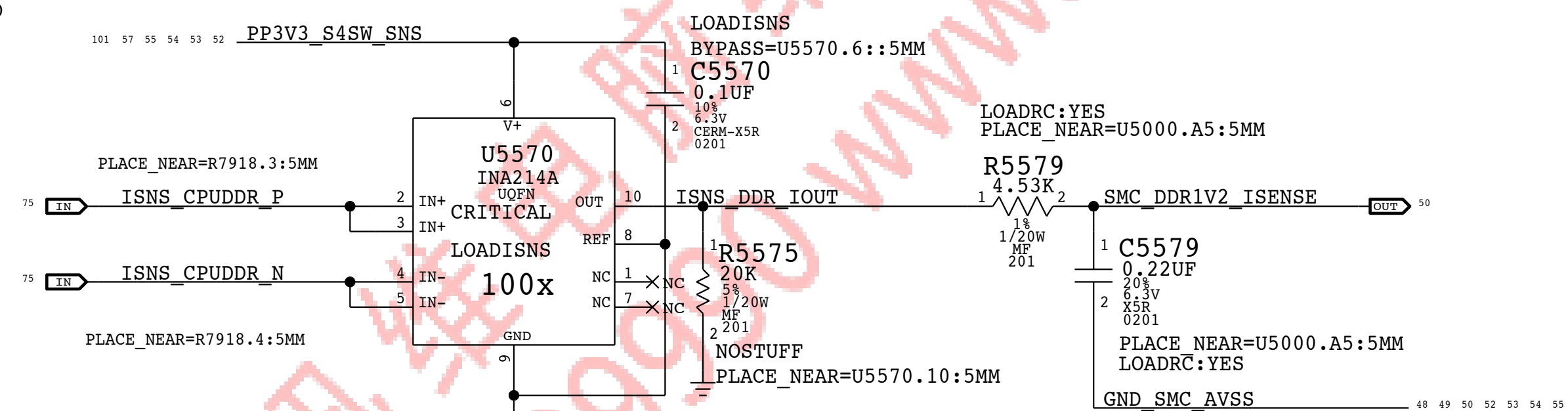
PCH 1.0V Current Sense (IS1C)

Gain: 200x, EDP: 3.29 A
Rsense: 0.003 (R8004) or Rsense SHORT
Vsense: 9.87 mV, Range: 5 A
SMC ADC: 11



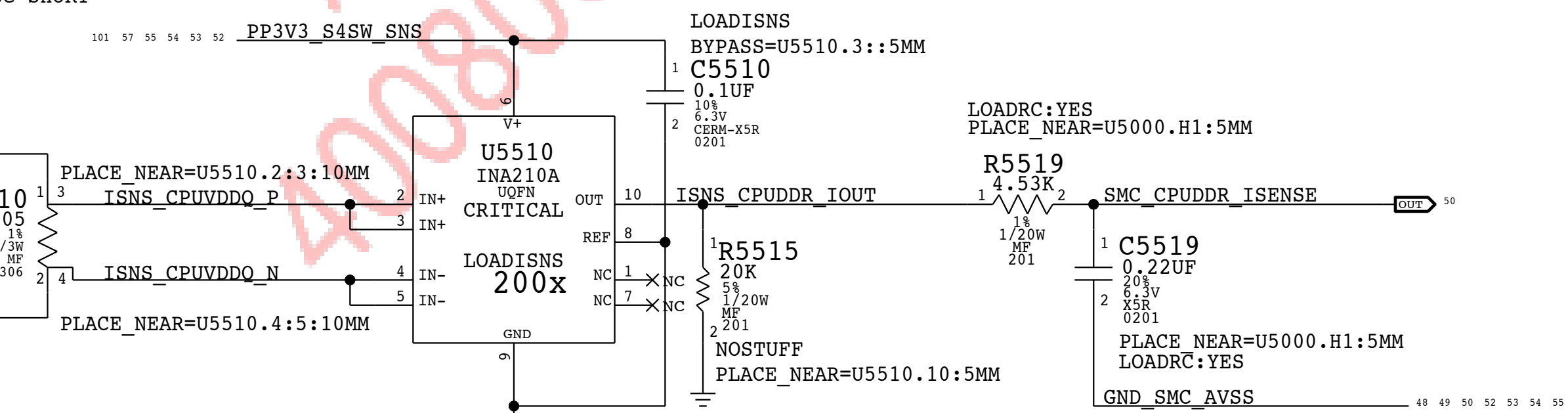
DDR 1.2V S3 (CPU & Memory) Current Sense (IM0C)

Gain: 100x, EDP: 8.21 A
Rsense: 0.003 (R7918) or XWTBD
Vsense: 24.63 mV, Range: 10 A
SMC ADC: 09



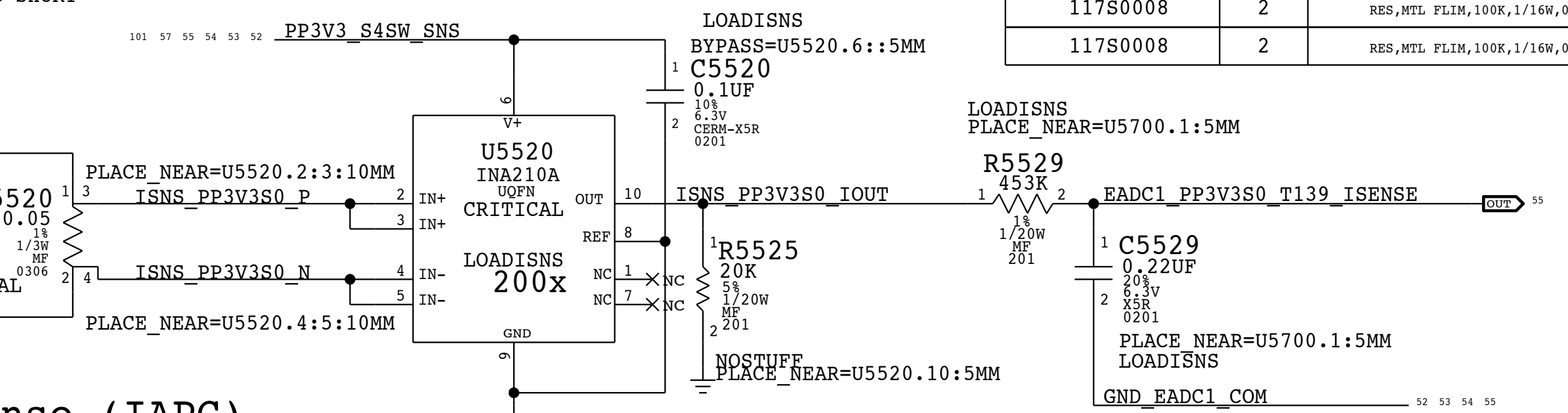
CPU DDR 1.2V S3 (CPU Only) Current Sense (IMCC)

Gain: 200x, EDP: 2 A
Rsense: 0.005 (R5510) or Rsense SHORT
Vsense: 10 mV, Range: 3 A
SMC ADC: 18



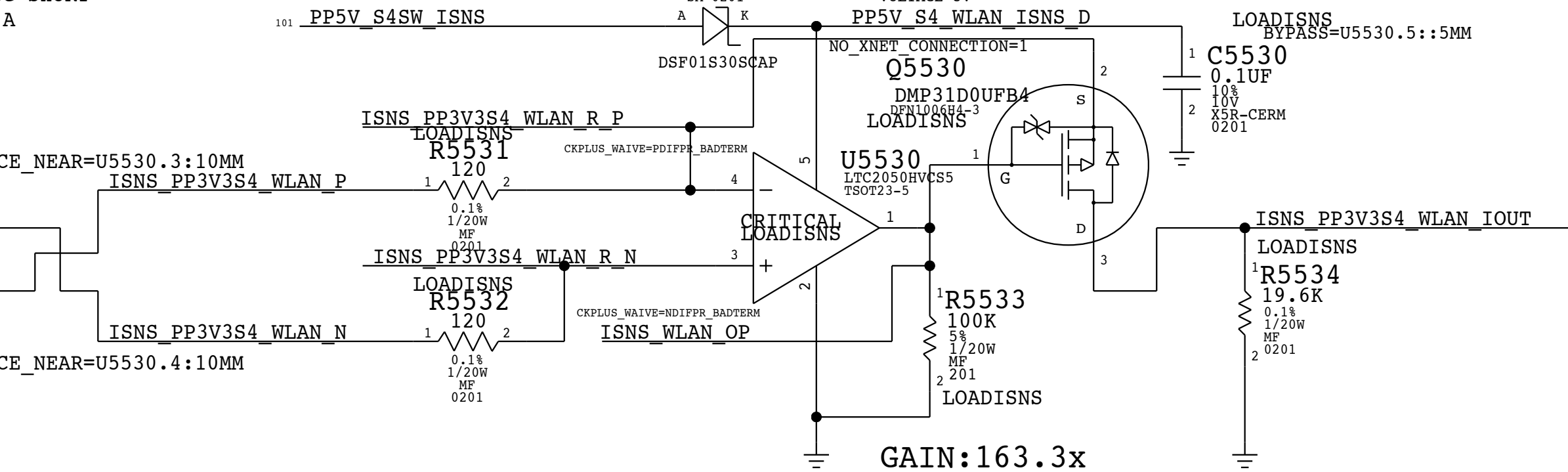
T139 Current Sense (IF3C)

Gain: 200x, EDP: 0.06 A
Rsense: 0.05 (R5520) or Rsense SHORT
Vsense: 3 mV, Range: 0.25 A
EADC1: CH3



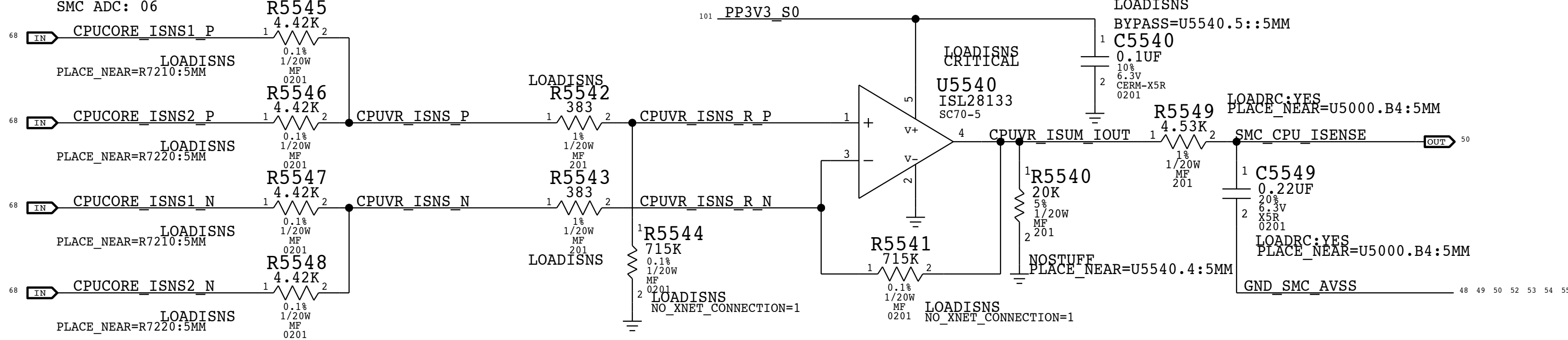
WLAN Current Sense (IAPC)

Gain: 163.3x, EDP: 1.67 A
Rsense: 0.015 (R5530) or Rsense SHORT
Vsense: 25.05 mV, Range: 1.67 A
EADC1: CH4



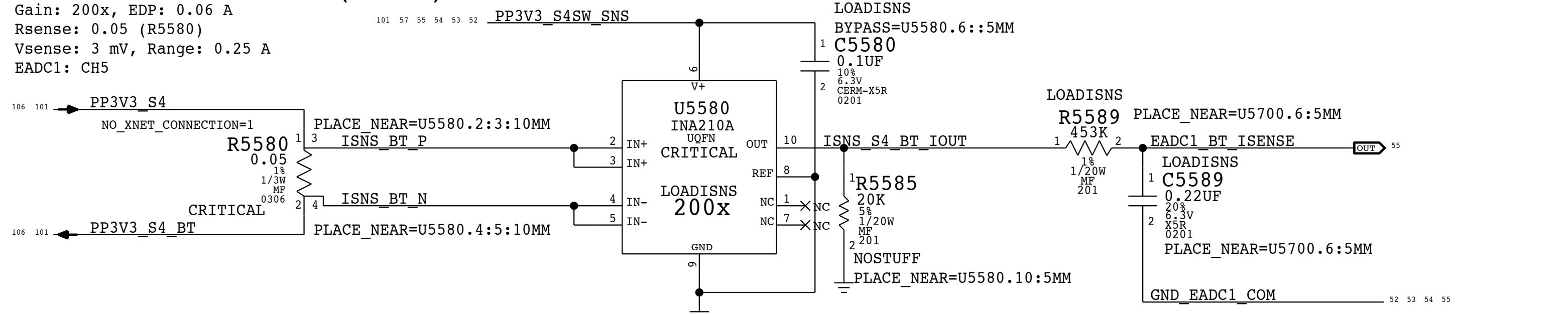
CPU Fixed Current Sense (ICAC)

Gain: 275.74x, EDP: 29 A
Rsense: 2x of 0.00075 (R7310, R7320), Rsum: 0.000375
Vsense: 10.875 mV, Range: 29.01 A
SMC ADC: 06



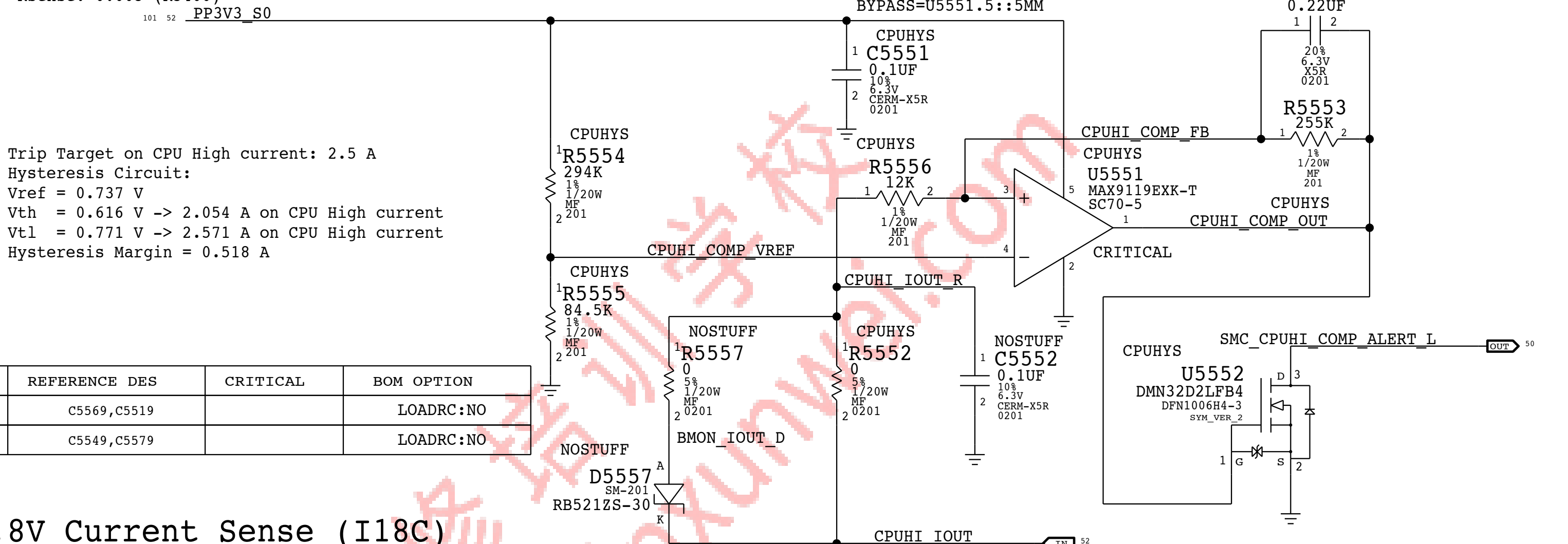
BT Current Sense (IBTC)

Gain: 200x, EDP: 0.06 A
Rsense: 0.05 (R5580)
Vsense: 3 mV, Range: 0.25 A
EADC1: CH5



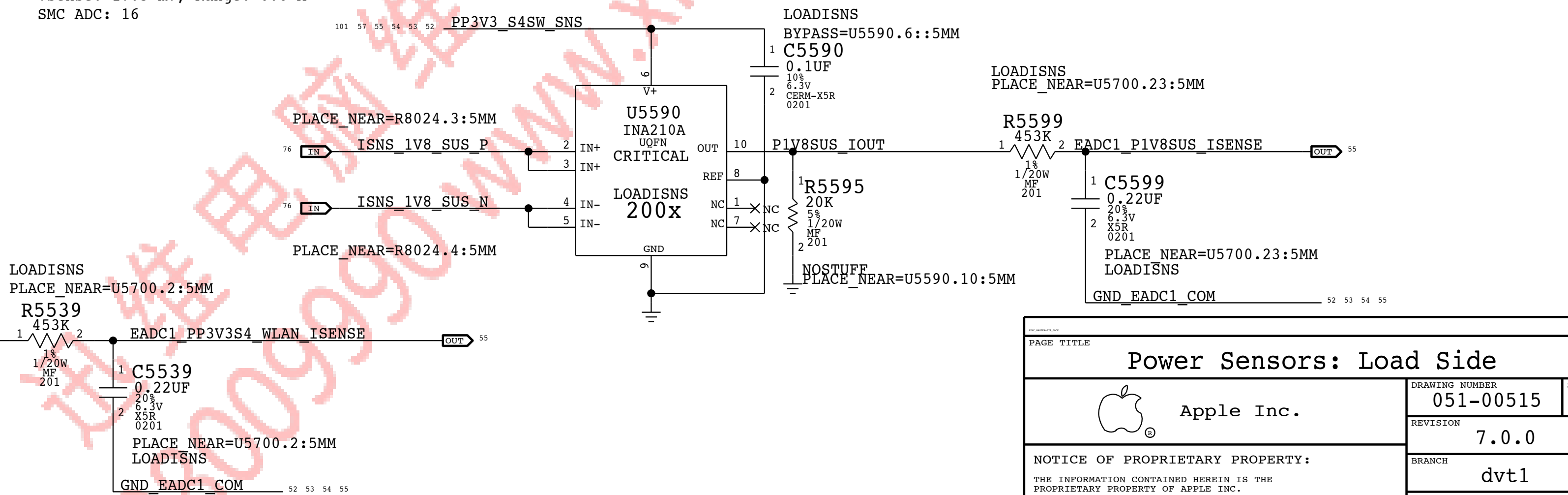
CPU High Side Current (IC0R) Threshold Alert

Gain: 100x
Rsense: 0.003 (R5400)



1.8V Current Sense (I18C)

Gain: 200x, EDP: 0.7 A
Rsense: 0.025 (R8024) or Rsense SHORT
Vsense: 17.5 mV, Range: 0.6 A
SMC ADC: 16



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5569,C5519		LOADRC:NO
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5549,C5579		LOADRC:NO

Power Sensors: Load Side

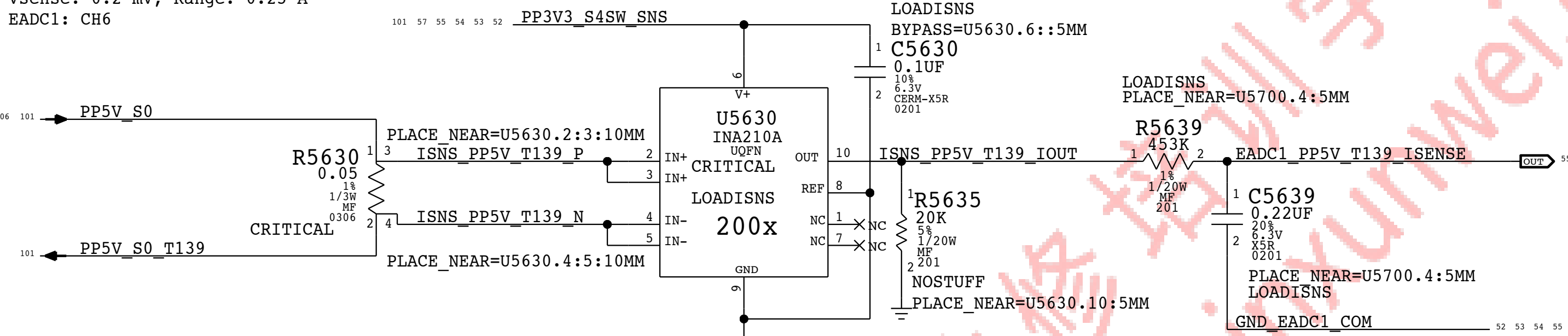
	DRAWING NUMBER	051-00515	STEP	D
	REVISION	7.0.0		
	BRANCH	dvt1		
	PAGE	55 OF 145		
NOTICE OF PROPRIETARY PROPERTY:		SHEET		
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BOM_COST_GROUP=SENSORS



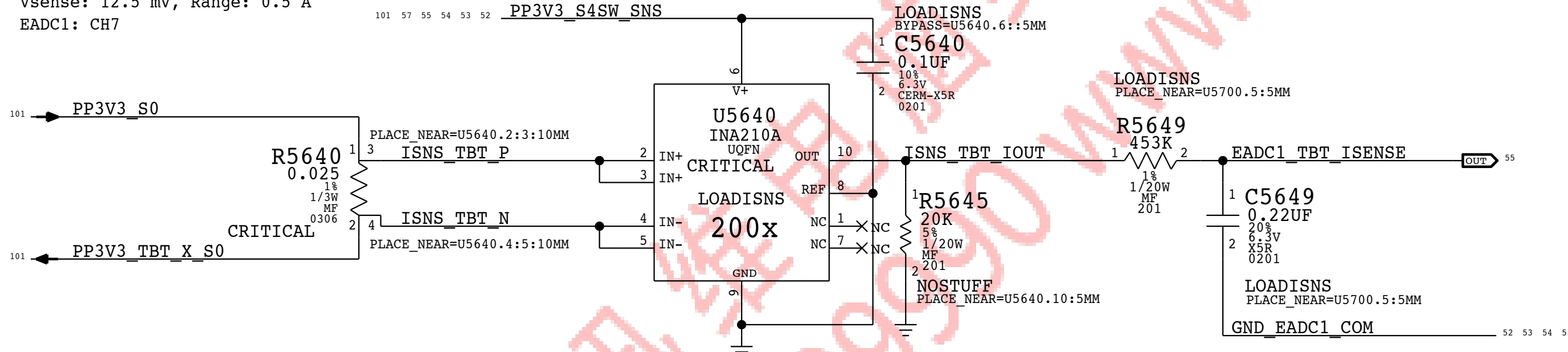
T139 5V Current Sense (IF5C)

Gain: 200x, EDP: 0.004 A
Rsense: 0.05 (R5630) or Rsense SHORT
Vsense: 0.2 mV, Range: 0.25 A
EADC1: CH6



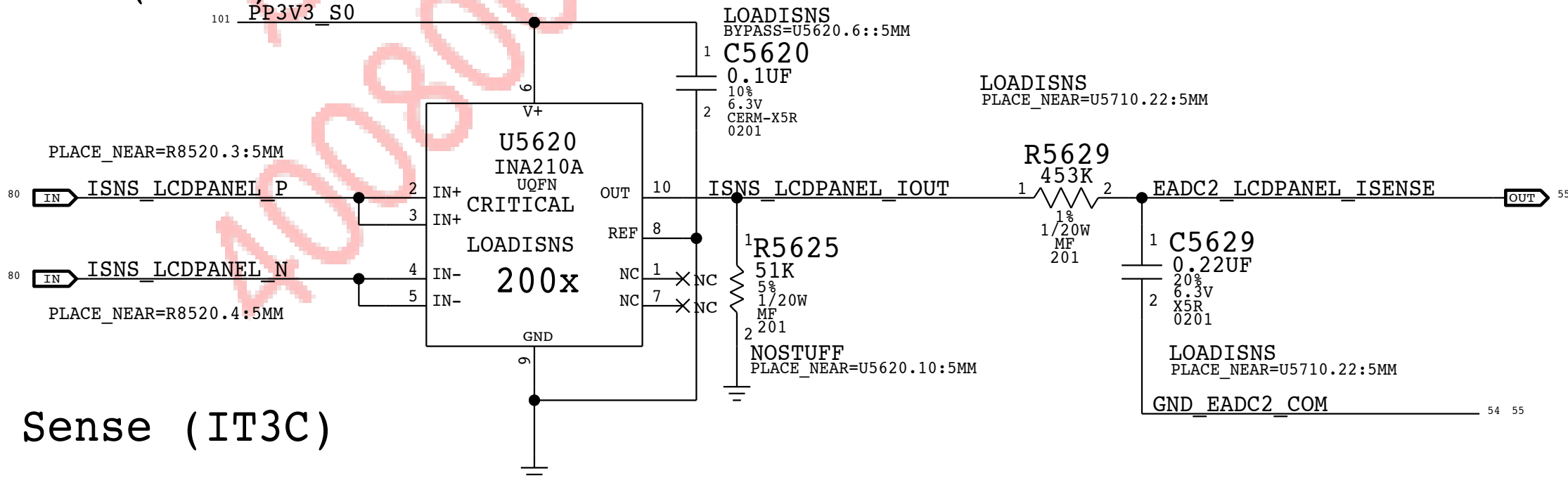
Thunderbolt TBT Current Left (IULC)

Gain: 200x, EDP: 0.5 A
Rsense: 0.025 (R5640) or Rsense SHORT
Vsense: 12.5 mV, Range: 0.5 A
EADC1: CH7



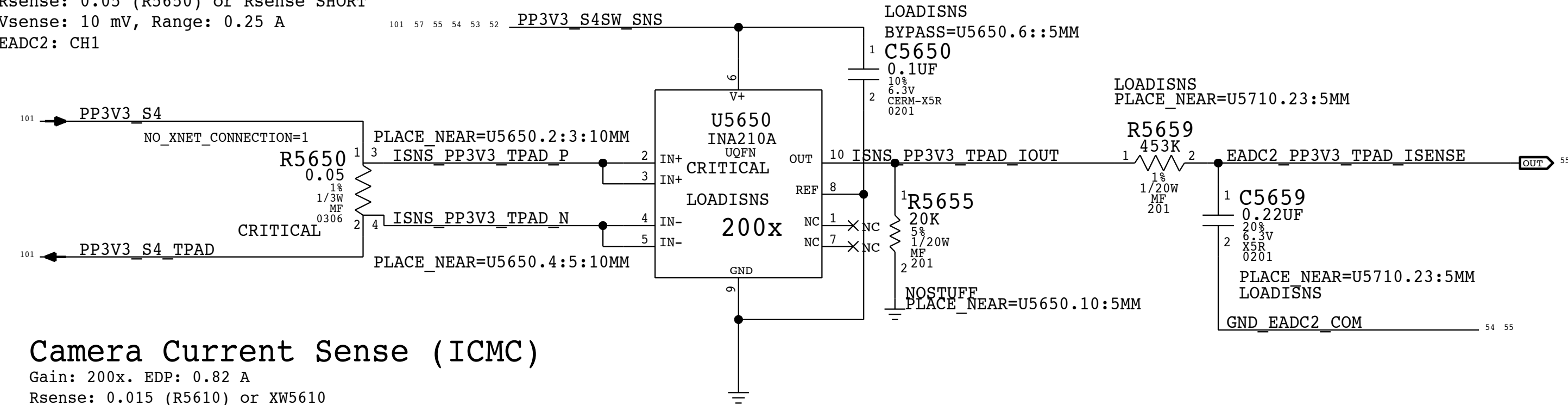
LCD Panel Current Sense (ILDC)

Gain: 200x, EDP: 1 A
RSENSE: 0.01 (R8520) or Rsense SHORT
Vsense: 5 mV, Range: 1.25 A
EADC2: CH0



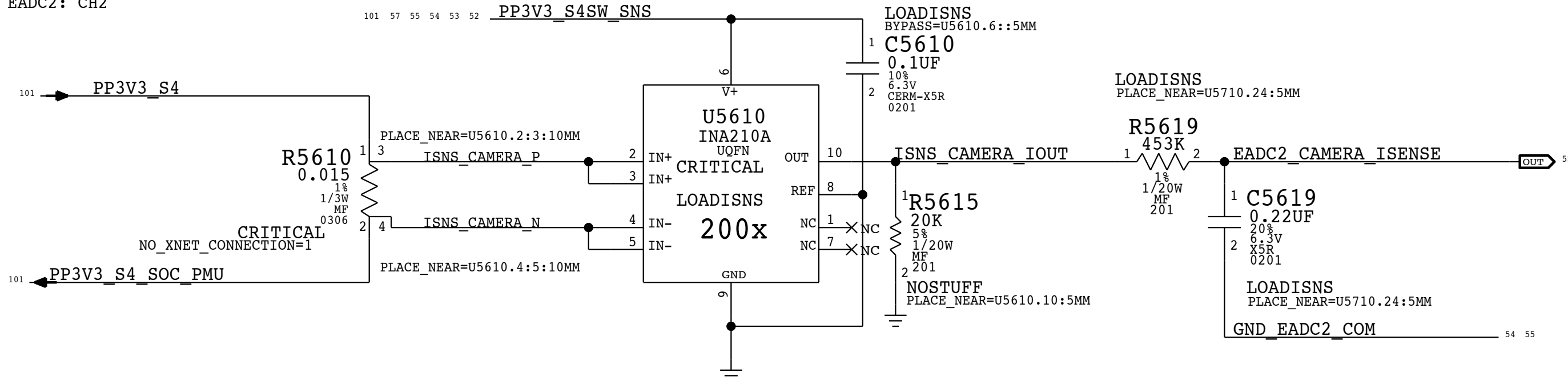
Trackpad 3V Current Sense (IT3C)

Gain: 200x, EDP: 0.2 A
Rsense: 0.05 (R5650) or Rsense SHORT
Vsense: 10 mV, Range: 0.25 A
EADC2: CH1

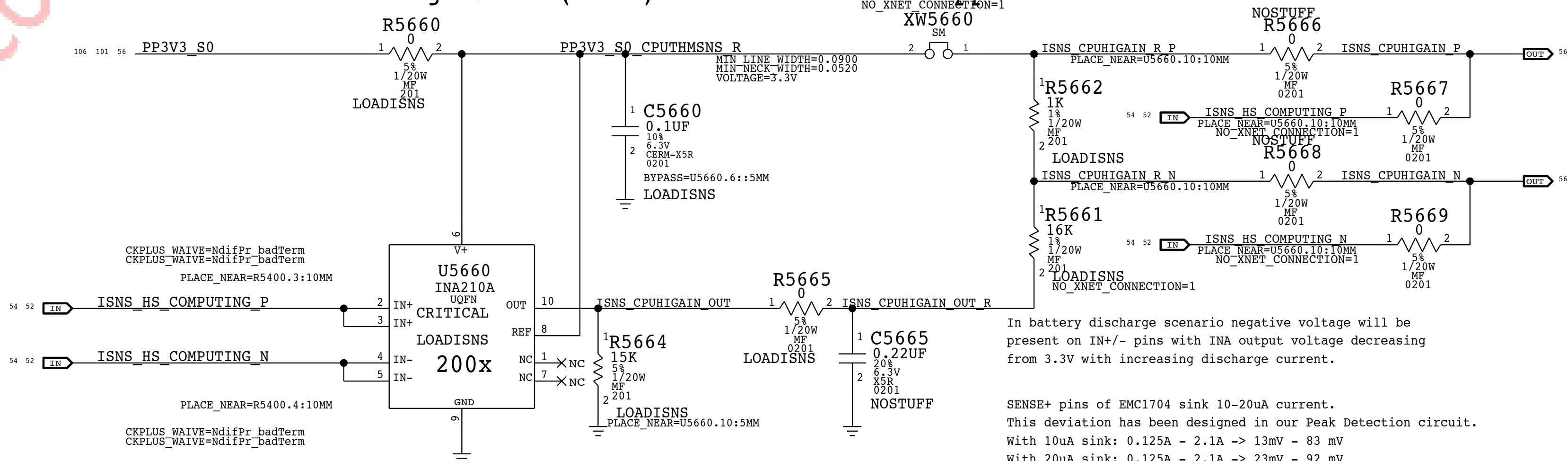


Camera Current Sense (ICMC)

Gain: 200x, EDP: 0.82 A
Rsense: 0.015 (R5610) or XW5610
Vsense: 12.3 mV, Range: 0.83 A
EADC2: CH2

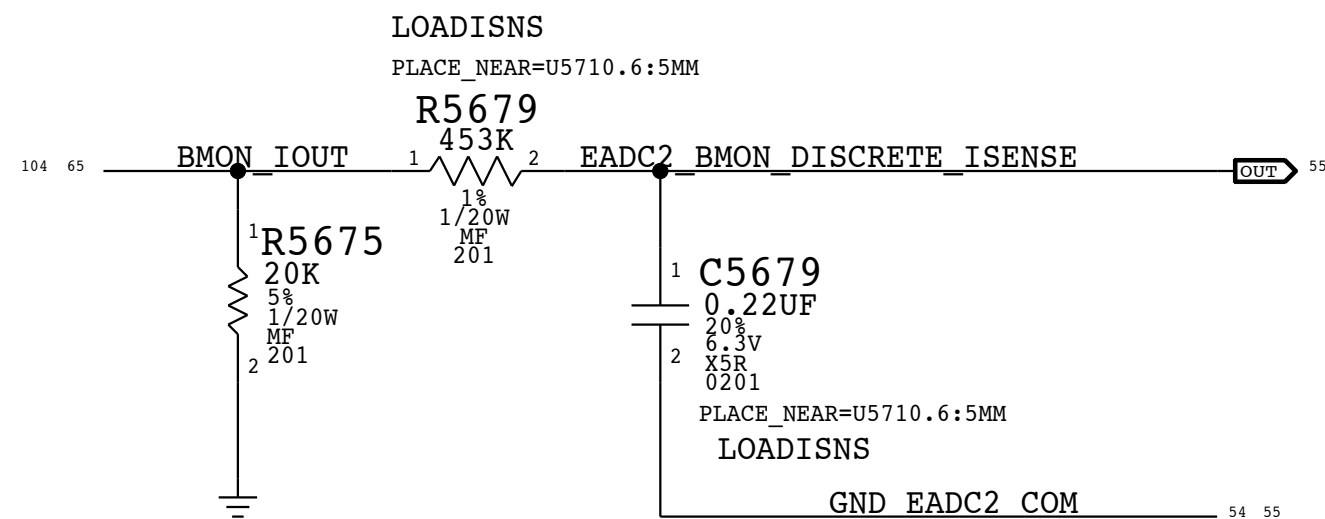


CPU High Side (IC0R) Peak Detection Support



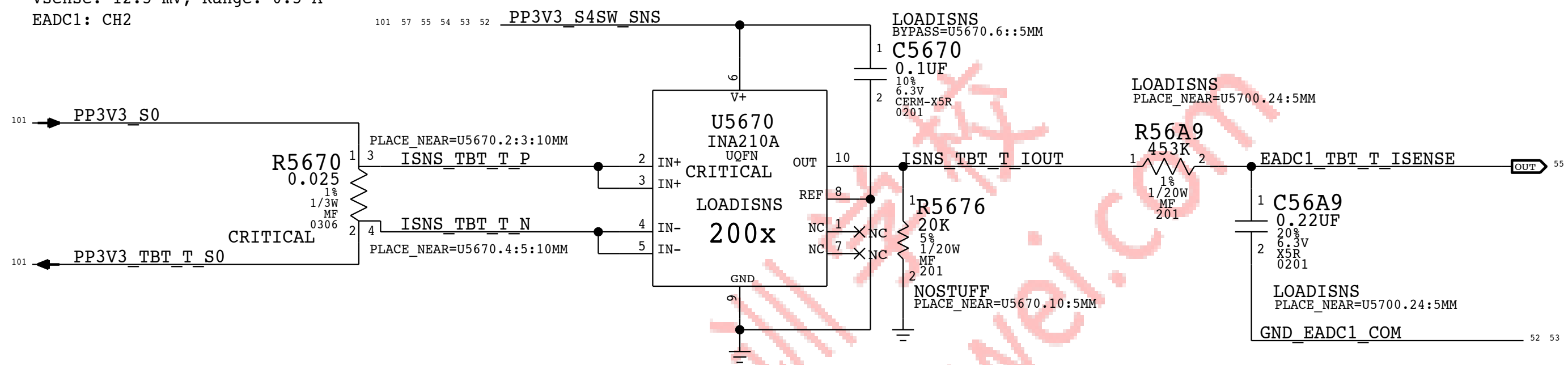
Battery Discrete Current Sense (IB0L)

Gain: 2940x, EDP: 8 A
Rsense: 0.003 (R501//R502)
Vsense: 24 mV, Range: 0.28 A
EADC2: CH5



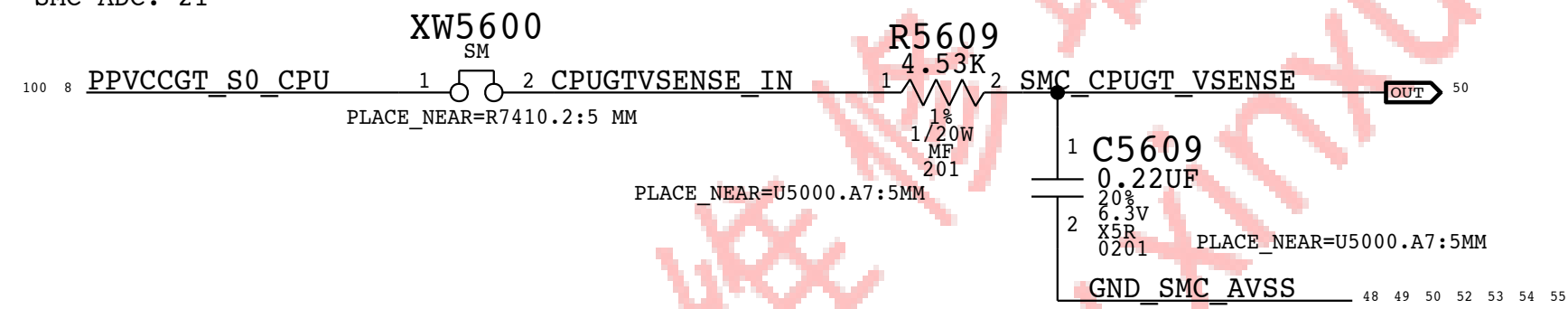
Thunderbolt TBT Current Right (IURC)

Gain: 200x, EDP: 0.5 A
Rsense: 0.025 (R5670) or Rsense SHORT
Vsense: 12.5 mV, Range: 0.5 A
EADC1: CH2



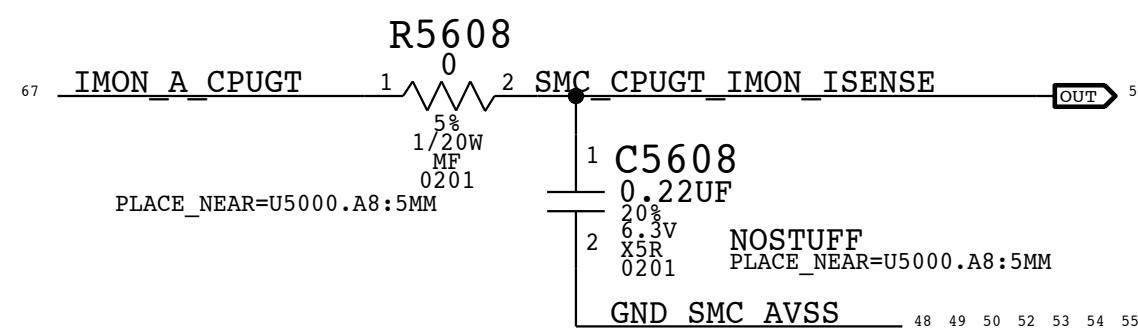
CPU GT Voltage Sense (VCGC)

SMC ADC: 21



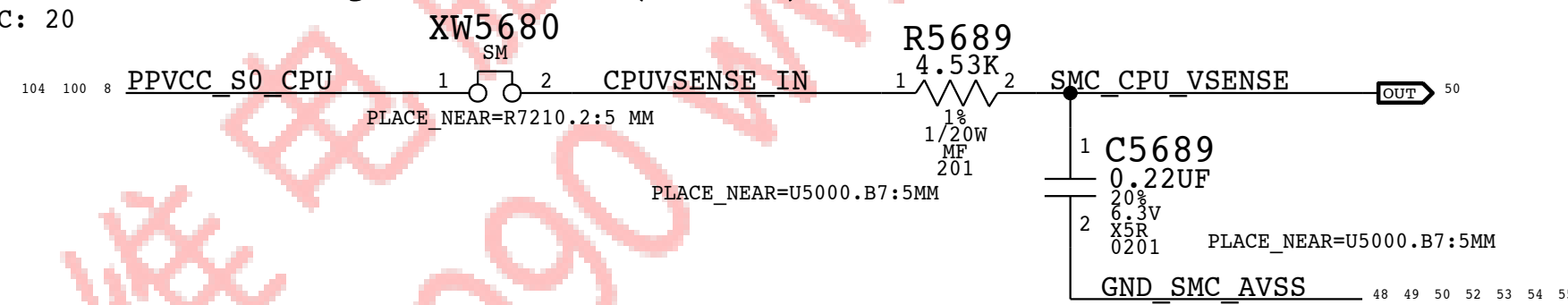
CPU GT IMON Current Sense (ICGM)

Gain: 1 A / 17.963 mV, Range: 64 A.
SMC ADC: 23



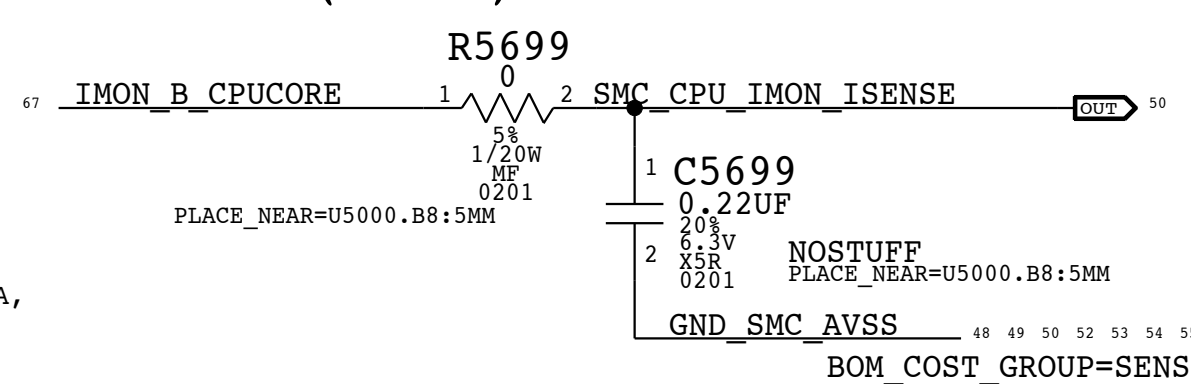
CPU Core Voltage Sense (VCAC)

SMC ADC: 20



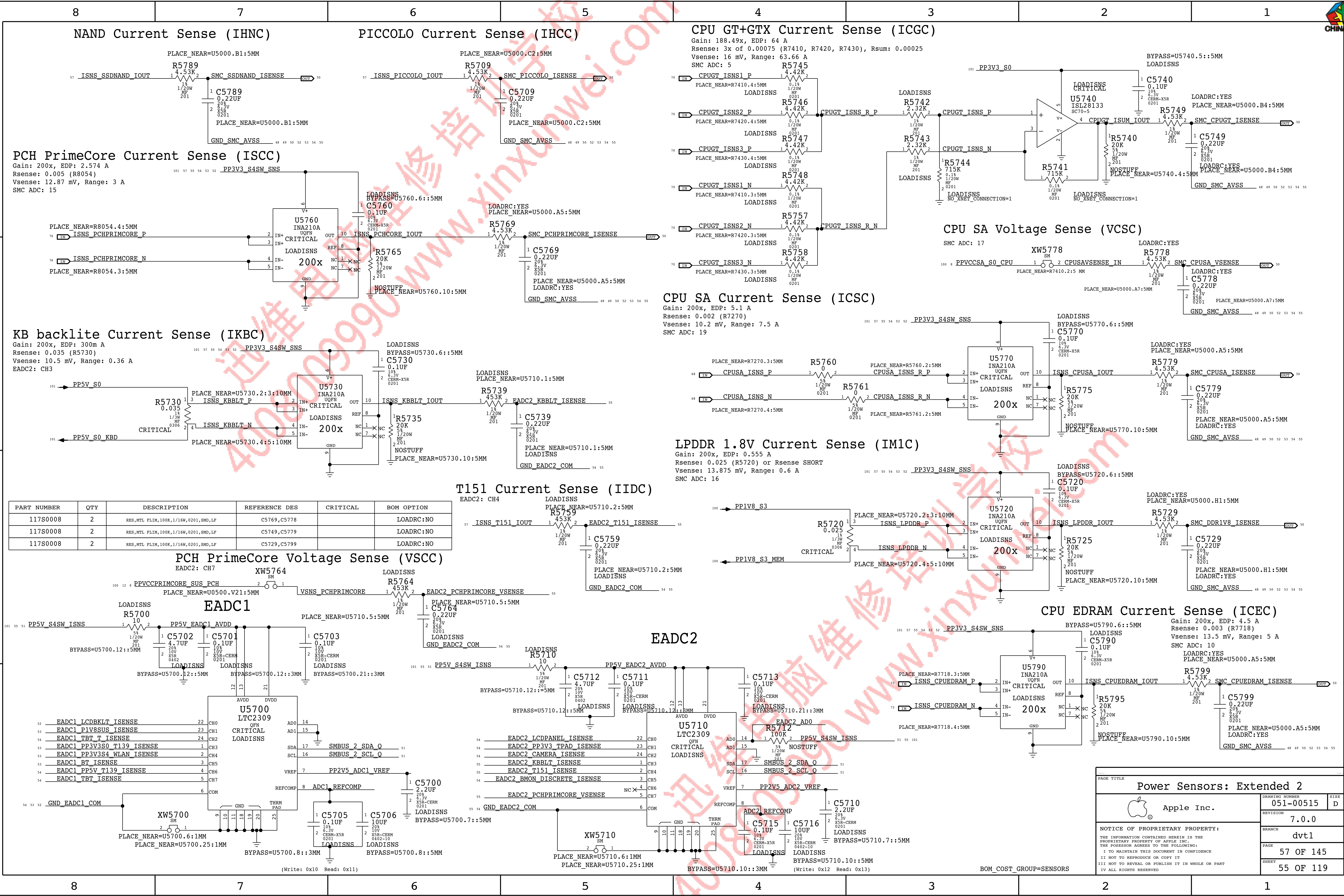
CPU Core IMON Current Sense (ICAM)

Gain: 1 A / 34.223 mV, Range: 29 A.
SMC ADC: 22



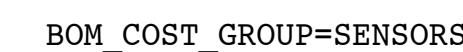
With R7150 (Ri) set to 226 Ohm,
R7210 (Rsen) set to 0.75 mOhm,
R7160 set to 82.5 kohm,
Num Phases (N) is 2, and Io (ICMax) is 29A,
then 1A of Io gives 34.223mV at the Vimon.

PAGE TITLE		
Power Sensors: Extended		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
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	PAGE	56 OF 145
	SHEET	54 OF 119



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5769,C5778		LOADRC:NO
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5749,C5779		LOADRC:NO
117S0008	2	RES,MTL FLIM,100K,1/16W,0201,SMD,LF	C5729,C5799		LOADRC:NO

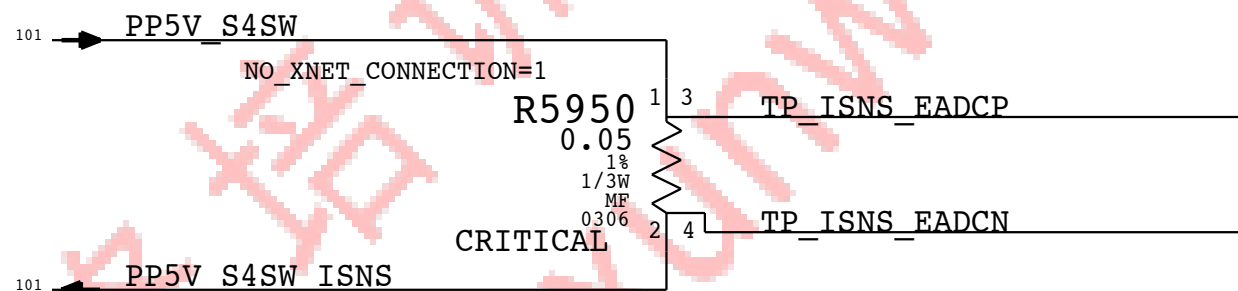
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Power Sensors: Extended 2		
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	PAGE	57 OF 145
	SHEET	55 OF 119



REV: 000001-1_0001 PAGE TITLE Thermal Sensors  Apple Inc.		DRAWING NUMBER 051-00515 REVISION 7.0.0		SHEET D
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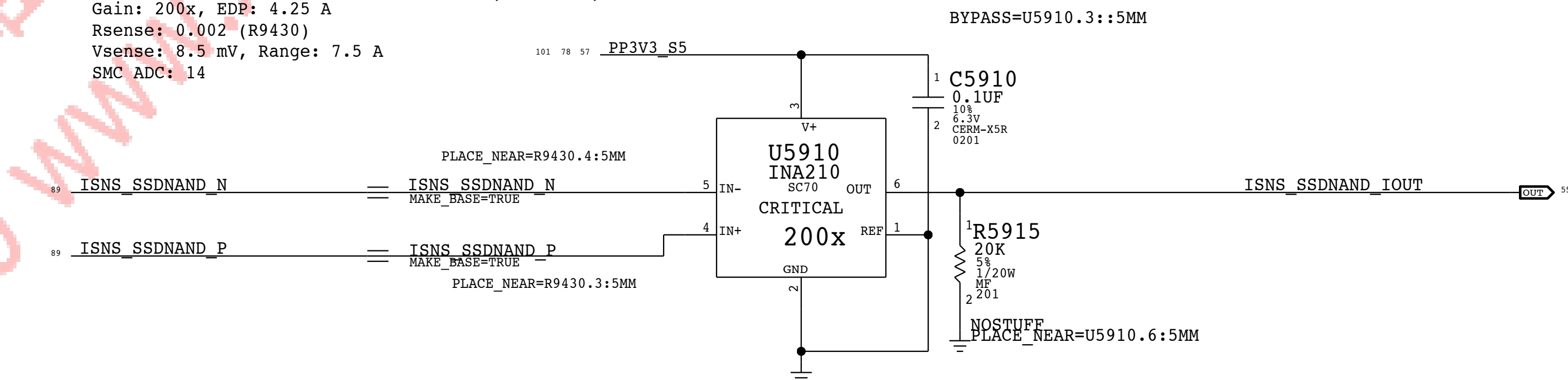
EADC Current Sense (IADC)

Gain: 200x, EDP: 200m A
Rsense: 0.050 (R5950) or Rsense SHORT
Vsense: 10 mV, Range: 0.25 A
EADC2: CH7



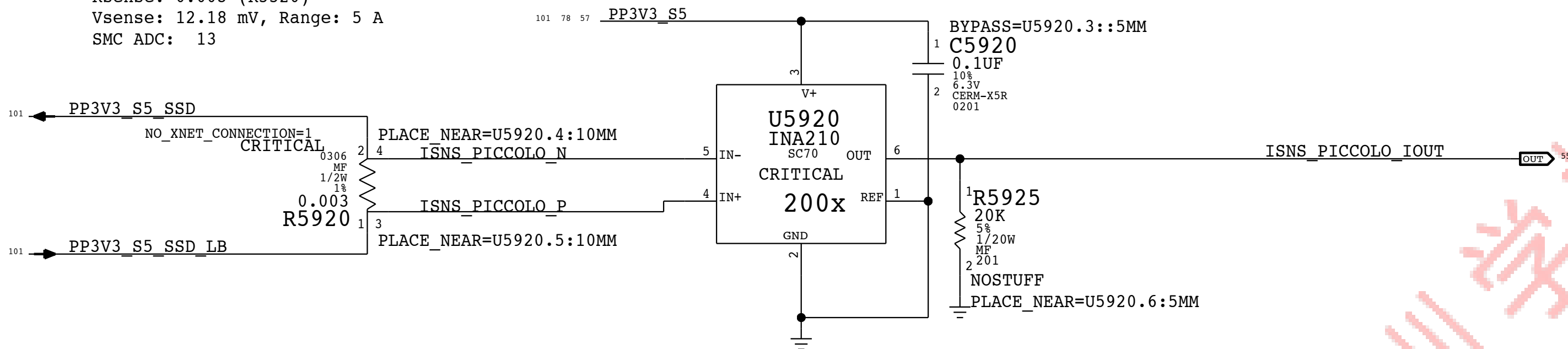
NAND Current Sense (IHNC)

Gain: 200x, EDP: 4.25 A
Rsense: 0.002 (R9430)
Vsense: 8.5 mV, Range: 7.5 A
SMC ADC: 14



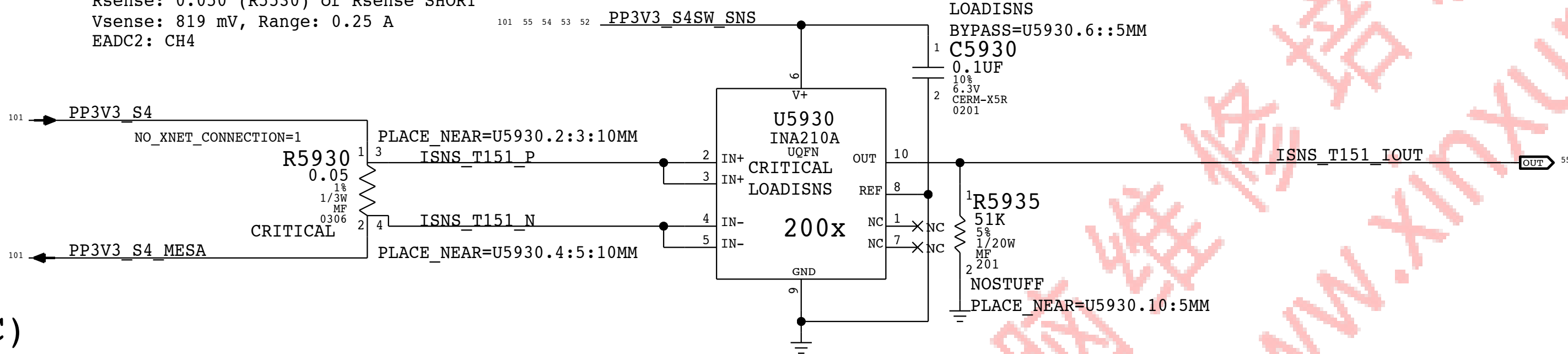
Piccolo Current Sense (IHCC)

Gain: 200x, EDP: 4.06 A
Rsense: 0.003 (R5520)
Vsense: 12.18 mV, Range: 5 A
SMC ADC: 13



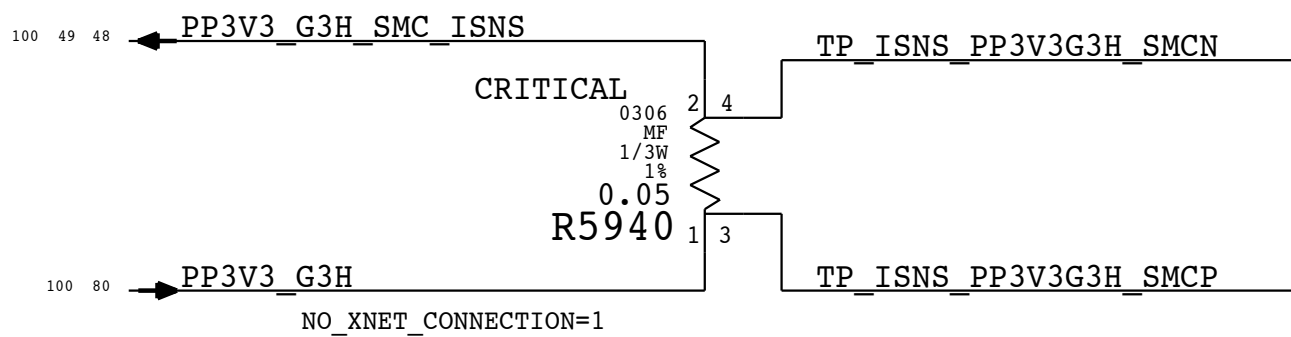
T151 Current Sense (IIDC)

Gain: 200x, EDP: 163.8m A
Rsense: 0.050 (R5530) or Rsense SHORT
Vsense: 819 mV, Range: 0.25 A
EADC2: CH4



SMC Current Sense (ISMC)

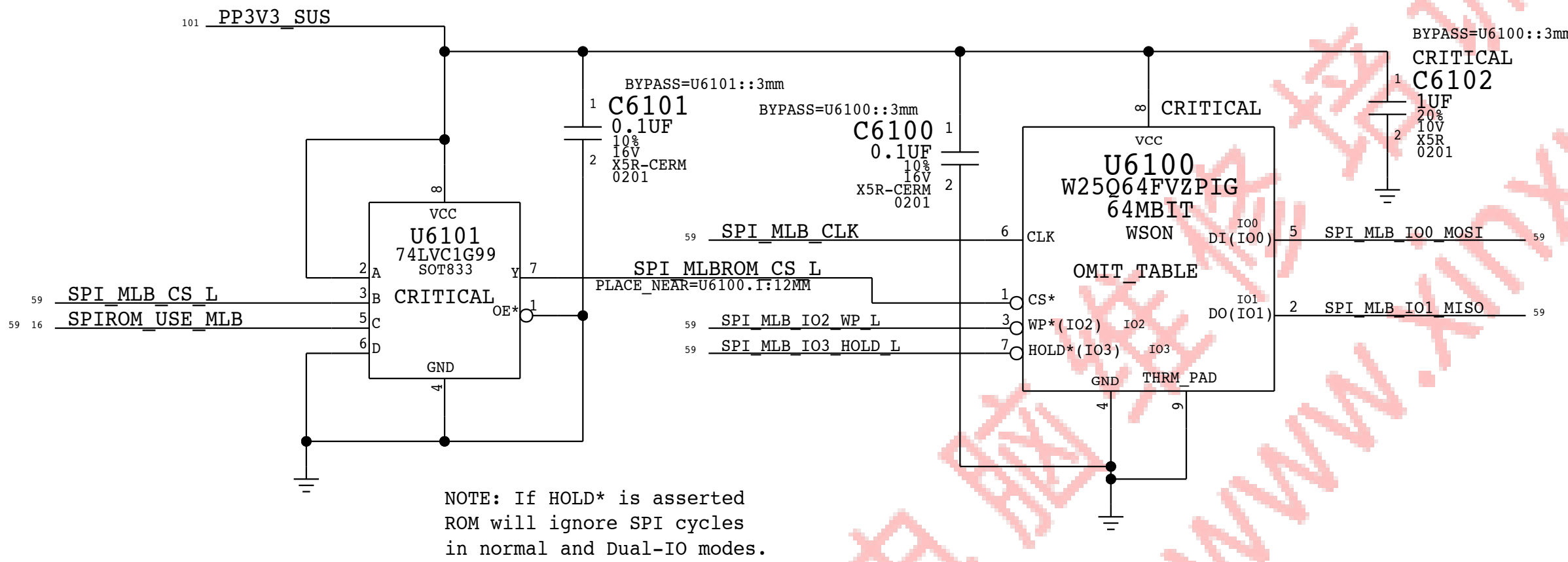
Gain: 391.67x, EDP: 0.2 A
Rsense: 0.05 (R5940) or Rsense SHORT
Vsense: 10 mV, Range: 0.21A
EADC2: CH6



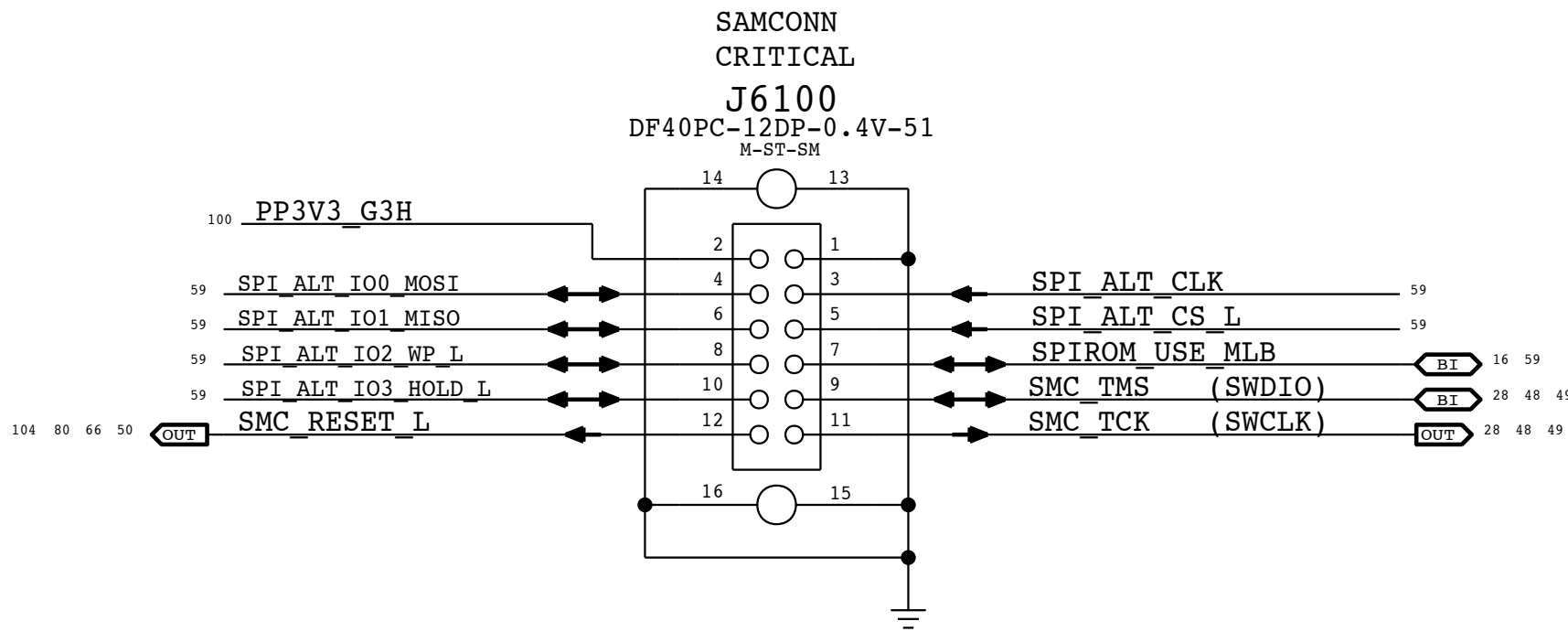
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	REVISION	7.0.0
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	PAGE	59 OF 145
	SHEET	57 OF 119

SPI ROM

Quad-I/O Mode (Mode 0 & 3) supported.
SPI Frequency: 50MHz for CPU, 20MHz for SMC.

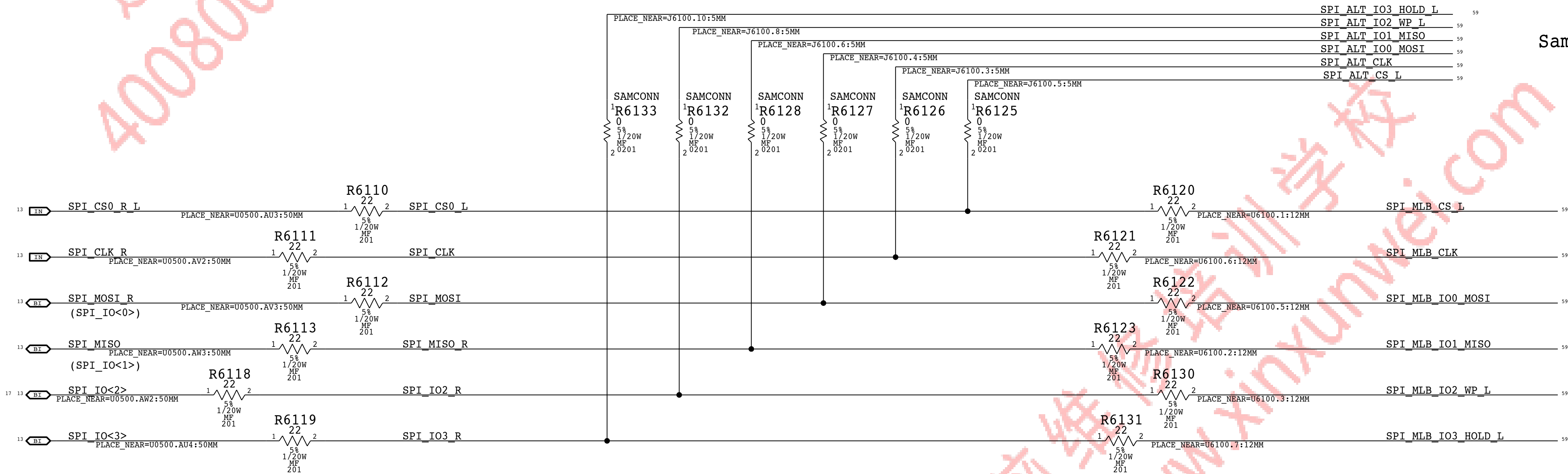


SPI+SWD SAM Connector



Quad SPI and QPI instructions require the non-volatile Quad Enable bit (QE) in Status Register-2 to be set. When QE=1, the /WP pin becomes IO2 and /HOLD pin becomes IO3.

SPI Bus Series Termination (Modified per PDG)



Sam Card ROM Slave

SPI ROM Slave

SYNC_MASTER=J52_MLB		SYNC_DATE=05/12/2015	
PAGE TITLE			
SPI Debug Connector			
 Apple Inc.	DRAWING NUMBER		SHEET
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	BRANCH		dvt1
	PAGE		61 OF 145
	SHEET		59 OF 119

BOM_COST_GROUP=CPU & CHIPSET

D

C

B

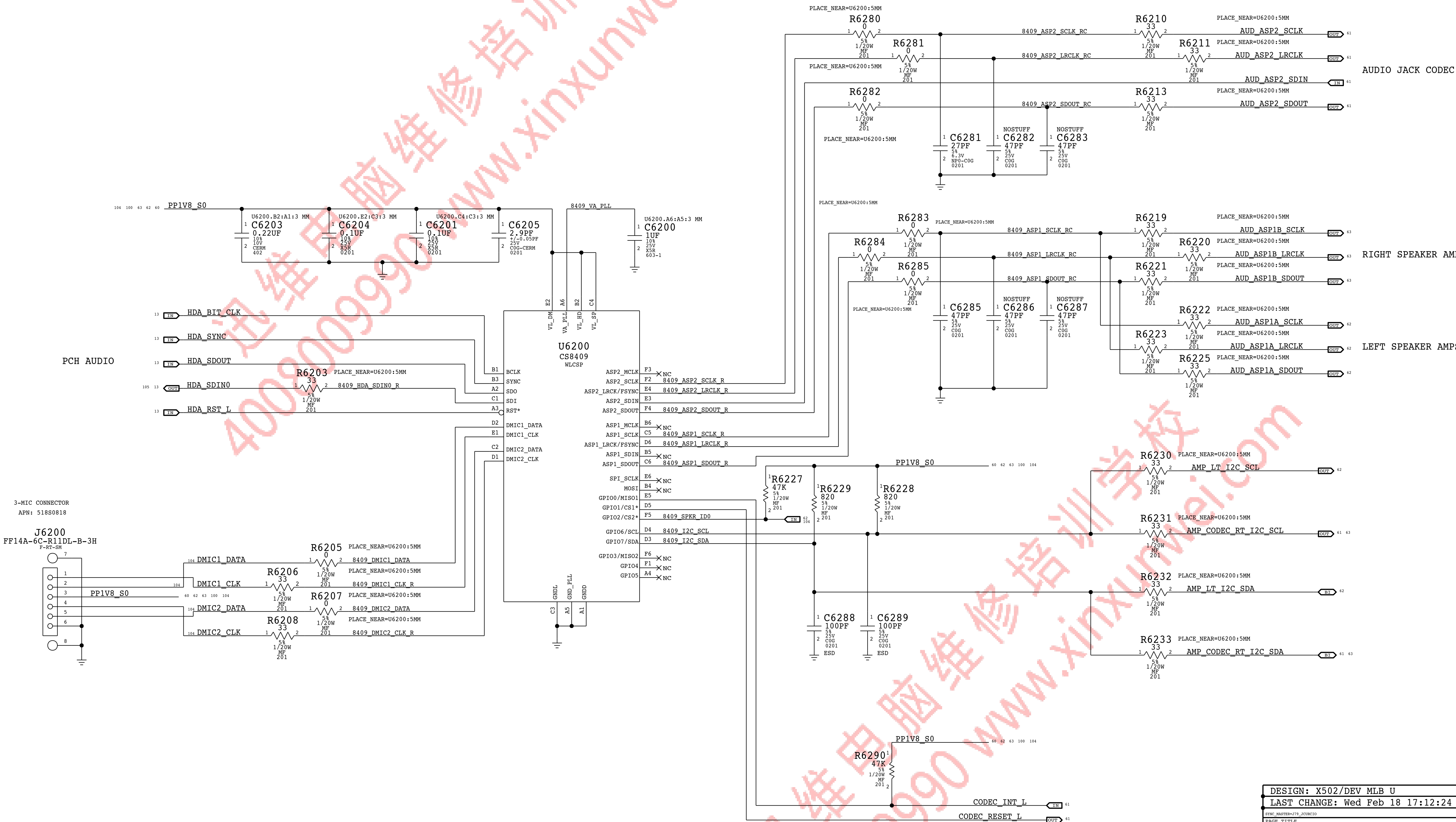
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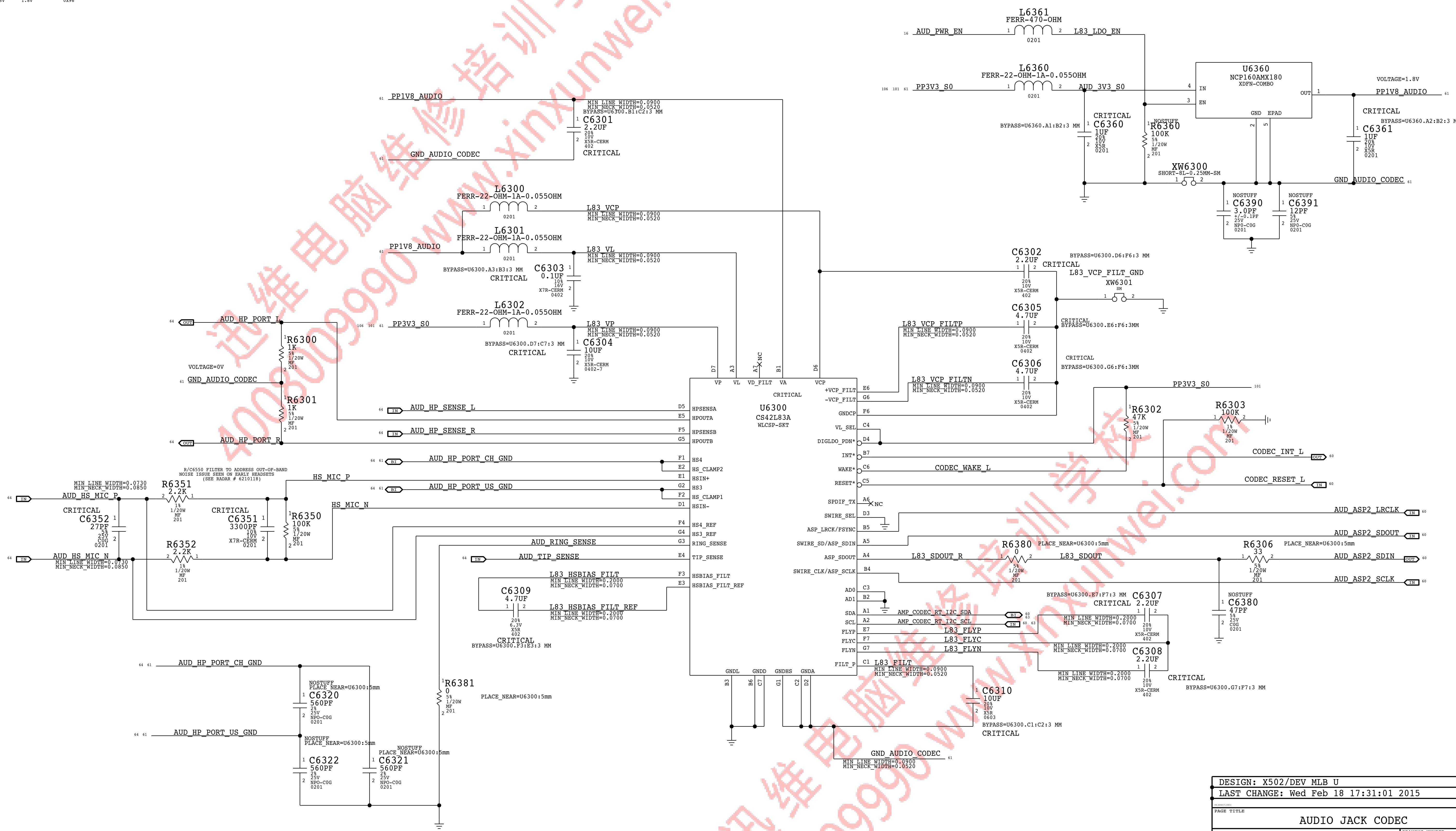
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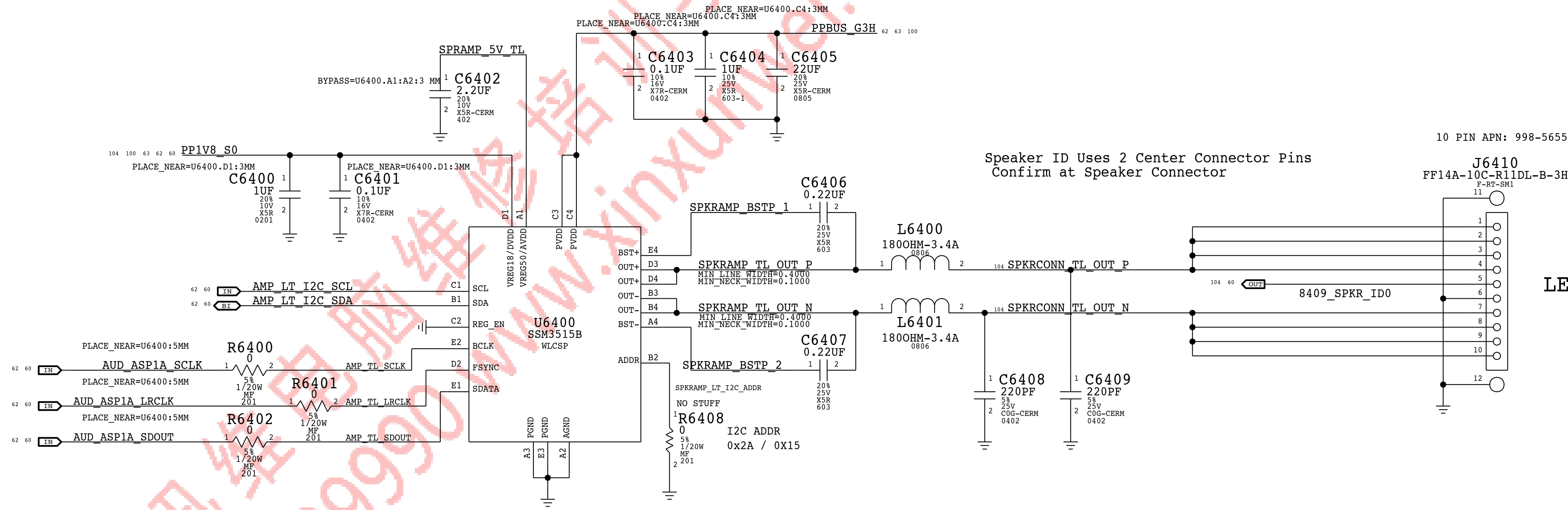
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DESIGN: X502/DEV MLB U		
LAST CHANGE: Wed Feb 18 17:12:24 2015		
PAGE TITLE		
HDA Bridge		
	DRAWING NUMBER	051-00515
	REVISION	7.0.0
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	PAGE	62 OF 145
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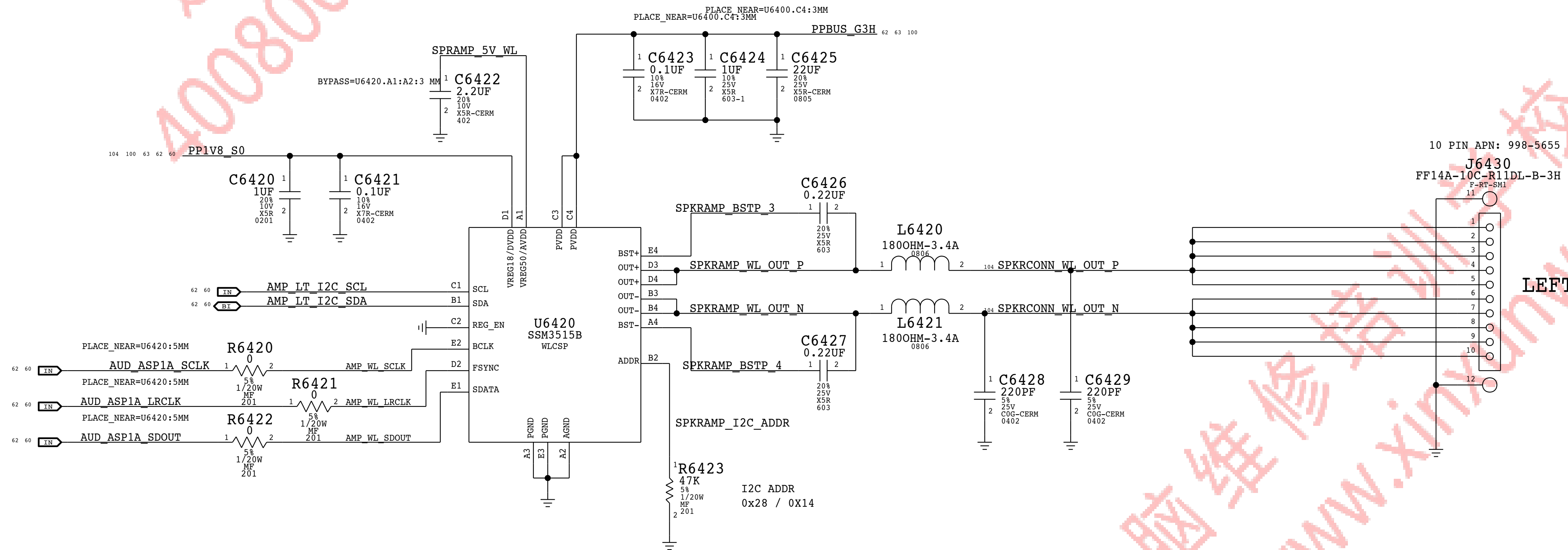
AUDIO JACK CODEC I2C ADDRESS		
AD1	AD0	ADDRESS
GND	GND	0x90 <--
GND	1.8V	0x92
1.8V	GND	0x96
1.8V	1.8V	0x96



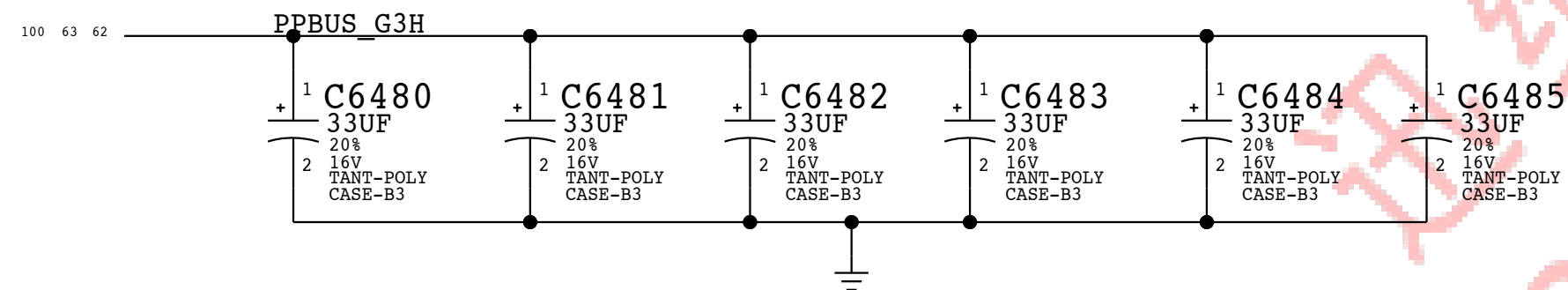
2X MONO SPEAKER AMPLIFIERS
APN: 35384074
GAIN = TBD



LEFT TWEETER SPEAKER CONNECTOR



LEFT WOOFER SPEAKER CONNECTOR



BOM_COST_GROUP=AUDIO

DESIGN: X502/DEV MLB U		
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PAGE TITLE		
Left Speaker Amps & Conn		
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	REVISION	7.0.0
	BRANCH	dvt1
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2X MONO SPEAKER AMPLIFIERS
APN: 35384073
GAIN = TBD

10-PIN APN: 998-5655

J6500
FF14A-10C-R11DL-B-3H
F-RT-SM1

RIGHT TWEETER SPEAKER CONNECTOR

10-PIN APN: 998-5655

J6550
FF14A-10C-R11DL-B-3H
F-RT-SM1

RIGHT WOOFER SPEAKER CONNECTOR

BOM_COST_GROUP=AUDIO

DESIGN: X502/DEV MLB U
LAST CHANGE: Wed Feb 18 17:12:24 2015

PAGE TITLE

Right Speaker Amps & Conn



Apple Inc.

DRAWING NUMBER
051-00515

STAR
D

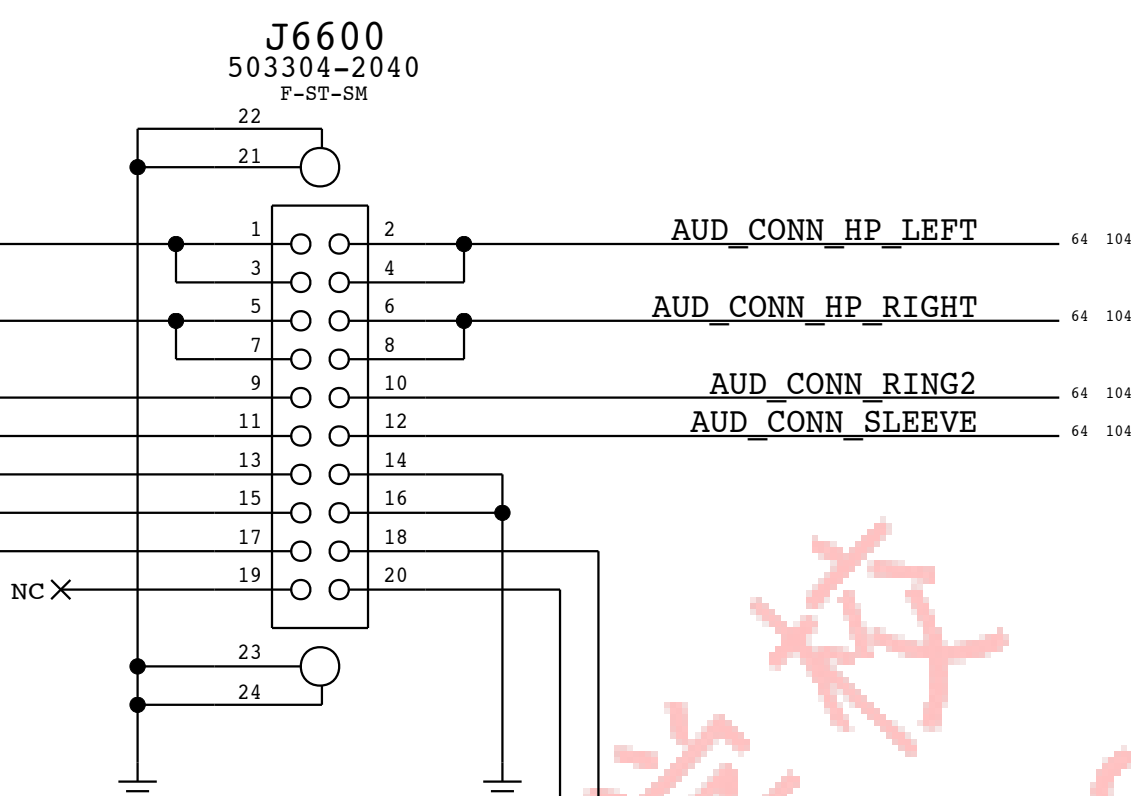
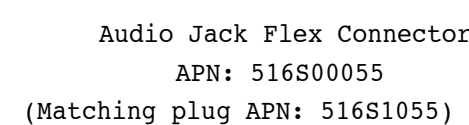
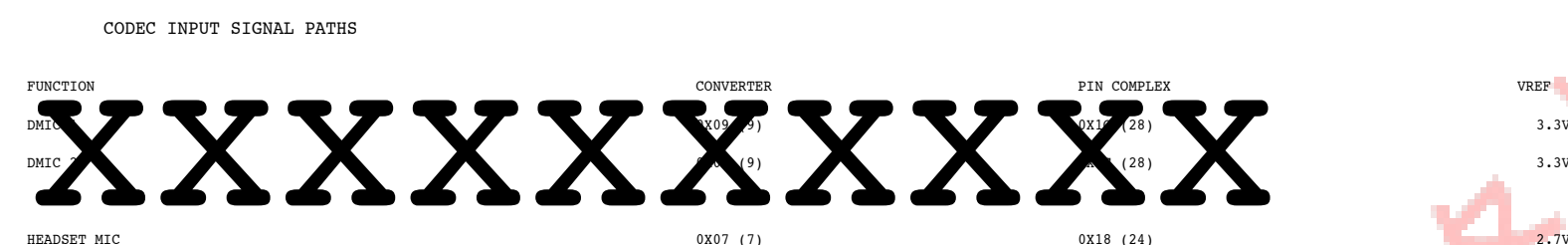
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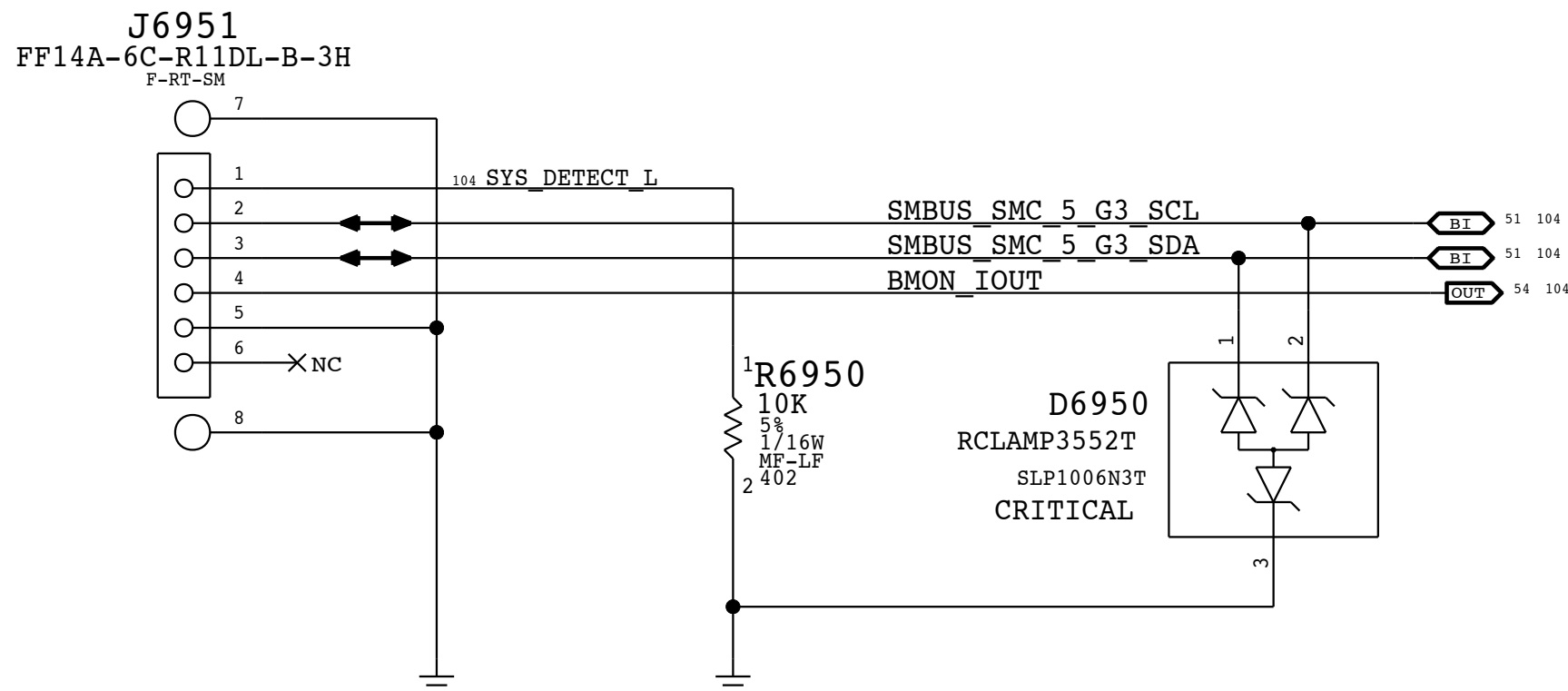
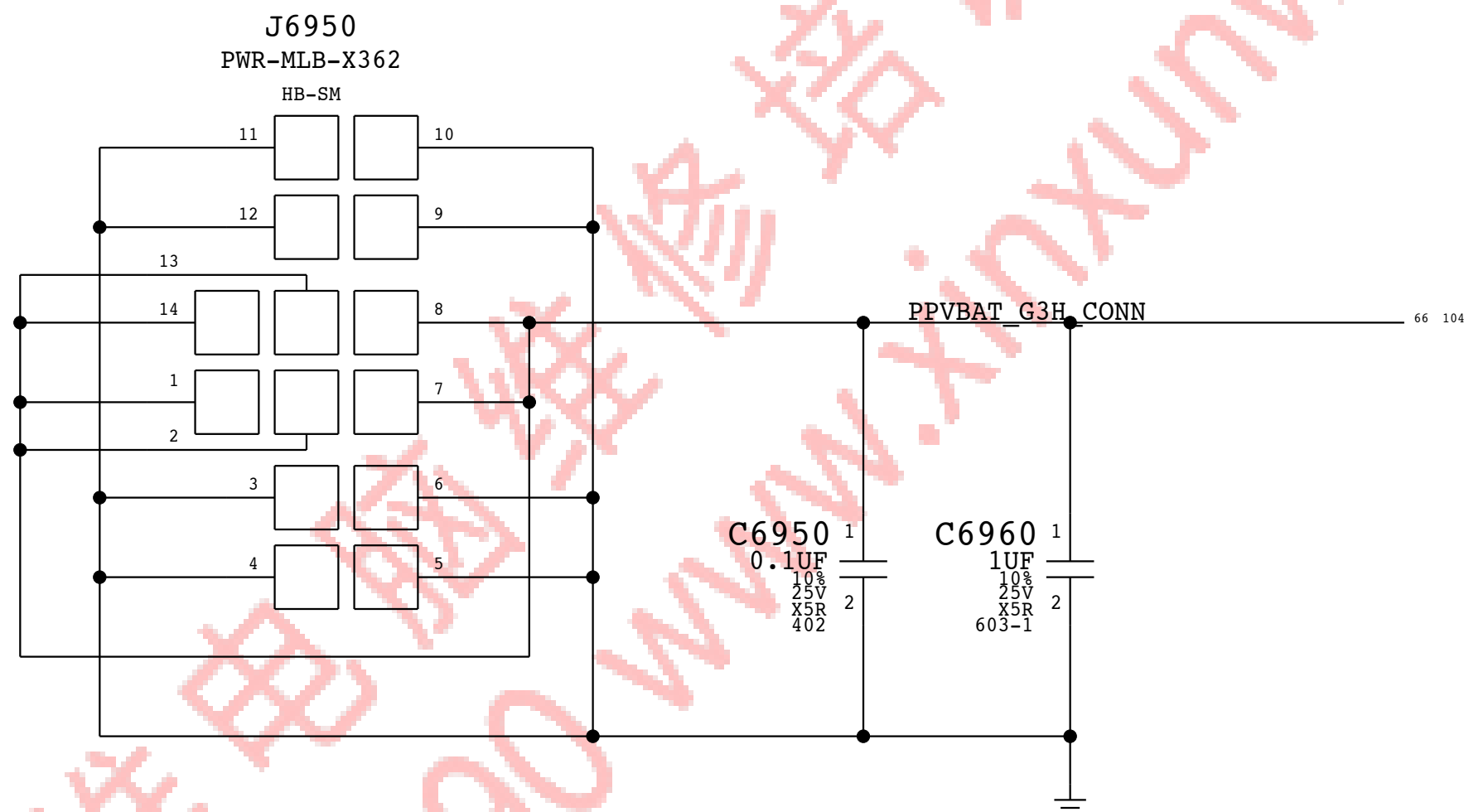
PAGE
65 OF 145

SHEET
63 OF 119

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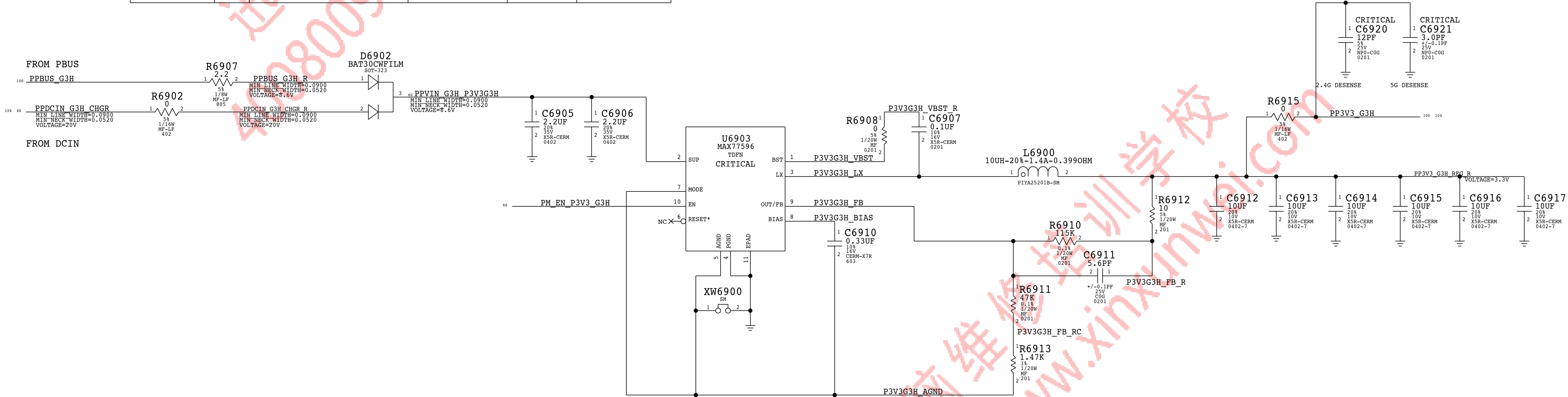


J79 Battery Hotbar Flex Pads

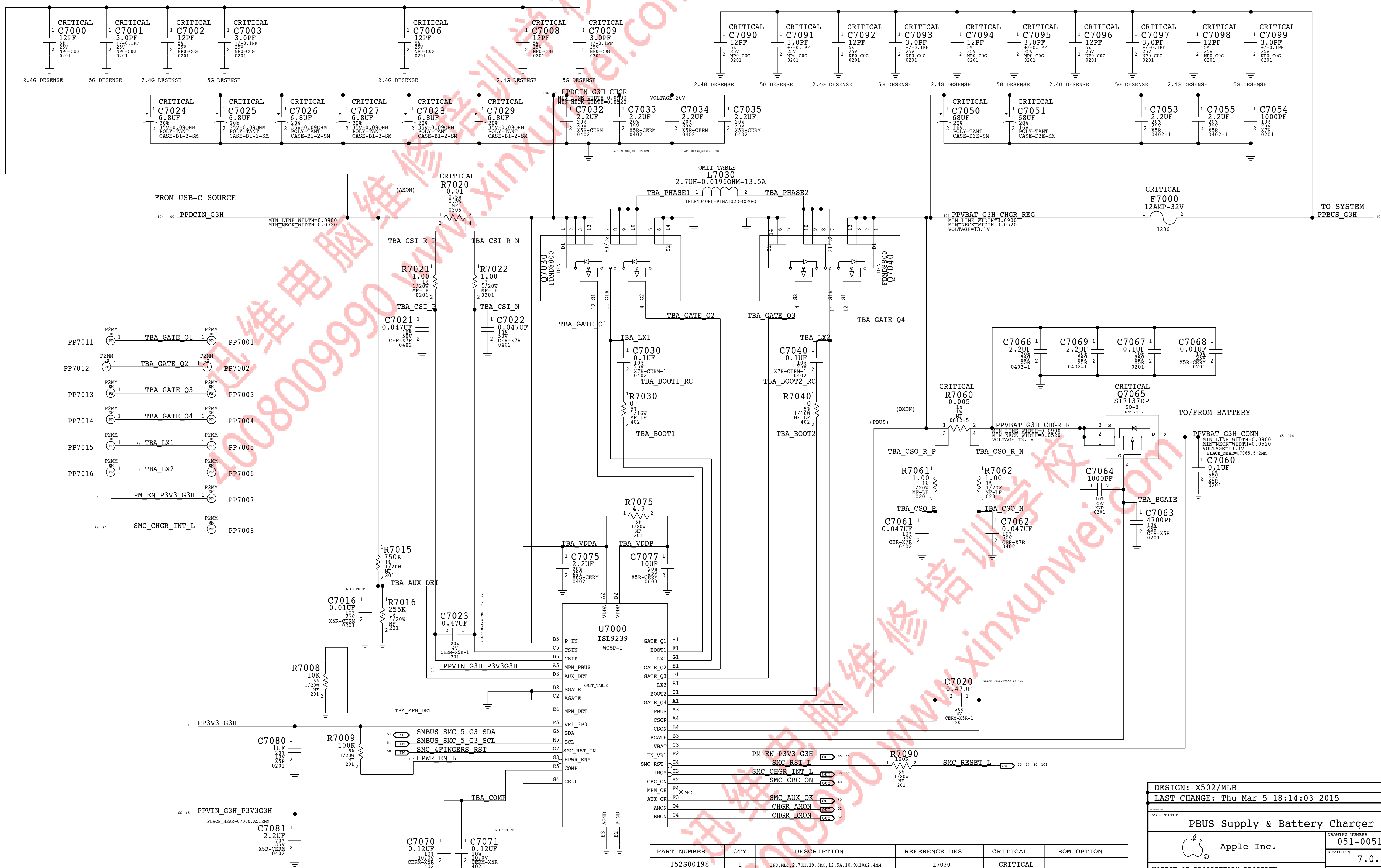


BMU POWER FLEX HOTBAR'd TO THE MLB:

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
632-00566	1	PCBA,FLEX,BMU PWR,X362	J6950	CRITICAL	

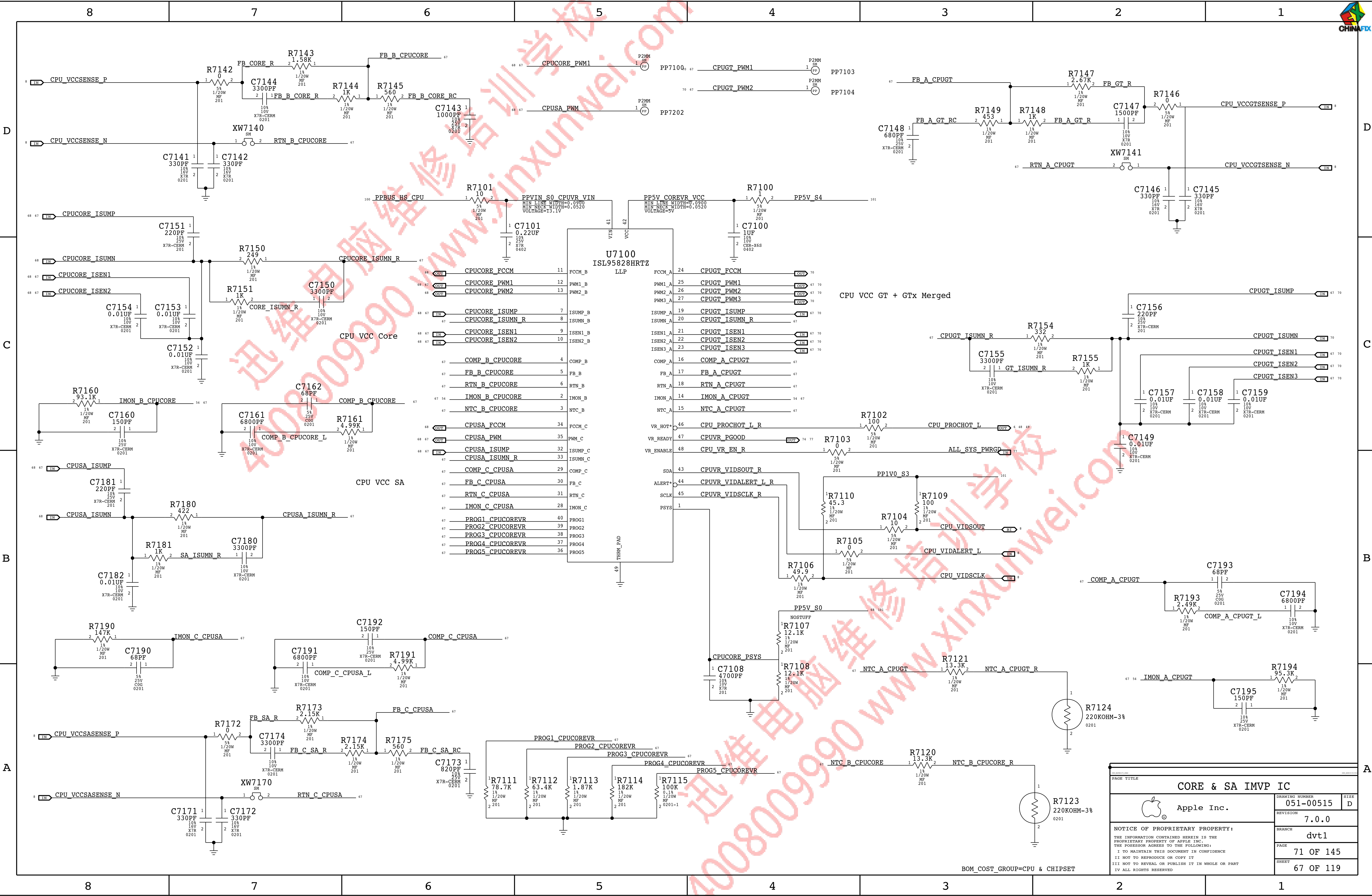


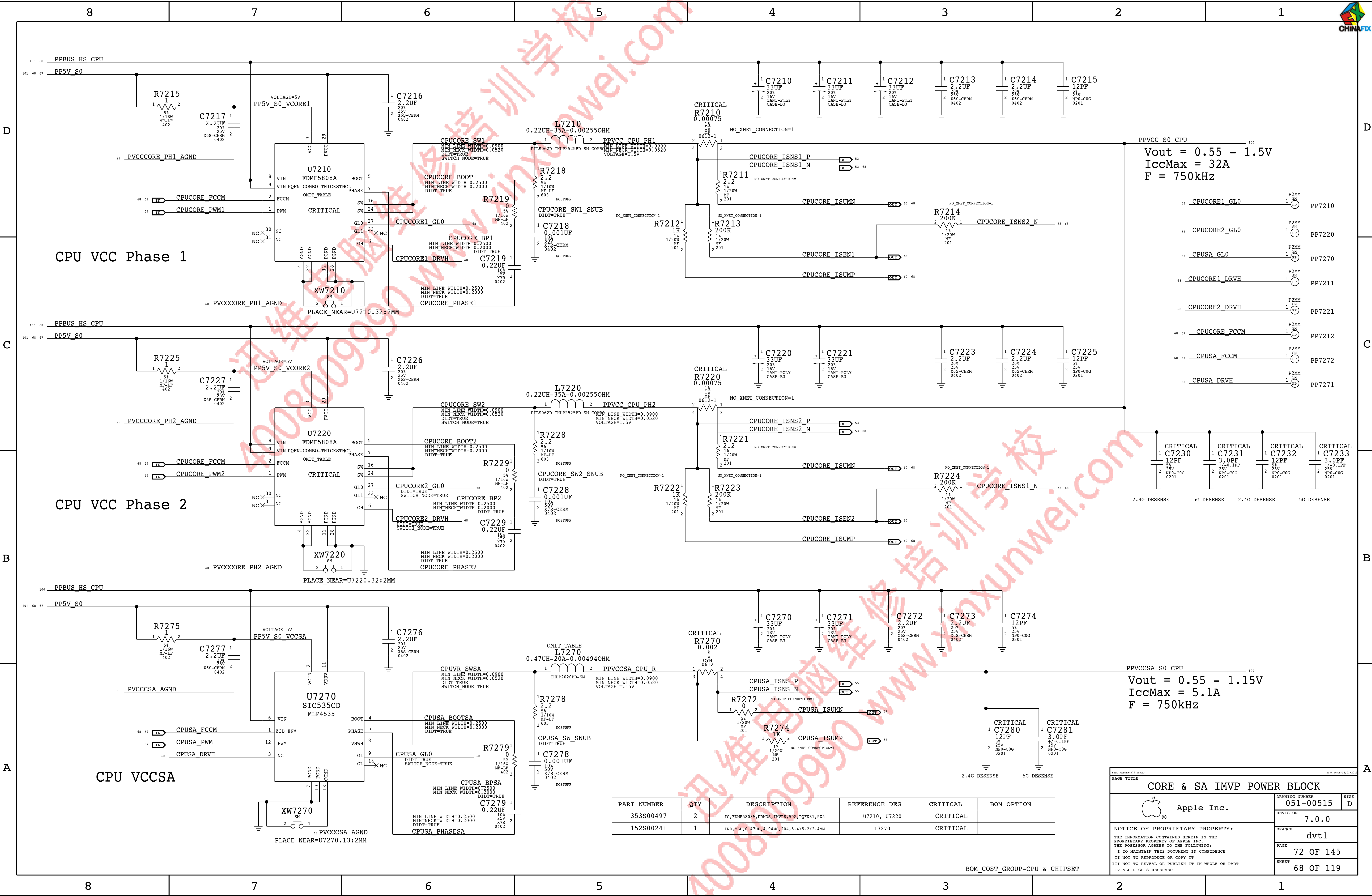
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DC-In & Battery Connectors		
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	REVISION	7.0.0
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	PAGE	69 OF 145
	SHEET	65 OF 119



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
152S00198	1	IND,MLD,2.7UH,19.6MO,12.5A,10.9K10X2.4MM	L7030	CRITICAL	

BOM_COST_GROUP=PLATFORM POWER

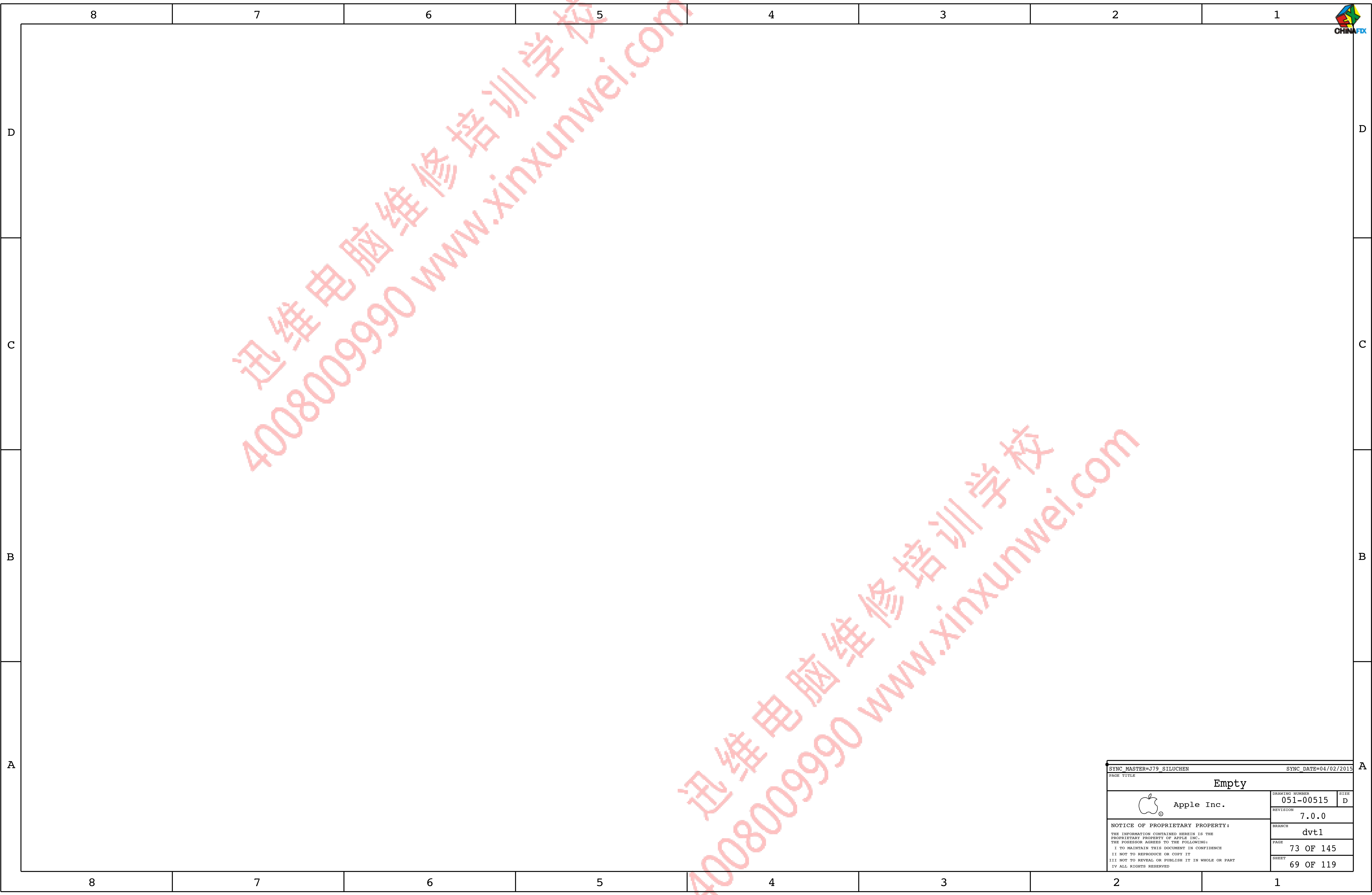




PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	2	IC,FDMF5808A,DRMOS,IMVP8,50A,PQFN31,5X5	U7210, U7220	CRITICAL	
152S00241	1	IND,MFD,0.47UH,4.94NO,20A,5.4X5.2X2.4MM	L7270	CRITICAL	

CORE & SA IMVP POWER BLOCK			DRWING NUMBER	051-00515	STAR
			REVISION	7.0.0	D
			BRANCH	dvt1	
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1111 NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART					

BOM_COST_GROUP=CPU & CHIPSET



SYNC_MASTER=J79_SILUCHEN		SYNC_DATE=04/02/2015	
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 Apple Inc.	DRAWING NUMBER		SIZE
	051-00515		D
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	7.0.0		
	BRANCH		
	dvt1		
	PAGE		
	73 OF 145		
	SHEET		
	69 OF 119		

D

D

C

C

B

B

A

A

CPU VCCGT/GTx Phase 1

PLACE_NEAR=U7410.32:2MM
PPVCCGT_PH1_AGND

Vout = 0.55 - 1.5V
IccMax = 64A
F = 750kHz

CPU VCCGT/GTx Phase 2

PLACE_NEAR=U7420.32:2MM
PPVCCGT_PH2_AGND

CPU VCCGT/GTx Phase 3

PLACE_NEAR=U7430.32:2MM
PPVCCGT_PH3_AGND

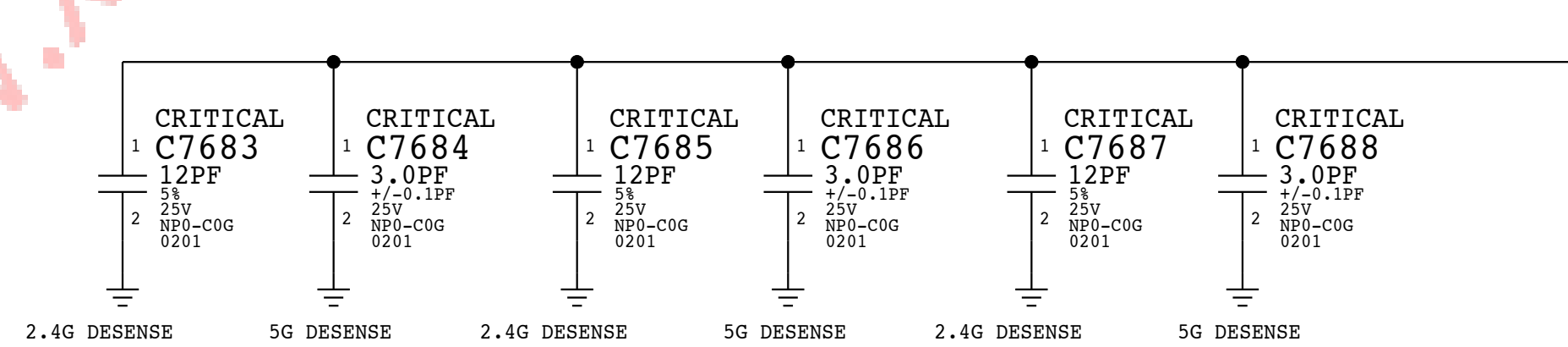
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
353S00497	3	IC,FDMF5808A,DRMOS,1MVP8,50A,PQFN31,5X5	U7410, U7420, U7430	CRITICAL	

GT & GTX IMVP POWER BLOCK		DRAWING NUMBER	051-00515	STEP	D
Apple Inc.		REVISION	7.0.0		
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		PAGE	74 OF 145		
		SHEET	70 OF 119		

BOM_COST_GROUP=CPU & CHIPSET

	8	7	6	5	4	3	2	1	
D									
C									
B									
A									
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	051-00515		D
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	7.0.0		
	BRANCH		
	dvt1		
		PAGE	
		75 OF 145	
		SHEET	
		71 OF 119	



REV 000001-1_0608		DATE: 09/07/2006	
PAGE TITLE			
Power - 5V 3.3V Supply			
 Apple Inc.	DRAWING NUMBER	SIZE	
	051-00515	D	
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	PAGE	76 OF 145	
SHEET	72 OF 119		

4



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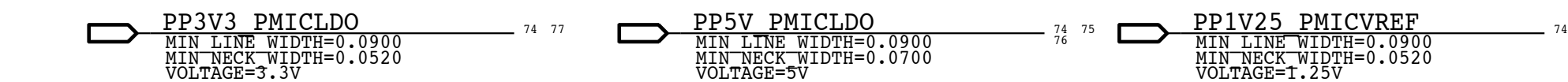


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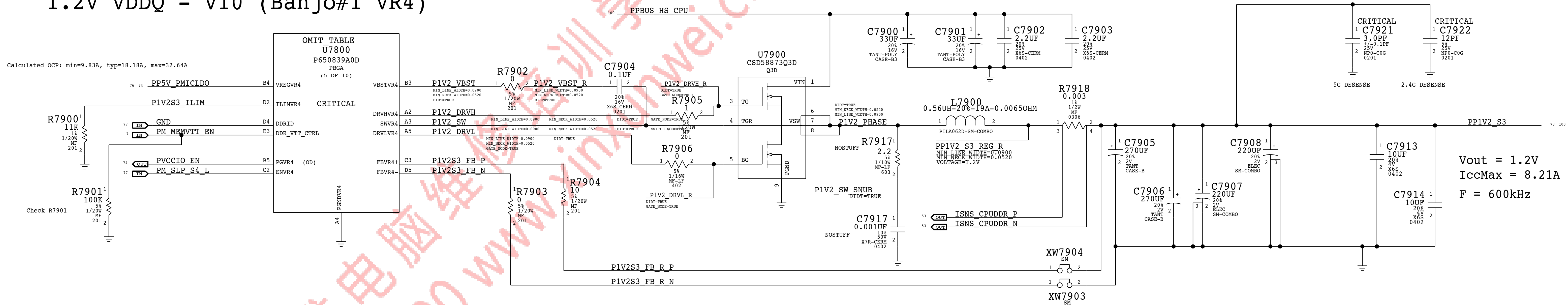
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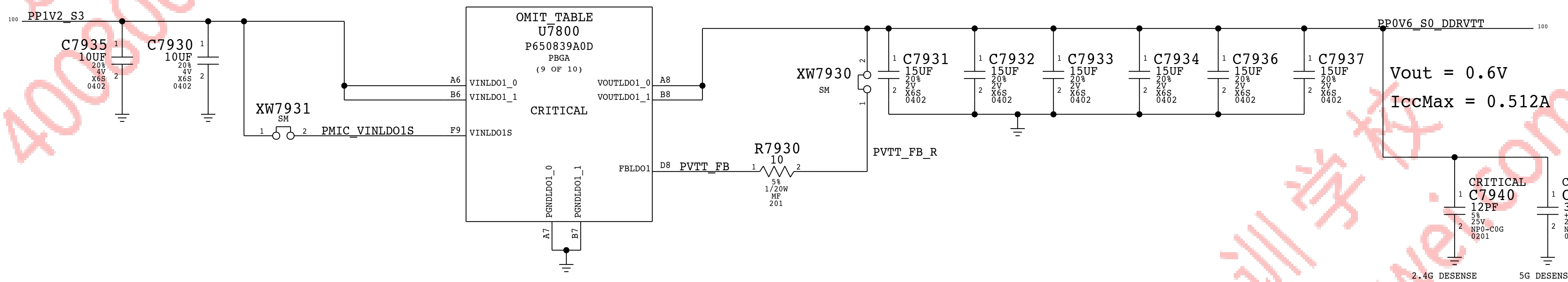
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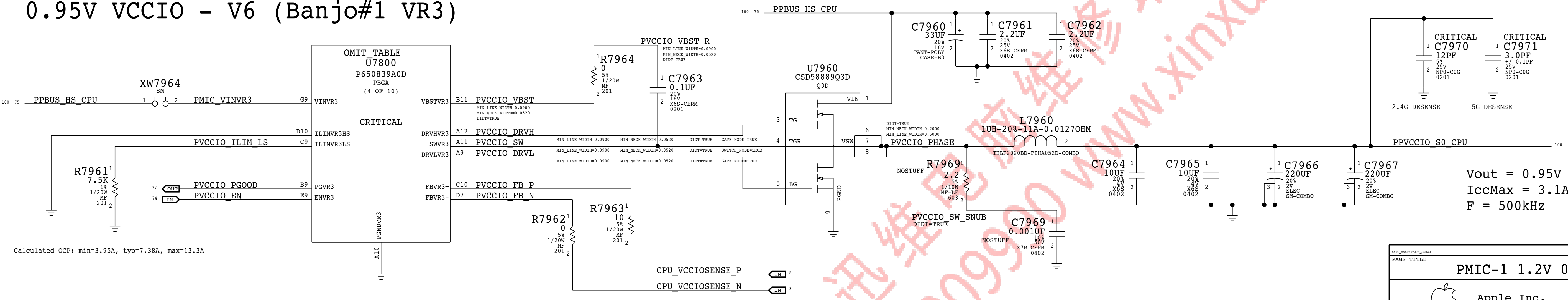
1.2V VDDQ - V10 (Banjo#1 VR4)



0.6V VTT - V13 (Banjo#1 LD01)

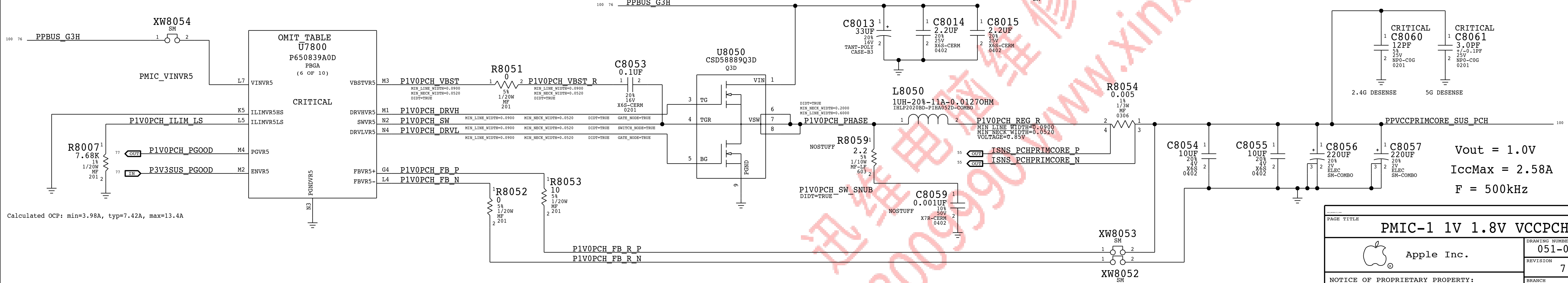
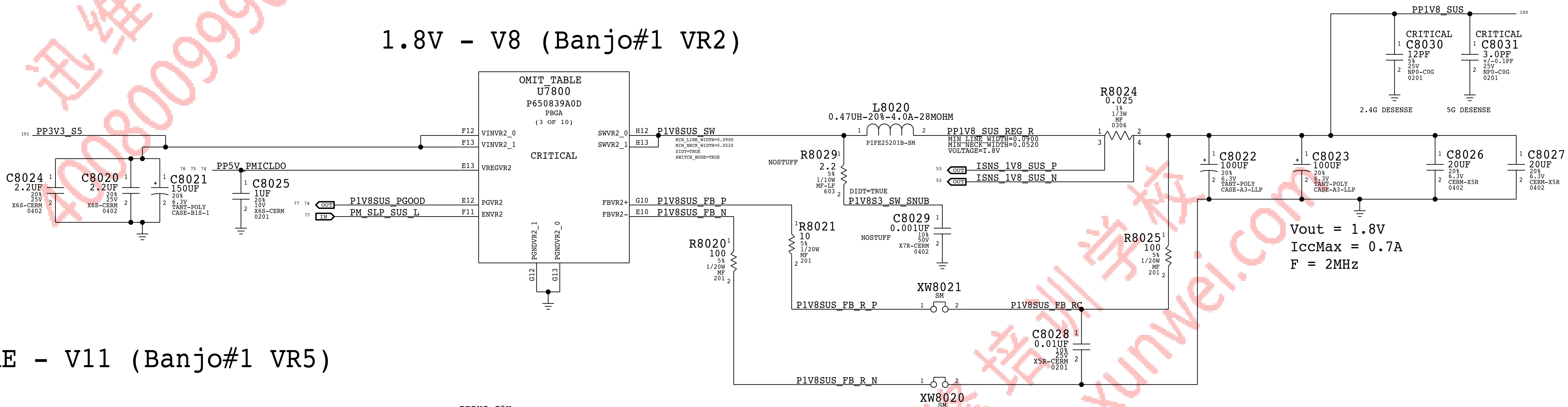


0.95V VCCIO - V6 (Banjo#1 VR3)

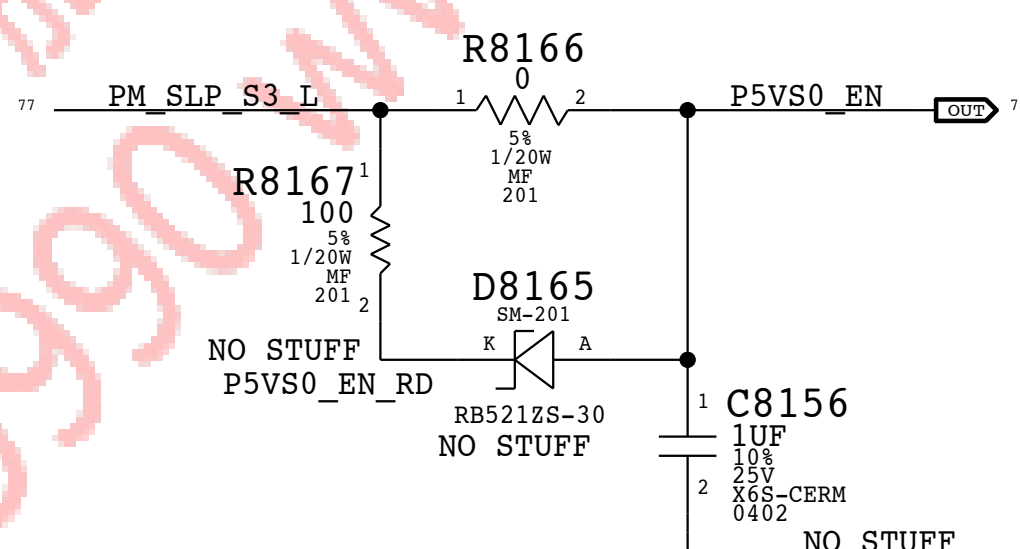
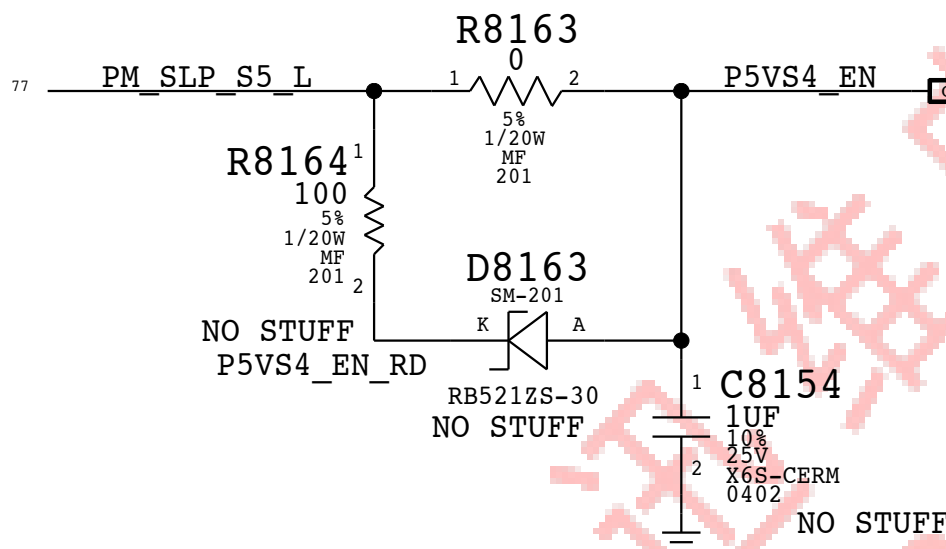
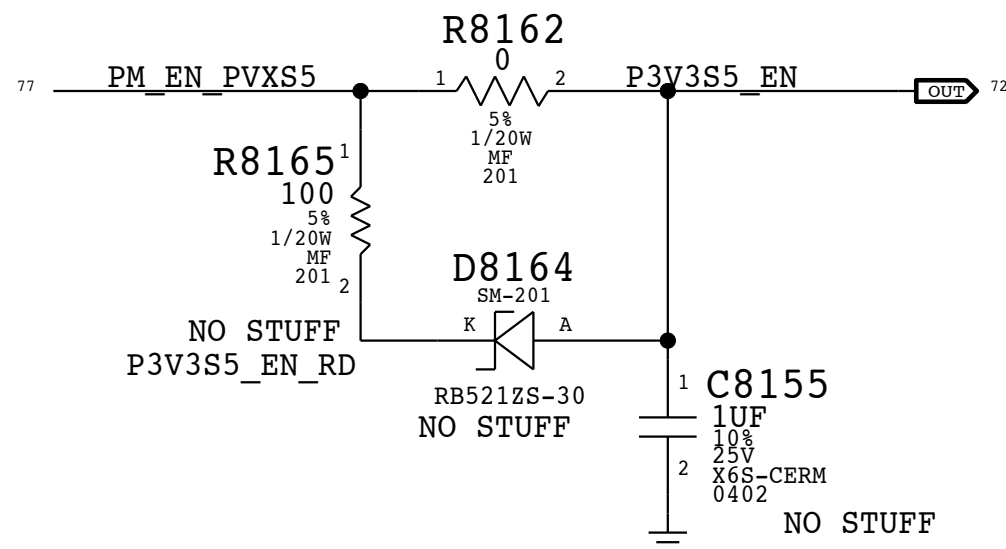
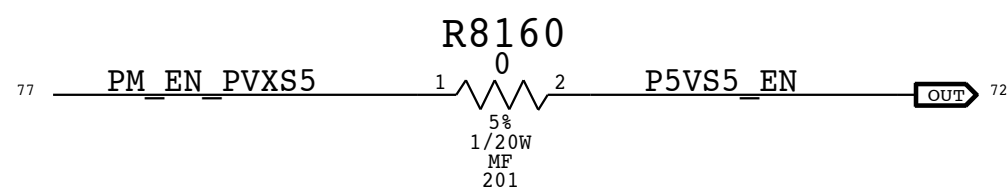
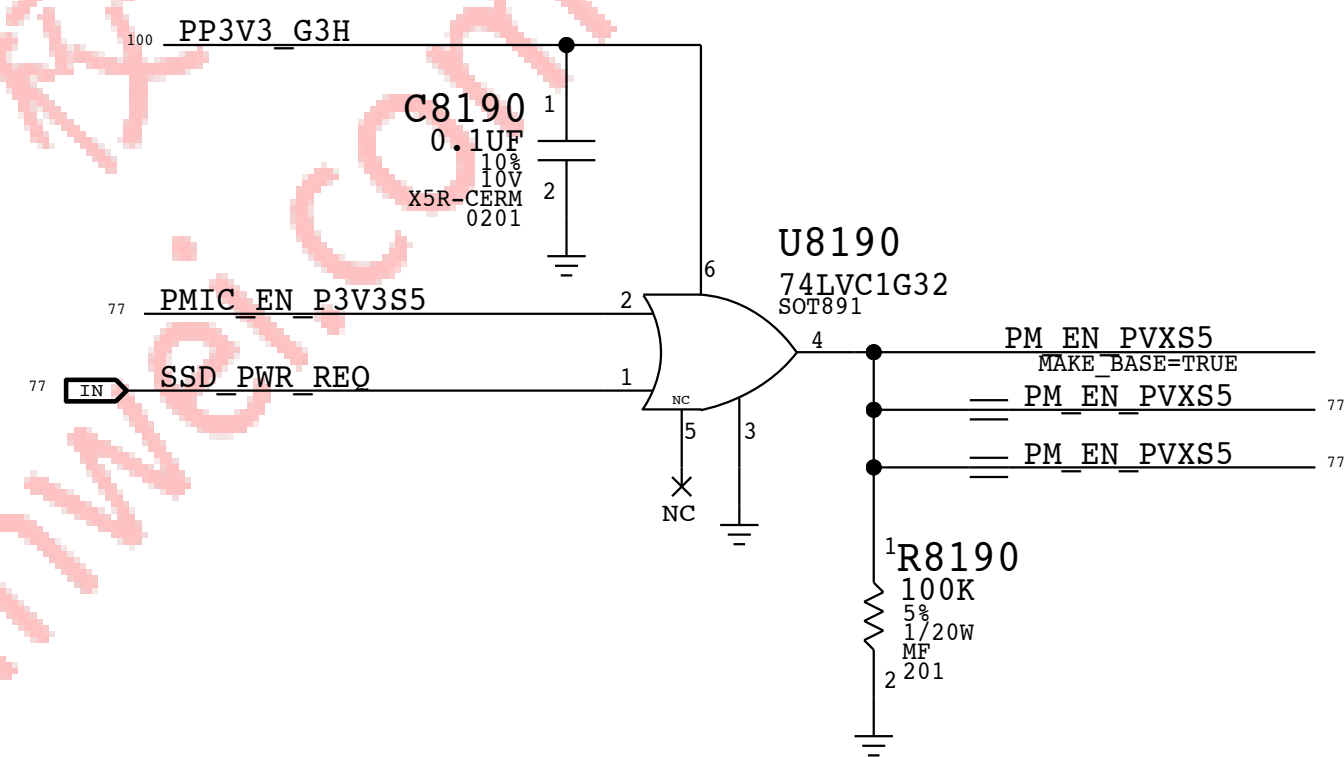
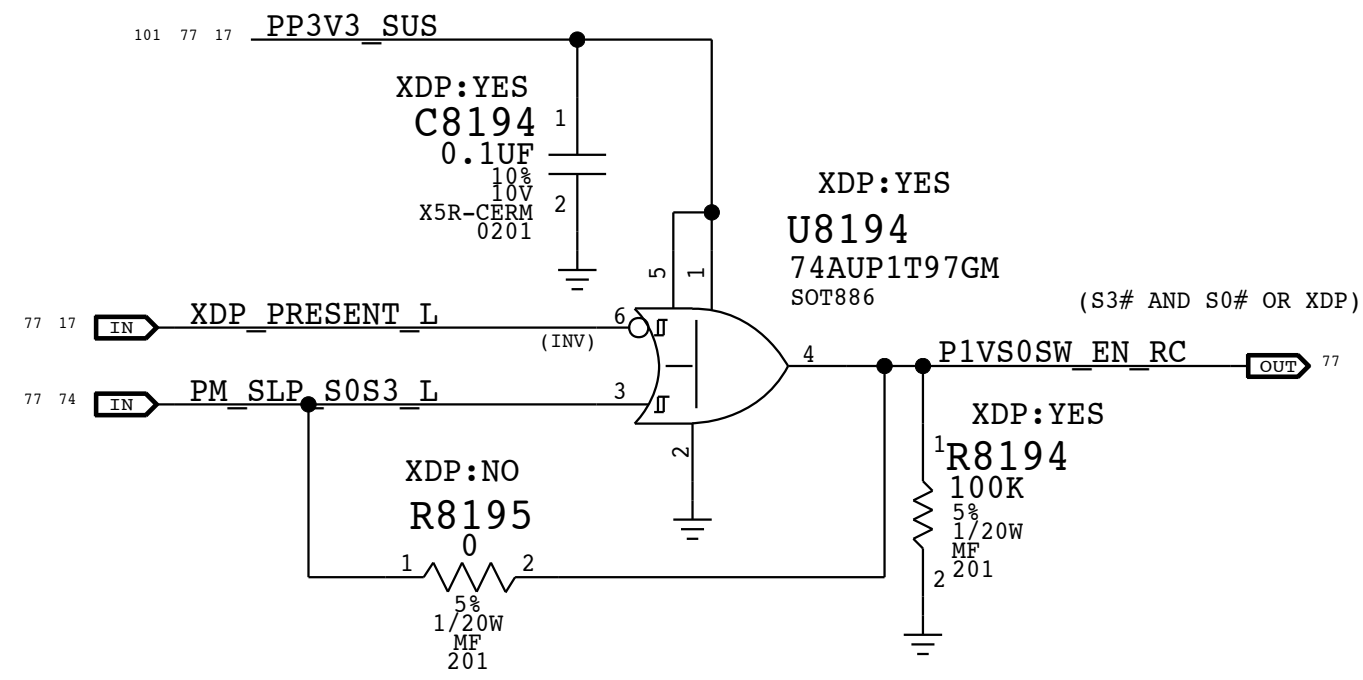
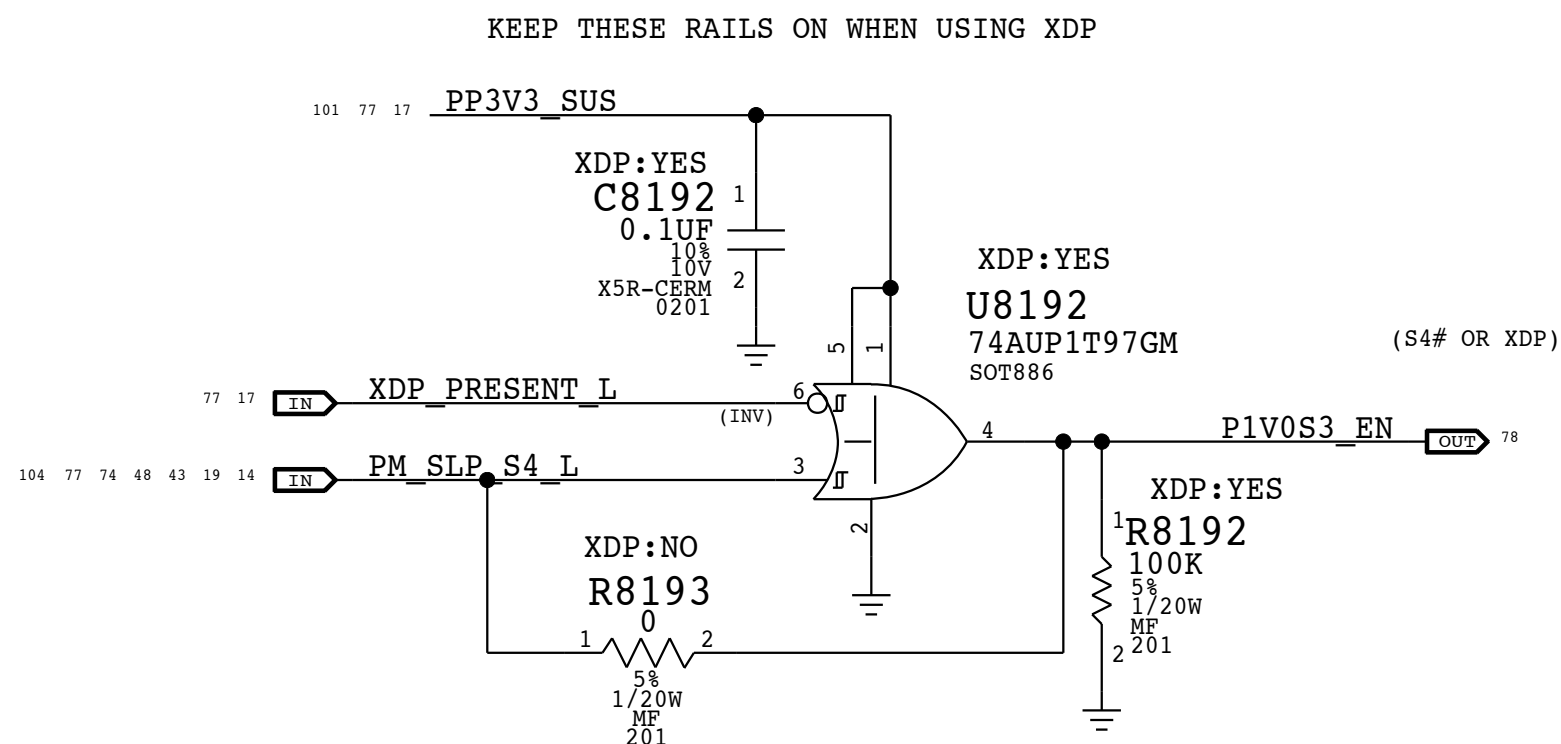


PMIC-1 1.2V 0.6V VCCIO			051-00515	D
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1.0V PCH CORE - V11 (Banjo#1 VR5)
0.7V LPM



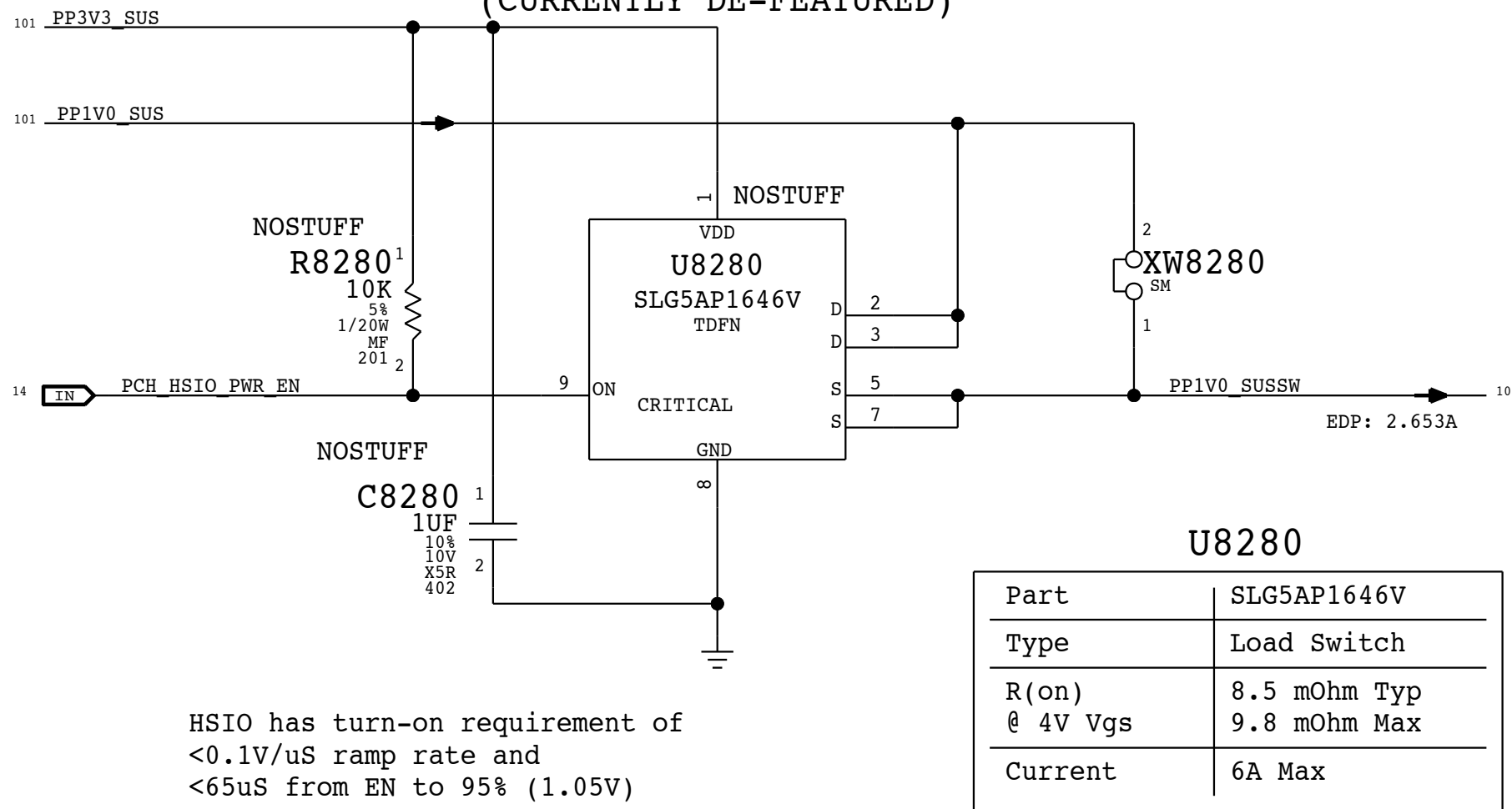
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			PAGE	
			SHEET	
			81 OF 145	
			77 OF 119	

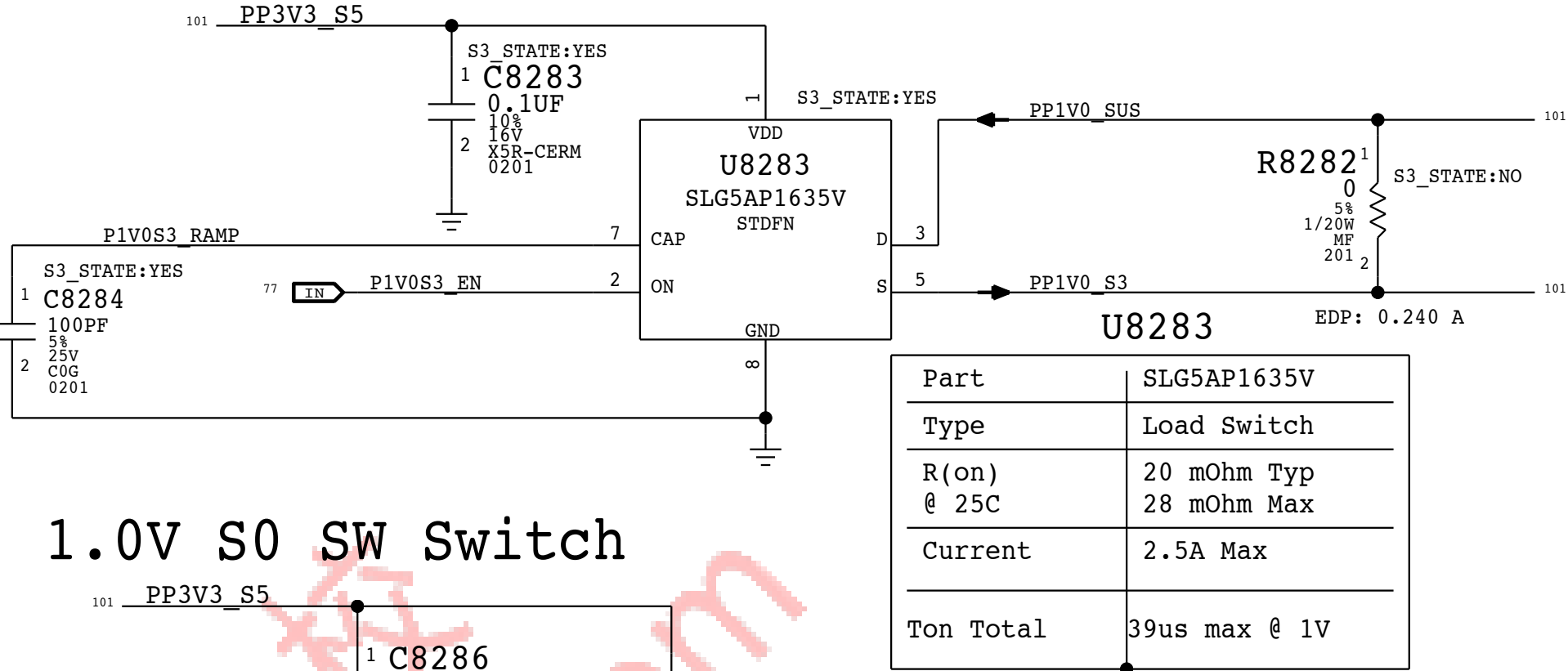
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1.0V SUS SW Switch
(CURRENTLY DE-FEATURED)



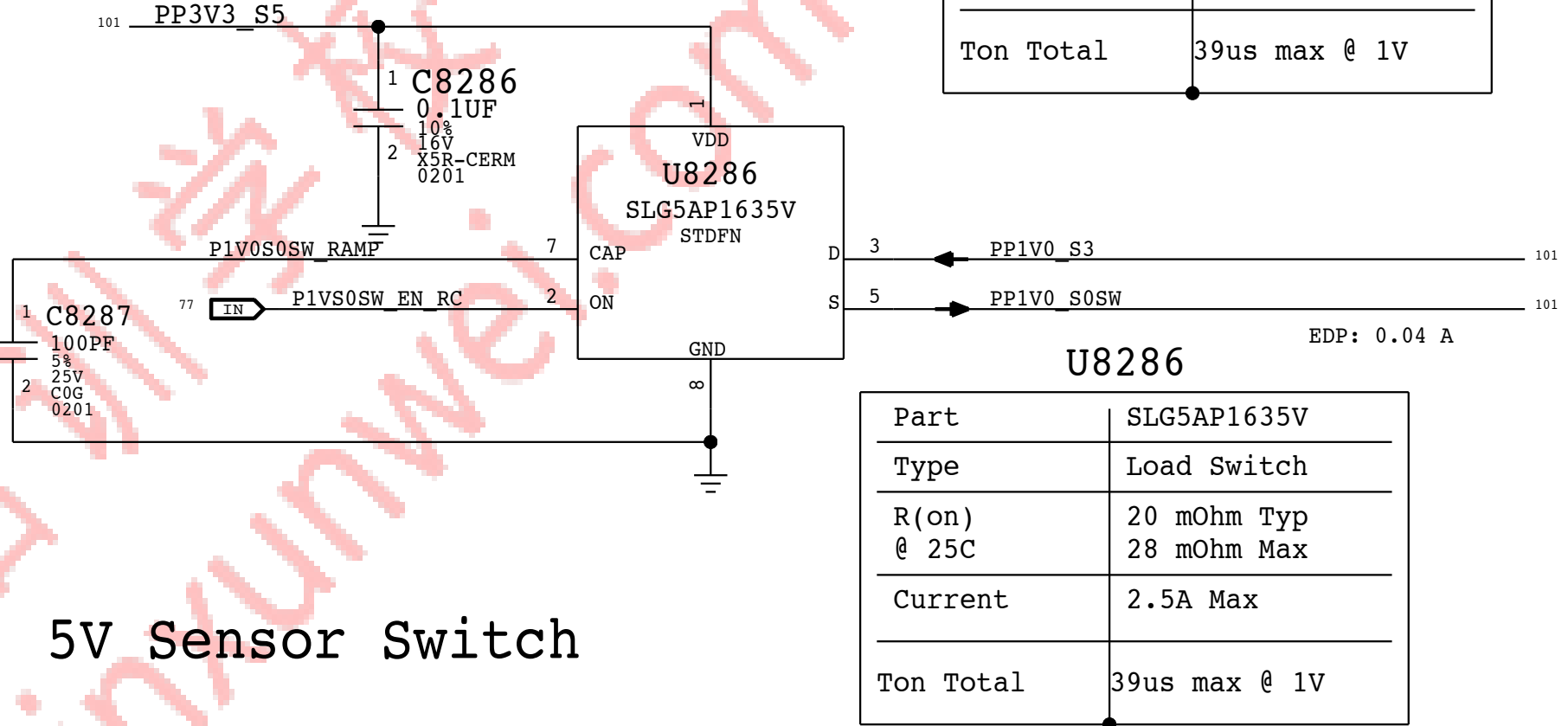
Part	SLG5AP1646V
Type	Load Switch
R(on) @ 4V Vgs	8.5 mOhm Typ 9.8 mOhm Max
Current	6A Max

1.0V S3 Switch



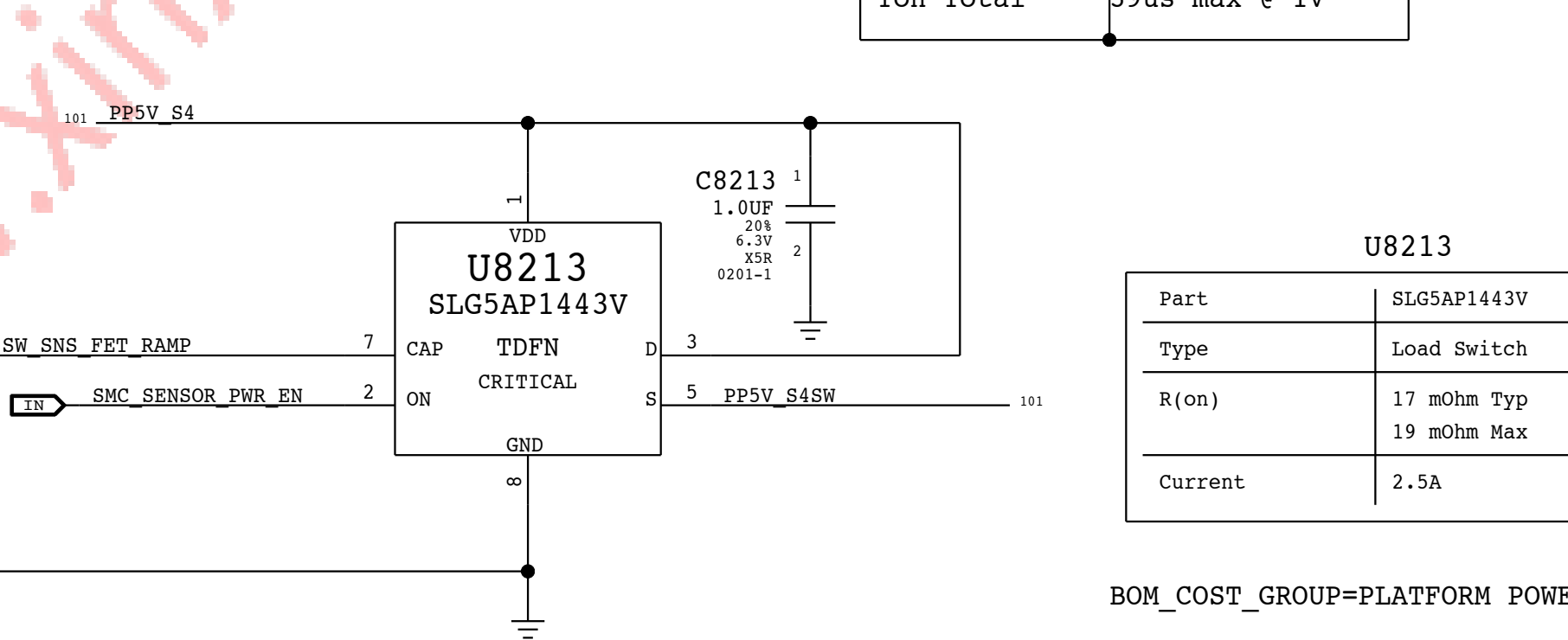
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Type	Load Switch
R(on) @ 25C	20 mOhm Typ 28 mOhm Max
Current	2.5A Max
Ton Total	39us max @ 1V

1.0V S0 SW Switch



Part	SLG5AP1635V
Type	Load Switch
R(on) @ 25C	20 mOhm Typ 28 mOhm Max
Current	2.5A Max
Ton Total	39us max @ 1V

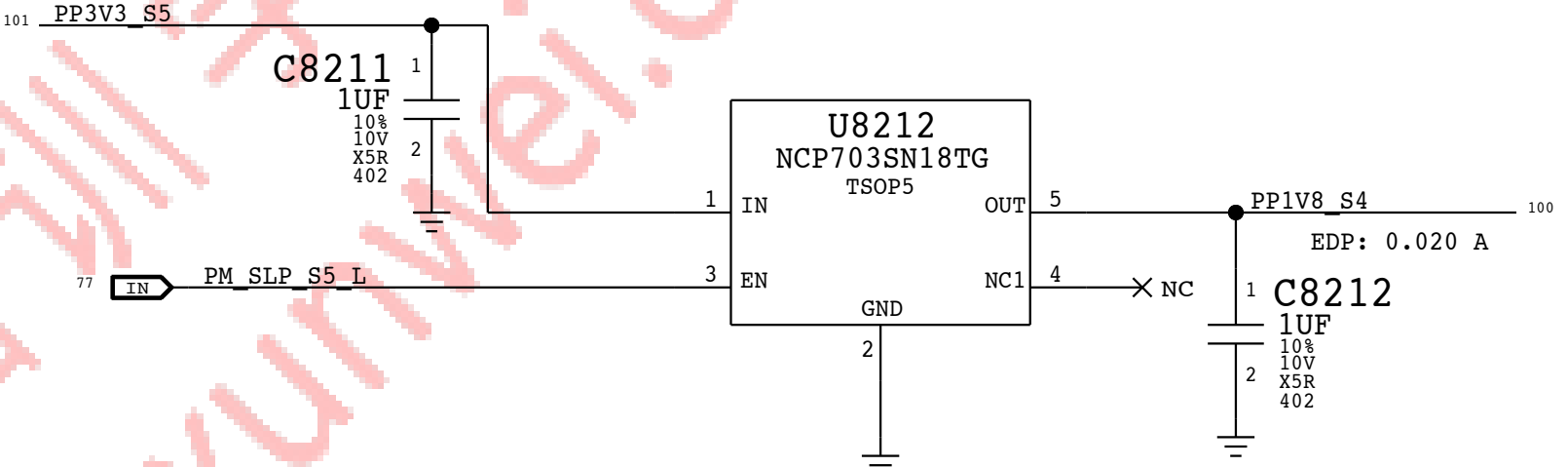
5V Sensor Switch



Part	SLG5AP1443V
Type	Load Switch
R(on) @ 25C	17 mOhm Typ 19 mOhm Max
Current	2.5A

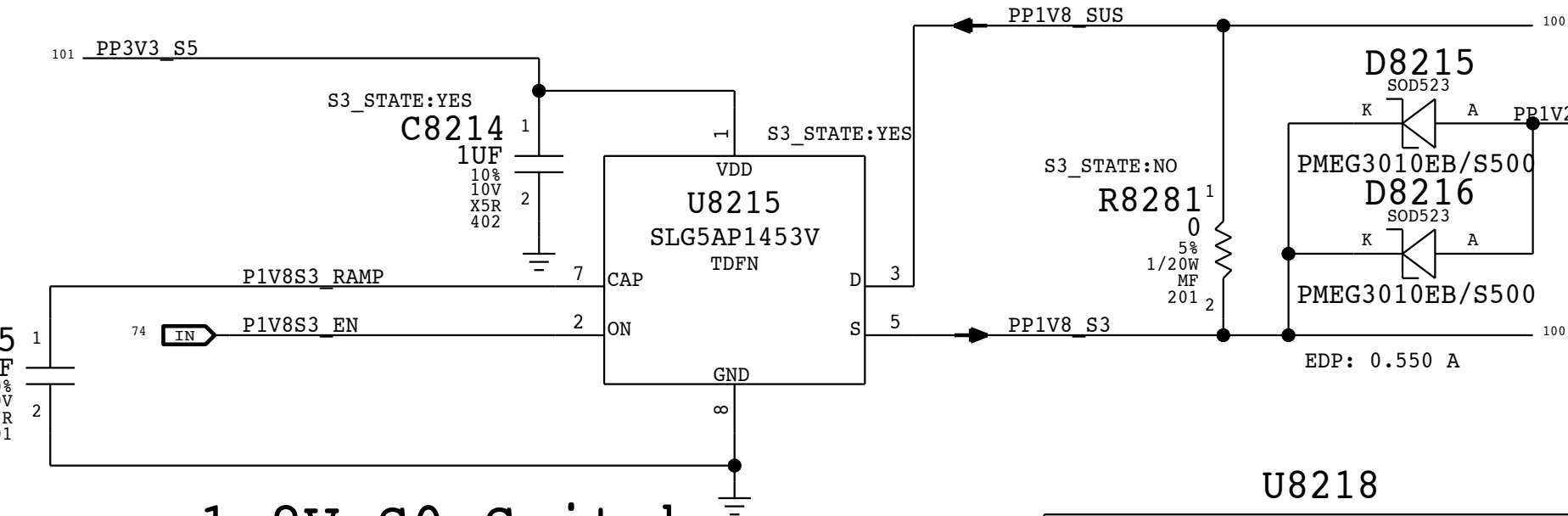
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1.8V S4 LDO



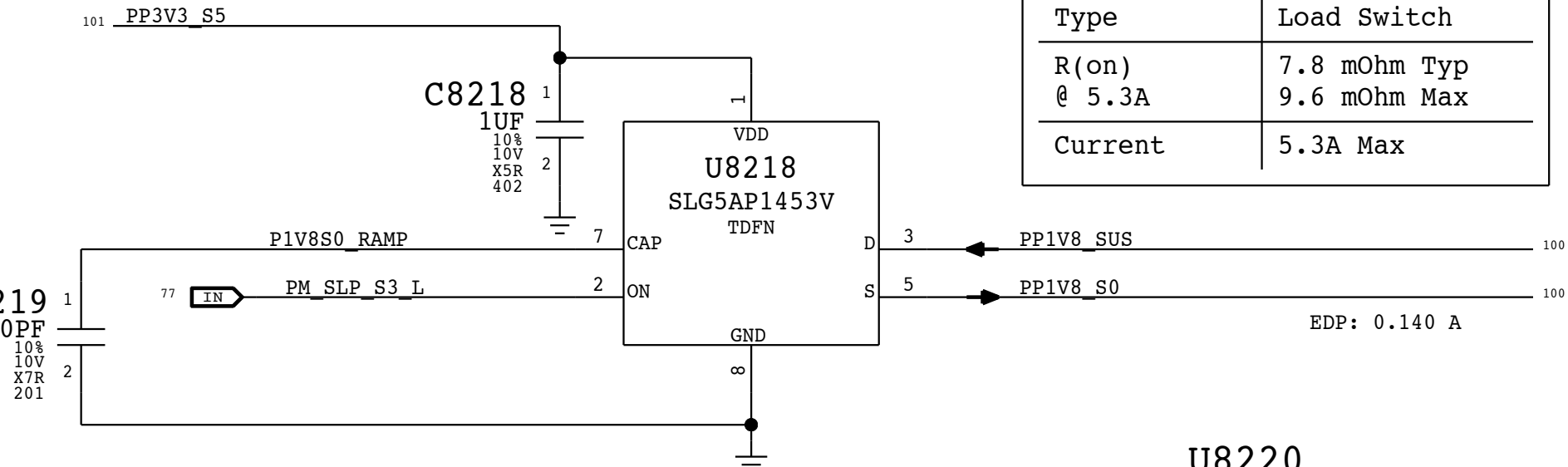
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R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

1.8V S3 Switch



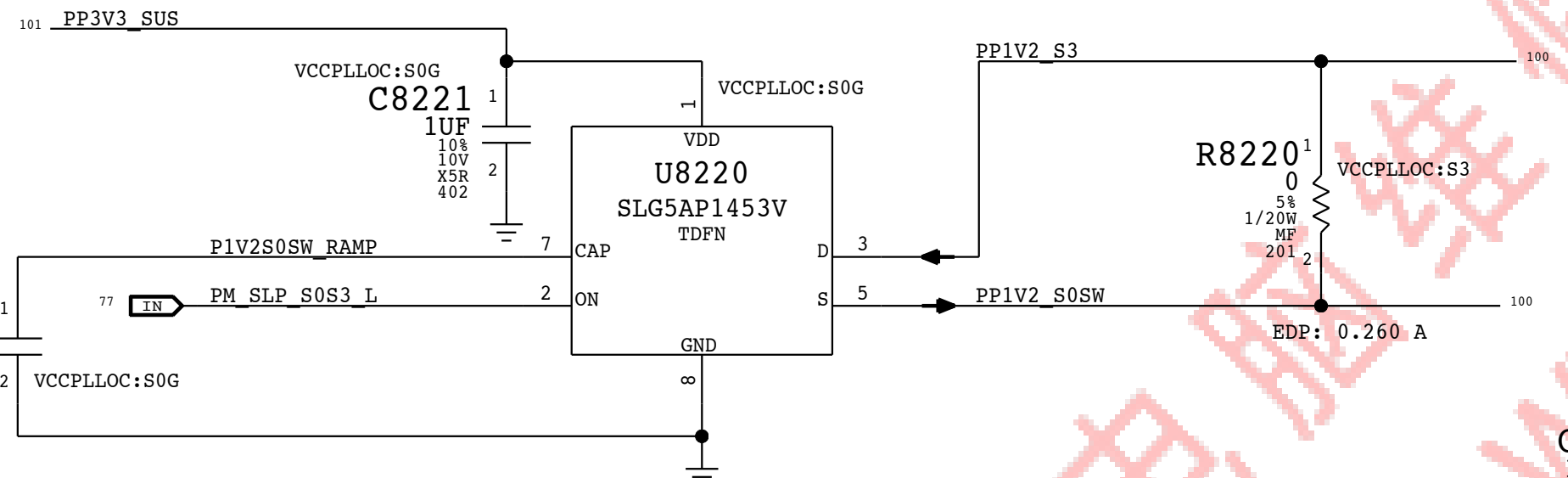
Part	SLG5AP1453V
Type	Load Switch
R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

1.8V S0 Switch



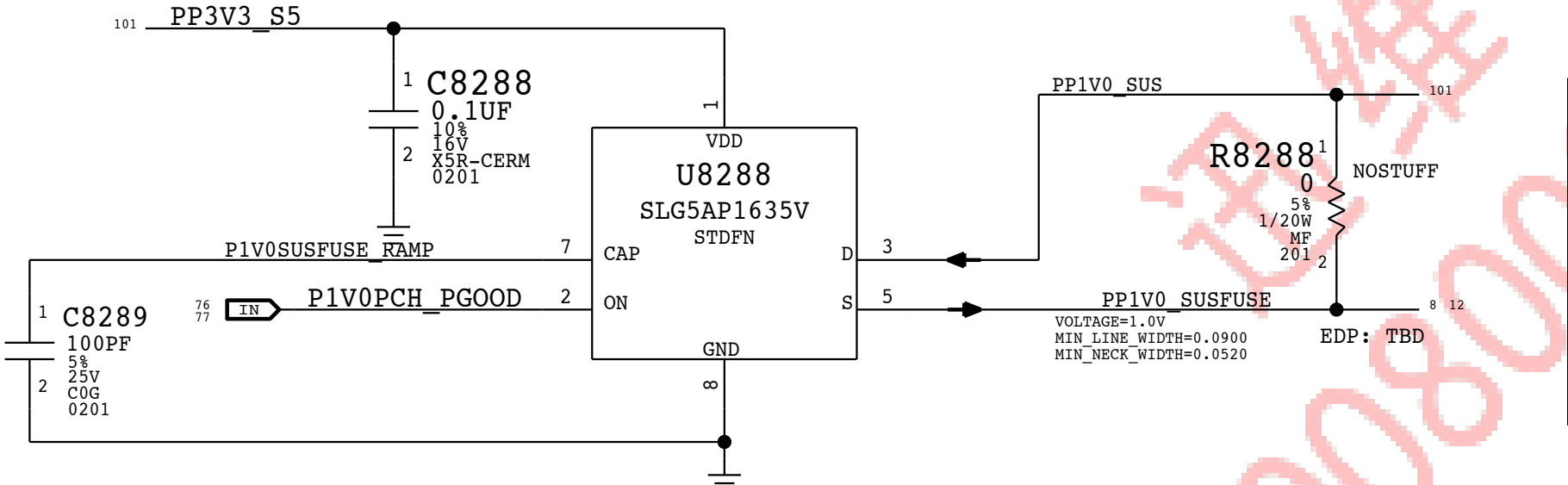
Part	SLG5AP1453V
Type	Load Switch
R(on) @ 5.3A	7.8 mOhm Typ 9.6 mOhm Max
Current	5.3A Max

1.2V S0 SW Switch

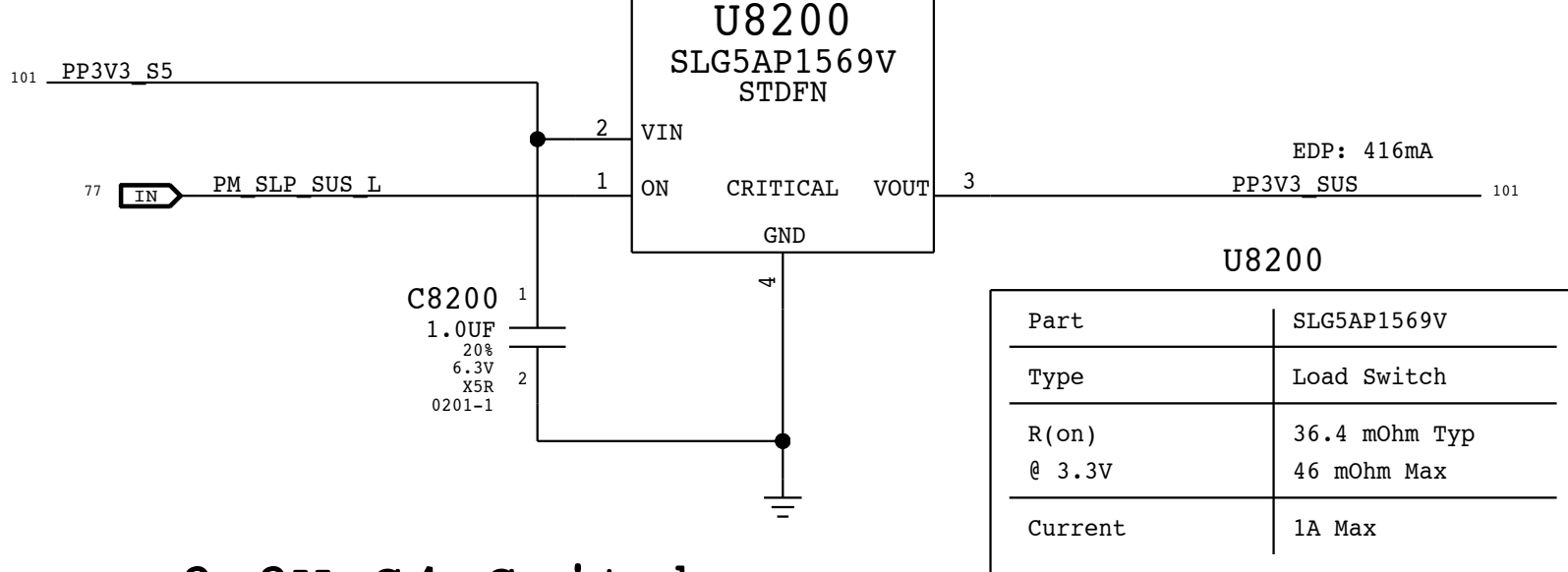


Part	SLG5AP1635V
Type	Load Switch
R(on) @ 25C	20 mOhm Typ 28 mOhm Max
Current	2.5A Max
Ton Total	39us max @ 1V
R(discharge)	300 Ohm Max

1.0V SUS FUSE Switch

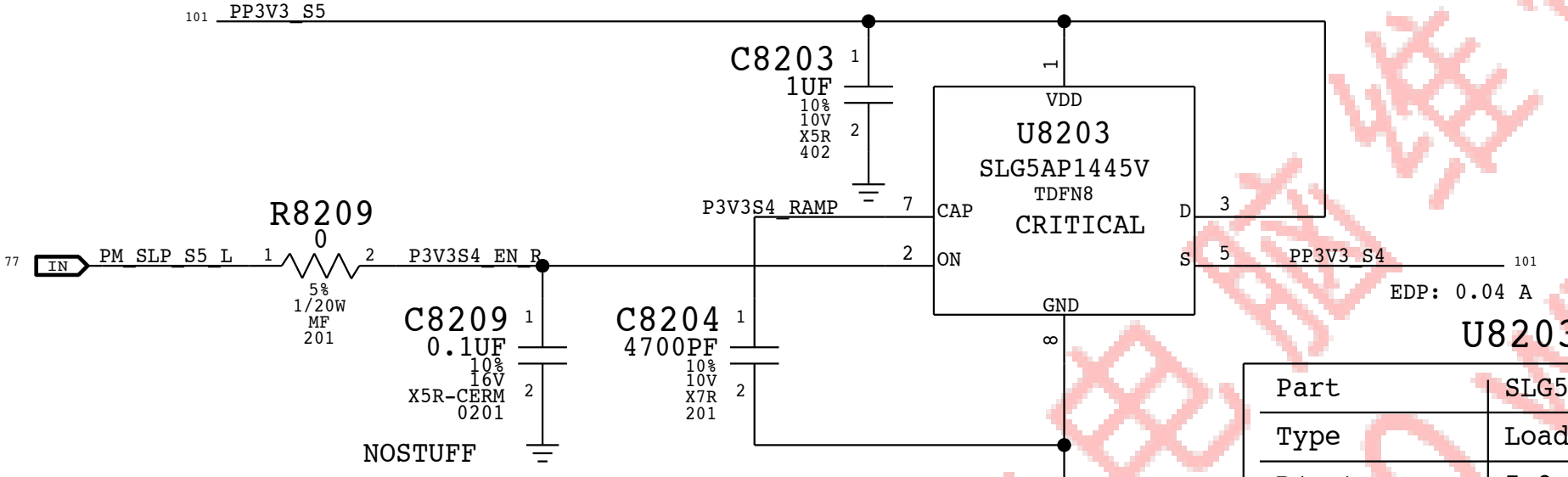


3.3V SUS Switch



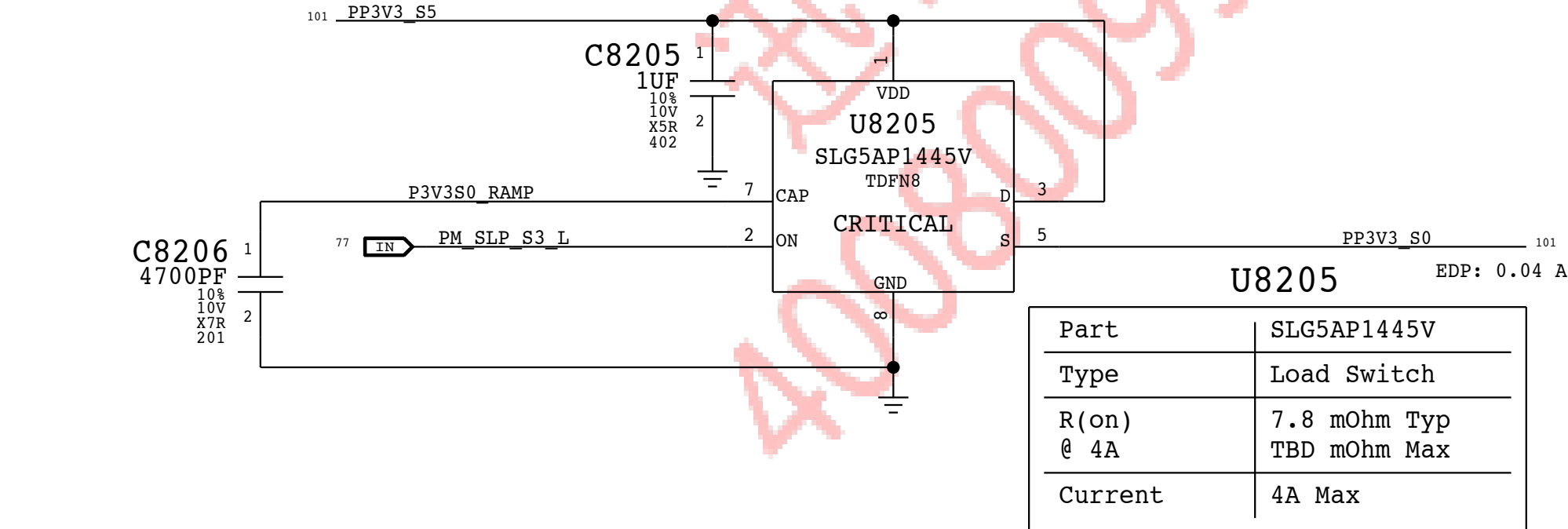
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Type	Load Switch
R(on) @ 3.3V	36.4 mOhm Typ 46 mOhm Max
Current	1A Max

3.3V S4 Switch



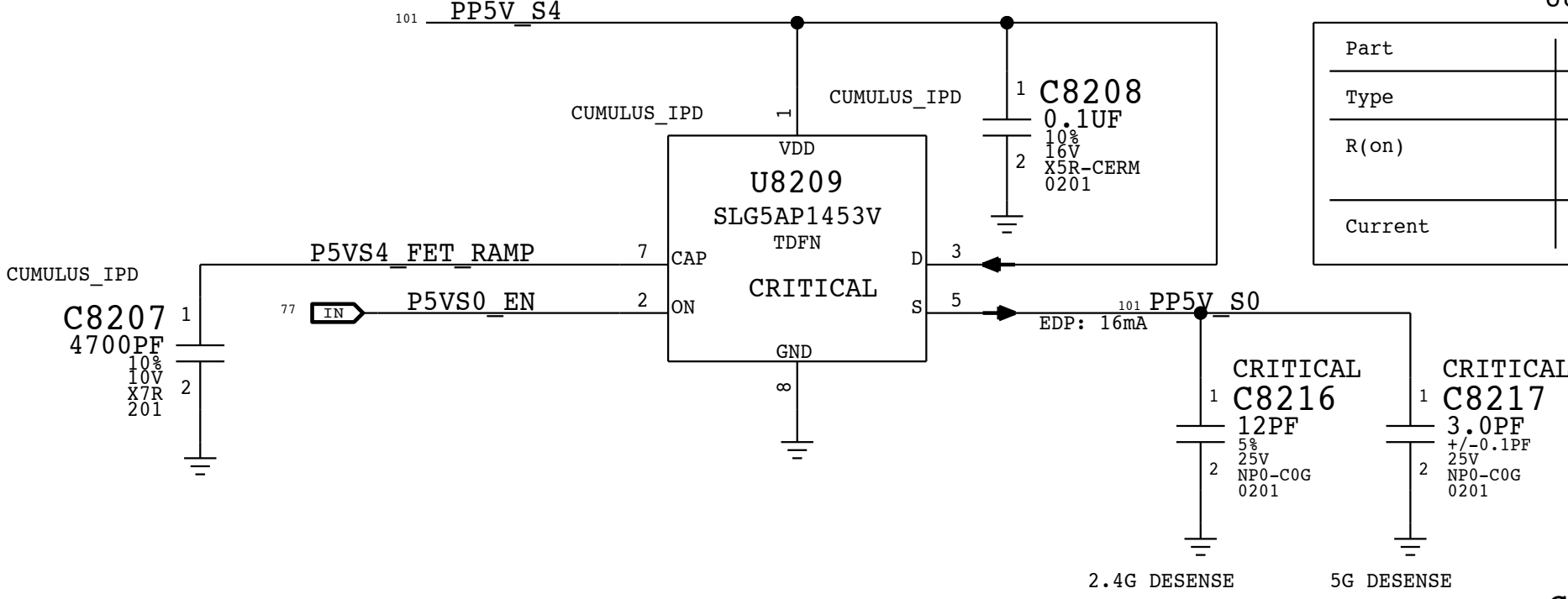
Part	SLG5AP1445V
Type	Load Switch
R(on) @ 4A	7.8 mOhm Typ TBD mOhm Max
Current	4A Max

3.3V S0 Switch



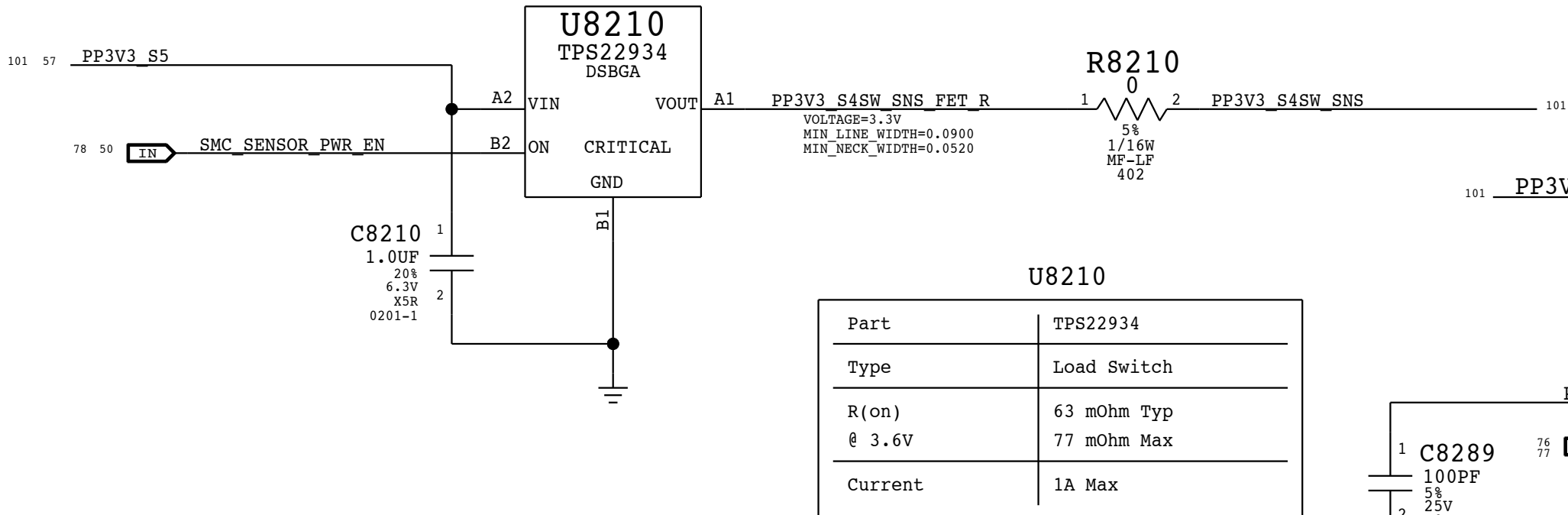
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Current	4A Max

5V S0 Switch (Cumulus vs Kona)



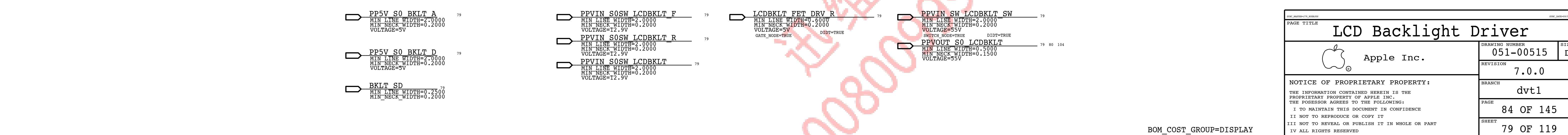
Part	SLG5AP1443V
Type	Load Switch
R(on) @ 5.3A	17 mOhm Typ 19 mOhm Max
Current	2.5A

3.3V Sensor Switch



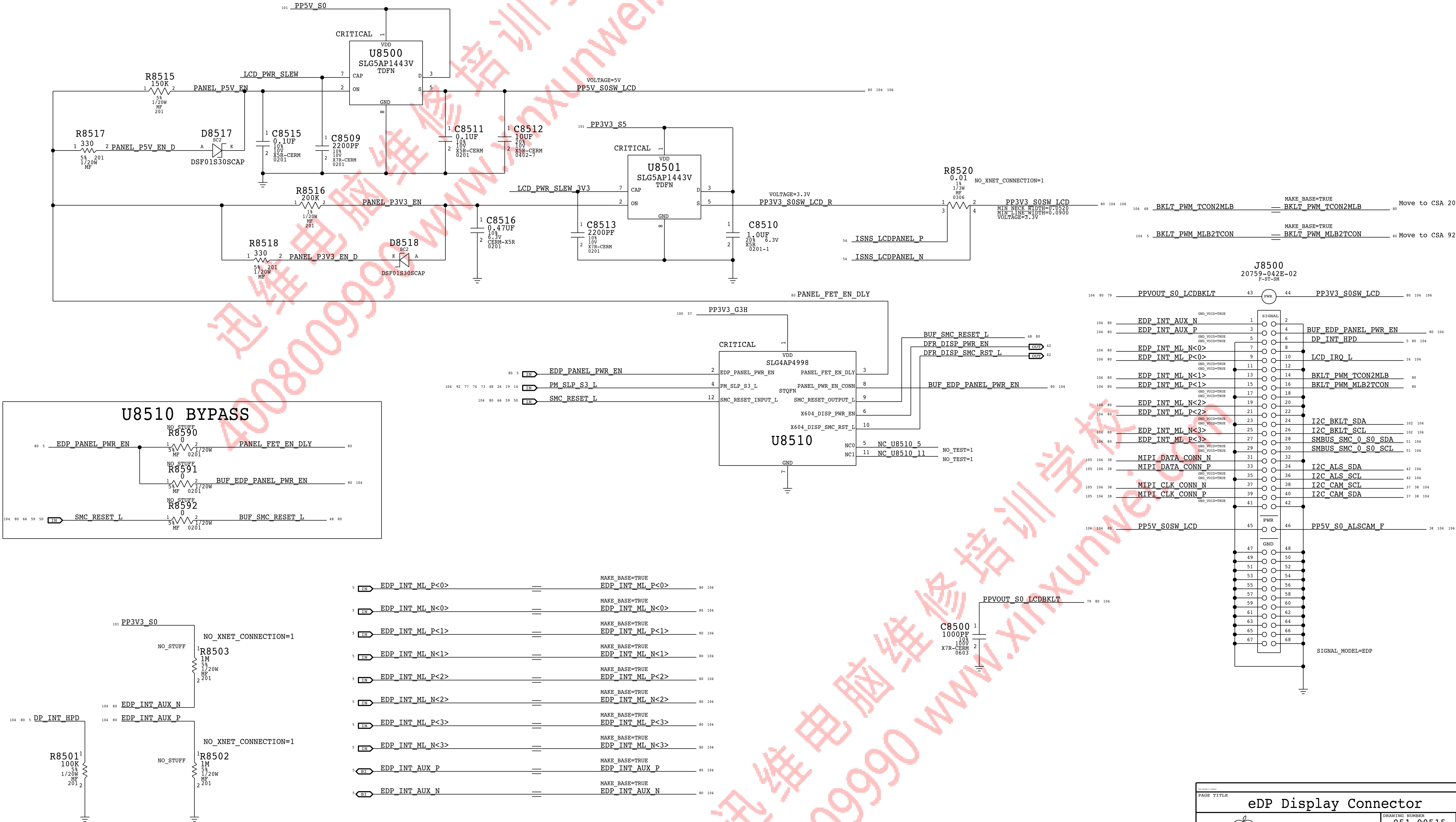
Part	TPS22934
Type	Load Switch
R(on) @ 3.6V	63 mOhm Typ 77 mOhm Max
Current	1A Max

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BOM_COST_GROUP=DISPLAY

LCD PANEL INTERFACE (eDP) + Camera (MIPI)

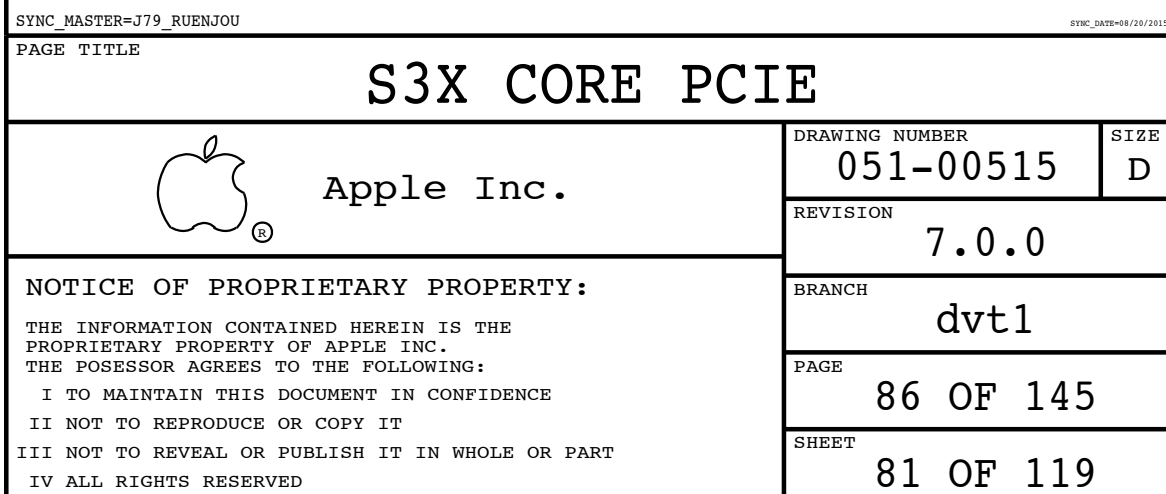


LCD Panel HPD & AUX strapping

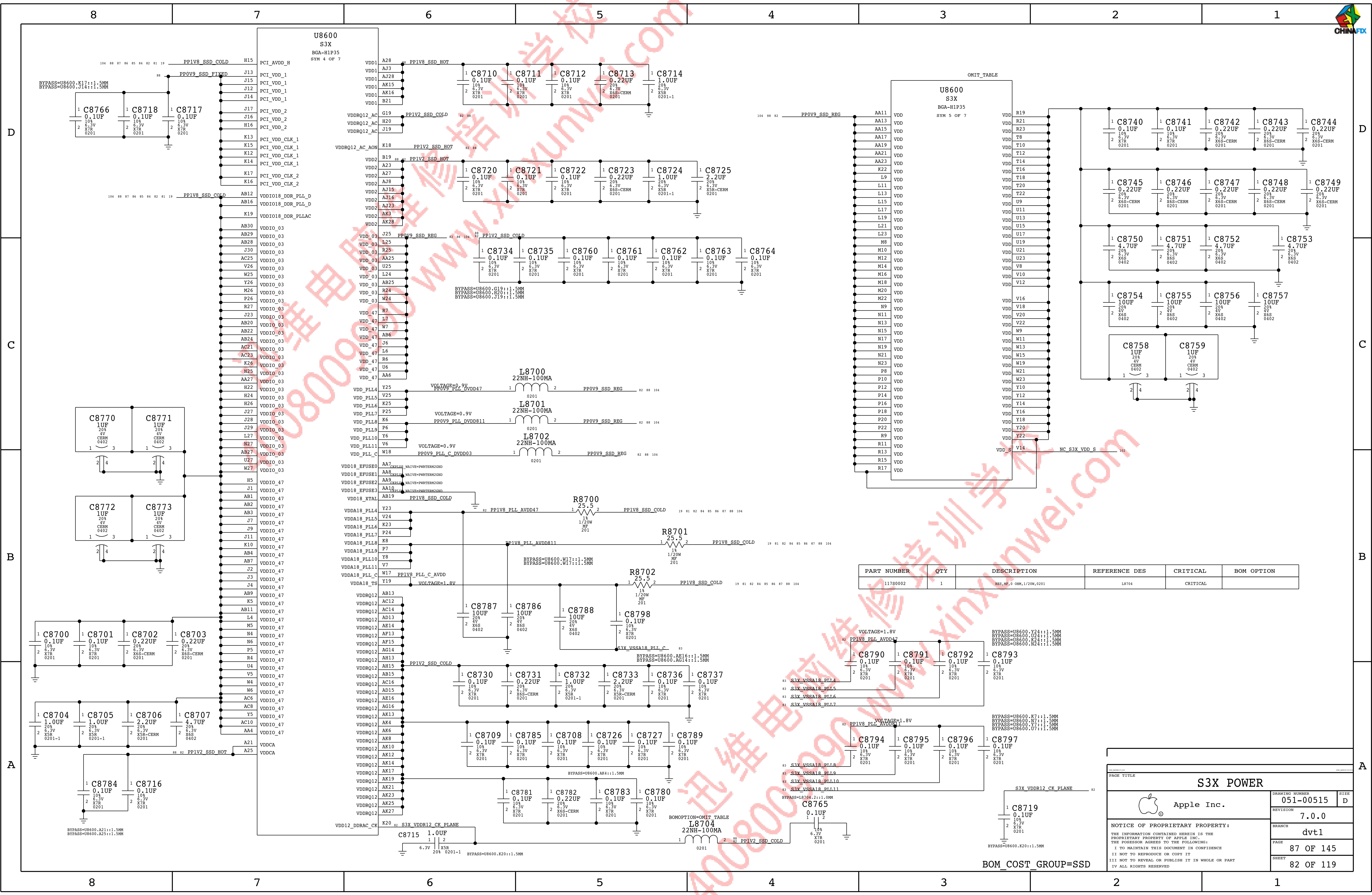
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	PAGE	85 OF 145
	SHEET	80 OF 119

BOM_COST_GROUP=DISPLAY

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
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117S00002	1	RES,MF,0 OHM,1/20W,0201	C8601	CRITICAL	S3XCLK:EXT

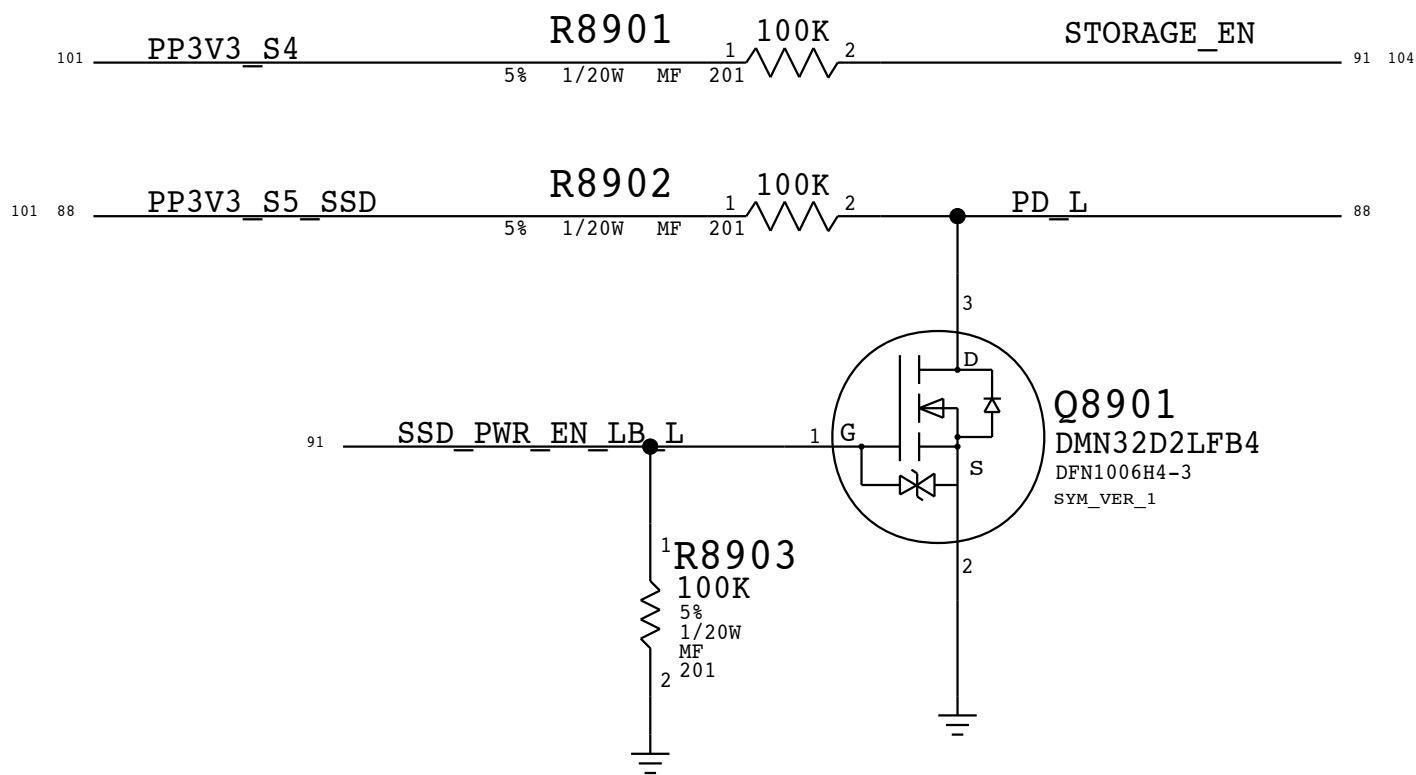
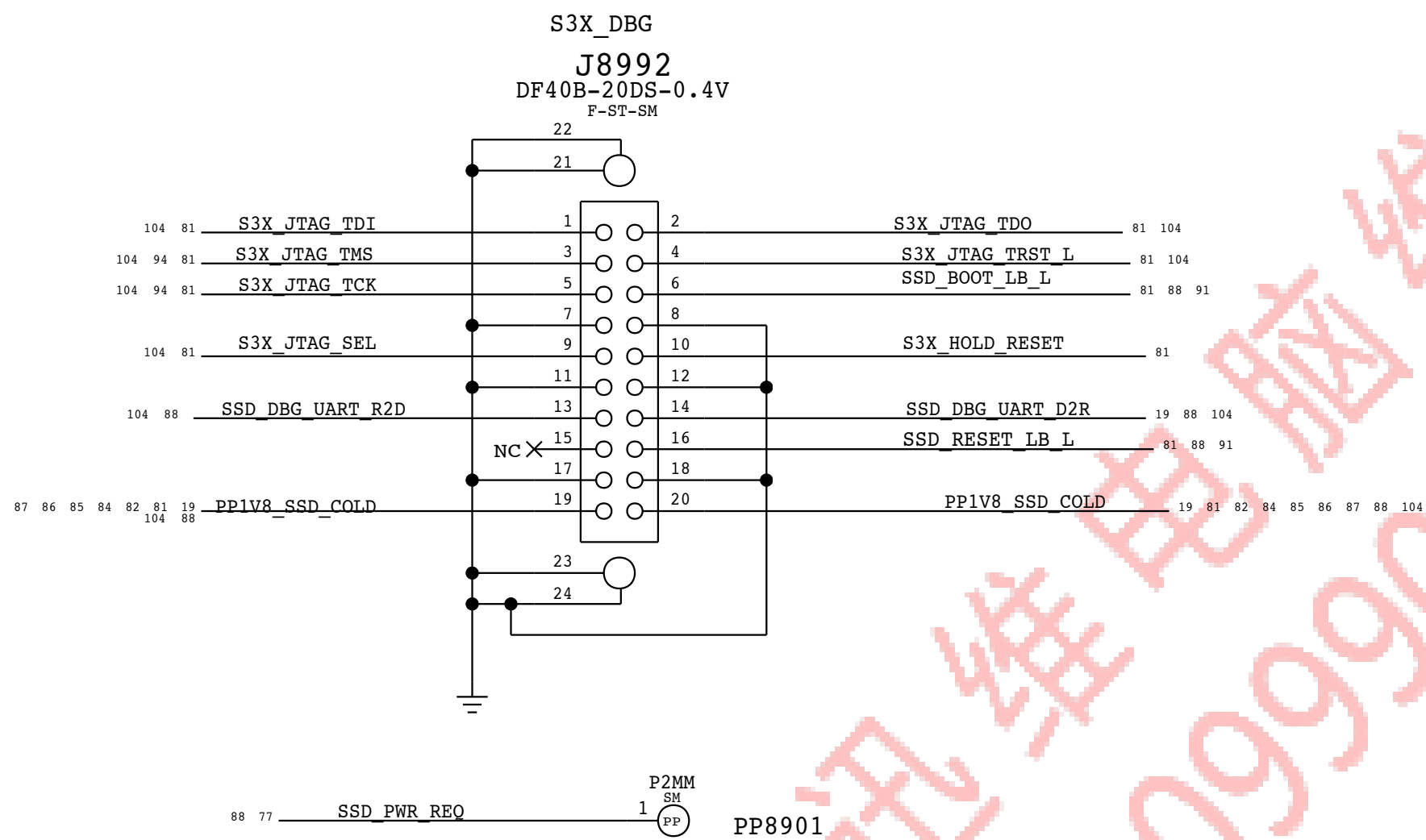


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	PAGE		
	88 OF 145		
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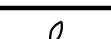
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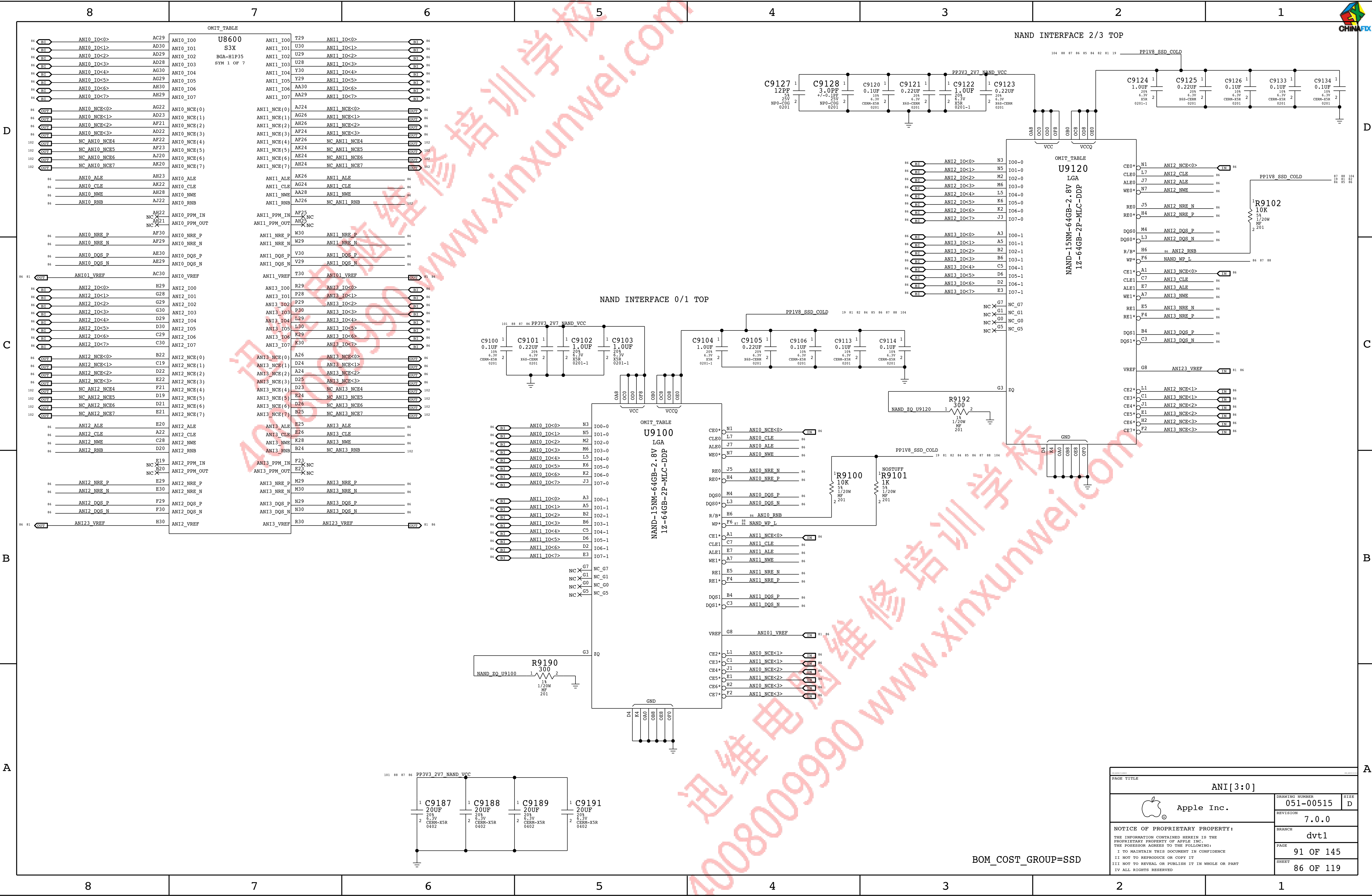
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Connector		
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	PAGE	89 OF 145
	SHEET	84 OF 119

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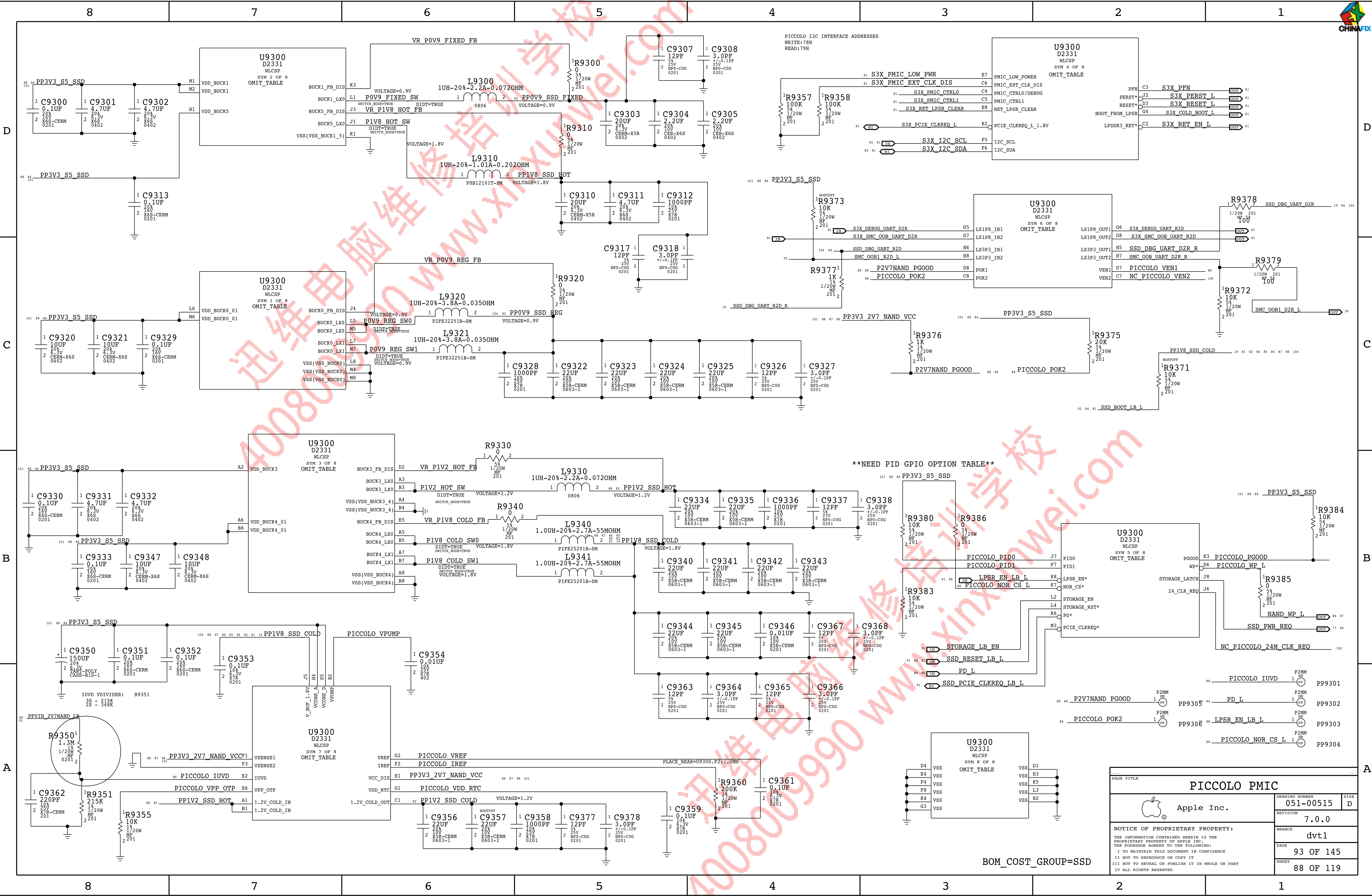
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DRAWING NUMBER	SITE						
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BRANCH	dvt1						
PAGE	90 OF 145						
SHEET	85 OF 119						





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BRANCH dvt1		PAGE 93 OF 145	
SHEET 88 OF 119			

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D

C

B

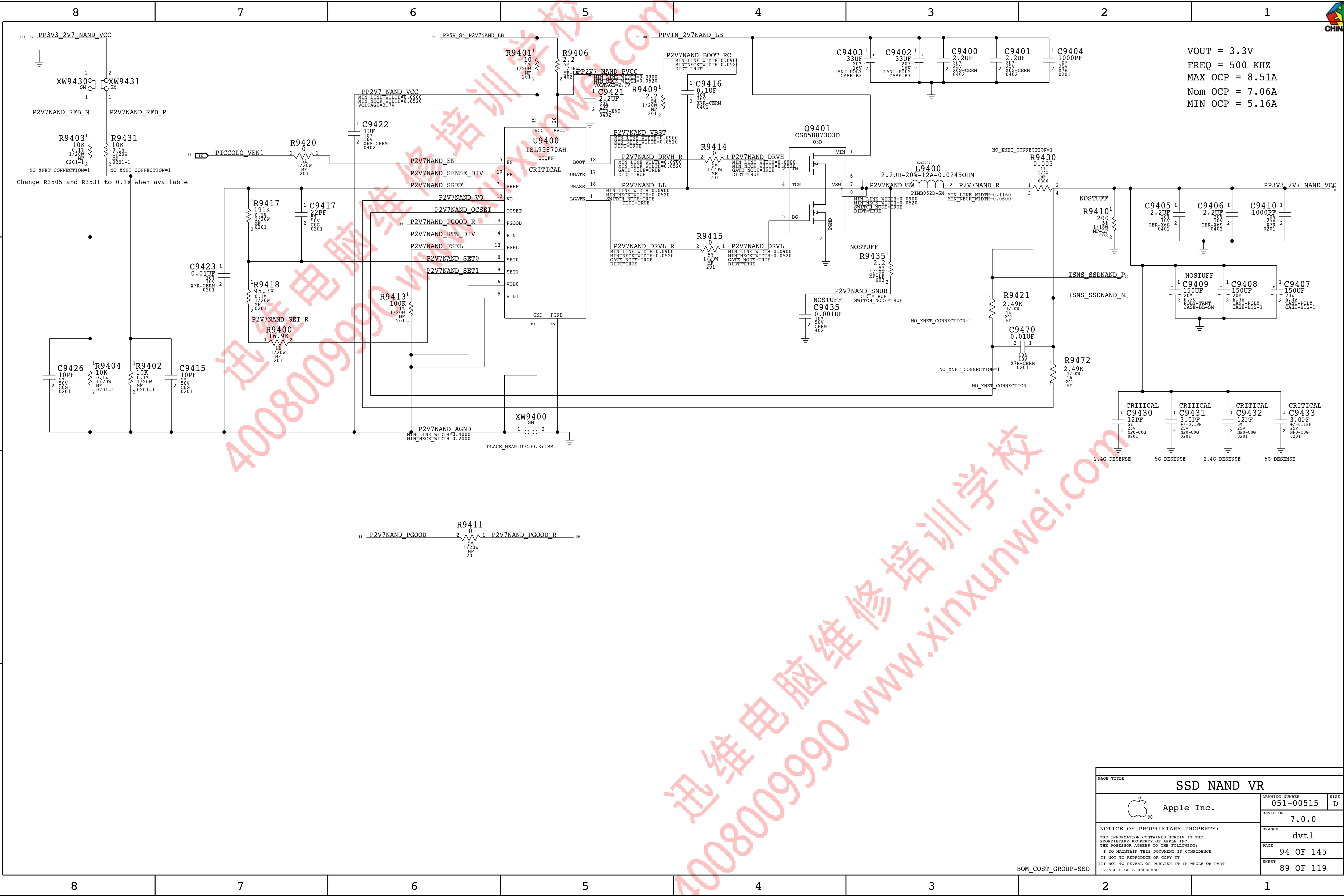
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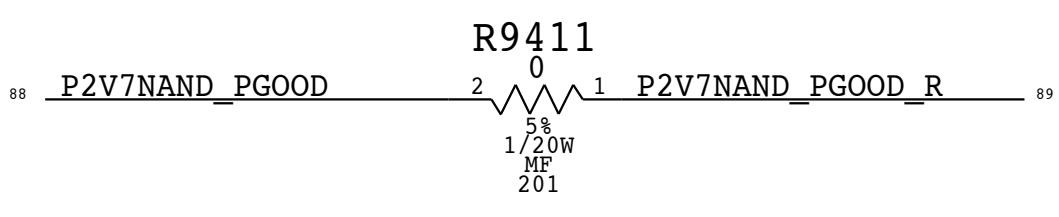
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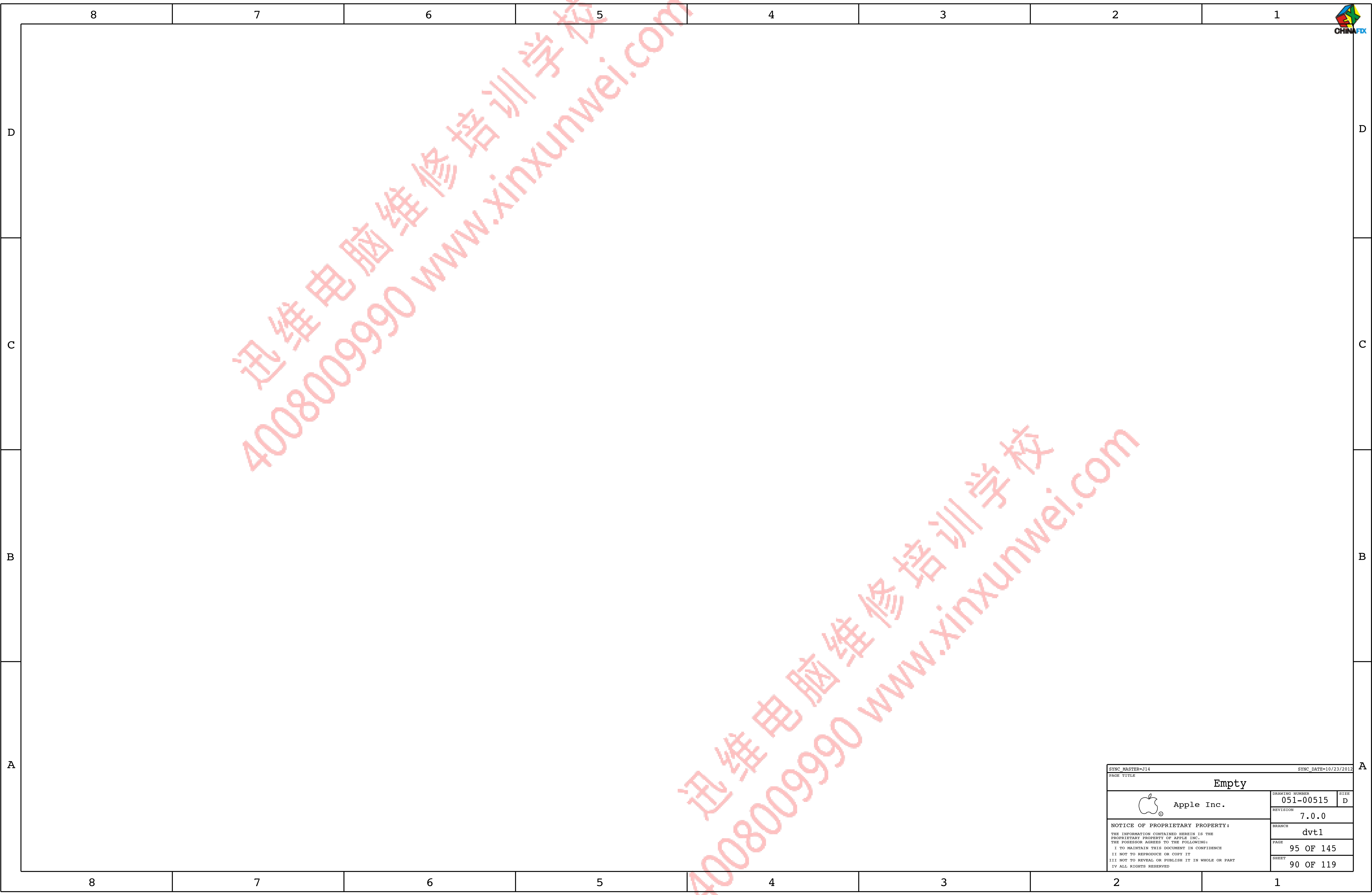


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FREQ = 500 KHZ
MAX OCP = 8.51A
Nom OCP = 7.06A
MIN OCP = 5.16A



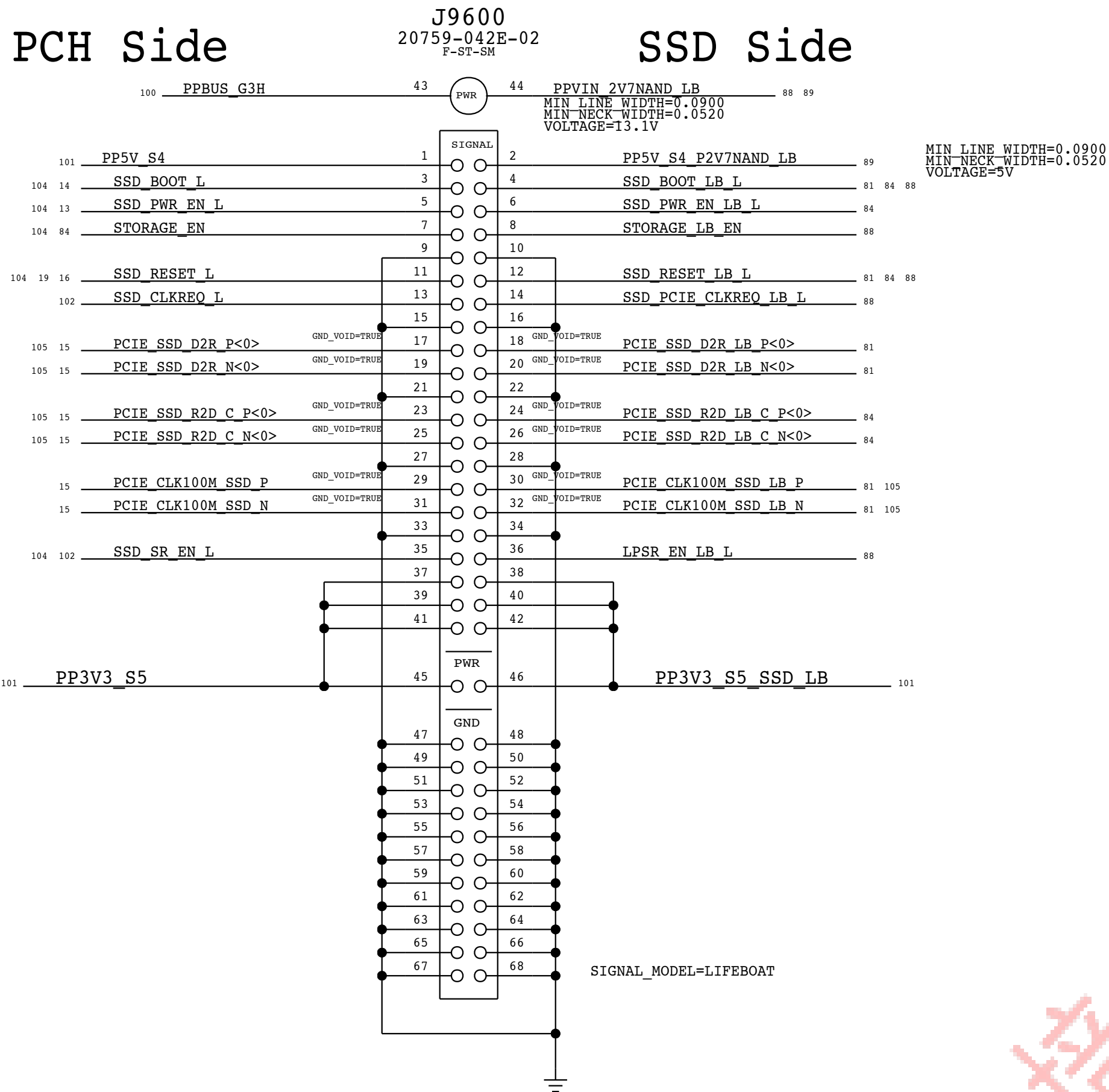
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	PAGE	94 OF 145
	SHEET	89 OF 119

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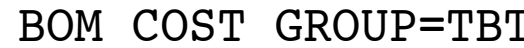
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	PAGE	95 OF 145	
	SHEET	90 OF 119	

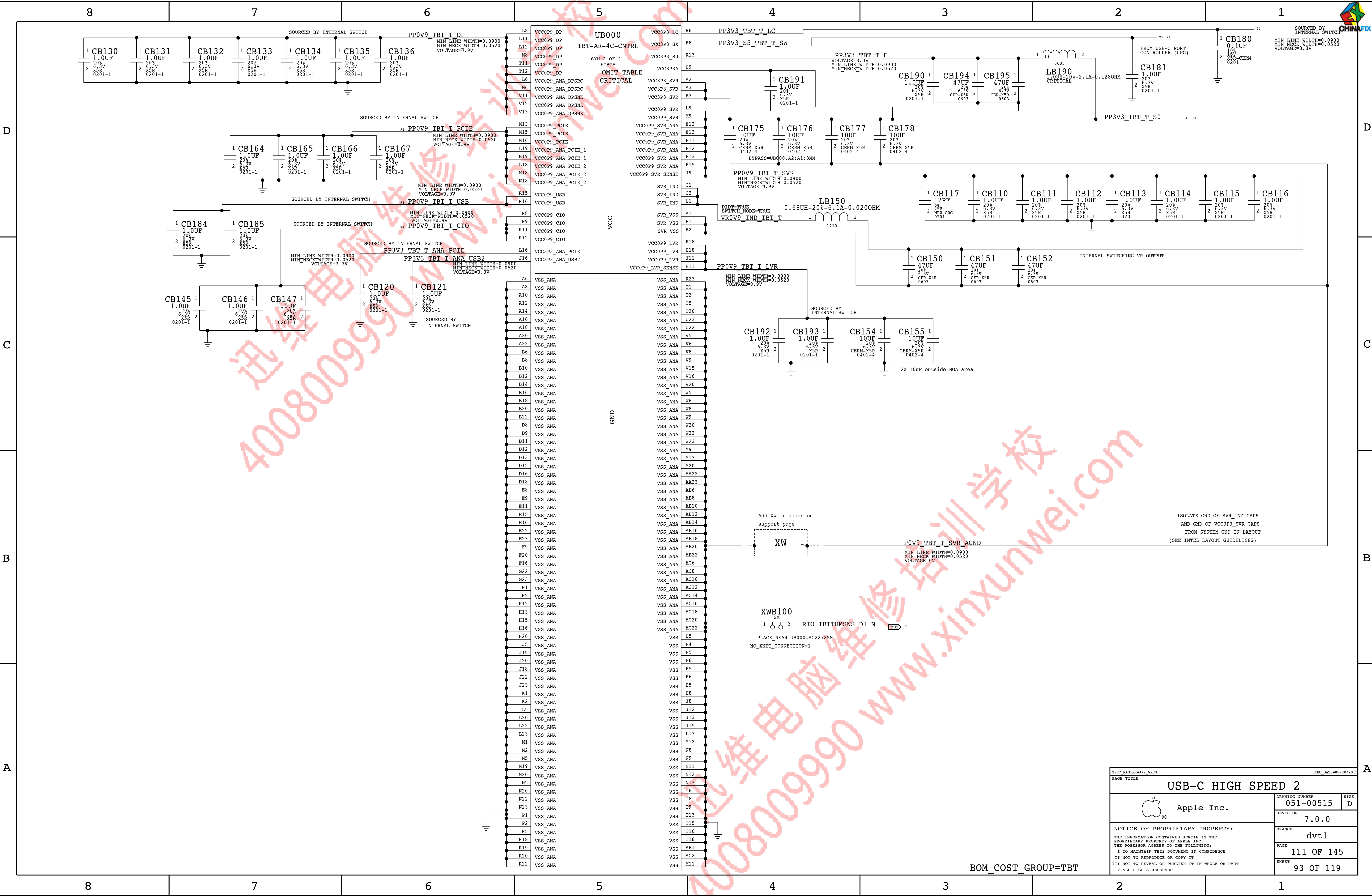
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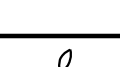


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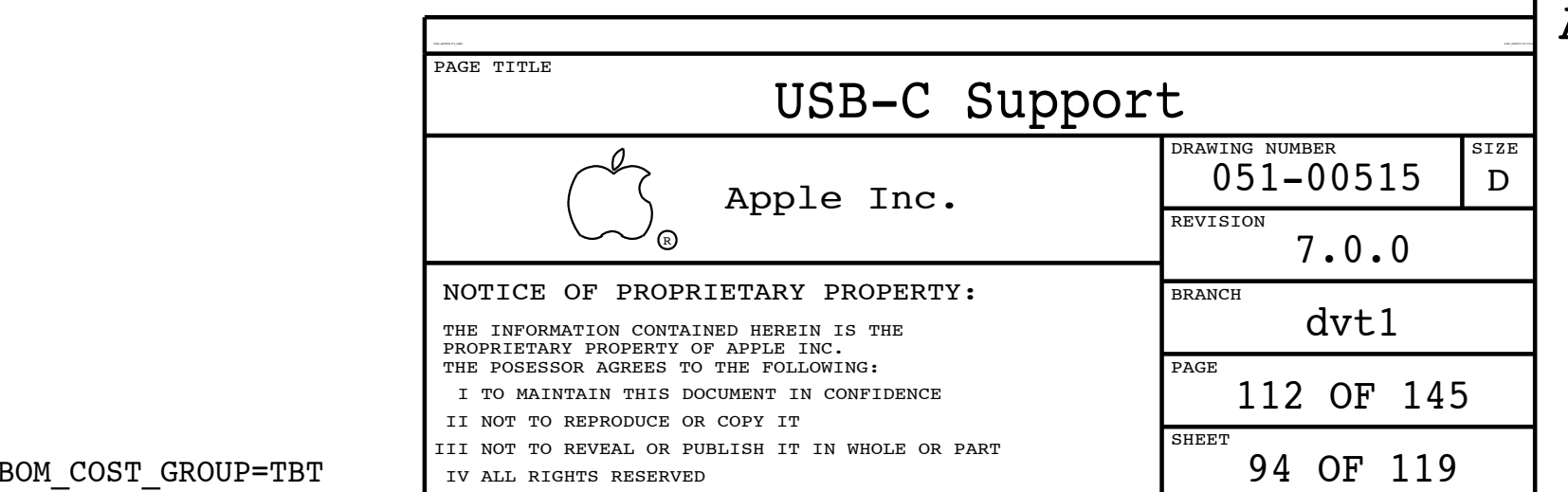
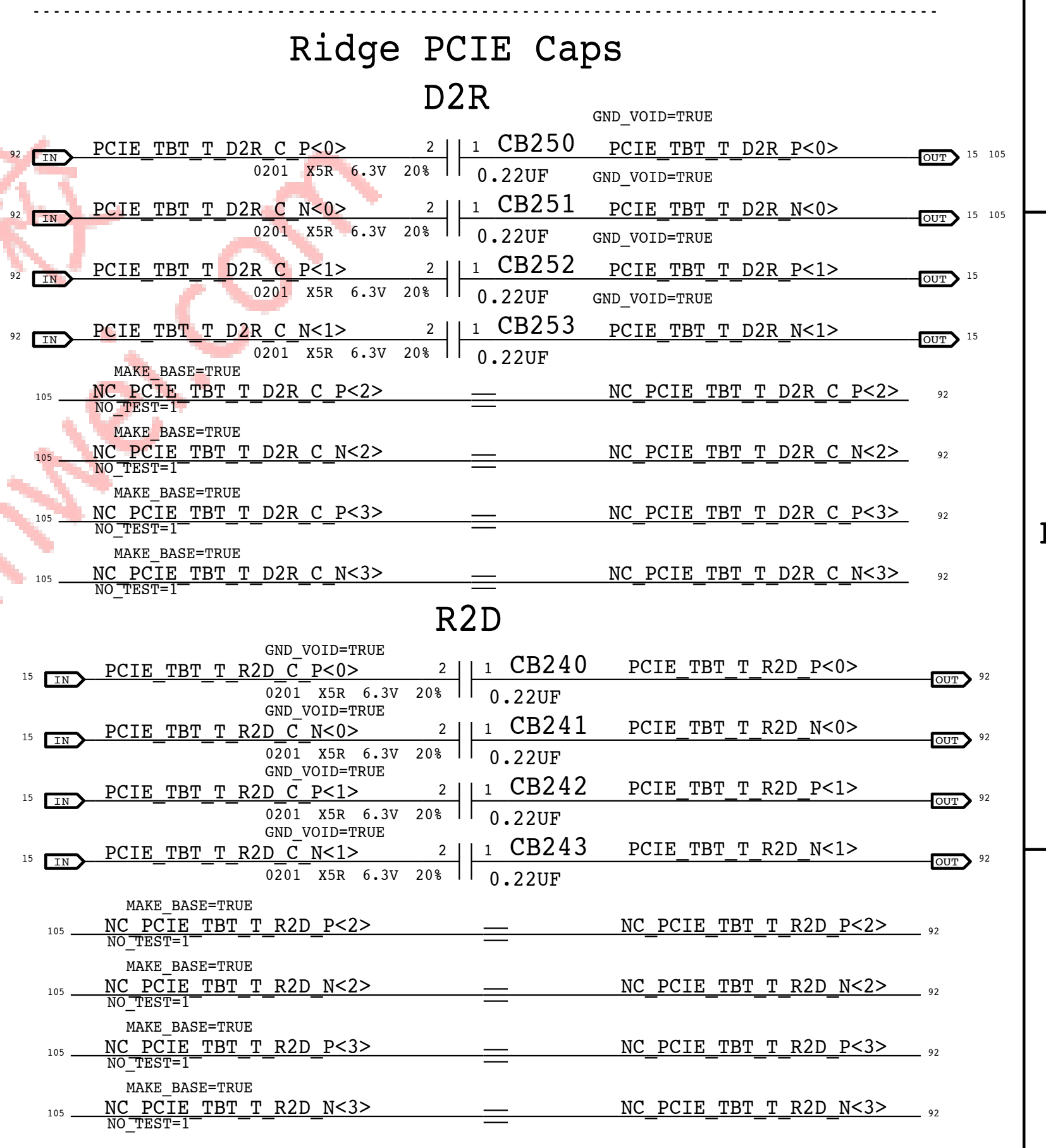
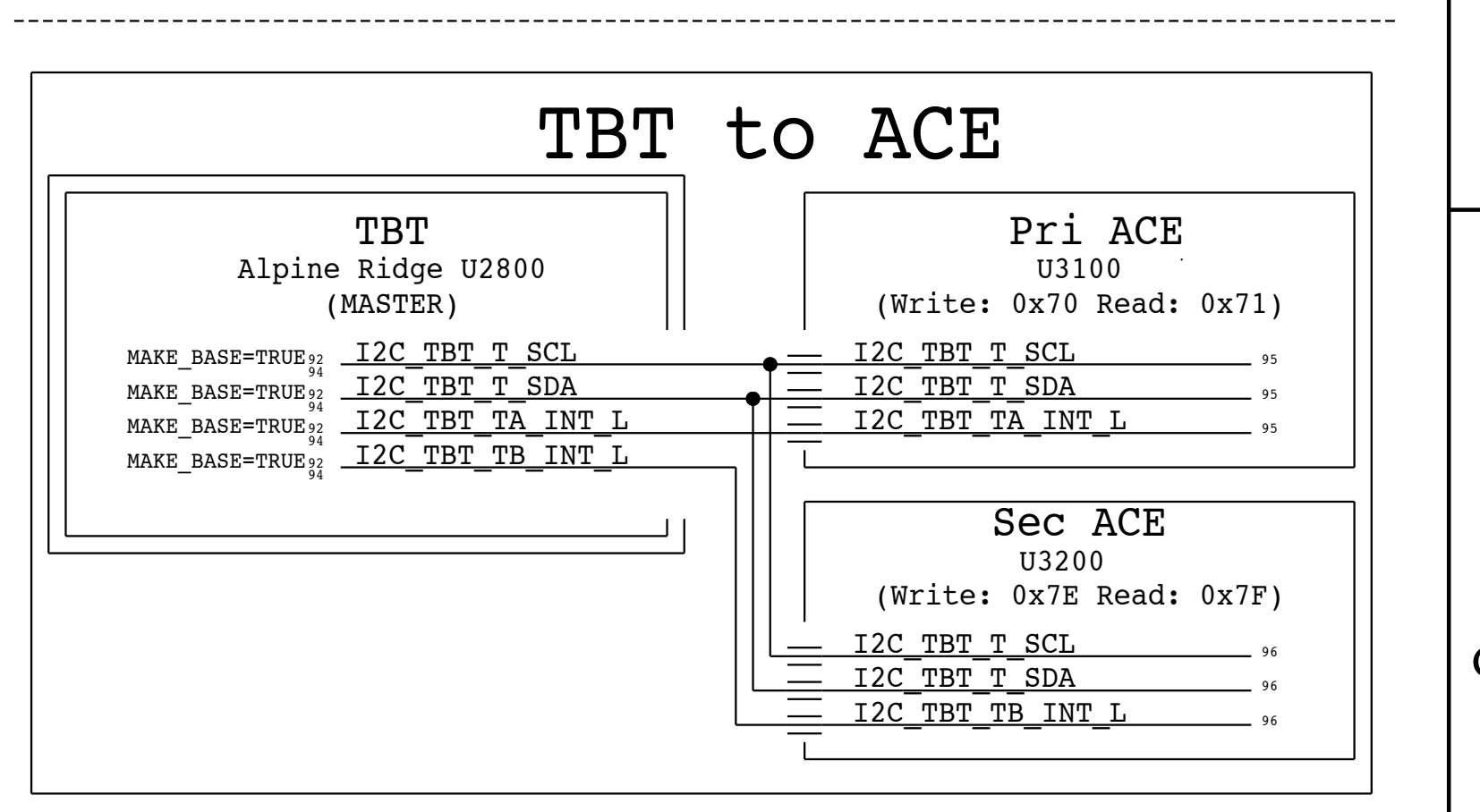
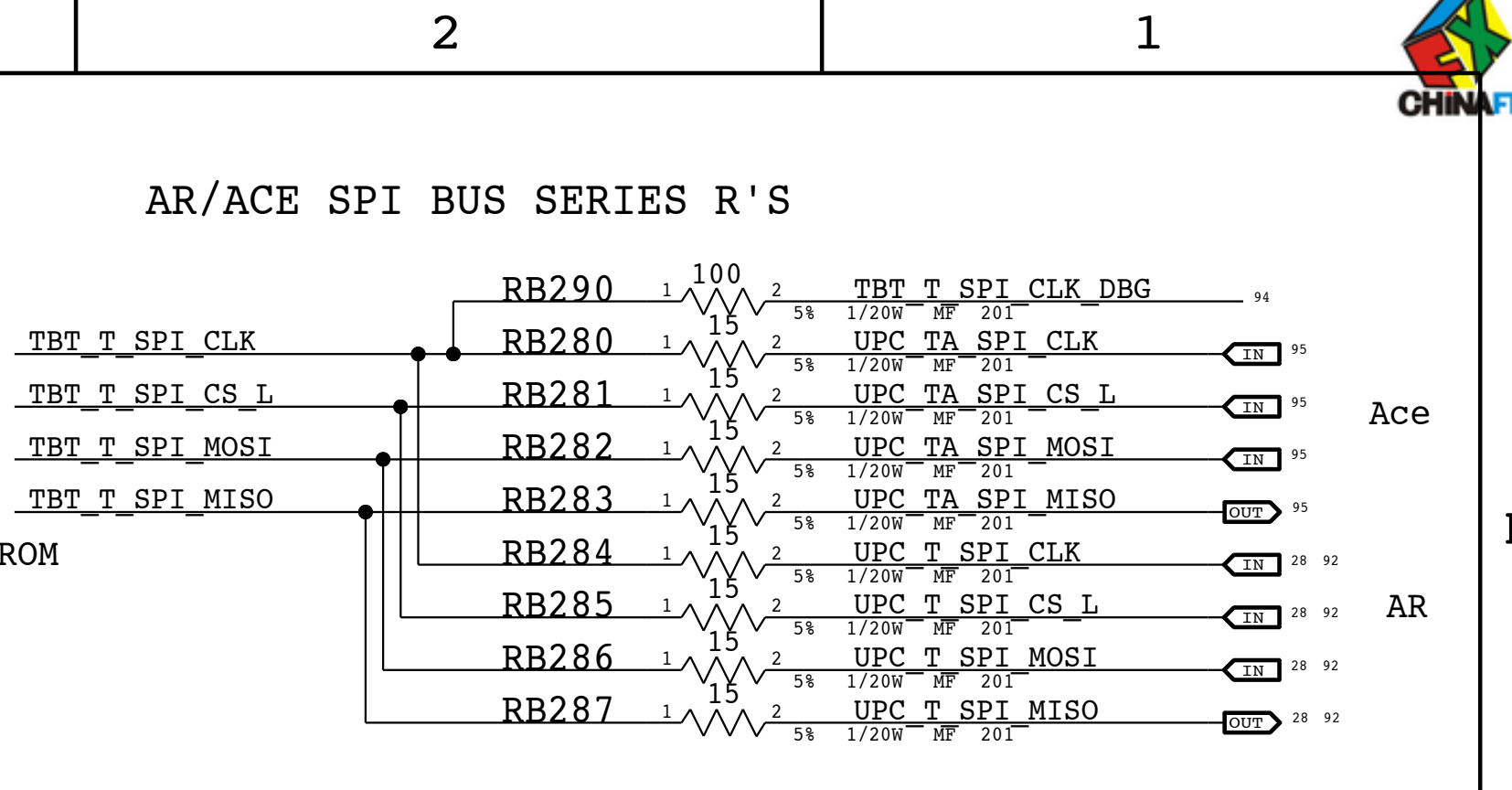
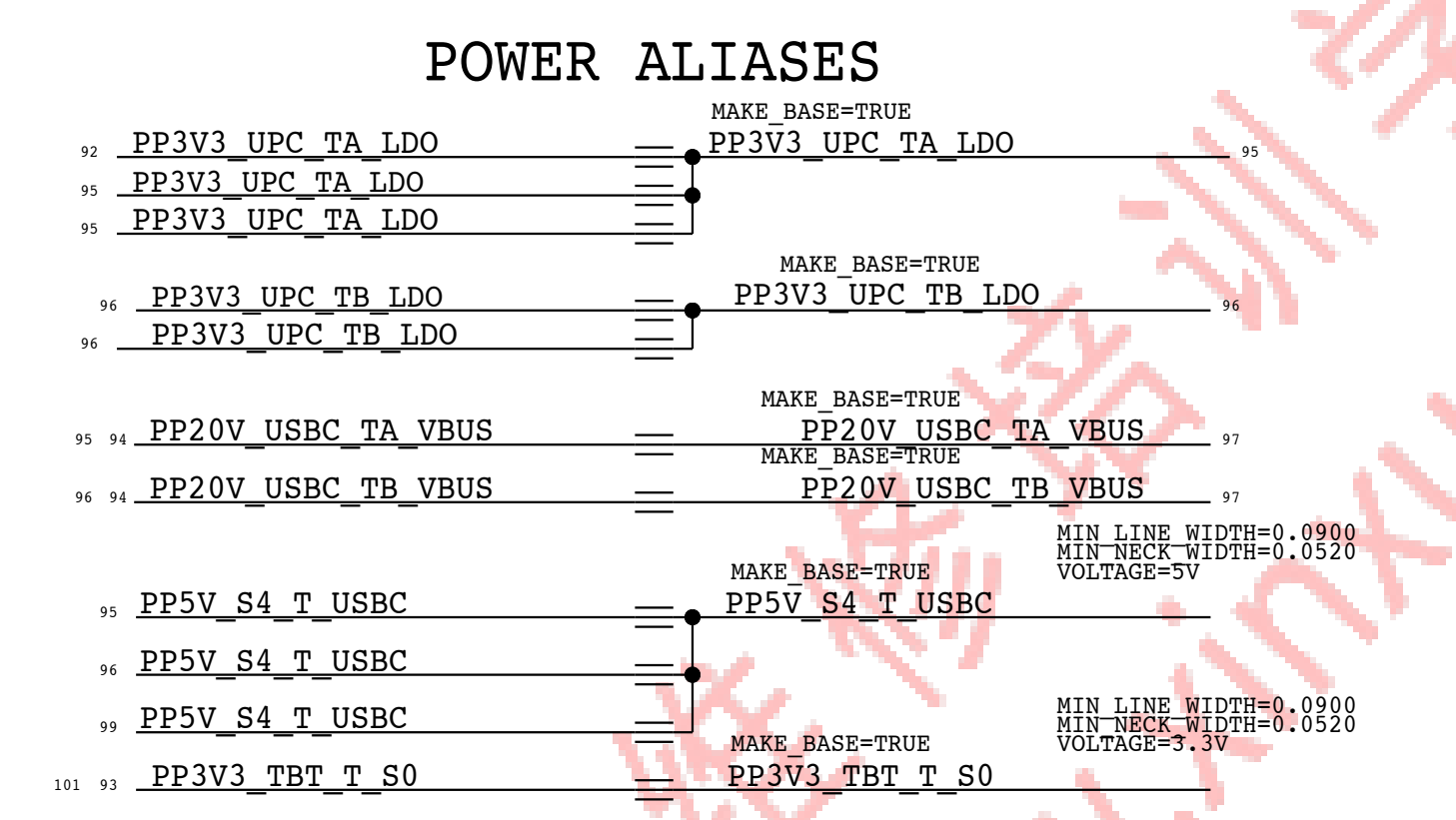
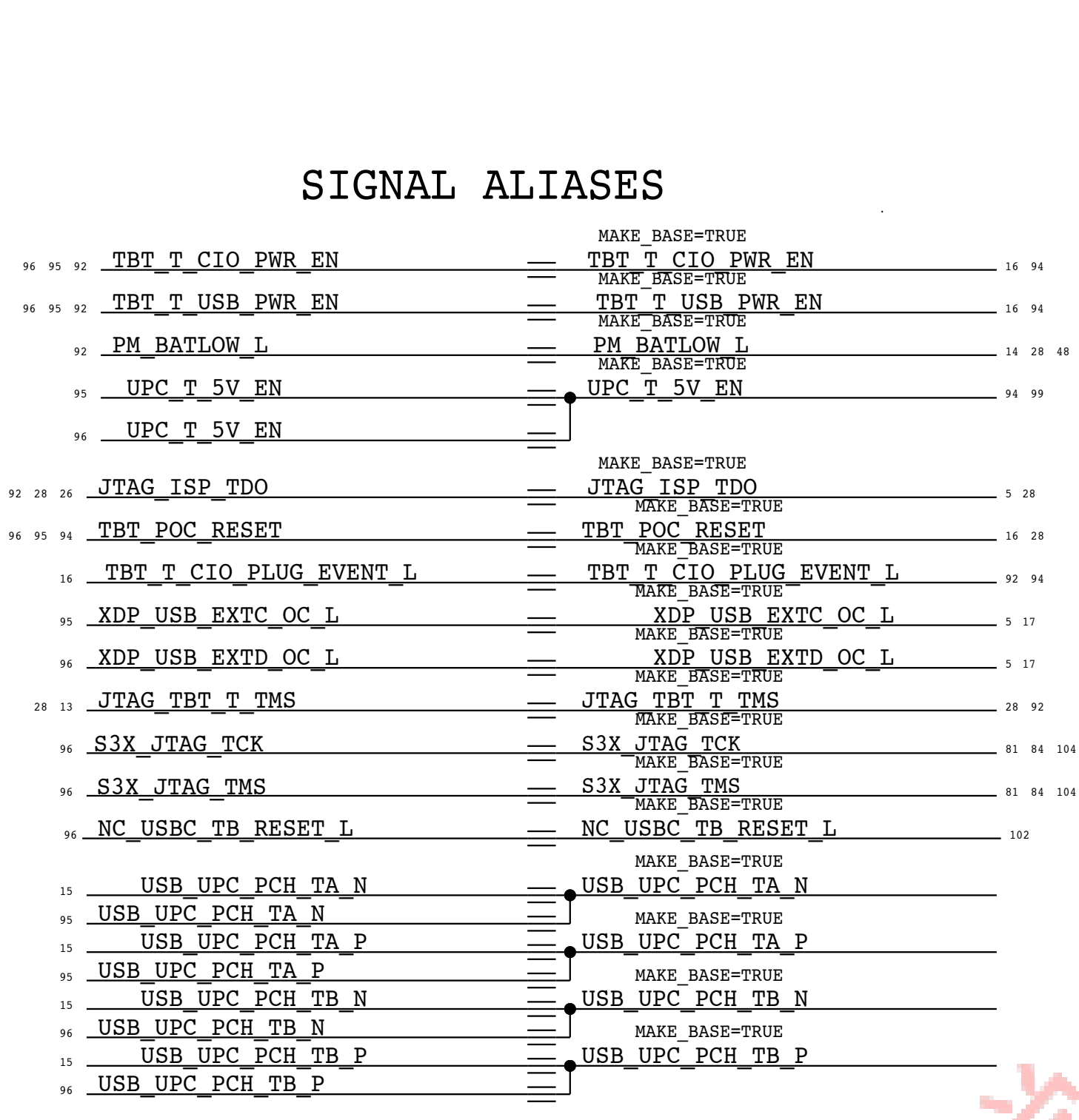
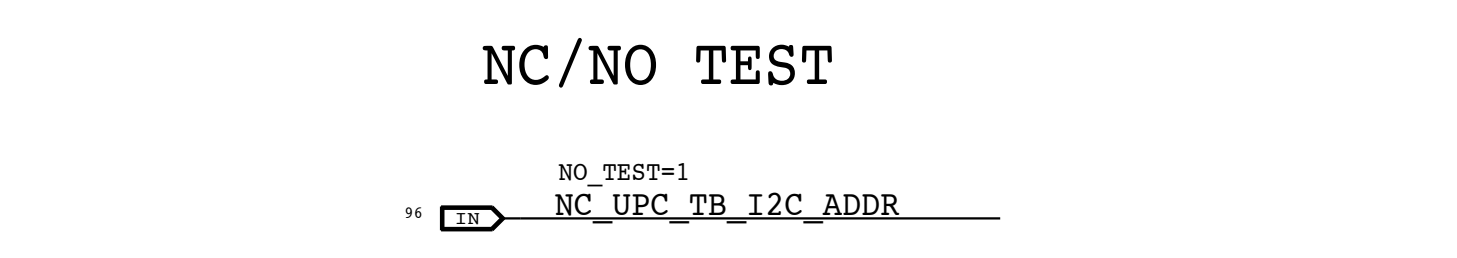
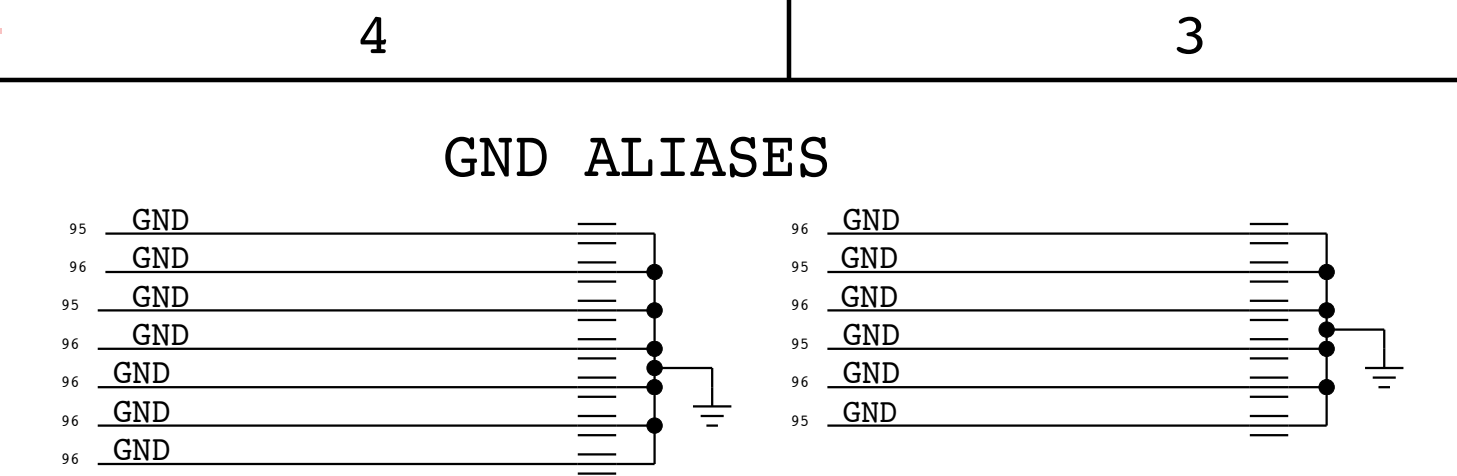
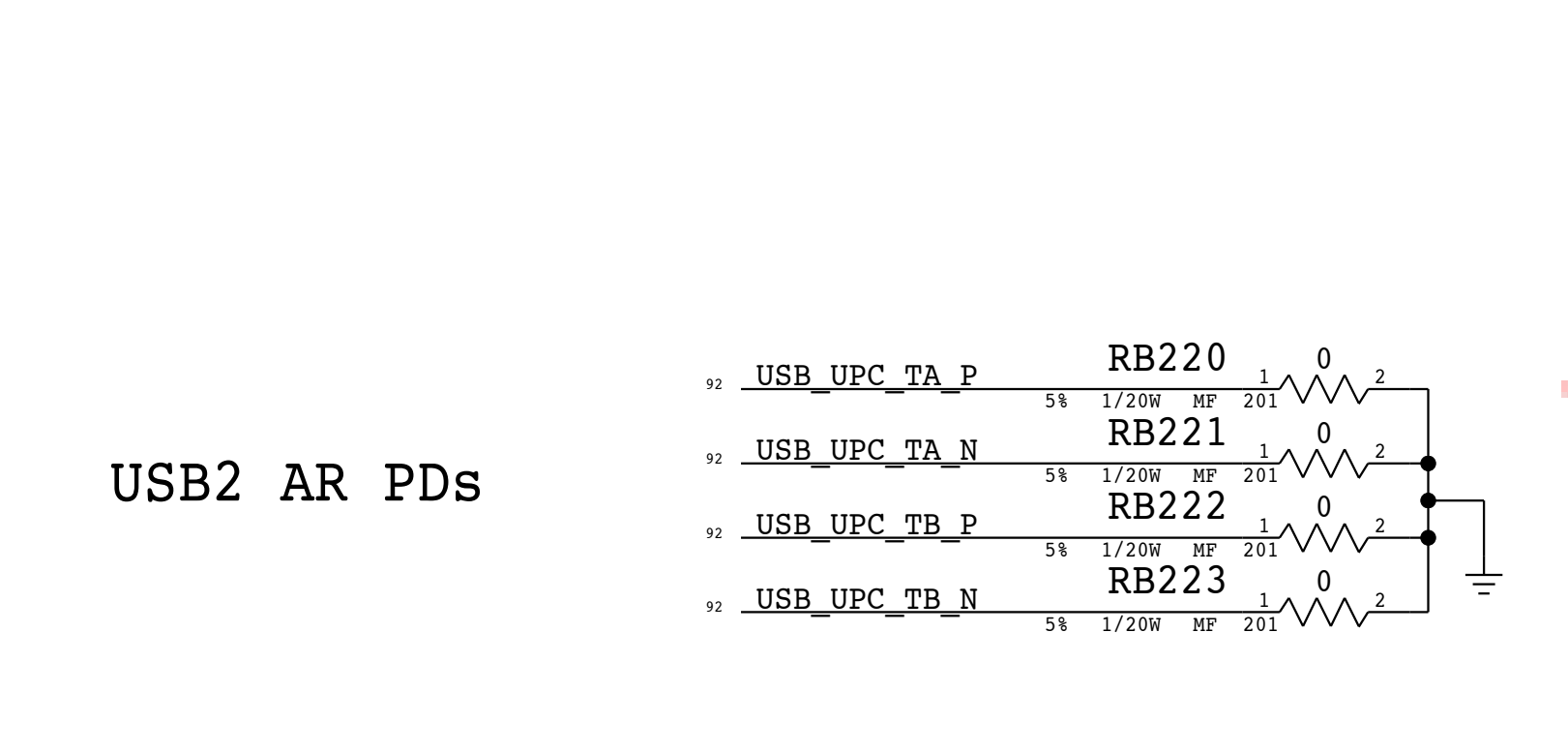
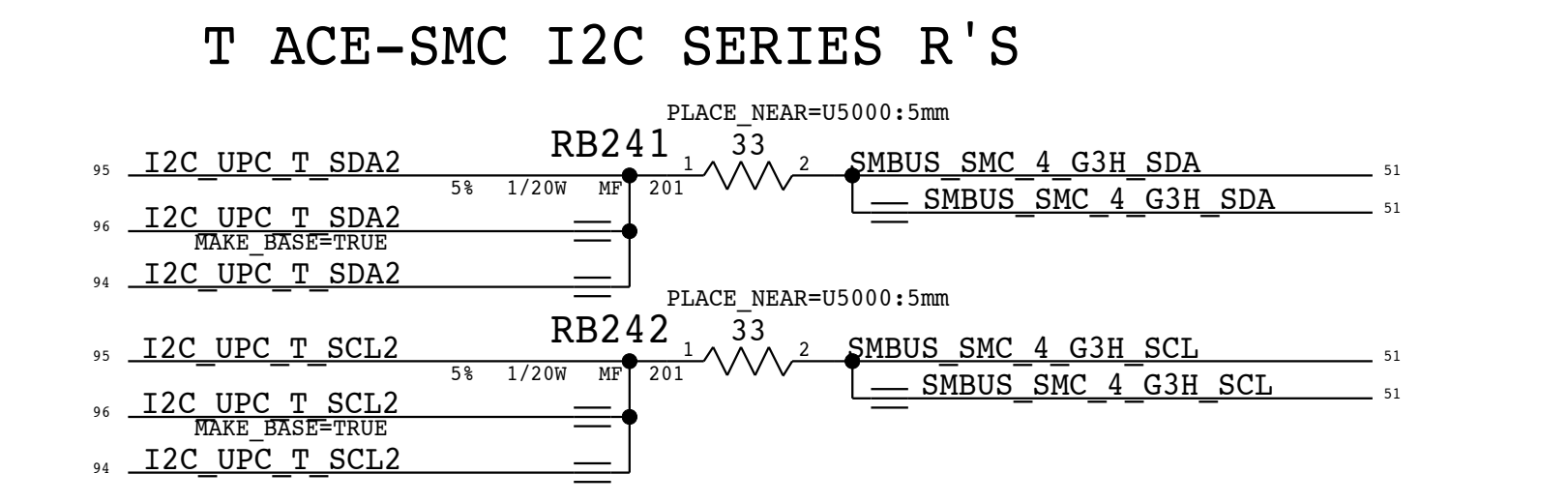
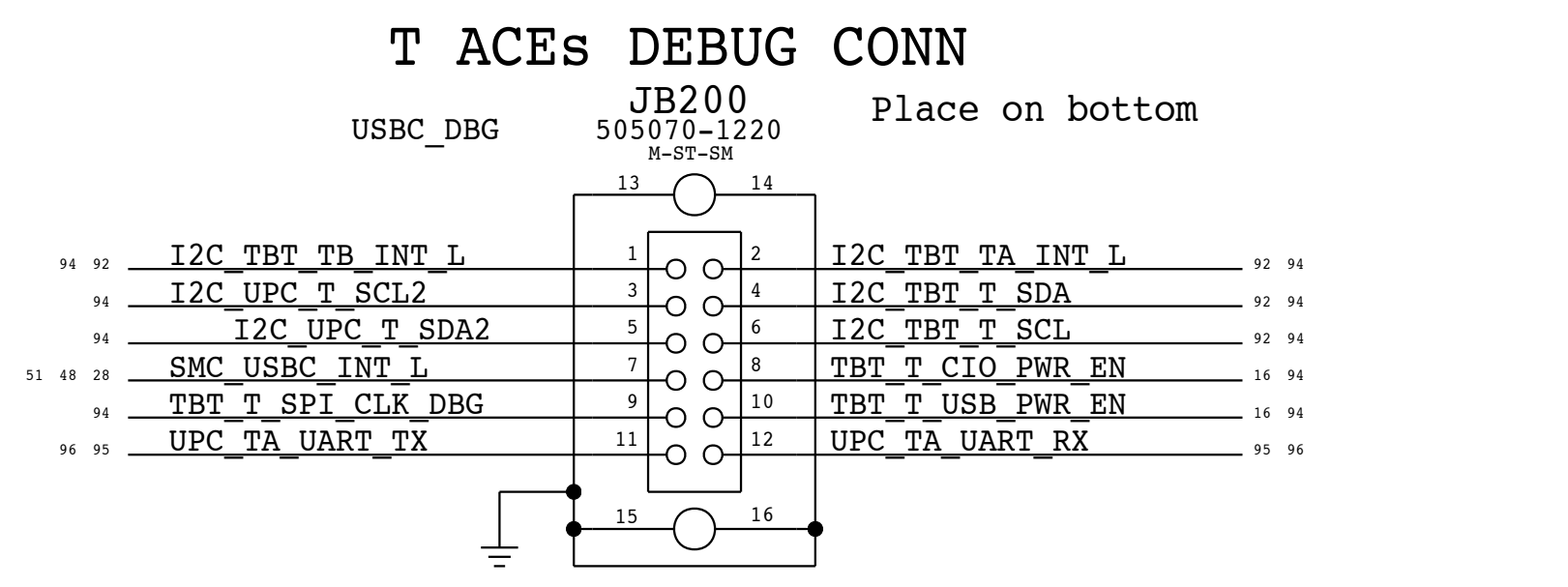
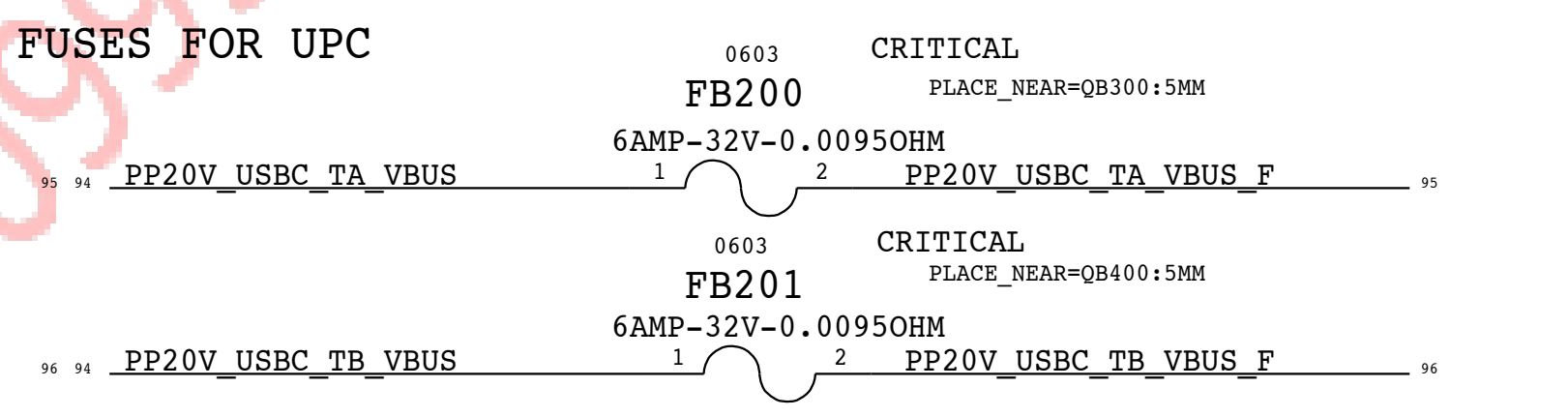
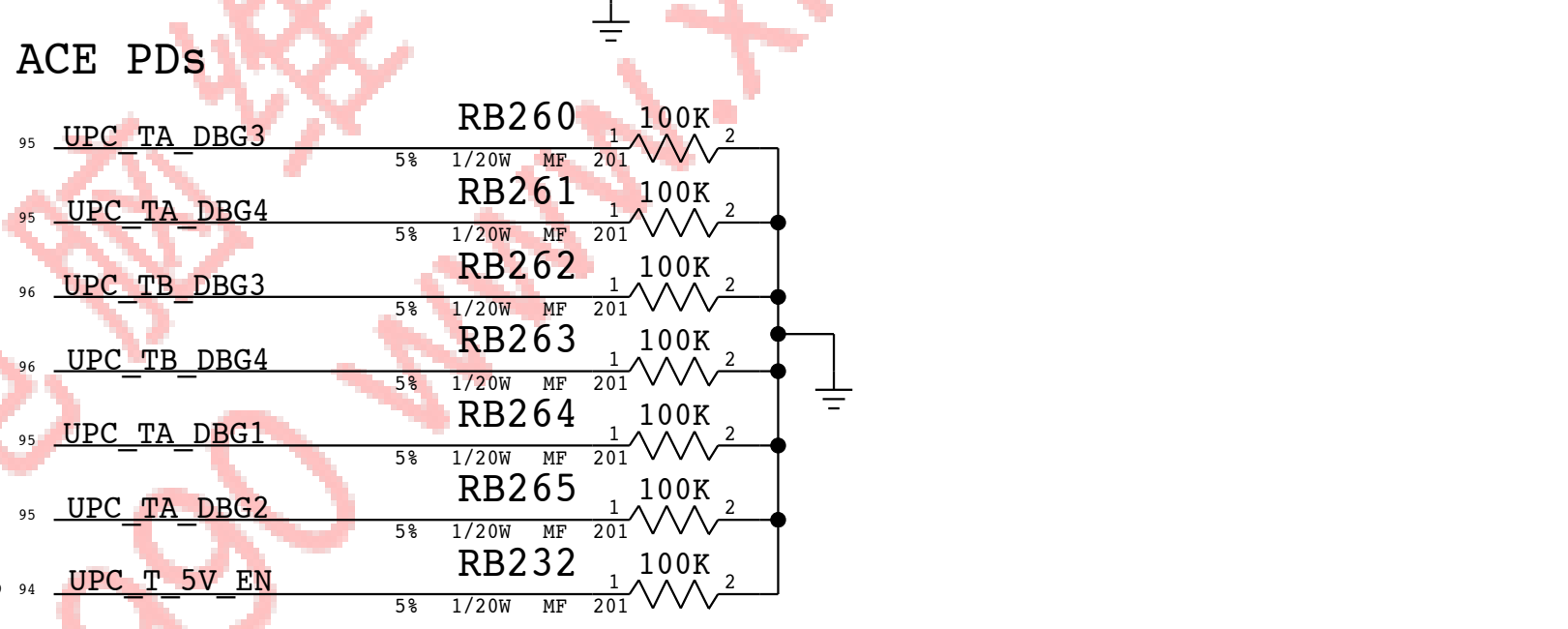
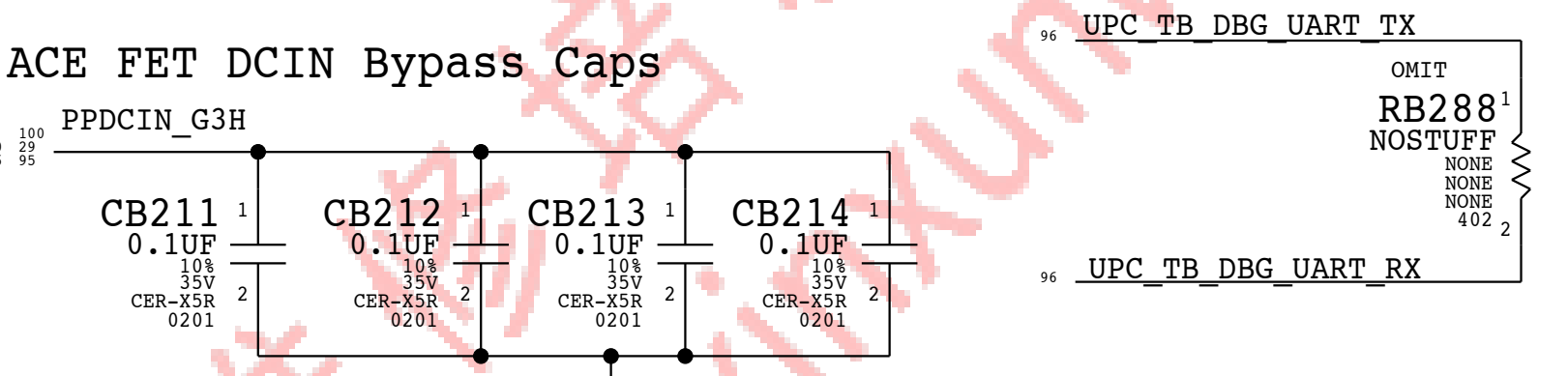
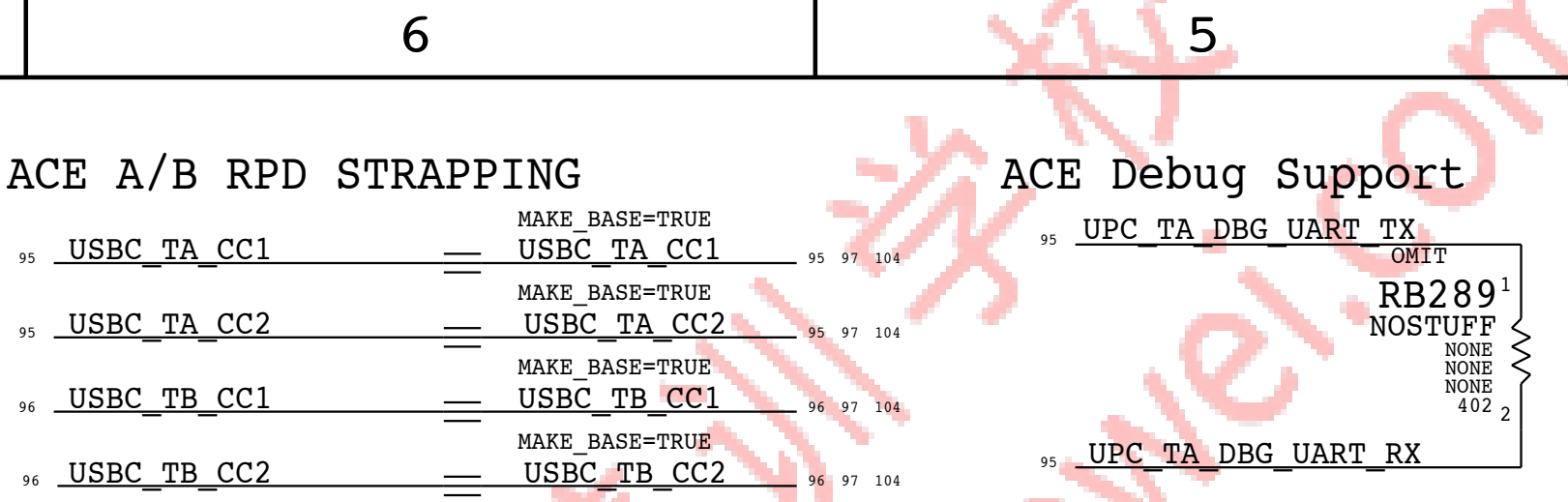
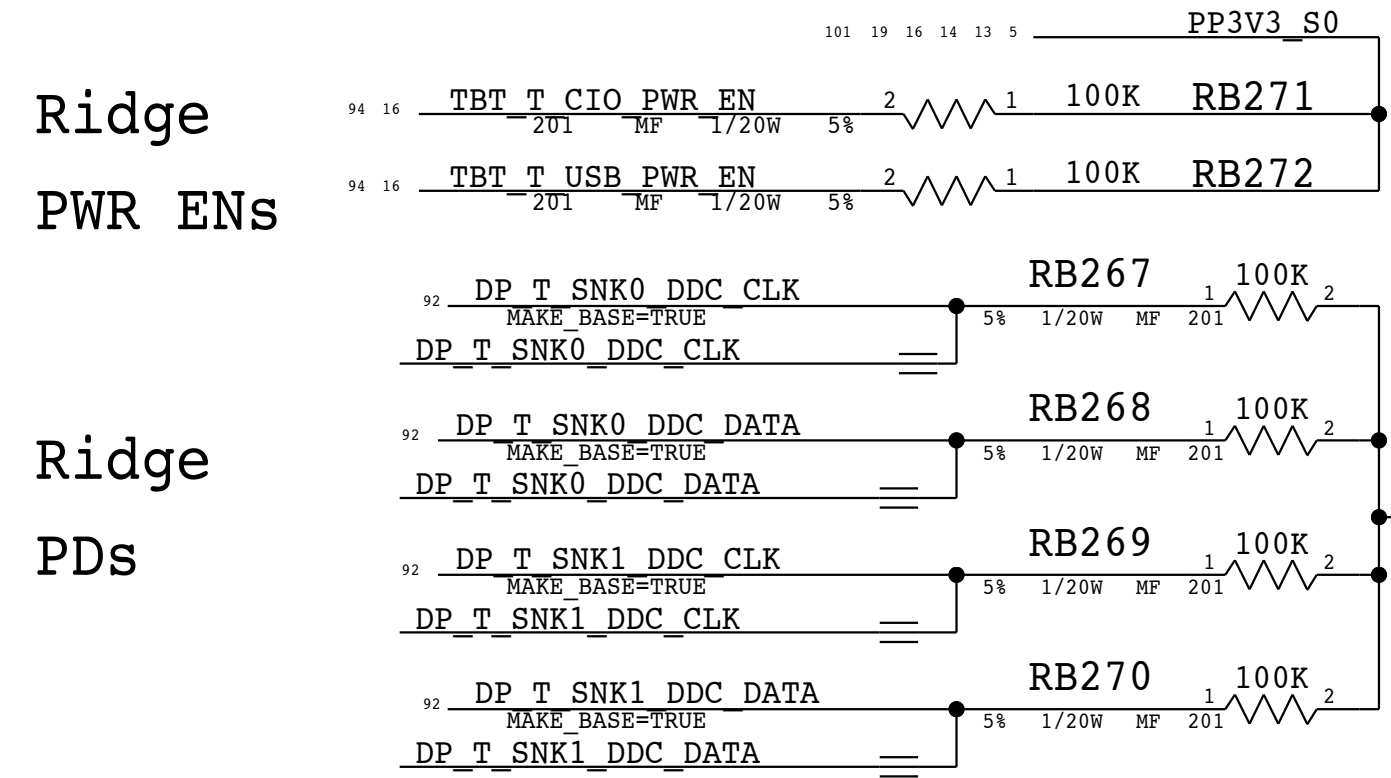
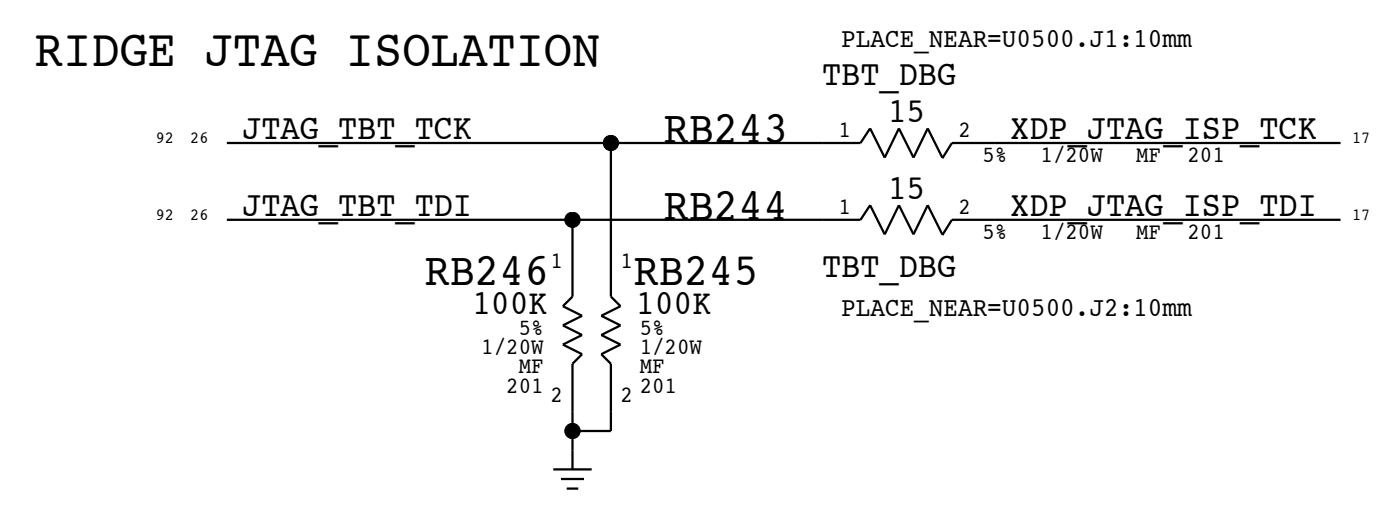
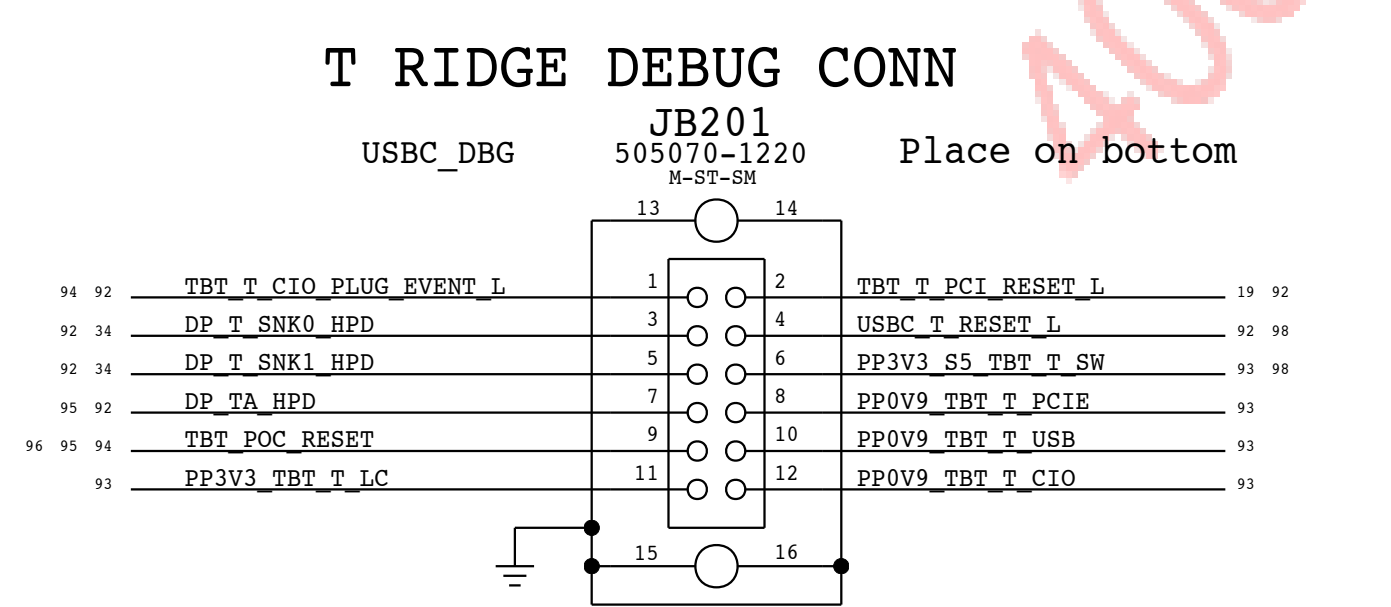
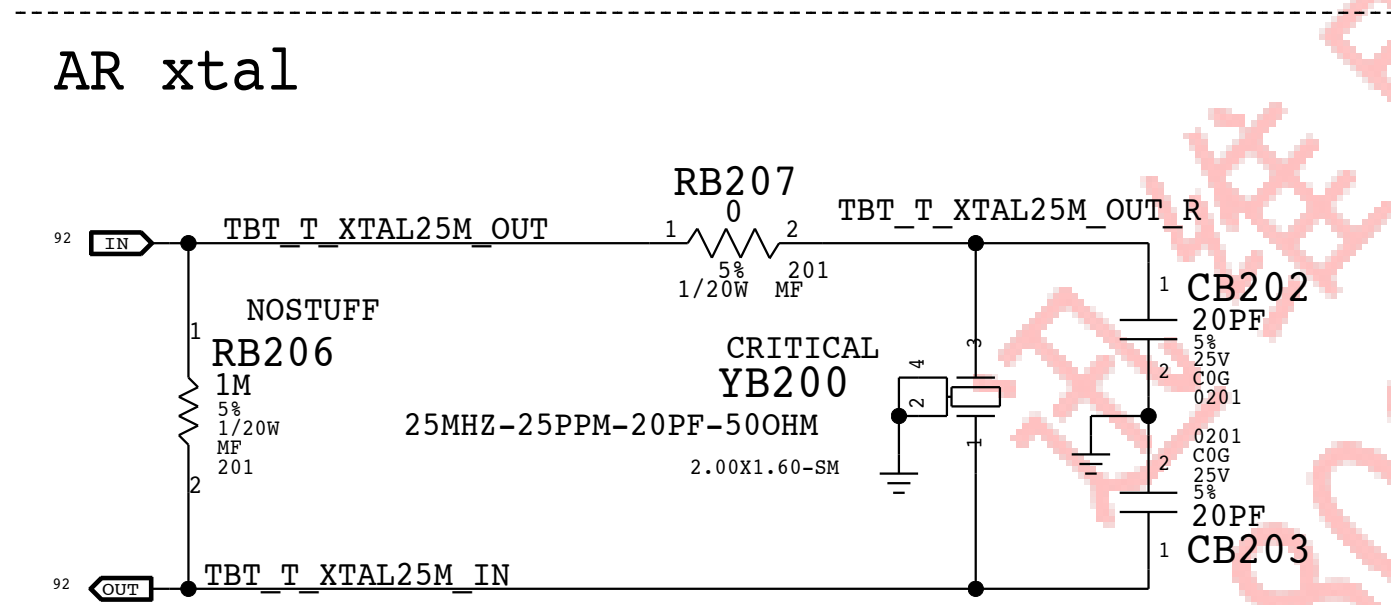
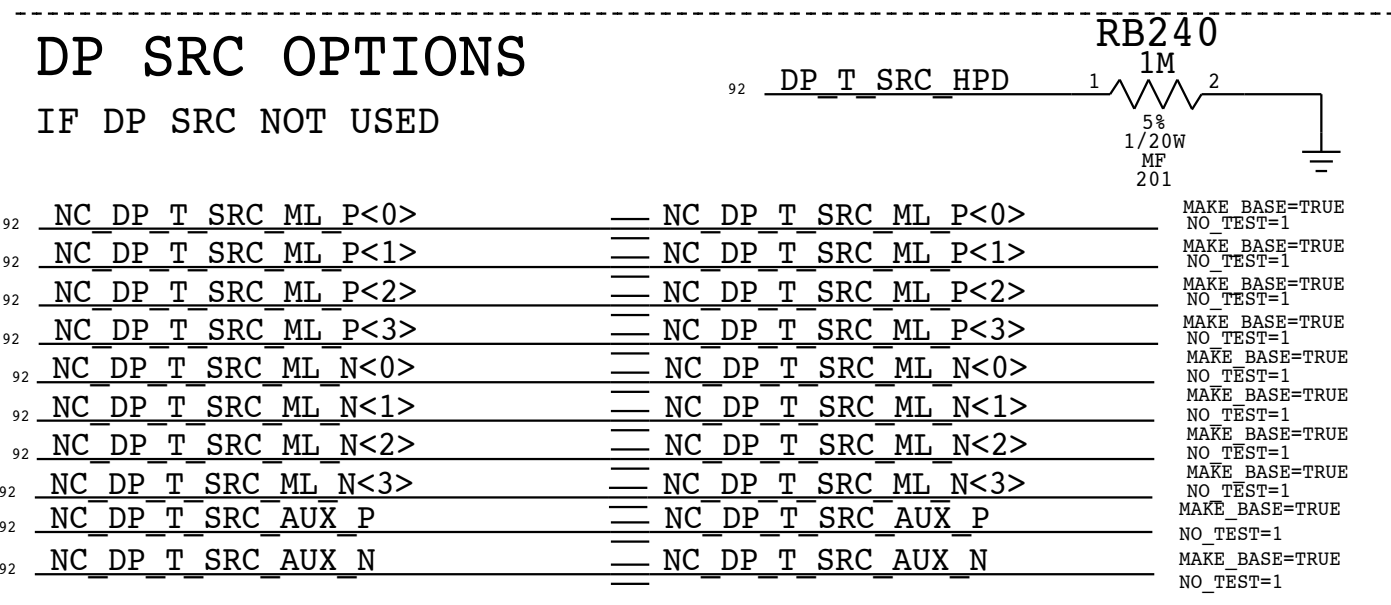
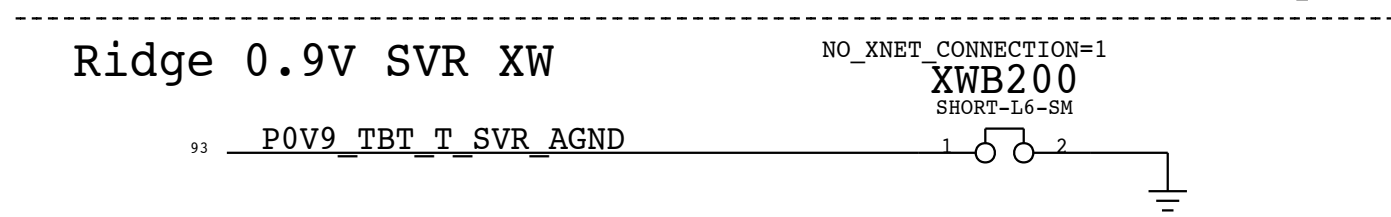
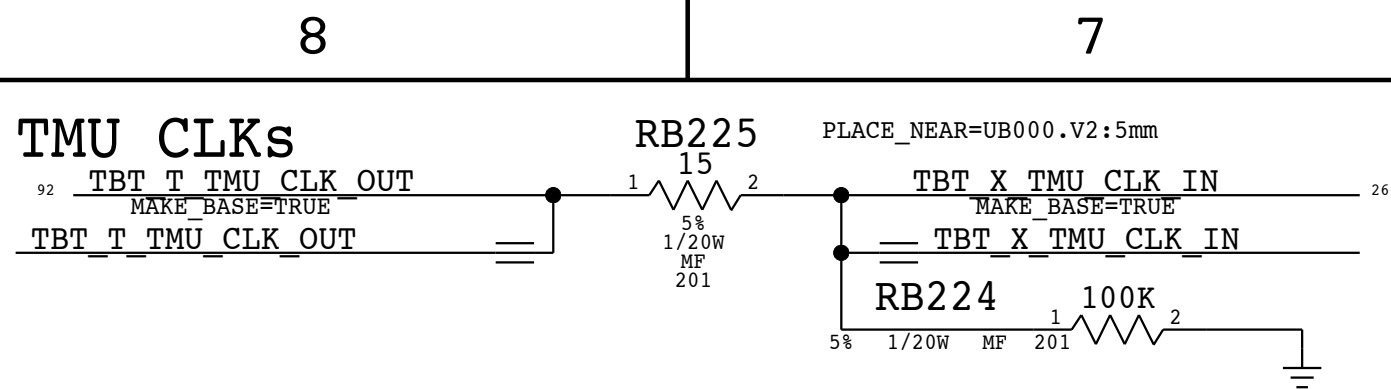
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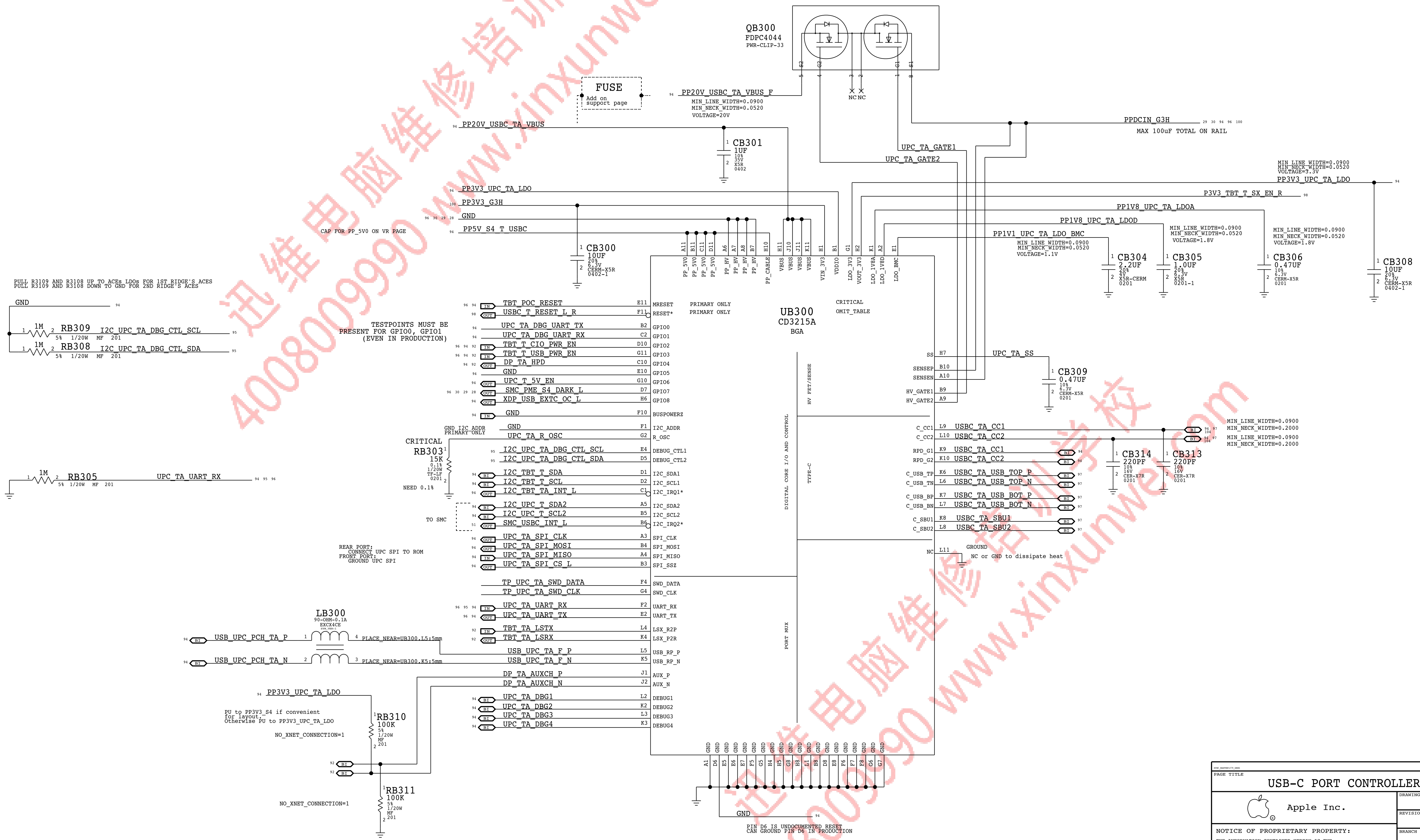


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		PAGE	111 OF 145
		SHEET	93 OF 119

BOM_COST_GROUP=TBT



PRIMARY ACE USB-C PORT CONTROLLER (UPC)



PAGE TITLE		
USB-C PORT CONTROLLER A		
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	REVISION	7.0.0
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	PAGE	113 OF 145
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BOM_COST_GROUP=USB-C

Diagram illustrating the USB-C Port Controller B circuit, showing connections for digital core I/O and control signals, and port mux signals.

Legend:

- TYPE-C
- PORT MUX

Pin Connections:

- Pin 1: L9 USBC_TB_CC1
- Pin 2: L10 USBC_TB_CC2
- Pin 3: K9 USBC_TB_CC1
- Pin 4: K10 USBC_TB_CC2
- Pin 5: K6 USBC_TB_USB_TOP_P
- Pin 6: L6 USBC_TB_USB_TOP_N
- Pin 7: K7 USBC_TB_USB_BOT_P
- Pin 8: L7 USBC_TB_USB_BOT_N
- Pin 9: K8 USBC_TB_SBU1
- Pin 10: L8 USBC_TB_SBU2
- Pin 11: L11 GROUND (NC or GND to dissipate heat)

Component Values:

- CB414: 220PF
- CB413: 220PF
- 101: 16V CER-X7R 0201
- 102: 16V CER-X7R 0201

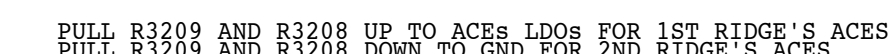
Notes:

- MIN_LINE_WIDTH=0.0900
- MIN_NECK_WIDTH=0.2000
- MIN_LINE_WIDTH=0.0900
- MIN_NECK_WIDTH=0.2000

Pin 11 is UNDOCUMENTED RESET CAN GROUND PIN D6 IN PRODUCTION

BOM COST GROUP=USB-C

DRAWING NUMBER		SIZE
051-00515		D
REVISION		
7.0.0		
BRANCH		
dvt1		
PAGE		
114 OF 145		
SHEET		
96 OF 119		



TESTPOINTS MUST BE
PRESENT FOR GPIO0, GPIO1
(EVEN IN PRODUCTION)

```

REAR PORT:
    CONNECT UPC SPI TO ROM
FRONT PORT:
    GROUND UPC SPI

```

PU to PP3V3_S4 if convenient
for layout.
Otherwise PU to PP3V3_UPC_TB_LDO

NO XNET CONNECTION=1

PIN D6 IS UNDOCUMENTED RESET
CAN GROUND PIN D6 IN PRODUCTION

USB-C PORT CONTROLLER B		DRAWING NUMBER 051-00515		SIZE D	
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		SHEET 96 OF 119			

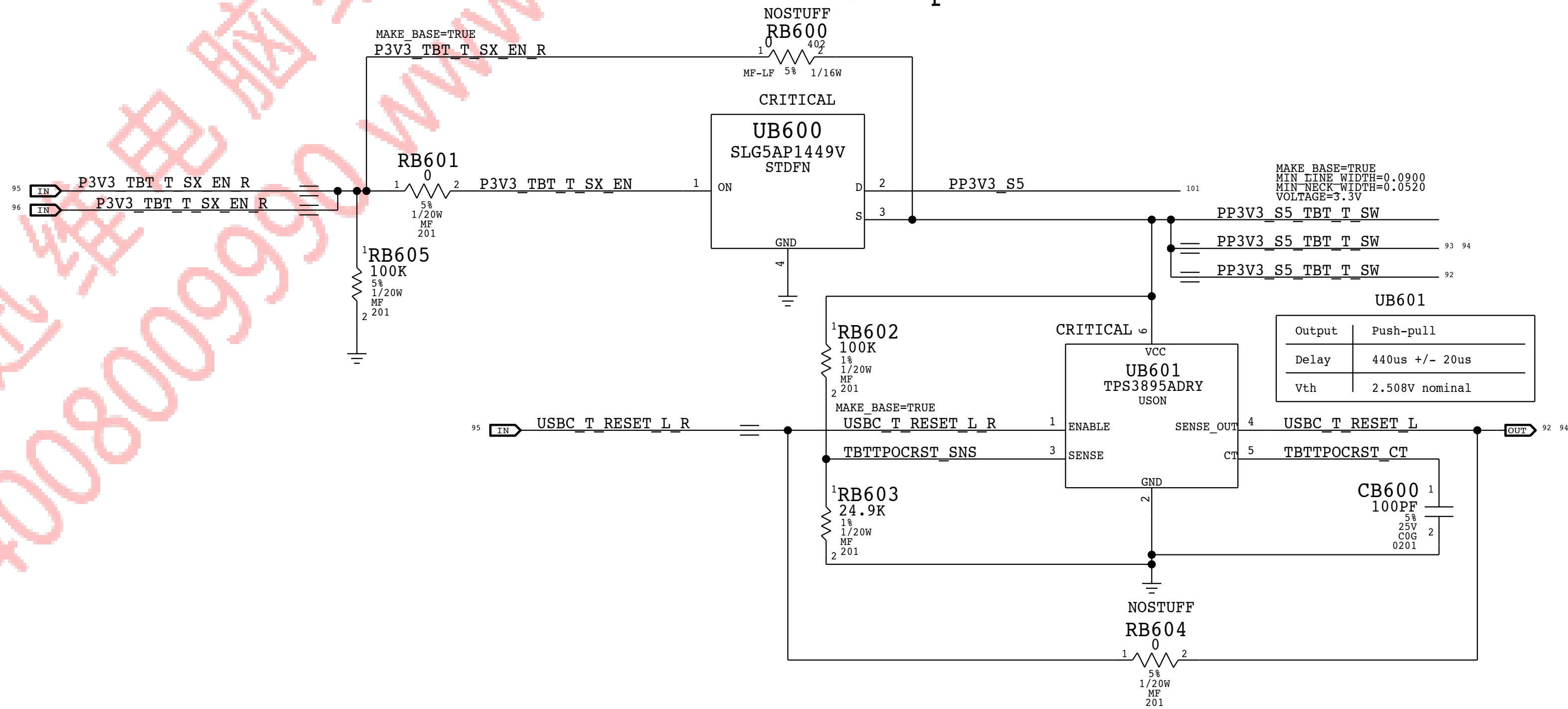
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			115 OF 145
		SHEET	
			97 OF 119

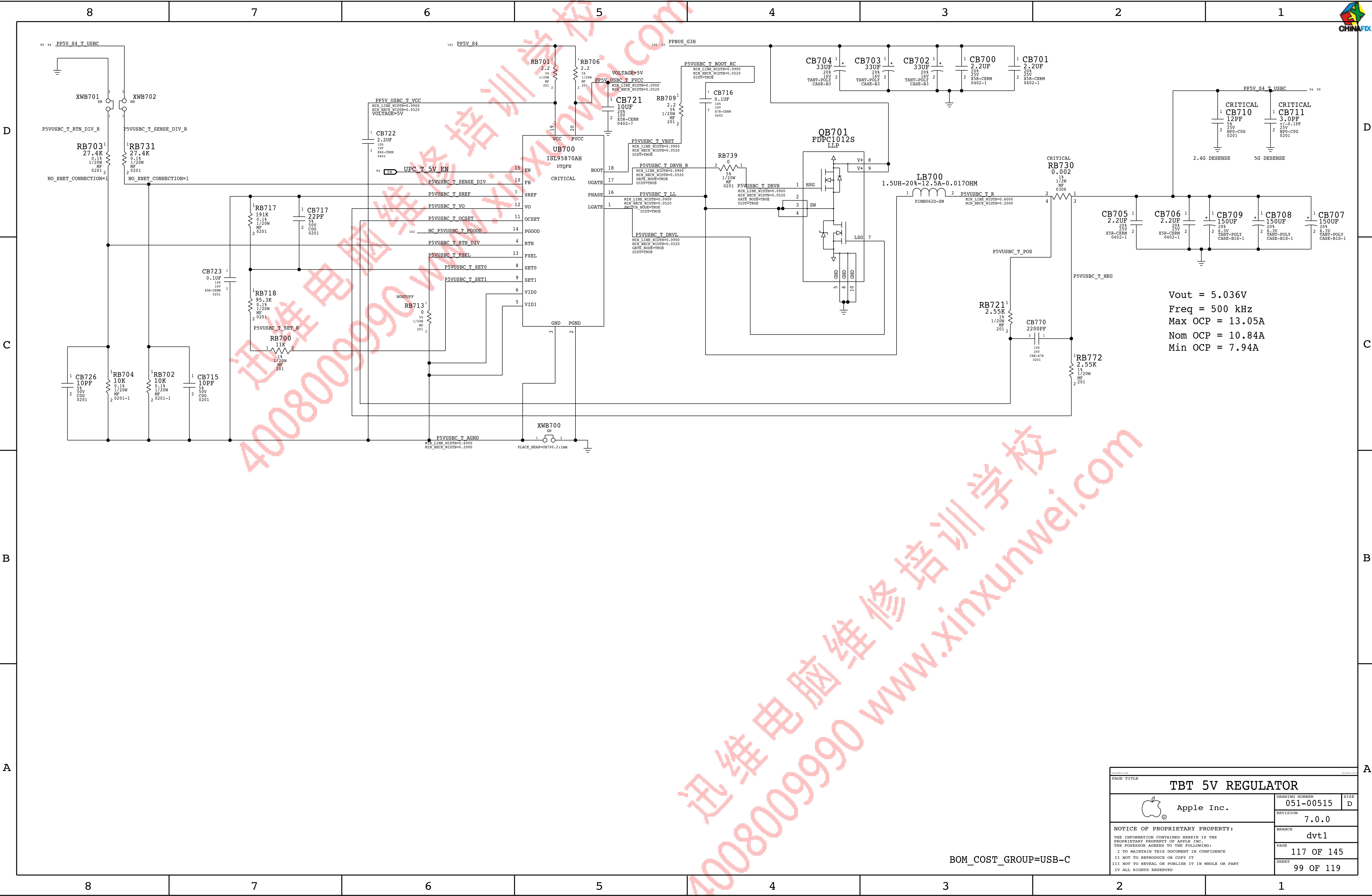
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TBT T "POC" Power-up Reset



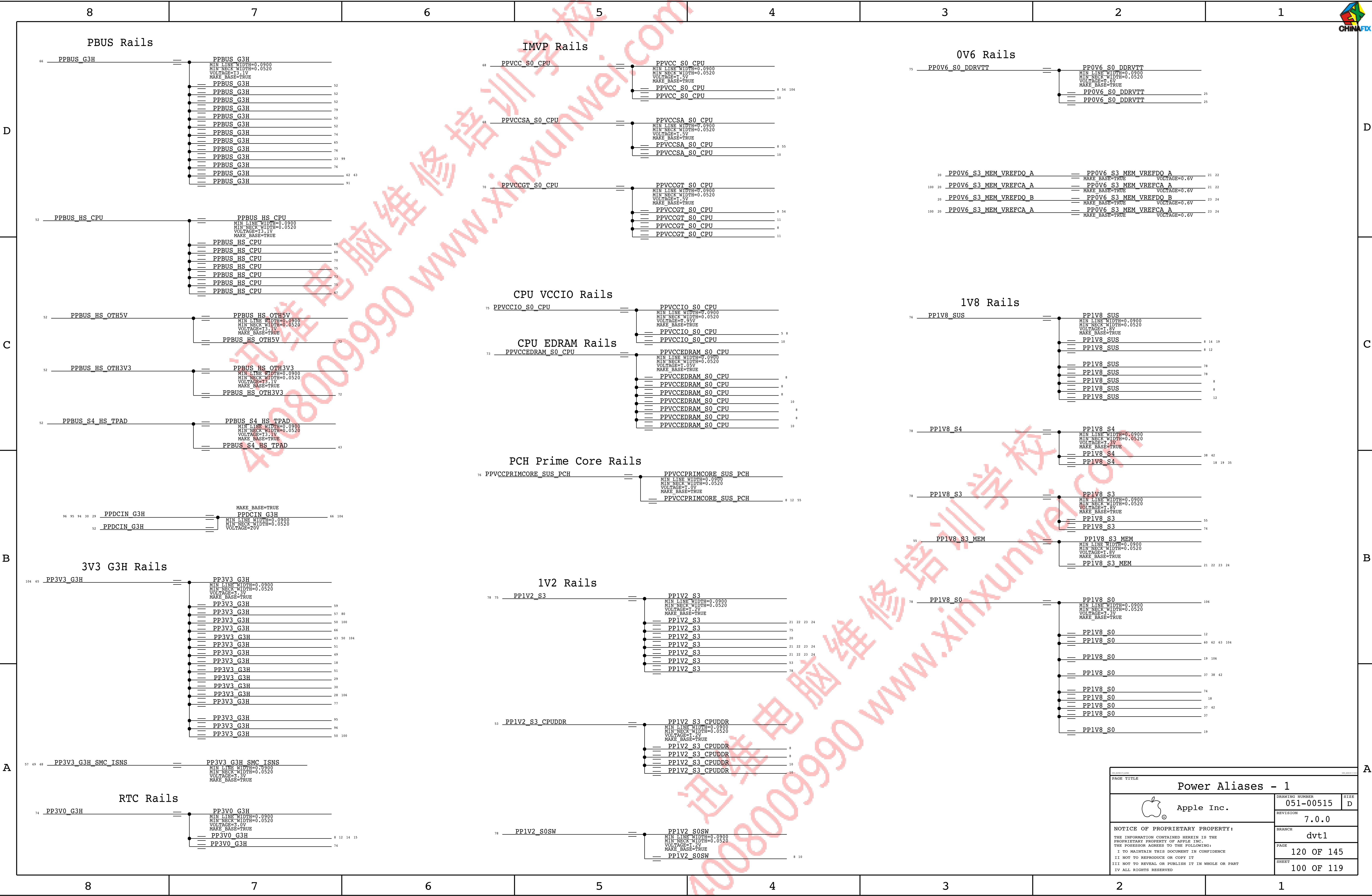
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	PAGE	116 OF 145
	SHEET	98 OF 119

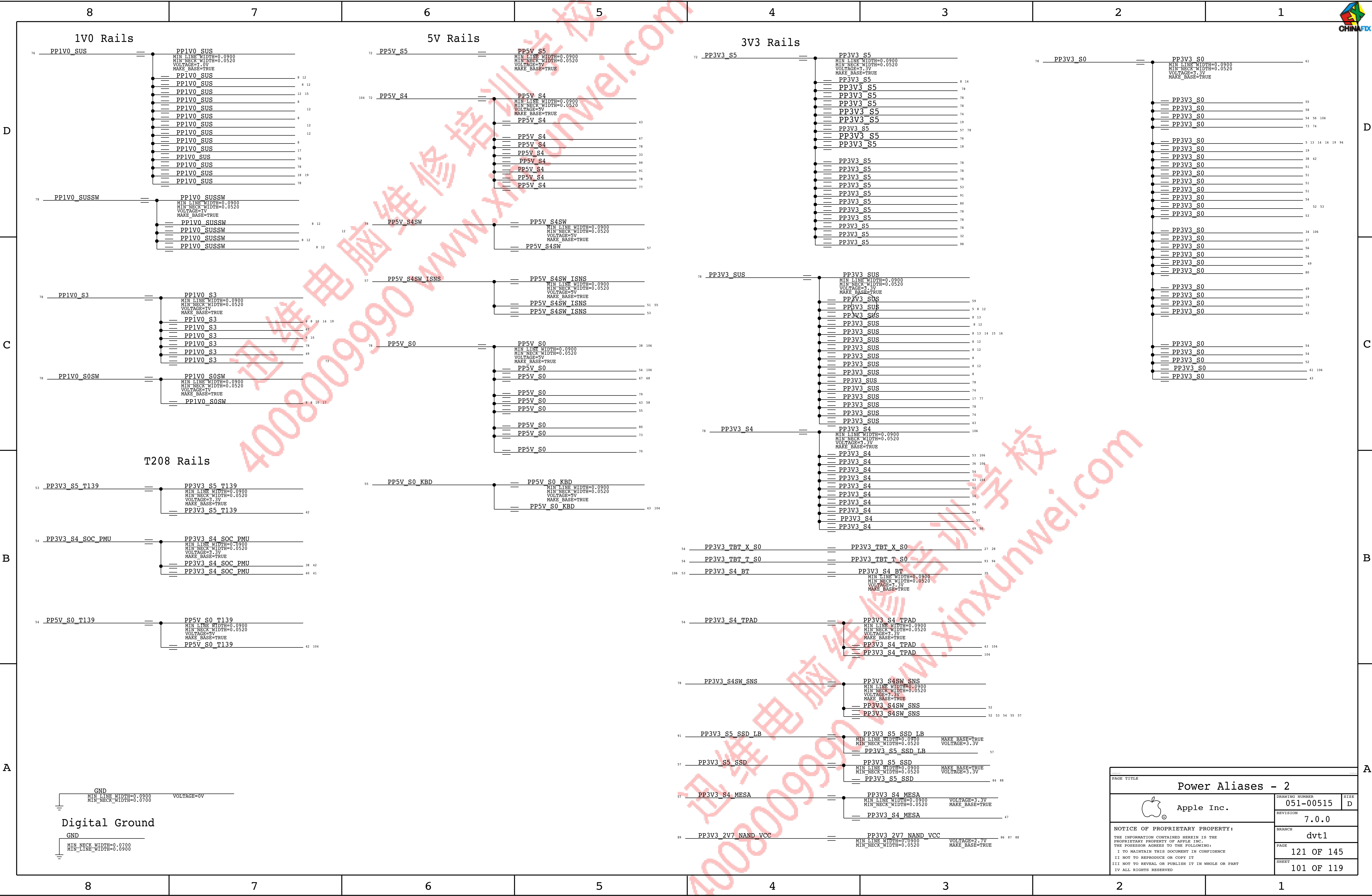


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	PAGE	117 OF 145
	SHEET	99 OF 119

BOM_COST_GROUP=USB-C



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Power Aliases - 1		
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	PAGE	120 OF 145
	SHEET	100 OF 119



PAGE TITLE			Power Aliases - 2	
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I9 ALL RIGHTS RESERVED				

Memory Bit & Byte Swizzle

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D

C

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				NC_FCH_SLP_WLAN_L NC_SPI_CS1_L NC_SPI_CS2_L NC_SPI_CS1_L NC_SPI_CS2_L NC_PCIE_CAMERA_D2R_N NC_PCIE_CAMERA_D2R_P NC_PCIE_CAMERA_R2D_C_N NC_PCIE_CAMERA_R2D_C_P NC_PCIE_CLK100M_CAMERA_N NC_PCIE_CLK100M_CAMERA_P NC_PICCOLO_VEN2 NC_PCIE_TBT_T_D2R_C_P<2> NC_PCIE_TBT_T_D2R_C_N<2> NC_PCIE_TBT_T_D2R_C_P<3> NC_PCIE_TBT_T_D2R_C_N<3> NC_PCIE_TBT_T_R2D_P<2> NC_PCIE_TBT_T_R2D_N<2> NC_PCIE_TBT_T_R2D_P<3> NC_PCIE_TBT_T_R2D_N<3> High Speed NO_TEST PCIE_SSD_R2D_C_P<3..0> PCIE_SSD_R2D_C_N<3..0> PCIE_SSD_D2R_P<3..0> PCIE_SSD_D2R_N<3..0>			
PP9919 PP9920	PCIE_TBT_X_D2R_N<0> PCIE_TBT_X_D2R_P<0>	MIPID_DATA_CONN_P MIPID_DATA_CONN_N MIPID_CLK_CONN_P MIPID_CLK_CONN_N					
PP9927 PP9928	PCIE_TBT_T_D2R_N<0> PCIE_TBT_T_D2R_P<0>	MIPI_DATA_CONN_P MIPI_DATA_CONN_N MIPI_CLK_CONN_P MIPI_CLK_CONN_N					
PP9933 PP9934	USB_CAMERA_DFR_N USB_CAMERA_DFR_P						
PP9941 PP9942 PP9943 PP9944 PP9945 PP9946 PP9947 PP9948	PCIE_CLK100M_SSD_LB_N PCIE_CLK100M_SSD_LB_P PCIE_CLK100M_TBT_X_N PCIE_CLK100M_TBT_X_P PCIE_CLK100M_TBT_T_N PCIE_CLK100M_TBT_T_P PCIE_CLK100M_AP_N PCIE_CLK100M_AP_P						
PP9801 PP9802 PP9803 PP9804 PP9808 PP9819	XDP_USB_EXT_A_OC_L XDP_USB_EXT_B_OC_L XDP_USB_EXT_C_OC_L XDP_USB_EXT_D_OC_L HDA_SDINO LPC_CLK24M_SMC						
TP9820	XDP_BPM_L<0>						

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		REVISION	7.0.0
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		PAGE	125 OF 145
		SHEET	105 OF 119



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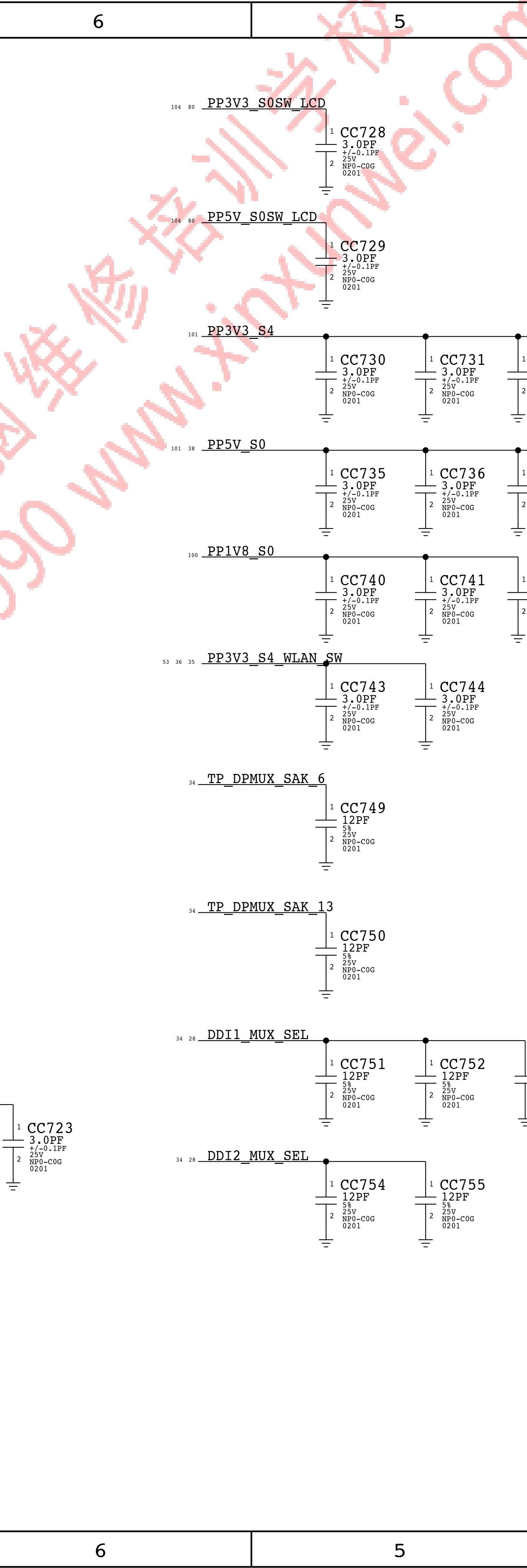
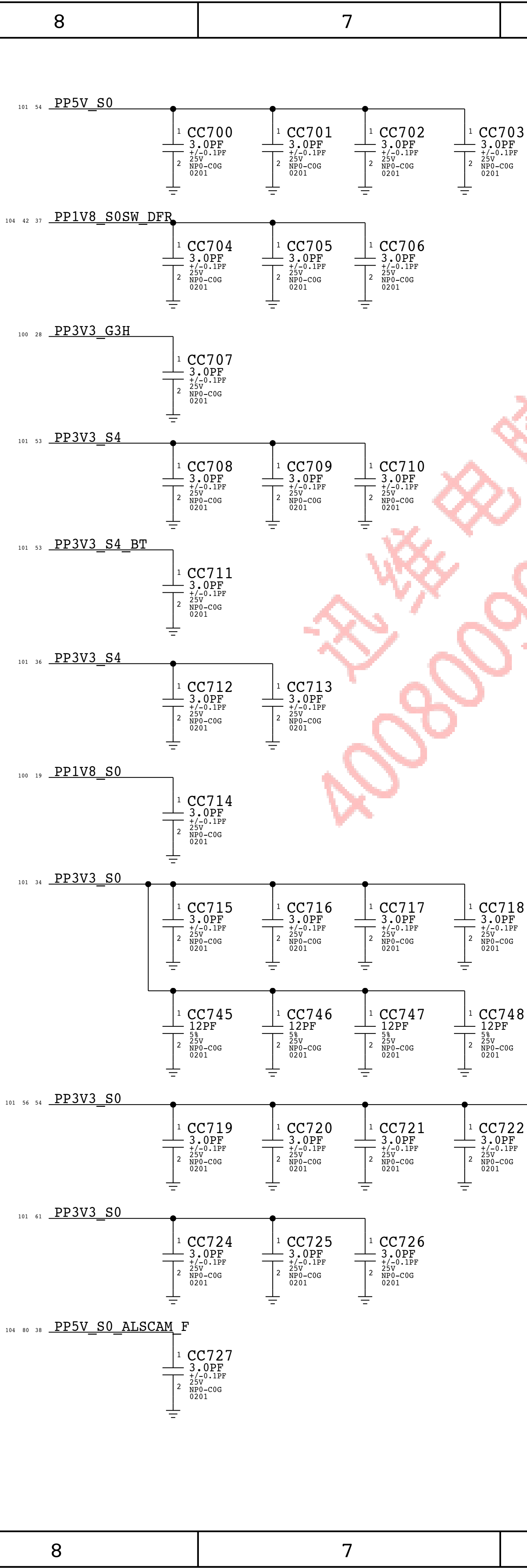
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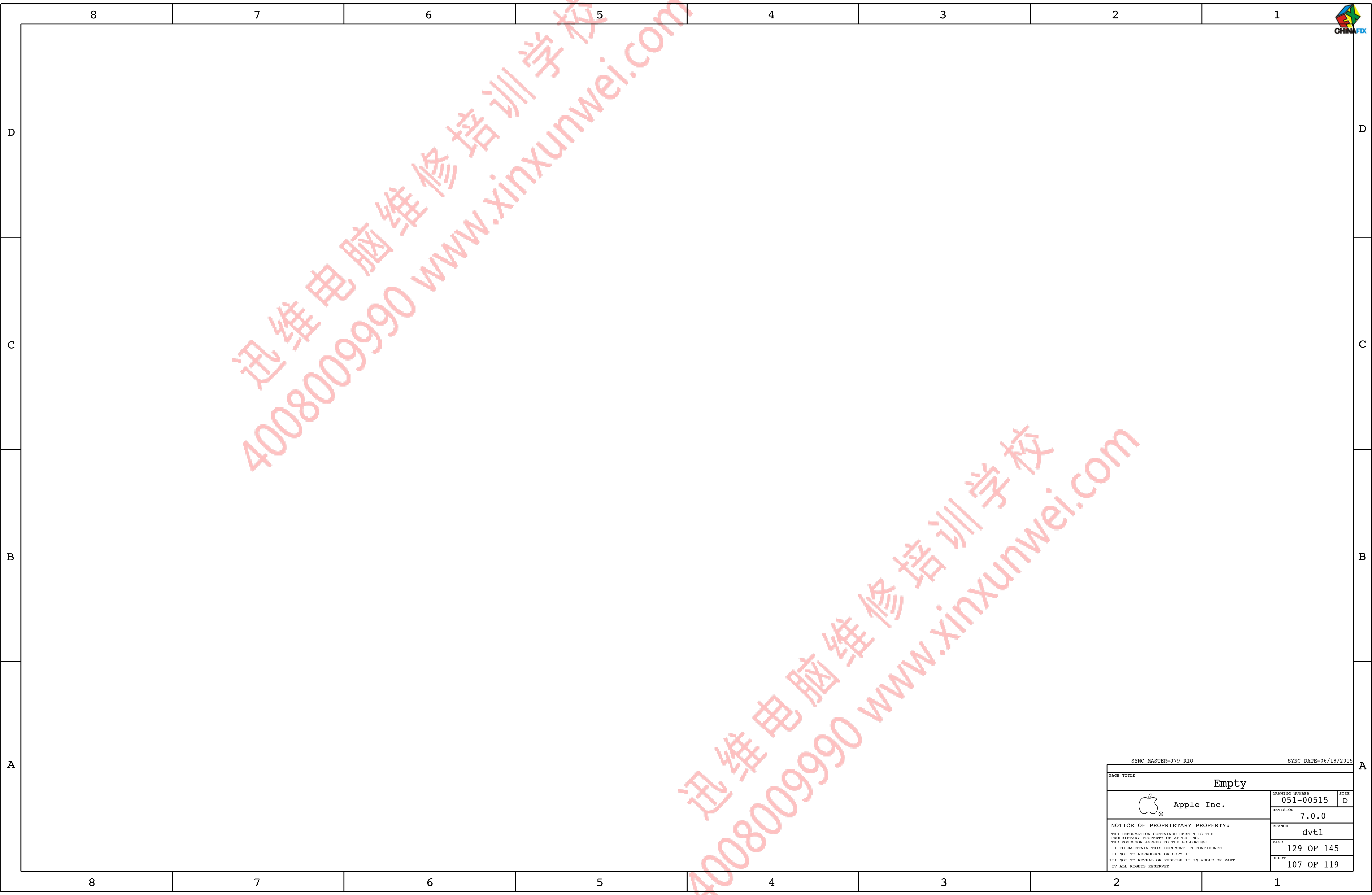
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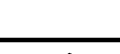
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	PAGE	127 OF 145	
	SHEET	106 OF 119	



SYNC MASTER=J79 RIO		SYNC DATE=06/18/2015	
PAGE TITLE			
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	051-00515		D
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BRANCH		dvt1	
PAGE		129 OF 145	
SHEET		107 OF 119	

	8	7	6	5	4	3	2	1	CHINAFOX
D	J79 BOARD-SPECIFIC SPACING & PHYSICAL CONSTRAINTS								D
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	PAGE	130 OF 145	
	SHEET	108 OF 119	

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NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_70D	BGA_MEM	MEM_80D
MEM_40S	BGA_MEM	MEM_45S

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE
MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
MEM_75D	*	=75_OHM_DIFF	=75_OHM_DIFF	=75_OHM_DIFF	=75_OHM_DIFF	=75_OHM_DIFF	=75_OHM_DIFF
MEM_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=4x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=6x_DIELECTRIC	?
MEM_CMD2CMD	*	=6x_DIELECTRIC	?
MEM_CMD2CTL	*	=6x_DIELECTRIC	?
MEM_CTL2CTL	*	=6x_DIELECTRIC	?
MEM_CLK2CLK	*	=12x_DIELECTRIC	?
MEM_DATA2OTHERMEM	*	=16x_DIELECTRIC	?
MEM_2OTHERMEM	*	=8x_DIELECTRIC	?
MEM_2PWR	*	=4x_DIELECTRIC	?
MEM_2GND	*	=4x_DIELECTRIC	?
MEM_2OTHER	*	=12x_DIELECTRIC	?
MEM_12MIL	*	0.305 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF_B	*	=1.5x_DIELECTRIC	?
MEM_DQS2OWNDATA_B	*	=1.5x_DIELECTRIC	?
MEM_CMD2CMD_B	*	=1.5x_DIELECTRIC	?
MEM_CMD2CTL_B	*	=1.5x_DIELECTRIC	?
MEM_CTL2CTL_B	*	=1.5x_DIELECTRIC	?
MEM_CLK2CLK_B	*	=1.5x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	TOP,BOTTOM	=10x_DIELECTRIC	?
MEM_DQS2OWNDATA	TOP,BOTTOM	=10x_DIELECTRIC	?
MEM_CMD2CMD	TOP,BOTTOM	=10x_DIELECTRIC	?
MEM_CMD2CTL	TOP,BOTTOM	=10x_DIELECTRIC	?
MEM_CTL2CTL	TOP,BOTTOM	=10x_DIELECTRIC	?
MEM_CLK2CLK	TOP,BOTTOM	=16x_DIELECTRIC	?
MEM_20THERMEM	TOP,BOTTOM	=16x_DIELECTRIC	?
MEM_2PWR	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2GND	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2OTHER	TOP,BOTTOM	=20x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2OTHERMEM_B	*	=1.5x_DIELECTRIC	?
MEM_2OTHERMEM_B	*	=1.5x_DIELECTRIC	?
MEM_2OTHER_B	*	=1.5x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	*	*	MEM_2OTHER
MEM_*_DQS_*	*	*	MEM_2OTHER
MEM_CMD	*	*	MEM_2OTHER
MEM_CTL	*	*	MEM_2OTHER
MEM_CLK	*	*	MEM_2OTHER
MEM_*	MEM_*	*	MEM_2OTHERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYT*_*	*	BGA_MEM	MEM_20THER_B
MEM_*_DQS_*	*	BGA_MEM	MEM_20THER_B
MEM_CMD	*	BGA_MEM	MEM_20THER_B
MEM_CTL	*	BGA_MEM	MEM_20THER_B
MEM_CLK	*	BGA_MEM	MEM_20THER_B
MEM_*	MEM_*	BGA_MEM	MEM_20THERMEM_B

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	=SAME	BGA_MEM	MEM_DATA2SELF_B

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	MEM_*	*	MEM_DATA20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	MEM_*	BGA_MEM	MEM_DATA2OTHERMEM_B

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTL	*	MEM_CMD2CTL
MEM_CTL	MEM_CTL	*	MEM_CTL2CTL
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	BGA_MEM	MEM_CMD2CMD_B
MEM_CMD	MEM_CTL	BGA_MEM	MEM_CMD2CTL_B
MEM_CTL	MEM_CTL	BGA_MEM	MEM_CTL2CTL_B
MEM_CLK	MEM_CLK	BGA_MEM	MEM_CLK2CLK_B

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DQBYTE_0	*	MEM_DQS20WINDATA
MEM_A_DQS_1	MEM_A_DQBYTE_1	*	MEM_DQS20WINDATA
MEM_A_DQS_2	MEM_A_DQBYTE_2	*	MEM_DQS20WINDATA
MEM_A_DQS_3	MEM_A_DQBYTE_3	*	MEM_DQS20WINDATA
MEM_A_DQS_4	MEM_A_DQBYTE_4	*	MEM_DQS20WINDATA
MEM_A_DQS_5	MEM_A_DQBYTE_5	*	MEM_DQS20WINDATA
MEM_A_DQS_6	MEM_A_DQBYTE_6	*	MEM_DQS20WINDATA
MEM_A_DQS_7	MEM_A_DQBYTE_7	*	MEM_DQS20WINDATA
MEM_B_DQS_0	MEM_B_DQBYTE_0	*	MEM_DQS20WINDATA
MEM_B_DQS_1	MEM_B_DQBYTE_1	*	MEM_DQS20WINDATA
MEM_B_DQS_2	MEM_B_DQBYTE_2	*	MEM_DQS20WINDATA
MEM_B_DQS_3	MEM_B_DQBYTE_3	*	MEM_DQS20WINDATA
MEM_B_DQS_4	MEM_B_DQBYTE_4	*	MEM_DQS20WINDATA
MEM_B_DQS_5	MEM_B_DQBYTE_5	*	MEM_DQS20WINDATA
MEM_B_DQS_6	MEM_B_DQBYTE_6	*	MEM_DQS20WINDATA
MEM_B_DQS_7	MEM_B_DQBYTE_7	*	MEM_DQS20WINDATA

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	MEM_A_DQBYTE_0	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_A_DQS_1	MEM_A_DQBYTE_1	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_A_DQS_2	MEM_A_DQBYTE_2	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_A_DQS_3	MEM_A_DQBYTE_3	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_A_DQS_4	MEM_A_DQBYTE_4	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_A_DQS_5	MEM_A_DQBYTE_5	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_A_DQS_6	MEM_A_DQBYTE_6	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_A_DQS_7	MEM_A_DQBYTE_7	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_B_DQS_0	MEM_B_DQBYTE_0	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_B_DQS_1	MEM_B_DQBYTE_1	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_B_DQS_2	MEM_B_DQBYTE_2	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_B_DQS_3	MEM_B_DQBYTE_3	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_B_DQS_4	MEM_B_DQBYTE_4	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_B_DQS_5	MEM_B_DQBYTE_5	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_B_DQS_6	MEM_B_DQBYTE_6	BGA_MEM	MEM_DQS2OWNDATA_B
MEM_B_DQS_7	MEM_B_DQBYTE_7	BGA_MEM	MEM_DQS2OWNDATA_B

ELECTRICAL CONST SET	NET TYPE	
	PHYSICAL	SPACING

SYNC MASTER=YBARTANTO J44		SYNC DATE=01/02/2013	
PAGE TITLE			
Memory Constraints			
	Apple Inc.		DRAWING NUMBER 051-00515
			SIZE D
		REVISION	7.0.0
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		SHEET	111 OF 119

Thunderbolt, DP, HDMI Constraints

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBT_SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt & DisplayPort Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
TBTD _P _85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
TBTDP_2SAME	*	=3X_DIELECTRIC	?
TBTDP_TXRX	*	=6X_DIELECTRIC	?
TBTDP_2OTHER	*	=4X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBTDP_*	*	*	TBTDP_2OTHER
TBTDP_*	=SAME	*	TBTDP_2SAME
TBTDP_TX	*_RX	*	TBTDP_TXRX
TBTDP_RX	*_TX	*	TBTDP_TXRX

DisplayPort & HDMI Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
HDMI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2SAME	*	=3x_DIELECTRIC	?	DP_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_2OTHER	*	=4x_DIELECTRIC	?	DP_2OTHER	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMICKL_2OTHER	*	=7x_DIELECTRIC	?	HDMICKL_2OTHER	TOP,BOTTOM	=10x_DIELECTRIC	?
HDMICKL_2DPHDMI	*	=4x_DIELECTRIC	?	HDMICKL_2DPHDMI	TOP,BOTTOM	=6x_DIELECTRIC	?
HDMIDATA_2SAME	*	=3x_DIELECTRIC	?	HDMIDATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
HDMIDATA_2OTHER	*	=4x_DIELECTRIC	?	HDMIDATA_2OTHER	TOP,BOTTOM	=6x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
HDMI_DATA	*	*	HDMIDATA_20THER
HDMI_DATA	=SAME	*	HDMIDATA_2SAME
HDMI_DATA	TBTDP_TX	*	HDMIDATA_2SAME
HDMI_DATA	TBTDP_RX	*	TBTDP_TXRX
HDMI_CLK	*	*	HDMICLK_20THER
HDMI_CLK	HDMI_DATA	*	HDMICLK_2DPHDMI
HDMI_CLK	DISPLAYPORT	*	HDMICLK_2DPHDMI
HDMI_CLK	TBTDP_TX	*	HDMICLK_2DPHDMI

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DISPLAYPORT	*	*	DP_20THER
DISPLAYPORT	=SAME	*	DP_2SAME
DISPLAYPORT	HDMI_DATA	*	DP_2SAME
DISPLAYPORT	TBTDP_TX	*	DP_2SAME
DISPLAYPORT	TBTDP_RX	*	TBTDP_TXRX

Thunderbolt, DP, HDMI Net Properties

ELECTRICAL CONST SET	NET TYPE	
	PHYSICAL	SPACING

SYNC_MASTER=J79 JACK		SYNC_DATE=05/19/2015	
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TBT DP HDMI Constraints			
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	051-00515		D
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		PAGE	
		134 OF 145	
		SHEET	
		112 OF 119	

	8	7	6	5	4	3	2	1	CHINAFOX
D	Change List: <RDAR://COMPONENT/XXXXXX> J79 HW EE SCHEMATIC PROTO 0 Kismet: AFP://KISMET.APPLE.COM/KISMET-PROJECTS/J79 Useful Wiki Links: Schematic Conventions - https://hmts.ecs.apple.com/wiki/index.php/User:Wferry/SchConventions Schematic Design Wiki - https://hmts.ecs.apple.com/wiki/index.php/Schematic_Design MobileMac HW Radar: <rdar://component/XXXXXX> MobileMac HW Task <rdar://component/XXXXXX> MobileMac HW Schematic <rdar://component/XXXXXX> MobileMac HW New Bugs <rdar://component/XXXXXX> MobileMac HW Layout <rdar://component/XXXXXX> MobileMac HW Investigation <rdar://component/XXXXXX> MobileMac HW Architecture Other Info: Page Allocations - <rdar://problem/19817053> 2015 Schematic Page Allocations								D
C									C
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A	879C_04878D-719_000000_0000000000 879C_04878D-719_000000_0000000000 PAGE TITLE References  Apple Inc. NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF APPLE INC. THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED DRAWING NUMBER 051-00515 REVISION 7.0.0 BRANCH dvt1 PAGE 140 OF 145 SHEET 118 OF 119 SIZE D								A
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Fuse</td></tr><tr><td>107S0249</td><td>107S0251</td><td></td><td>ALL</td><td></td></tr><tr><td>107S00015</td><td>107S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>107S00071</td><td>107S00053</td><td></td><td>ALL</td><td></td></tr><tr><td>128S0364</td><td>128S0264</td><td></td><td>ALL</td><td>Kemet w/ Panasonic</td></tr><tr><td>128S0325</td><td>128S0397</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00009</td><td>128S00007</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00029</td><td>128S00007</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00070</td><td>128S00007</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00010</td><td>128S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00031</td><td>128S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00026</td><td>128S00011</td><td></td><td>ALL</td><td></td></tr><tr><td>132S00064</td><td>132S0409</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0614</td><td>138S0578</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0703</td><td>138S0648</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00032</td><td>138S0831</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00049</td><td>138S0831</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0863</td><td>138S0853</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0775</td><td>138S0860</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00084</td><td>138S00060</td><td></td><td>ALL</td><td></td></tr><tr><td>152S00369</td><td>152800268</td><td></td><td>ALL</td><td>Cyntec w/ NEC</td></tr><tr><td>155S00188</td><td>155S0275</td><td></td><td>ALL</td><td>Murata w/ Taiyo</td></tr><tr><td>155S0694</td><td>155S0387</td><td></td><td>ALL</td><td></td></tr><tr><td>155S0660</td><td>155S0513</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00018</td><td>155S0664</td><td></td><td>ALL</td><td>Murata w/ Taiyo</td></tr><tr><td>155S00007</td><td>155S0667</td><td></td><td>ALL</td><td></td></tr></table>	PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	107800033	107800034		ALL		138S0738	138S1101		ALL	Samsung alt to Murata	138S0846	138S0811		ALL	Samsung alt to Murata	152800359	152800253		ALL	Chillisin alt to Cyttec	371S0704	371S00077		ALL	NXP alt to Diodes	376S1053	376S0604		ALL	Diodes alt to Fairchild	376S1106	376S0678		ALL	Fairchild alt to Vishay	740S00027	740S0159		ALL	Bourns alt to Little Fuse	107S0249	107S0251		ALL		107S00015	107S00011		ALL		107S00071	107S00053		ALL		128S0364	128S0264		ALL	Kemet w/ Panasonic	128S0325	128S0397		ALL		128S00009	128S00007		ALL		128S00029	128S00007		ALL		128S00070	128S00007		ALL		128S00010	128S00011		ALL		128S00031	128S00011		ALL		128S00026	128S00011		ALL		132S00064	132S0409		ALL		138S0614	138S0578		ALL		138S0703	138S0648		ALL		138S00032	138S0831		ALL		138S00049	138S0831		ALL		138S0863	138S0853		ALL		138S0775	138S0860		ALL		138S00084	138S00060		ALL		152S00369	152800268		ALL	Cyntec w/ NEC	155S00188	155S0275		ALL	Murata w/ Taiyo	155S0694	155S0387		ALL		155S0660	155S0513		ALL		155S00018	155S0664		ALL	Murata w/ Taiyo	155S00007	155S0667		ALL		BLC	<table><tr><th>PART NUMBER</th><th>ALTERNATE FOR PART NUMBER</th><th>BOM OPTION</th><th>REF DES</th><th>COMMENTS:</th></tr><tr><td>197800046</td><td>197800036</td><td></td><td>ALL</td><td>Epson w/ TXC</td></tr><tr><td>197800047</td><td>197800036</td><td></td><td>ALL</td><td>Kyocera w/ TXC</td></tr><tr><td>197800048</td><td>197800036</td><td></td><td>ALL</td><td>Murata w/ TXC</td></tr><tr><td>197800053</td><td>197800050</td><td></td><td>ALL</td><td>Kyocera w/ TXC</td></tr><tr><td>197800054</td><td>197800050</td><td></td><td>ALL</td><td>NDK w/ TXC</td></tr><tr><td>197800055</td><td>197800050</td><td></td><td>ALL</td><td>Murata w/ TXC</td></tr><tr><td>311S0596</td><td>311S0593</td><td></td><td>ALL</td><td>NXP w/ Diodes</td></tr><tr><td>107S0276</td><td>107800020</td><td></td><td>ALL</td><td>Cyntec w/ TFT</td></tr><tr><td>107800021</td><td>107S0284</td><td></td><td>ALL</td><td>TFT w/ Yageo</td></tr><tr><td>152800343</td><td>152S1682</td><td></td><td>ALL</td><td>NXP w/ Diodes</td></tr><tr><td>107S00087</td><td>107800029</td><td></td><td>ALL</td><td>TFT w/ Yageo</td></tr><tr><td>128S00058</td><td>128S00018</td><td></td><td>ALL</td><td>NEC w/ Rohm</td></tr><tr><td>138S0706</td><td>138S0739</td><td></td><td>ALL</td><td>NEC w/ Vishay</td></tr><tr><td>138S0945</td><td>138S0739</td><td></td><td>ALL</td><td>NEC w/ Rohm</td></tr><tr><td>152S00358</td><td>152800208</td><td></td><td>ALL</td><td>Murata w/ Chillisin</td></tr><tr><td>152S00400</td><td>152800361</td><td></td><td>ALL</td><td>Murata w/ Cyntec</td></tr><tr><td>152S1872</td><td>152800361</td><td></td><td>ALL</td><td>Murata w/ Cyntec</td></tr><tr><td>155S00034</td><td>155S0706</td><td></td><td>ALL</td><td>Taiyo w/ Murata</td></tr></table>	PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:	197800046	197800036		ALL	Epson w/ TXC	197800047	197800036		ALL	Kyocera w/ TXC	197800048	197800036		ALL	Murata w/ TXC	197800053	197800050		ALL	Kyocera w/ TXC	197800054	197800050		ALL	NDK w/ TXC	197800055	197800050		ALL	Murata w/ TXC	311S0596	311S0593		ALL	NXP w/ Diodes	107S0276	107800020		ALL	Cyntec w/ TFT	107800021	107S0284		ALL	TFT w/ Yageo	152800343	152S1682		ALL	NXP w/ Diodes	107S00087	107800029		ALL	TFT w/ Yageo	128S00058	128S00018		ALL	NEC w/ Rohm	138S0706	138S0739		ALL	NEC w/ Vishay	138S0945	138S0739		ALL	NEC w/ Rohm	152S00358	152800208		ALL	Murata w/ Chillisin	152S00400	152800361		ALL	Murata w/ Cyntec	152S1872	152800361		ALL	Murata w/ Cyntec	155S00034	155S0706		ALL	Taiyo w/ Murata	<table><tr><td>353800711</td><td>353S2073</td><td></td><td>ALL</td><td>On Semi w/ TI</td></tr></table>	353800711	353S2073		ALL	On Semi w/ TI	<table><tr><td>740S00019</td><td>740S00007</td><td></td><td>ALL</td><td>Bourns w/ Polytronics</td></tr><tr><td>155S00189</td><td>155S0342</td><td></td><td>ALL</td><td>Murata w/ Taiyo</td></tr><tr><td>138S0714</td><td>138S0713</td><td></td><td>ALL</td><td>Murata w/ Samsung</td></tr><tr><td>138S0715</td><td>138S0732</td><td></td><td>ALL</td><td>Murata w/ Samsung</td></tr><tr><td>107S00086</td><td>107800056</td><td></td><td>ALL</td><td>TFT w/ Cyntec</td></tr><tr><td>138S0875</td><td>138S0678</td><td></td><td>ALL</td><td>Taiyo w/ Mur&SS</td></tr><tr><td>138S0786</td><td>138S0705</td><td></td><td>ALL</td><td>Murata w/ Samsung</td></tr><tr><td>152S2052</td><td>152S1954</td><td></td><td>ALL</td><td>Taiyo w/ Cyntec</td></tr><tr><td>152S2015</td><td>152S1958</td><td></td><td>ALL</td><td>Taiyo w/ Cyntec</td></tr><tr><td>138S0789</td><td>138S0941</td><td></td><td>ALL</td><td>Murata w/ SS</td></tr><tr><td>107S00101</td><td>107800005</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S00102</td><td>107800017</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S00100</td><td>107800057</td><td></td><td>ALL</td><td>Cyntec w/ TFT</td></tr><tr><td>107S00103</td><td>107800058</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S00104</td><td>107800061</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>107S00105</td><td>107800062</td><td></td><td>ALL</td><td>Cyntec w/ Yageo</td></tr><tr><td>152S00403</td><td>152800322</td><td></td><td>ALL</td><td>Murata w/ Chillisin</td></tr></table>	740S00019	740S00007		ALL	Bourns w/ Polytronics	155S00189	155S0342		ALL	Murata w/ Taiyo	138S0714	138S0713		ALL	Murata w/ Samsung	138S0715	138S0732		ALL	Murata w/ Samsung	107S00086	107800056		ALL	TFT w/ Cyntec	138S0875	138S0678		ALL	Taiyo w/ Mur&SS	138S0786	138S0705		ALL	Murata w/ Samsung	152S2052	152S1954		ALL	Taiyo w/ Cyntec	152S2015	152S1958		ALL	Taiyo w/ Cyntec	138S0789	138S0941		ALL	Murata w/ SS	107S00101	107800005		ALL	Cyntec w/ Yageo	107S00102	107800017		ALL	Cyntec w/ Yageo	107S00100	107800057		ALL	Cyntec w/ TFT	107S00103	107800058		ALL	Cyntec w/ Yageo	107S00104	107800061		ALL	Cyntec w/ Yageo	107S00105	107800062		ALL	Cyntec w/ Yageo	152S00403	152800322		ALL	Murata w/ Chillisin	<table><tr><td>138S00104</td><td>138S0978</td><td></td><td>ALL</td><td>Murata w/ Taiyo</td></tr><tr><td>311S00072</td><td>311S0657</td><td></td><td>ALL</td><td>NXP w/ On Semi</td></tr><tr><td>311S00090</td><td>311S00028</td><td></td><td>ALL</td><td>On Semi w/ TI</td></tr><tr><td>353S00854</td><td>353S4342</td><td></td><td>ALL</td><td>TI w/ ST</td></tr><tr><td>377S0077</td><td>377S0183</td><td></td><td>ALL</td><td>Infineon w/ ST</td></tr></table>	138S00104	138S0978		ALL	Murata w/ Taiyo	311S00072	311S0657		ALL	NXP w/ On Semi	311S00090	311S00028		ALL	On Semi w/ TI	353S00854	353S4342		ALL	TI w/ ST	377S0077	377S0183		ALL	Infineon w/ ST	<table><tr><td>138S0700</td><td>138S0641</td><td></td><td>ALL</td><td>Murata w/ SS&Taiyo</td></tr><tr><td>152S00363</td><td>152S00048</td><td></td><td>ALL</td><td>Cyntec w/ Vishay</td></tr><tr><td>155S0659</td><td>155S0382</td><td></td><td>ALL</td><td>Murata w/ TDK</td></tr><tr><td>376S00146</td><td>376S1061</td><td></td><td>ALL</td><td>NXP w/ Diodes</td></tr></table>	138S0700	138S0641		ALL	Murata w/ SS&Taiyo	152S00363	152S00048		ALL	Cyntec w/ Vishay	155S0659	155S0382		ALL	Murata w/ TDK	376S00146	376S1061		ALL	NXP w/ Diodes	<table><tr><td>128S00062</td><td>128S00067</td><td></td><td>ALL</td><td></td></tr><tr><td>128S00069</td><td>128S00067</td><td></td><td>ALL</td><td></td></tr><tr><td>132S00012</td><td>132S0401</td><td></td><td>ALL</td><td></td></tr><tr><td>138S0660</td><td>138S0684</td><td></td><td>ALL</td><td></td></tr><tr><td>138S1103</td><td>138S0719</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00097</td><td>138S0750</td><td></td><td>ALL</td><td></td></tr><tr><td>138S00111</td><td>138S00036</td><td></td><td>ALL</td><td></td></tr><tr><td>155S00203</td><td>155S0894</td><td></td><td>ALL</td><td></td></tr><tr><td>197S00082</td><td>197S00081</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00105</td><td>311S0233</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00060</td><td>311S0273</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00118</td><td>311S0489</td><td></td><td>ALL</td><td></td></tr><tr><td>311S0271</td><td>311S00008</td><td></td><td>ALL</td><td></td></tr><tr><td>311S00104</td><td>311S00091</td><td></td><td>ALL</td><td></td></tr><tr><td>311S0437</td><td>311S00112</td><td></td><td>ALL</td><td></td></tr><tr><td>335S00213</td><td>335S0888</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00880</td><td>353S3452</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00878</td><td>353S00599</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00879</td><td>353S00754</td><td></td><td>ALL</td><td></td></tr><tr><td>353S00750</td><td>353S00877</td><td></td><td>ALL</td><td></td></tr><tr><td>371S00089</td><td>371S00085</td><td></td><td>ALL</td><td></td></tr><tr><td>377S0178</td><td>377S00031</td><td></td><td>ALL</td><td></td></tr></table>	128S00062	128S00067		ALL		128S00069	128S00067		ALL		132S00012	132S0401		ALL		138S0660	138S0684		ALL		138S1103	138S0719		ALL		138S00097	138S0750		ALL		138S00111	138S00036		ALL		155S00203	155S0894		ALL		197S00082	197S00081		ALL		311S00105	311S0233		ALL		311S00060	311S0273		ALL		311S00118	311S0489		ALL		311S0271	311S00008		ALL		311S00104	311S00091		ALL		311S0437	311S00112		ALL		335S00213	335S0888		ALL		353S00880	353S3452		ALL		353S00878	353S00599		ALL		353S00879	353S00754		ALL		353S00750	353S00877		ALL		371S00089	371S00085		ALL		377S0178	377S00031		ALL	
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145 OF 145

119 OF 119